THE DURGAPUR PROJECTS LIMITED

NOTICE INVITING TENDER (NIT)

e-Procurement

NIT No: WBDPL/PP/T/PP-435/2021/21-22/E-163 , Dated:-22/12/2021

National Competitive Bidding
For

E-tender cum Reverse Auction for "Design & Engineering, Manufacture/ Procurement, Supply, Erection, Testing and Commissioning of 5 MW Phase-II Grid Connected Ground Mounted Solar Photovoltaic Power Plants at Durgapur Projects Limited(DPL), Durgapur, West Bengal on Turnkey basis with 05 (Five) years comprehensive Operation and Maintenance".

THE DURGAPUR PROJECTS LIMITED
(A Govt. of West Bengal Enterprise)
CIN: "U40102WB1961SGC025250"

Dr Bidhan Chandra Roy Avenue, Industrial Area, Durgapur, West Bengal-713201

SECTION – I

Instruction to Bidder (ITB)

BID INFORMATION SHEET

S. No	Aspect	:	Description of Aspect
1.	**Title of the NIT**	:	**E-tender cum Reverse Auction** for "Design & Engineering, Manufacture/Procurement, Supply, Erection, Testing and Commissioning of 5 MW Phase-II Grid Connected Ground Mounted Solar Photovoltaic Power Plants at Durgapur Projects Limited(DPL), Durgapur, West Bengal on Turnkey basis with 05 (Five) years comprehensive Operation and Maintenance".
2.	**NIT NO. & Date**	:	**WBDPL/PP/T/PP-435/2021/21-22/E-163 dated 22/12/2021**
3.	Publishing Date		22/12/2021
4.	Document Download start date		23/12/2021
5.	Pre-bid queries submission end date		15/01/2022
6.	Pre-Bid Meeting		18/01/2022 ; 11:30 Hrs. If any party wants to attend the meeting via online, please click on the following link:- https://meet.google.com/idd-rgji-kvw
7.	Bid submission start date		21/01/2022

S. No	Aspect	:	Description of Aspect
8.	Bid submission end date		05/02/2022
9.	Submission of hard copy of Technical and Financial QR document, Statutory documents, balance sheet etc along with e-challan receipt of online submission of EMD		Upto 08-02-2022; 16:00 Hrs.
10.	**Technical Bid opening date**		09/02/2022; 13:00 Hrs
11.	Uploading of Technical Bid Evaluation sheet		To be notified through system generated message
12.	Financial Bid opening date		To be notified through system generated message
13.	Uploading of Financial Bid evaluation sheet		To be notified through system generated message
14.	**Scope of Work**	:	Clause No.ITB1.3
15.	**Qualifying Requirement**		**Technical:** **Job Experience :** a) At least **one (01) number 5(Five) MW** capacity or **two(2) number Three(3)MW** Capacity or higher capacity Solar PV project installation at a single location during preceding 07 (Seven) years. b) Minimum **01 (one)** number **5(Five) MW or two(2) number Three(3)MW** or **higher** capacity Solar PV Power

S. No	Aspect	:	Description of Aspect
			Plant Operation and Maintenance (O&M)
			Financial : **MAAT : Last 3 FY shall be Rs. 7.59 Cr** **1 Order : INR 20.26 Crores** **2 Order: INR 12.67 Crores** **3 Order : INR 10.13 Crores**
16.	**Cost of the Tender**	:	**Not applicable**
17.	**Bid Security/ EMD**	:	**Rs. 50,66,000/- (Rupees Fifty Lakhs Sixty six thousand only) to be submitted via online**
18.	**Name, Designation, Address and other details**	:	Partha De, SM (Contract Cell) Durgapur Projects Limited
19.	**For any Bid quarry contact Person**		1. Partha De, SM (Contract Cell) , 9434713057,email: contractcelldpl@gmail.com 2. Sudipto Pal,DGM(Projects) 9434735849,email:sudiptopal1971@gmail.com
20.	**Date of Commencement**	:	From the date of issuance of Letter Of Award (LOA).
21.	**Time for Completion**	:	**12 (Twelve) Months from date of LOA**
22.	**Possession of Site**	:	Immediately on receipt of LOA from Purchaser (DPL).
23.	**Performance guarantee**	:	i. 10% of the Project Cost **plus GST** in the form of bank guarantee valid for a period of Seventy Two (72) months with further claim period for ninety (90) days thereafter and as per clause from 3.14.2 & 4.2 of general instructions from Scheduled bank

S. No	Aspect	:	Description of Aspect
			as per DPL approved format. ii. Should be submitted within 21 days from the date of LOA.(BG Format **Annexure-2**)
24.	**Defects Liability Period each contract (**	:	Sixty (60) calendar months up to successful completion of Comprehensive O&M job from the date of issue of final completion certificate of first contract of each project (**Annexure-6**).
25.	**Minimum value of work for each RA Bill of each contract**	:	Rs.20.00 Lakhs. (Rupees Twenty lakhs only)
26.	**Performance Warranty**	:	5% of the **Project Cost plus GST** will be retained by the Purchaser for the performance warranty and the same will be paid in 1% yearly after successful completion of the work i.e. up to defect liability period.
27.	**Integrated project performance of Net Minimum Guaranteed Generation (NMGG)**	:	Integrated project performance of minimum solar energy generation at the rate of 1.6 MU annually per MW AC with degradation of 0.7% for any reason, from second year onwards.
28.	**Liquidated damages for Delay (LD Clause)**	:	a. **Time Delay:** 0.5 % of Project Cost plus GST for per week delay or part there of subject to a maximum of 10% of the Project Cost **plus GST**. b. **Milestone Delay:** Delay in attaining the milestones by the contractor shall lead to

S. No	Aspect	:	Description of Aspect
		:	imposition of intermediary Liquidated damages @0.25% per week of delay upto the maximum extent of 5(Five) Percent of the Project Cost **plus GST**. c. **Total LD value (LD for Time Delay + LD for Milestone Delay) shall not exceed 10% of total Project Cost plus GST.**
29.	**Performance Penalty**	:	Penalty shall be levied for performance shortfall on PG test. Maximum penalty for failure of PG test shall be 5% of Project cost plus GST.
30.	**Statutory Taxes**	:	**Bidder shall submit the base price excluding GST:** i) GST will be reimbursed at actual by the DPL to the bidder on submission of appropriate supporting documents. ii) In case of any change in custom duty, entry tax. etc. during the currency of the contract, the same shall be borne by the bidder. No reimbursement shall be allowed.
31.	**Operation and Maintenance Cost (Including all Consumables)**	:	**Operation and Maintenance Cost (Including all Consumables and spares for Routine and preventive maintenance, Break down maintenance, capital maintenance)**: **7.5 %** of the Project cost (excluding Taxes and Duties) for 5(five) years. Clause No. GCC 3.8.4
32.	**Insurance**	:	Insurance of supply, erection work and workmen including third party insurance of each equipment are applicable and to be borne by the Contractor until

S. No	Aspect	:	Description of Aspect
			final handover of the project upto defect liability period including successful completion of O&M period. Contractor will be responsible for maintaining the Insurance Policy for the complete Plant and Facilities during the O&M period also.
33.	**Mode of Tendering**	:	**E-tendering cum Reverse Auction**
34.	**Payment Terms**		**Supply**: 70% after delivery (Prorata basis, Mobilization Advance paid as per Clause 4.04.02 will be recovered proportionately from bills under this phase of payment) 15% after Erection (Prorata basis) 5% after commissioning (Prorata basis) 5% after PG Test 5% will be retained for performance securities **Service:** 65% of the contract value will be given pro rata basis after erection of the material (Mobilization Advance paid as per Clause 4.04.02 will be recovered proportionately from bills under this phase of payment). 20% after Testing & Commissioning 5% after PG Test 5% after successful completion 5% will be retained for performance securities **Mandatory Spares:** 70% of the Supply price production of MDCC 30% of the Supply price production of MRC

S. No	Aspect	:	Description of Aspect

Note : GST rate shall be considered at actual according to directive during actual execution time

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INSTRUCTION TO THE BIDDER (ITB)

A. Scope of Work & Qualifying Requirements

1.1.Project particulars

Particulars	**Description**
Design & Engineering	
AC capacity of the solar power plant	5 MW AC
Technology	(Si Mono crystalline)
O&M Period	05 Years
Estimated life of PV Power plant	25 years
Area available for Plant	**17.6 Acres** (COGP Area of DPL)
Location/Site Details	
Location	The Durgapur Projects Limited premises
Type of Land	Government Land
Ownership	The Durgapur Projects Limited
District	Paschim Bardhaman
State	West Bengal
Electrical Interconnection details	
Power Evacuation & Nearest Substation details	The inverter outputs are transformed at 6.6kV. 6.6 kV indoor type switchgear to be installed in the Power plant control room. It will have two number of outgoing feeder; 6.6kV 400Ams, 3C Al Armoured cable will be used for these feeders and it will be connected with the existing 6.6kV bus of Solar PV Plant Switchgear room (beside Administrative building) aprox2.2 Km away from site
Performance Parameters	
Minimum values of Performance Ratio(PR) and CUF of the plant after netting off the auxiliary consumption.	PR : 80% CUF : 18.2%

Other Details	
Water and Power for Construction	To be arranged by the Contractor, however tapping will be provided within 100 metrs from site boundary by the Owner.

1.1.2. Name of Work : *E-tender cum Reverse Auction* for "Design & Engineering, Manufacture/Procurement, Supply, Erection, Testing and Commissioning of 5 MW Phase-II Grid Connected Ground Mounted Solar Photovoltaic Power Plants at Durgapur Projects Limited(DPL), Durgapur, West Bengal on Turnkey basis with 05 (Five)years comprehensive Operation and Maintenance".

1.1.3 . Nature of Job : Design & Engineering, Manufacture , Supply, Erection, Testing and Commissioning of 5 MW AC Grid Connected Ground Mounted Solar Photovoltaic Power Plant at Durgapur Projects Limited , Durgapur, West Bengal including warrantee obligation with 05 (Five) years comprehensive Operation and Maintenance.

1.2. SOURCE OF FUND

The Durgapur projects Ltd (hereinafter referred to as **DPL** or "Purchaser") intends to finance the Work covered under these Bidding Documents from the Source as mentioned in the Bid Data Sheet (BDS).

1.3. SCOPE OF WORK:

The brief scope of work covered under this Tender shall be included but not limited to the following:-

1.3.1. The scope of work for each project shall be on the basis of single source responsibility, completely covering all the Equipment/Material specified under the **Technical Specifications**. The work is to be executed on turnkey basis. The Purchaser will not supply any material departmentally. It shall include the following:

a. Detailed Site Survey of the proposed solar plant area of DPL for Designing and Engineering.

b. Submission of Detailed Design Report indicating technical suitability of site for installation of the Power Plant with layout plan.

c. Detail calculation of Solar Energy generation (MWp and MW ac) and selection of Module considering Net Minimum Guaranteed Generation (NMGG) stipulation for the first five year as well as 25 years of life.

d. Detailed Design of the Equipment/ Materials and Submission of Billing Breakup (BBU) with matching the project cost.

e. Obtaining approval of engineering drawing, technical data, operational manual etc. and necessary inspection from the Purchaser.

f. Complete manufacturing including shop testing.

g. Procurement, Packing, forwarding, transportation and insurance of Equipment/ Material from the manufacturer's works to the Site.

h. Material Supply, Receipt, storage, preservation, insurance and conservation of Equipment/ Material at the Site.

i. Grading of site to even out the surface and provide a solid foundation , clearing of vegetation of the Site .

j. Design and assemble of Module mounting structure (MMS). Detail GA and data sheet of each type of MMS with its fixing arrangement to be provided by the vendor before job execution.

k. Design calculation of total generation including NMGG requirement for next five years shall be substantiated with latest version of PV system for approval with maximum accurate weather data **like latest version of Solargis.**

l. Providing power supply for construction purposes.

m. Construction of RCC/PEB type Inverter room with Power conditioning unit and associated RCC type LT and HT switchgear cum control room.

n. Construction of Equipment for switchgear room, **SCADA room**, store room, battery & Battery Charger room with all electrical fitting and Control room with Central Monitoring and Control Station, security cabin etc.

o. Installation and commissioning of equipment as per technical design.

p. All associated electrical and civil works required for interfacing with grid i.e. transformer(s), breakers, isolators, panels, protection system, cables. metering, earthing etc.

q. Power evacuation : The solar modules produce Direct Current (DC), which would be converted to Alternating Current (AC) through the use of inverters. The inverters used as a part of the Solar power plant is envisaged to produce current at 415V,which would be further stepped up to 6.6kV for integration with the 6.6kv bus.

The DC from modules is converted to AC by inverters. The inverter outputs are fed into a junction box, which in turn is connected to the common 415 V AC bus bar and further transformed at 6.6kV. 6.6 kV indoor type switchgear to be installed in the Power plant control room. The 5 MW power would be evacuated through two number 6.6kV 400Ams,Armoured underground XLPE cable from this indoor switchgear; and it will be connected with the 6.6kV bus of Solar PV Plant (Located beside Administrative building) which is aprox 2000m from the plant site. The LV and the HT panels will have all necessary metering and protection.

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r. Water supply arrangement for Control Building, cleaning water for PV modules etc.

s. Preparation of Land, Grading of Land ,covering the land with CLSM.

t. Construction of roads, walkways and drainage system of Control Building.

u. Modules should be so mounted and foundation structure should be such that it can withstand wind-speed of 180 kmph as well as seismic effects

v. SCADA system for remote monitoring and control of SPV panels with all hardware & software.

w. Low lying area below HFL(High flood level) to be converted into Reservoirs with garlanding for supplying cleaning water of the PV modules

x. Boundary wall of the site

y. Till the successful completion of O&M activities of the plant and handing over the same to the Purchaser, the necessary security arrangement of all the materials and equipment will be the sole responsibility of the Contractor from the date of site hand over.

z. Final check-up of equipment, installation, and commissioning of power plant and putting the system into successful functional operation.

aa. Reliability tests, performance and guarantee tests, wherever applicable, on completion of commissioning.

bb. Insurance of all the Equipment/ Materials.

cc. Supply of Mandatory Spares.

dd. O&M for 5 years

ee. Providing training up to the satisfaction of the purchaser operating personnel at Manufacturers works, Operating Solar power stations at site.

ff. Other Services:

a. Providing training material to the end users during onsite training for end users.
b. Preparing commissioning certificate and documentation as per MNRE, GoI.
c. Handing over of power plant.
d. Operation & maintenance manpower of SPV Plant along with electrical equipment, consumables and spare parts for a period of five years from the date of successful completion of trial run and completion of all facilities.
e. Providing of routine and break down maintenance of grid connected ground solar PV power plants during comprehensive maintenance period.
f. Fulfilment of guarantee obligation.

Note: All the engineering drawing, documents, design, sizing calculation, layout etc. shall be submitted for approval from DPL or Consultant of DPL.

1.3.2. SUPPLY:

a) The Supply scope includes the following but not limited to:

Manufacture , Supply and delivery of all the materials like PV Module **(Solar Monofacial/Bifacial crystalline modules having capacity minimum 500 Wp and MNRE approved) with minimum 15% DC margin**, Module Mounting Structure, Grid Connected Inverters, Station Aux Transformer, Inverter Voltage

/6.6 KV transformer, String Monitoring Boxes, HT panel, LT Panel, other LT boards, Web based monitoring systems & SCADA, Weather Monitoring Systems, PLC/ DCS, Cables, System protection and other accessories conforming to the Technical Specification as required for successful Installations & commissioning of 5 MW AC Grid Connected Ground Mounted Solar PV Power Plants on turnkey basis at DPL.

b) Mandatory Spares as per list Volume V.

1.3.3. SURVICE, ERECTION & OTHER SEVICES:

The Erection and Commissioning scope includes the following but not limited to:

Survey, area grading, Installation, Testing and Commissioning, PG test etc. Including completion of all facilities of 5 MW AC Grid connected Ground Mounted Solar Photovoltaic Power Plants at DPL, Durgapur, West Bengal, India in turnkey basis including warrantee obligation with 05 (Five) years comprehensive Operation and Maintenance.

1.3.4. OPERATION AND MAINTENANCE:

Five (5) years comprehensive Operation & Maintenance(O&M) of the Solar PV Plant including power evacuation along with electrical equipment, consumables and spare parts along with regular module cleaning from the date of successful completion of trial run and completion of all facilities of each project. **PG test shall be completed within six months from the Start date of O&M Contract**

1.3.5. PROJECT SCHEDULE:

Zero date: Date of issue of LOA

Total time for competition: **365** days from zero date from each project.

1.3.6. **MODE OF EXECUTION**

The entire work shall be executed on Lump sum turnkey basis. Any item(s) not included in the specification / schedule but required for completion of the work shall have to be carried out/supplied without any extra cost. Such works, not listed in the schedule of works but elaborately described to perform or to facilitate particular operation(s) required for completion of the project shall be deemed to have been included in the scope of this work and the bidder shall supply, install the same without any extra cost to DPL.

The work shall be executed in conformity with the relevant standard of Bureau of Indian Specification (or equivalent International Standard), Indian Electricity Rules, 1956 (as amended up to date), Indian Electricity Act 2003(as amended up to date), BARC/DAE rules, Explosive Act 1948, Petroleum Act 1934, National Building Code and relevant Rules in vogue at the time of execution including operation & maintenance period.

The bidder shall comply with all applicable laws or ordinances, codes, approved standards, rules, and regulations and shall procure all necessary Panchayat / Municipal and Government permits & licenses etc at his own cost.

All sub systems /components such as cables, connectors, Junction boxes, surge protection devices, etc., shall conform to the relevant international and national standards for electrical safety besides that for Quality required for ensuring Expected service life and Weather resistance.

The bidder to provide full time round the clock watches and ward to protect the material from theft and pilferage.

1.3.7.SITE INSPECTION

The bidder is advised to visit and examine project site and its surroundings and obtain for himself, on his own responsibility, all information that may be necessary for entering into contract. The bidder will assess and satisfy himself as to the adequacy of the local conditions such as approach roads to the site, adequacy of existing culverts/bridges/roads for the expected traffic, water

and power supply, nature of ground and sub soil condition, water table level, accommodations required during the contract, climatic conditions, local terrain, availability of labour and construction materials, details of taxes, duties and levies as applicable and any other information required. The cost of visiting the site shall be at the bidder’s own expenses.

1.3.8.FACILITIES AT SITE

The Bidder will be provided with maximum 100 A capacity 415V, 3 Phase, 50 HZ power sources at one point near DPL Administrative Building boundary for construction and fabrication purpose but cabling to be done by the Bidder, Control, metering and distribution inside the solar power plant area shall be made by the bidder at his own cost. Cost of the construction power will be charged basis with the rate of WBSEDCL. Bidder shall also ready with sufficient number alternative power supply arrangements like DG set etc.

Water supply shall be provided to the contractor at 500 mtr. Distance from site for construction and maintenance purpose. The Bidder shall arrange for pumps and distribution piping to various locations within the solar power plant depends on the requirement.

1.3.9. System Description

1.3.9.1. LAYOUT:

a) **Capacity of the Plant**: 5000 kW (AC)

b) **Area of the Land:**

Approximate Land area and expected estimated power plant capacity:-

Land area	Capacity (MW) AC
17.6 Acre approx	5
Location : Area of under demolition Coke Oven Group of Plants (COGP) , DPL	

c) **Tentative topography of the Site:**

Please check NIT drawing

<u>Physical demarcation between Solar Plant and remaining DPL Premises:</u>

Physical demarcation by erecting precast pole (1200 mm x 200mm x 200 mm). Height of each pole shall be 1000 mm high from FGL. Distance between two poles shall be maximum 15 mtrs. This is in the scope of the Bidder.

d) **INVERTERS/POWER CONDITIONING UNIT (PCU)**

The DC power shall be converted to AC by PCU to supply AC loads. Sizing of Inverter of the Plant will be decided during designing & engineering.

e) **SWITCHGEAR CUM CONTROL ROOM**

Total 5 MW Solar plant to be installed on the area mentioned as per the clause no.1.3.9.1. (b) above. This switchgear cum **control room shall be constructed at a suitable location to be decided during designing & engineering.**

This building shall be two parts, one part for Equipment Room for housing Switchgear, electrical equipment/panels and other part for Control Room for Control operation. The control room (must) and equipment room (if required) be provided with AC environment for the operation & maintenance of Solar Plant. Capacity of the AC system shall be as per design of the room. SCADA panel, UPS and UPS DB etc. shall be placed inside the Control room.

Dry Type inverter transformers (400 V/6.6kV) to be placed near to the Inverter room. However LV side of Transformer voltage may be varied according to the Inverter AC voltage.

In Equipment Room at least the following electrical equipment shall be present-

i. HT & LT switchgear

i. AC Distribution Panel

ii. Inverters

iv. 220 V DC Battery and Battery Charger, DCDB;

v. UPS Battery and

vi. Other equipment which are required for better operation of the Plant.

Equipment Room to be designed in such a way that HT/LT Switchgear could accommodate the future extension.

Space provision for total two numbers equivalent extension panel for future project to be considered on both the sides of 6.6 kV HT panel.

LT switchgear shall have atleast 2.5 mtr. additional space to be kept, apart from its original size for future use.

In the Control Room at least the following electrical equipment should be present-

i. Computer table and chair with capacity of at last three persons.

ii. One Meeting room.

iii. Engineering Work Station

iv. UPS

v. DCS / PLC

vi. Fire Alarm Panel

vii. Others equipment which requires AC environment

f) **GRID CONNECTIVITY**:

- Dry Type inverter transformers (400 V/6.6kV) to be Used. HV side volatage should be 6.6 KV , LV side of Transformer voltage may be varied according to the Inverter AC voltage.
- The HV side of the Inverter Transformers should terminate at 6.6 KV BUS through Breakers , protection switchgears etc , located at the Ground floor of the Control room building constructed for this purpose.
- Space provision for total two numbers equivalent extension panel for future project to be considered on both the sides of 6.6 kV HT panel.
- **From the 6.6 KV Site Bus , entire power to be transferred to 6.6 KV Bus of Solar PV Plant Switchgear room (Beside Administrative building) of the Owner, located aprox 2000 metres from the site.**
- Three run 3 Core 400 sq mm Armoured aluminum Power cable to be layed for this purpose . The cable should be layed underground through route arrived upon by mutual discussion during actual engineering. The scope work of cable laying should be approved by DPL.

- Fitting of 6.6 KV Breakers including supply of CT and relays etc at both end (4 Circuit breakers) need to be carried out by the Bidder. Scope of service of retrofitting job of breaker should be approved by DPL. Necessary modification of Bus Bar including change of Bus Separator and post-insulator are also in the scope of bidder, which should be approved by DPL.

g) **THREE WINDING TRANSFORMER**

Three winding Dry type step up transformer (0.415kV/0.415kV/6.6kV) for this Ground Solar shall be connected to the outputs of two inverters as input and the 6.6kV HV output sides will be connected to the 7 MW Solar PV Plant (Beside Adm bldg) 6.6kV bus through Protection system, VCB etc.

h) **STATION AUXILIARY TRANSFORMER (SAT)**

Two numbers 6.6 kV/415V Dry type Station transformer having a minimum capacity of 250 kVA each has to be considered for light and other auxiliary purposes. HT side of the transformer shall be connected to the 6.6 kV Bus through a VCB and LT side shall be terminated to the Station Service Board (SSB)/ 415 V LT Switchgear. Successful bidder shall consider entire LT load under this Aux. Transformers during detail engineering.

i) **HT SWITCHGEAR**

Inside control room building, 6.6 kV switchgear should have outgoing and incoming feeders. Outgoing feeders should be connected with two Station Service Transformers for lighting and other auxiliary consumption and feeder is used for evacuation of power. Sufficient number of incoming other feeders shall be provided for incoming power from Inverter Transformer . One feeder for Inverter transformer another feeder for Inverter Transformer shall be kept as spare with all VCB and protection system equivalent.

Power transmission/evacuation capacity through Feeder should be minimum 5 MW AC plus 10% higher margin, so bidder has to design the system accordingly. Outgoing evacuation cable should have 5 MW plus 10% design margin.

Spare feeder should be equipped with all existing facility like protection, control, and indication etc. of active breaker.

j) **LT SWITCHGEAR:**

One number indoor type 415V LT switchgear has been considered for supply of 415 V (3 phases, 1 neutral and single phase for lighting etc). The said LT bus has to be made with proper protection system with two Incomers and one Bus-coupler philosophy. Incomer and Bus-coupler shall be Air Circuit Breaker of similar rating. Two spare out going feeders of capacity 200A MCCB also to be considered under this LT Board from both bus-sections. Control Voltage 220 V DC.

Bidder shall consider following Air Circuit Breaker feeders:

Sl. No	Description	Quantity
1.	Incomer feeder ACB	Two (02 nos.) (630 A capacity each)
2.	Bus coupler ACB	One (01 no.) (630 A capacity)
3.	Spare MCCB	200 A capacity each, two (02 nos.)

Two Incomer and one Bus-coupler with a castle key interlock.

1.4 QUALIFYING REQUIREMENT FOR BIDDERS:

The Bidder shall meet the following minimum qualification for 5 MW grid connected Ground Solar PV Power Plant:

1.4.1GENERAL :

The Bidder, who intends to participate in the Bid, must have to meet the following criteria:

1.4.1.1. The Bidder shall be a – Firm/Company incorporated in India under Companies Act, 2013 /Indian Partnership act 1932/LLP act 2008.

1.4.1.2. Bidders shall have to submit:

a) Audited Balance Sheet and Statement of Profit and Loss Account of Last three (3) consecutive financial years for which the audited accounts are available

b) Acknowledgement of Income Tax Return for the last three Assessment Years

c) The above financial QR certificate must be from registered Chartered Accountant with seal and signature and membership number.

1.4.1.3. The bidder must have valid GST, PF Registration, Return cum Challan (latest available) for Provident fund, ESI registrations; and these are to be submitted along with the bid.

1.4.1.4. If the Bidder is **Bidding Consortium** then-

Bidding Consortium/ Joint Venture Bidders shall comply with the following requirements

i **Number of members in a Joint Venture shall not exceed 3 (three)**;

ii Subject to the provisions of clause (i) above, the Bid should contain the detail information required for each member of the Joint Venture, viz. Financial Capacity, Technical capacity etc of each member;

iii Members of the Joint Venture shall nominate one member as the lead member (the "Lead Member"). Lead Member shall meet at least 50% requirement of Financial Capacity and at least 30% of Technical Capacity. The nominated Lead member shall remain unchanged during the entire period of project execution including the Defects Liability Period. The nomination(s) shall be supported by a Power of Attorney, as per the format at (**Form- 12**) duly Signed by all the other Members of the Joint Venture. **Each of the other Member(s) shall meet at least 30% of the required Technical Capacity and 20% of the required Financial Capacity** iv The duties, responsibilities and powers of Lead Member shall be specifically included in the Joint Bidding Agreement or Memorandum of Understanding. The Lead Member shall be authorized to incur liabilities and to receive instructions and payments for and on behalf of the Consortium. The Lead Member should have entire responsibility pertaining to execution of the Project;

v All the Consortium Member should fulfill the criteria as per clause No.1.4.1.

vi The Bid should include a brief description of the roles and responsibilities of individual members, particularly with reference to financial, technical and defect liability obligations which will satisfy the sub-clause 1.4.1.4(iii) above;

vii All the members of the bidding consortium after the award and signing of the EPC Contract Agreement shall be obliged to continue to discharge their responsibility as the "members" of the consortium for a period covering the entire project completion period including defect liability plus Five (5) years of the Operation & Maintenance period of the project. This five (5) years period shall be deemed to be effective from the date of commencement of the O&M period this project.

viii **Conflict of interest**-An individual Bidder cannot at the same time be a member of a Consortium applying for the Project. Further, a member of a particular Bidder Consortium cannot be member of any other Bidder Consortium applying for the Project;

ix No Change in the composition of the Consortium will be allowed to be permitted by the Client during the Selection Process and during the subsistence of the Contract (in case the successful Bidder is a consortium).

x Members of the Consortium shall enter into a binding JV Bidding Agreement duly registered (herein after called as "JV/Consortium Agreement"), for the purpose of submitting a Bid. The registered JV/Consortium Agreement, to be submitted along with the Bid as per format Form-11 of the NIT.

xi The award of the contract will be conferred on the Lead Member only(refer clause-3.2.3)

1.4.2.TECHNICAL REQUIREMENT:

The Bidder, who intends to participate in the Bid, must have to meet the following criteria:

i. The Bidder shall have experience of satisfactorily execution of contracts in Planning, Designing, Supply, Installation, Testing & Commissioning of Grid connected Rooftop and/or Ground Mounted and/or Floating Solar PV Power plants at any organization /PSU/ Government Organization having capacity of at least **one (01) number 5 (Five) MW** capacity or higher capacity project at a single location or **two(2) number 3(Three) MW** Capacity during preceding 07 (Seven) years. This plant(s) should be in successful operation since their commissioning. A certificate to this effect issued by the concerned authority is to be submitted. The bidder shall furnish documentary evidences of satisfactory performances of the said solar power plants by way of submission of monthly generation data on annual basis along with performance certificates issued by the purchaser for minimum SIX(06) months.

ii. The bidder should submit a list of contracts of similar nature already executed and presently under execution giving details of client, completion time, scope and value of work.

iii. The Bidder should have minimum **01 (one)** number **5 (Five) MW** or **two(2) number 3 (Three) MW** Capacity **or higher** capacity Solar PV Power Plant Operation and Maintenance (O&M) experience which is in operational in India or abroad for last one year ending at last date of Bid submission.

iv. The Bidder shall submit a latest O&M certificate issued by Project Developer of such plant.

v. If the Bidder is Bidding Consortium then the combined technical capability of those Members in such consortium should satisfy the clause 1.4.1.4 (iii) conditions of eligibility.

1.4.3.FINANCIAL REQUIREMENT:

The Bidder, who intends to participate in the Bid, must have to meet the following criteria:

i. **Minimum Average Annual Turnover (MAAT) of the Bidder during the last 3 (three) financial years ending 31st March of the previous financial year shall be Rs. 7.59 Cr.**

ii. **Minimum one Order of similar job of INR 20.26 Crores / two Order of similar job of INR 12.67 Crores each / three Order of similar job of INR 10.13 Crores each during last seven years**

iii. If the Bidder is bidding Consortium then the combined MAAT of those Members in such consortium should satisfy the above conditions of eligibility.

iv. Net worth as per the financial statement of the last financial year should not be less than **100%** of the paid share capital of the bidder.

v. If the Bidder is Bidding Consortium then combined Net Worth of the both partner should not be less than **100% of** the paid share capital of the bidder.

vi. **"Net worth"** should be positive for the last three years

1.4.4. OTHER QUALIFICATION REQUIREMENT

1.4.4.1. Bidder shall have adequate design, manufacturing and/or fabrication capability and capacity available to perform the work properly and expeditiously within the time period specified. The evidence shall specifically cover, with written details, the installed manufacturing and/or fabrication capacities and present commitments (excluding those anticipated under these bidding documents). If the present commitments are such that the installed capacity results in an inadequacy of manufacturing and/or fabrication capacities to meet the requirements appropriate to the works covered in his bid, then the details of alternative arrangements to be organized by the bidder and/or his collaborator/associate for this purpose and which shall meet the Purchaser's approval, shall also be furnished.

1.4.4.2. Bidder shall have an adequate project management organization covering the areas related to engineering of Equipment/Materials, interface engineering, procurement of equipment and the necessary field services required for successful construction, testing and commissioning of all the Works covered in the scope of work for this package and as required by the bidding documents

1.4.4.3.Bidder shall have established quality assurance systems and organization designed to achieve high levels of system reliability, both during his manufacturing and/or fabrication and field installation activities

1.4.4.4. Notwithstanding anything stated herein, DPL reserves the right to inquire and review the bidder's capability and capacity to perform the work at the time of evaluation.

1.5.RESPONSIBILITY OF BIDDERS

1.5.1. The DPL will not assume any responsibility regarding information gathered, interpretations or conclusions made by the bidder or regarding information, interpretations or deductions the bidder may derive from the data furnished by the DPL. Verbal agreement or conversation with any employee of the DPL either before or after the submission of bid shall not affect or modify any of the terms or obligations contained herein.

1.5.2.It shall be the sole responsibility of bidders to determine and to satisfy themselves by such means as they consider necessary or desirable as to all matters pertaining to this bidding process including in particular all factors that may affect the cost, duration and execution of the work.

1.5.3. It must be understood and agreed by the bidders that factors which may affect the cost, duration and execution of the works have properly been investigated and considered while submitting the bid. Claims whatsoever including those for financial adjustment in the price of the Contract awarded in accordance with these bidding documents will not be entertained by the Purchaser. Neither any change in time schedule of Contract nor any financial adjustments arising thereof shall be permitted by the Purchaser, which are based on the lack of investigation or its effect on the cost of the Contract to the bidder.

1.5.4. If the Bidder did not execute Electrical Sub-station of 6.6 kV or above voltage level then the Bidder have to engage DPL approved vendor for 6.6 kV Substation work.

1.5.5. Notwithstanding anything stated herein, DPL reserves the right to inquire and review the bidder's capability and capacity to perform the work at the time of evaluation.

1.6 COST/**EXPENDITURE** OF BIDDING

The bidder shall bear all costs associated with the preparation and submission of his bid and DPL in no case shall be responsible or liable for these costs, regardless of the conduct or outcome of the bidding process.

B. THE BIDDING DOCUMENTS

1.7. CONTENTS OF BIDDING DOCUMENTS

The scope of work, bidding procedures, Contract terms and conditions and technical specifications are prescribed in the bidding documents. The set of bidding documents uploaded for the purpose of bidding includes the sections stated below together with any addendum/amendment (Clause No. **1.11)** to be issued**.**

Section I	:	Instructions To Bidders (**ITB**)
Section II	:	Bid Data Sheet (**BDS**)
Section III	:	General Conditions of Contract (**GCC**)
Section IV	:	Special Conditions of Contract (**SCC**)
Section V		Technical Specification
Section VI	:	**Form**

		1	Check List
		2	Forwarding Letter for submission of Bid Security
		3	Bid Form/Undertaking
		4	Bid Security (Bank Guarantee format)
		5	Summary Statement of Yearly Turnover and Net Worth
		6	Capability Status
		7	Statement of Similar type of order, Orders executed as on date of issuance of NIT
		8	Curriculum Vitae of Key Personnel
		9	Format for Submission of Pre-Bid Queries
		10	Format for Proposed modifications
		11	JV/Consortium Agreement
		12	Power of Attorney
		13	Declaration for Net Minimum Guaranteed Generation
Section VII		**Annexure**	
		1	Proforma of Contract Agreement
		2	Proforma of Bank Guarantee for Mobilisation Advance
		3	Proforma of Bank Guarantee for Contract Performance
		4	Proforma for extension of Bank Guarantee
		5	Proforma of Indemnity Bond
		6	Completion Certificate
		7	Application for Payments
		8	Taking-Over Certificate

		9	No-Claim Certificate
		10A	Indemnity for Equipment
		10B	Application for Material Gate Pass
		11	Application for material gate pass
		12	Authorization letter

The bidder is expected to examine all instructions, forms, terms, conditions, specifications and other information in the bidding documents. Failure to furnish all information required as per the bidding documents or uploading of a bid not substantially responsive to the bidding documents in every respect will be at the bidder's risk and may result in rejection of his bid.

1.8. SITE VISIT

The bidder is advised to visit and examine project site and its surroundings and obtain for himself, on his own responsibility, all information that may be necessary for entering into contract. The bidder will assess and satisfy himself as to the adequacy of the local conditions such as approach roads to the site, adequacy of existing culverts/bridges/roads for the expected traffic, water

and power supply, nature of ground and sub soil condition, water table level, accommodations required during the contract, climatic conditions, local terrain, availability of labour and construction materials, details of taxes, duties and levies as applicable and any other information required. The cost of visiting the site shall be at the bidder's own expenses.

1.9.CLARIFICATIONS ON BIDDING DOCUMENTS

1.9.1. A prospective bidder requiring any clarification on bidding documents may notify the DPL by uploading the same in the e-tendering portal, which shall be available to all the participant bidders, as per Standard Format enclosed with this document Form 9 and Form 10 not later than the date and time specified in NIT. The soft copy of the same must be sent in Excel format at the mail address : Email: contractcelldpl@gmail.com, sudiptopal1971@gmail.com

1.9.2. The DPL will issue clarification(s) as it may think fit after pre-bid meeting prior to the deadline/ extended deadline for submission of bids prescribed by the DPL. Written copies of the DPL's response will be uploaded in the e-tendering portal in the corrigendum folder which shall be available to all the participant bidders

1.9.3. Any queries sent by the bidders after the date and time notified in NIT or any extended date, if any, shall not be entertained.

1.10. 0.PRE-BID MEETING

1.10.1. The bidder or its authorized representative is invited to attend pre-bid meeting online to be held on the date and time specified in **NIT or any specific change, which will be uploaded before the meeting date**. The purpose of the meeting will be to clarify the exact scope of work, and any issues regarding the bidding documents and the technical specifications for its clarification, if raised at this stage by the bidders. The Purchaser shall not be under any obligation to entertain /respond to the suggestions made or to incorporate modifications sought for by the prospective bidders.

1.10.2. Any modification/amendment of the bidding documents shall be made by the Purchaser exclusively through the issue of an amendment pursuant to **ITB. 1.11**

1.10.3. Non-attendance at the pre-bid meeting will not be a cause for disqualification of bidders but at the same time shall not entitle them to raise any query at a later date.

1.10.4. Any essential requirement not included in the Price Schedules but required for successful commissioning and operation of Works as per scope of Contract shall be indicated by the bidders as per **Form: 9** of Section VI and submitted before the pre-bid meeting by the date specified in the NIT in line with **ITB. 1.9.1.** The Purchaser shall make related modifications/ amendments as may be considered necessary based on this form in the bidding documents as per provisions mentioned in this clause.

1.10.5. Bidders shall not be permitted to indicate any additional requirements in the bid for any reason whatsoever after the Purchaser has considered such amendments.

1.10.**6. Venue of Pre bid meeting: DPL Technical Building, Durgapur.**

1.11. AMENDMENT OF BIDDING DOCUMENTS

1.11.1. At any time, but not later than seven (7) days prior to the deadline for submission of bids, the Purchaser may, for any reason, whether at its own initiative or in response to a clarification request by a prospective bidder, modify the bidding documents by issue of an addendum/amendment.

1.11.2. The addendum/amendment will be intimated through e-tendering portal at corrigendum folder. The Purchaser shall assume that the information contained therein have been taken into account by the bidder in its bid. The Purchaser will bear no responsibility or liability arising out of non- cognizance of the same in time or otherwise by the bidder.

1.11.3. In order to afford prospective bidders reasonable time in which to take the addendum/amendment into account in preparing their bids, Purchaser may, at its discretion, extend the deadline for the submission of bids.

1.11.4. DPL has the liberty to modify the bidding documents by issue of an addendum/amendment or to cancel the bid at any time.

For the information of bidders, the addendum/ amendments, if any, shall be uploaded on the e-tendering portal **https://wbtenders.gov.in.**

C. PREPARATION OF BIDS

1.12.0 ABOUT THE BID

1.12.1. Mode of Tendering: E-Tender cum Reverse Bidding

1.12.2. Collection of Bid Document

The bidder can search & download NIT & Bid Document(s) electronically from e-tender portal **https://wbtenders.gov.in** once he/she logs on to the portal using the Digital Signature Certificate (DSC). This is the only mode of collection of Bid Documents.

1.12.3. Language of the bid

The bid prepared by the bidder and all correspondences and documents relating to the bid, exchanged between the bidder and the DPL shall be written in the **English language**, provided that any printed literature furnished by the bidder may be written in another language so long as the bid is accompanied by an English translation of its pertinent passages. Failure to comply with this may disqualify a bid. For purposes of interpretation of the bid, the English translation shall govern.

1.12.4. The bidder is expected to examine all instructions, forms, terms, conditions, specifications and other information in the bidding documents. Failure to furnish all information required as per the bidding documents or uploading of a bid not substantially responsive to the bidding documents in every respect will be at the bidder's risk and may result in rejection of his bid.

1.13.0.DEVIATION

This tender is a '**No Deviation**' tender.

Request for any deviation may be considered only if pointed out by any bidder in the Pre Bid meeting. The queries and proposed modification regarding tender must be submitted by writing as per format (**Vide Form -9 and Form-10**) before pre bid meeting (**ITB. 1.10)**

1.14.0. GENERAL GUIDANCE FOR E- TENDER

Instructions/Guidelines for electronic submission of the tenders have been mentioned below for assisting the bidders to participate in e-Tendering.

1.14.1. Registration of Bidder:

Any bidder willing to take part in the process of e-Tendering will have to be enrolled & registered with the Government e-Procurement System, by logging on to **https://wbtenders.gov.in** The contractor is to click on the link for e-Tendering site as given on the web portal.

1.14.2. Digital Signature certificate (DSC):

Each bidder is required to obtain a Class-II or Class-III Digital Signature Certificate (DSC) for submission of tenders from the approved service provider of the National Informatics Centre (NIC) on payment of requisite amount. Details are available at the Web Site stated above. DSC is given as a USB e-Token.

1.15.0.BID PRICES

1.15.1. Unless otherwise specified in the Technical Specification, Bidders shall quote for the entire works on a "**Single Responsibility**" basis such that the total bid price covers all the Contractor's obligations mentioned or to be reasonably inferred from the bidding documents in respect of design, manufacture, including procurement, packing, forwarding transportation, handling, insurance, delivery, installation, testing, pre-commissioning, commissioning, completion of the work and conductance of guarantee tests for the work including supply of spare (if any). This includes the acquiring of all permits, approvals and licenses etc as may be specified in the bidding documents. The bidder shall quote in the appropriate schedule for the proposed bid price for the entire scope of work covered under the bidding documents

1.15.2. PRICE SCHEDULE OF THE BIDDING

1.15.2.1. Price Schedule-1(Supply Schedule): Price Schedule-1 will consist of price of Equipment / Materials, including type tests, charges to be manufactured within/outside India i.e. basic cost (ex-factory, ex-works, ex-warehouse, or off-the-shelf, as applicable), then transport, loading, unloading, insurance charge. This base price shall be inclusive of Customs related Duties, BOCW, entry tax (if any) etc. payable on components and raw materials incorporated or to be incorporated in the goods. Bidder shall submit the base price only. Local transportation including transit insurance, and Taxes to delivery of Equipment / Materials to the Site shall also be included in Price Schedule -1 i.e. Supply Schedule.

1.15.2.2.Price Schedule-2 (Erection Schedule): Price for Installation and Erection service shall be quoted in the Price Schedule -2 (Service Schedule) and shall include the rates and prices for all labour, Contactor's Equipment Supply of consumables Materials and all matters and things of whatsoever nature, charges for site insurance covers other than transit insurance. The price schedule shall include the provision of operation and maintenance manuals, training of Purchaser and their nominated personnel and other services, as identified in the bidding documents and necessary for the proper execution of Installation and Erection Services. GST, Customs related duties and other tax and duties shall not be included in the service and consumables materials price but shall be quoted separately in this Price Schedule.

1.15.2.3. Price Schedule-3(Operation and Maintenance): Price for O&M contract shall be **7.5 % of the basic project cost (excluding GST) for 5(five) years** which is predefined and O&M contract shall be placed on this basis(refer clause-3.8.4.6) after completion of the project.

1.15.3. The taxes, duties and levies shall be indicated by the bidder in the respective price schedule and shall be quoted as per the rates in force seven (7) days prior to the last date of submission of bids with respect of direct transaction between Contractor and Purchaser. Details of Tax and Duties will be guided by the **clause no. GCC 3.15**.

1.15.4. The bidder shall fill in price for all items described in the price schedules. Item against which no price is entered by the bidder will not be paid for by DPL when executed and shall be deemed to have been covered in other prices in the

Price Schedule where the evaluation is being done on the basis of total prices quoted for all the Price Schedules.

1.15.5.All the prices shall be quoted in INR (Indian rupees) only. Foreign exchange component or foreign exchange variation will not be entertained for any reason whatsoever.

1.15.6.If any rebate/discount is offered, the price after overall discount shall be brought out in the Price Schedule. Conditional rebates/discount, if any, offered by any bidder shall not be considered during bid evaluation.

1.15.7. The financial proposal to be submitted in Finance Cover and should contain the following document in one cover. The bidder should fill up the BOQ with quoted value. Once completion of quoting rates, the bidder must encrypt the rates and upload the same with digitally signed. (Only downloaded copies of the above document are to be uploaded, virus scanned and digitally signed by the bidder). **Then, reverse e-auction process will be executed among all the eligible bidders.**

1.15.8. **PRICE ADJUSTMENT**

Price quoted by the bidder shall be firm during the entire period of contact and Bid evaluation for Phase I and Phase II will be done on the quoted base price only.

1.16.0. PERIOD OF VALIDITY OF BIDS

1.16.1. The bids submitted by the bidder shall remain valid for a minimum period of **180 days** from the next day of opening of Technical bid. A bid valid for a shorter period than 180 days shall be rejected by the DPL.

1.16.2. In exceptional circumstances, DPL may solicit the bidder’s consent to an extension of bid validity for a further period without any change in the terms and conditions of the NIT. The request and response thereto shall be made in writing by post or e-mail followed by post confirmation. The bidder may refuse the request without having his bid security forfeited. Bidders agreeing to the request will neither be required nor permitted to modify their respective bids, but will be required to extend the validity of their bid securities correspondingly. The provisions of ITB.1.17 regarding discharge and forfeiture of bid security shall continue to apply during the extended period of bid validity.

1.17.A. BID SECURITY

1.17.1. Bid Security /Ernest Money Deposit (EMD)

Earnest Money Deposit (EMD) – Print-out copy of E-challan of EMD has to be submitted within the Technical Cover.

i) **Rs.50,60,000/- (Rupees Fifty Lakhs and sixty thousand only)**

1.17.2. Bid security of the unsuccessful bidders will be discharged / returned as promptly as possible after the expiration of the validity of bid security or after the date of signing of Contract Agreement with the successful bidder whichever is earlier.

1.17.3. The bid security of the successful bidder will be discharged on furnishing the Performance Guarantee as per **ITB. 1.34** and execution of the Contract Agreement by the bidder.

1.17.4. If the bid security is not in adequate value the bid will be rejected by the DPL and returned to the bidder with in thirty (30) days of the bid opening date.

1.17.5. The bid security shall be forfeited in the following circumstances:

a) If the bidder withdraws its bid as a whole or in part as per **ITB.1.21,** during the period of bid validity specified by the bidder in its bid.

b) If the bidder deviates from any clarification/confirmation given by him subsequent to submission of his bid.

c) If the bidder does not accept the correction of its bid price pursuant to ITB. 1.26

1.17.6. If the successful bidder fails, within the specified time limit either to accept the Letter of Award (LoA) and sign the Contract Agreement unconditionally or, to furnish the Contract Performance Guarantee, in accordance with **ITB. 1.34.** DPL may cancel the bid and no interest shall be paid by the Purchaser on the bid security.

1.17.B **Methodology for ONLINE submission of EMD [as per order no- 3975-F(Y) Dated-28/07/2016 of Finance Dept , Govt of W.B.]**

A. Login by bidder:

a) A bidder desirous of taking part in a tender invited by a State Government Office/PSU/Autonomous Body/Local Body/ PRls, etc shall login to the e-Procurement portal of the Government of West Bengal https: wbtenders.gov.in using his login ID and password.
b) He will select the tender to bid and initiate payment of pre-defined EMD / Tender Fees (if any) for that tender by selecting from either of the following payments modes:
i) Net banking (any of the banks listed in the ICICI Bank Payment gateway) in case of payment through IClCI Bank Payment Gat way ;
ii) RTGS/NEFT in case of payment through bank account in any Bank.

B. Payment procedure:

a) Payment by Net banking (any listed bank) through ICICI Bank Payment Gateway :

i. On selection of net banking as the payment mode, the bidder will be directed to IClCI Bank Payment Gateway webpage (along with a string containing an Unique ID) where he will select the Bank through which he wants to do the transaction.

ii. Bidder will make the payment after entering his Unique ID and password of the bank to process the transaction.

iii. Bidder will receive a confirmation message regarding success/failure of the transaction.

iv. If the transaction is successful, the amount paid by the bidder will get credited in the respective Pooling account of the State Government /PSU/Autonomous Body/local Body/PRIs, etc maintained with the Focal Point Branch of ICICI Bank at R.N. Mukherjee Road, Kolkata for collection of EMD/Tender Fees(if any).

v. If the transaction is failure, the bidder will again try for payment by going back to the first step.

b) Payment through RTGS/NEFT:

i) On selection of RTGS/NEFT as the payment mode, the e-Procurement portal will show a pre-filled challan having the details to process RTGS/NEFT transaction.

ii) The bidder will print the challan and use the pre-filled information to make RTGS/NEFT payment using his Bank account.

iii) Once payment is made, the bidder will come back to the e-Procurement portal after expiry of a reasonable time to enable the NEFT/RTGS process to complete, in order to verify the payment made and continue the bidding process.

iv) If verification is successful, the fund will get credited to the respective Pooling account of the State Government /PSU/ Autonomous Body/Local Body/PRIs, etc maintained with the Focal Point Branch of ICICI Bank at R.N. Mukherjee Road, Kolkata for collection of EMD/Tender Fees.

v) Hereafter, the bidder will go to e-Procurement portal for submission of his bid.

vi. But if the payment verification is unsuccessful, the amount will be returned to the bidder's account.

C. Refund/Settlement Process:

i. After opening of the bids and technical evaluation of the same by the tender inviting authority through electronic processing in the e-Procurement portal of the State Government, the tender inviting authority will declare the status of the bids as successful or unsuccessful which will be made available, along with the details of the unsuccessful bidders, to ICICI Bank by the e-Procurement portal through web services.

ii. On receipt of the information from the e-Procurement portal, the Bank will refund, through an automated process, the EMD of the bidders disqualified at the technical evaluation to the respective bidders' bank accounts from which they made the payment transaction. Such refund will take place within T+2 Bank Working Days where T will mean the date of which information on rejection of bid is uploaded to the e-Procurement portal by the tender inviting authority.

iii. Once the financial bid evaluation is electronically processed in the e- Procurement portal, EMD of the technically qualified bidders other than that of the L1 and L2 bidders will be refunded, through an automated process, to the respective bidders' bank accounts from which they made the payment transaction. Such refund will take place within T+2 Bank Working Days where T will mean the date on which information on rejection of financial bid is uploaded to the e-Procurement portal by the tender inviting authority. However, the L2 bidder should not be rejected till the LOI process is successful.

iv. If the L1 bidder accepts the LOI and the same is processed electronically in the e- Procurement portal, EMD of the L2 bidder will be refunded through an automated process, to his bank account from which he made the payment transaction. Such refund will take place within T+2 Bank Working Days where T will mean the date on which information on Award of Contract (AOC) to the Ll bidder is uploaded to the e-Procurement portal by the tender inviting authority

v. As soon as the L1 bidder is awarded the contract (AOC) and the same is processed electronically in the e-Procurement portal -
a) EMD of the L1 bidder for tender of State Government offices will automatically get transferred from the pooling account to the State Government deposit head "8443-00 -103-001-07" through GRIPS along with the bank particulars of the L1 bidder.

b) EMD of the L1 bidder for tenders of the State PSUs/Autonomous Bodies/Local Bodies/PRIs, etc will automatically get transferred from the pooling account to their respective inked bank accounts along with the bank particulars of the Ll bidder.

In both the above cases, such transfer will take place within T+1 Bank Working Days where T will mean the date on which the Award of Contract (AOC) is issued.

vi. The Bank will share the details of the GRN No. generated on successful entry in GRIPS with the E-Procurement portal for updating.

vii. Once the EMD of the Ll bidder is transferred in the manner mentioned above, Tender fees, if any, deposited by the bidders will be transferred electronically from the pooling account to the Government revenue receipt head " 0070-60-800-013-27" through GRIPS for Government tenders and to the respective linked bank accounts for State PSU/Autonomous Body/Local Body/PRIs, etc tenders.

viii. All refunds will be made mandatorily to the Bank A/c from which the payment of EMD & Tender Fees (if any) were initiated.

1.18. SIGNING OF BIDS

All documents should be digitally signed by the bidders and uploaded by them.

D. SUBMISSION OF BID

1.19.0. SUBMISSION OF BID

1.19.1. Tenders are to be submitted through online to the website stated above in two folders at a time for each work, one in **Techno-commercial Proposal** & the other is **Financial Proposal** before the prescribed date & time using the Digital Signature Certificate (DSC). Virus scanned copy of the documents are to be uploaded duly digitally Signed. The documents will get encrypted (transformed into non readable formats).

1.19**.2. General process of submission:**

Bids are to be submitted online through the website **https://wbtenders.gov.in**. All the documents uploaded by the Tender Inviting Authority form an integral part of the contract. Bidders are required to upload all the Bid Documents along with the other documents, as asked for in the tender, through the above website within the stipulated date and time as given in the Tender. Tenders are to be submitted in two folders - one is **Techno-commercial Proposal** i.e. technical bid and the other is **Financial Proposal** i.e. financial bid. The bidder shall carefully go through the documents and prepare the required documents and upload the scanned documents in Portable Document Format (PDF) to the portal in the designated locations of Technical Bid.

The bidder needs to download the Forms / Annexure, fill up the particulars in the designated cell and upload the same in the designated location of Technical Bid. The bidder needs to download the BOQ, fill up the BOQ in the designated Cell and upload the same in the designated location of Financial Bid in Excel.

1.19.3. It is a two part bidding process so the offer contains two Proposals: a) Technical Proposal and b) Financial Proposal.

1.19.3.1.TECHNO-COMMERCIAL PROPOSAL:

The Technical Proposal shall contain scanned copies and/or declarations in the following standardized formats in two covers/folders:

1. Statutory Cover(**C1**) &
2. Non- Statutory Cover(**C2**)

C1. STATUTORY COVER:

Statutory contain three folders:

1) "**EMD**" folder
2) "**NIT**" Folder and
3) "**Form**" Folder.

1) "EMD" folder:

i. Earnest Money (EMD)/Bid Security

Earnest Money Deposit (EMD) – Print-out copy of E-challan of EMD has to be submitted within the Technical Cover

2) "NIT" Folder

i. Addenda /Corrigenda: if published

Note: Bidders are to keep track of all the Addendum / Corrigendum issued with a particular tender and upload all the above digitally signed along with the NIT. Tenders submitted without the Addendum / Corrigendum will be treated as informal and liable to be rejected.

3) "Forms" folder:

i. This folder will contain all the following forms given is **section-VI** of this documents
ii. Check List **(Form – 1)-document should submit accordingly,**
iii. Forwarding Letter for submission of Bid Security and Tender Fee **(Form – 2)**
iv. Bid Form **(Form – 3),**
v. Summary statement of yearly turnover and net worth **(Form – 5)**
vi. Capability Status **(Form – 6)**
vii. Statement of Similar Type of Order. Orders Executed as on date of issuance of NIT **[Applicability up to the extent of meeting Technical QR]. (Form – 7).**
viii. Curriculum Vitae of Key Personnel ***(*Form – 8*)*.**
ix. JV/Consortium Agreement(**Form-11**) –if JV/Consortium
x. Power of Attorney**(Form-12)-** if JV/Consortium
xi. Net Minimum Guaranteed Generation(NMGG) –**Form-13**

*(**Only downloaded copies of the above documents duly filled up and are to be uploaded, virus scanned and digitally signed by the bidder**).*

C2. NON STATUTORY COVER:

Sl. No.	Category Name	Detail(s)
A	Certificate(s)	**1.** Copy of the GST Certificate
		2. Copy of the PAN certificate/ PAN Card
		3. Declaration of PF Registration Number **or** Proof of PF Registration, Last paid PF, ESI Challan etc.
		4. Copy of the ESIC registration.
B	Company Detail(s)	**5.** Copy of the Registration Certificate under Company Act (Company Incorporation Certificate) o**r** copy of the Registered Deed for Partnership Firm
C	Credential	**6.** Copy of the Order(s)/ Contract Agreement(s) with the Purchaser / any other Proof of Purchase, as primary agency [Applicability up to the extent of meeting Technical QR]. **AND** Corresponding Copy of the Completion Certificate(s) /Commissioning report signed by the Purchaser / Ordering Authority to substantiate the proof of completion of the Solar PV Power Plant(s). [Applicability up to the extent of meeting Technical QR].
D		**7**. Audited Balance Sheet & Statement of Profit & Loss A/c. [Applicability as per Financial capability].

		8. Copy of Acknowledgement of Income Tax returns [Applicability as per Financial capability].

Bidders are requested to submit all the documents as per the same serial in the above table given.

1.19**.3.2. <u>FINANCIAL PROPOSAL</u>**

The Financial Proposal shall contain Price Bid and Mode of Transaction in the following standardized format i.e. file named BOQ –in Excels format.

<u>BOQ</u>

i The BOQ to be filled up and upload is in form of Excel file in the BOQ folder (Cover).

ii **In BOQ, Bidders will quote total Project Cost (excluding O&M job) and Total GST in INR for 5 MW Phase -2. Evaluation of L1 will be done based on minimum base price of total project cost of 5 MW Phase II (i.e. excluding O&M job & total GST) quoted by the bidder during reverse e-auction.**

iii BOQ file consist of one worksheet with two part i.e. 1) Supply Schedule & 2) Erection Schedule

iv Filling up procedure-

a) **Supply Schedule:** To be filled up by the bidders. It is related to the supply items as per **ITB. 1.15.2.1. It is inclusive of Local Transportation, Inland transit insurance, and other local cost including Mandatory Spares.**

b) **Erection Schedule:** To be filled up by the bidders. It is related to the corresponding erection of the equipment and other service item of the project as per **ITB.1.15.2.2 including site insurance.**

c) Bidders are advised to price their bids in such a manner that Installation Price Component of the bid price (excluding Civil/Structural works price) should not be less than 7% of total supply price of Main Equipment.

d) Bidders are advised to price their bids in such a manner that the Civil Works Price Component of the bid price (including Site Fabricated Structural works price) should not be less than 12% of total supply price of Main Equipment.

e) Mandatory spares component of each project should **not be less than 1.0% of total project cost .**

- **Sample calculation of estimated Project cost and O&M cost:**

Total estimated Project cost and Comprehensive O&M including GST = Rs. 10626.00 Lakh.

Say, estimated basic cost of project (excluding GST) = P (assume);

Estimated project cost including GST = 1.089P, where GST is 8.9% (GST will be changed accordingly as per changes in guidelines)

Basic estimated comprehensive O&M cost for 5 years = 0.075P, (7.5% of basic project cost)

Total estimated comprehensive O&M cost including GST = 0.075P x 1.18 = 0.0885P (Where GST is 18%)

Total estimated Project cost including O&M and GST = 1.089P + 0.0885P = 1.1775P

Therefore, $\boxed{\textbf{1.775P = 10626.00}}$

i.e. P = Rs. 9024.20 Lakh

Total estimated Project cost including GST = Rs. 9827.35 Lakh

Total estimated comprehensive O&M cost including GST = Rs. 798.64 Lakh

1.20. DEADLINE FOR SUBMISSION OF BIDS

1.20.1. The original Demand Draft/B.G against Earnest Money Deposit (EMD) must be submitted physically in the tender box at the office of the GM (M&C), Corporate office, DPL, under sealed cover super-scribing the name of the work with NIT no., name of the bidder, name of the work etc. on or before the date & time mentioned in the bid data sheet or any extension of date & time . If the bidder fails to submit the original copies within the due date and time his tender will not be opened and his bid will be rejected.

1.20.2. Bids must be received by DPL at the online e-tendering portal address specified in **NIT**, no later than the time and date mentioned in **NIT.**

1.20.3. The DPL may, at its discretion, under intimation to the bidders who have downloaded the bidding documents, extend the deadline for the submission of bids / opening of bids by issuing an addendum and hosting the same on the e-tender portal, in which case all rights and obligations of DPL and bidders previously subject to the original deadline shall thereafter subject to the deadline as extended.

1.20.4. In the event, the deadline for uploading of bid is extended by the DPL, the bidders who have already uploaded their bids within the original deadline of submission shall have the option to upload their revised bid in substitution either in full or in part of earlier bid. In the absence of a revised bid, the original bid shall be considered for opening and subsequent evaluation. Wherever, the bidder has submitted the revised bid in full, in modification of earlier bid, the earlier bid shall be returned unopened to the bidder.

1.20.5. Submission of original Bid Security (**EMD**) (Offline Submission)

a) The original copies of the DD/BC towards DD/BC/BG towards EMD (Bid security) as per NIT shall be submitted along with a forwarding letter **(Form –2**) within the date and time as specified in the bids.

b) If the bidder fails to submit the original copies of the Bid Security within the due date and time, his tender will not be opened and his bid will stand rejected.

1.21.WITHDRAWAL OF BIDS

The bidder shall not be permitted to withdraw their bid during the interval between bid submission deadline (as mentioned in NIT) and the period of bid validity as per **ITB.1.15.** If any withdrawal of bid is made by the bidder during the above period, it shall result in the forfeiture of the bid security

E. OPENING AND EVALUATION OF TENDER

1.22 BID OPENING

1.22.1 As it is a two part Bidding so DPL will open the bids electronically at e-tendering portal by the authorized personnel(s) using their Digital Signature Certificate (DSC), at the scheduled date & time for opening of bids as mentioned in NIT for techno-commercial bid(first part) and Financial bid (second part) will be opened on the date and time as intimated to the bidder on successful completion of evaluation of techno-commercial bids. The bidders' representatives who desire may attend/witness the bid opening event through e-tendering portal at their respective end. In the event of the specified date for the opening of bids being declared a holiday for the DPL or suspended for any involuntary reasons, the bids will be opened at the appointed time & date which shall be intimated/ communicated to all the intending bidders.

1.22.2 Bids that are not opened at bid opening will not be considered for further evaluation, regardless of the circumstances. The reason for which bids are not being opened will be notified to all the bidders through e-tendering portal.

1.22.3 The bidders' names, bid withdrawal and the presence or absence of the requisite bid security and such other details which DPL at his discretion may consider appropriate will be notified in the e-tendering portal at the bid opening date.

1.22.4 In this case of Single stage two part bids, on the date of opening of bid, the techno-commercial bid shall only be opened. The date for opening of the Price bid shall be intimated electronically at the appropriate time to the bidders whose bid is found responsive in the techno-commercial evaluation.

1.22.5 In the event, Purchaser, in its discretion, decides not to open the bid for want of adequate response to the bidding, the Purchaser may either extend the bid submission deadline or cancel the bidding process any time before issuance of Letter of Award(LOA).

1.23 PROCESS TO BE CONFIDENTIAL

1.23.1 Subject to ITB. 1.24, no bidder shall contact the Purchaser on any matter related to its bid from the time of opening of the bids to the time the Contract is awarded.

1.23.2 Any effort by a bidder to influence Purchaser or others connected in the process of examination, clarification, evaluation and comparison of bids, and in decisions concerning the award of Contract, may result in the rejection of his bid.

1.24 CLARIFICATION OF BIDS

1.24.1 During bid evaluation, Purchaser may, at its discretion and if so required, ask the bidders for any clarification in support of their compliance to stipulated Qualifying Requirements (QR) or any other matter related to its bid except to the extent in ITB.1.24.2. The request for clarification required from the bidder and the response thereto shall be in writing and shall be delivered by registered speed post/email/courier / hand delivery under acknowledgement / email / fax so as to reach the Purchaser within the time specified in the request for clarification issued by Purchaser.

1.24.2 Any post-bid change in the price or substance (techno-commercial) of the bid shall not be sought, offered or accepted.

1.25 DETERMINATION OF RESPONSIVENESS

1.25.1 The Purchaser will examine the bids to determine whether they are complete, whether any computational errors have been made, whether required securities have been furnished, whether power of attorney of signatory of the bid has been submitted, whether the documents have been properly signed and whether the bids are generally in order and substantially responsive to the requirements of the bidding documents.

1.25.2 For the purpose of this clause, a substantially responsive bid is one which conforms to all the Terms, Conditions and Specifications of the bidding documents without material deviation or reservation. The Purchaser's determination of a bid's responsiveness shall be based on the contents of the bid itself without recourse to extrinsic evidence.

1.25.3 Any material information/ data/ document required to be submitted by the bidders as per provisions of bidding documents, if not submitted by the bidder, may render the bid to be non-responsive provided such information/ data/ documents is such that it may adversely affect the evaluation.

1.25.4 The Purchaser may waive any minor infirmity, non-conformity or irregularity in a bid that does not constitute a material deviation, and that does not prejudice or affect the relative ranking of any bidder, as a result of the technical and commercial evaluation pursuant to **ITB. 1.28** & **ITB. 1.29**.

1.25.5 If a bid is not substantially responsive to the requirements of the bidding documents, it may be rejected by Purchaser and the same cannot subsequently be made responsive by the bidder by correction.

1.25.6 Conditional bid shall not be accepted by Purchaser

1.26 TIME SCHEDULE

The basic consideration and the essence of the Contract shall be the strict adherence to the time schedule specified in the Bids and NIT after the Commencement Date of the Contract as incorporated in the Contract Agreement for completion of Works. Bidders are required to base their prices on the time schedule mention in Clause no. **GCC 3.21**. No credit will be given for earlier completion for the purpose of evaluation.

1.27 PROCEDURE OF EVALUATION OF BIDS

1.27.1 The Purchaser will carry out a detailed evaluation of the bids determined to be substantially responsive as per clause no. **ITB 1.25** in order to determine whether the technical aspects are in accordance with the requirements set forth in the bidding documents. **Bids submitted by bidders with any deviations shall be rejected.**

1.27.2 The Purchaser will determine to its satisfaction whether the bidder selected as having submitted the lowest evaluated responsive bid is qualified to satisfactorily perform the Contract in terms of the qualifying requirements stipulated in NIT.

1.27.3 The determination will take into account the bidder's financial, technical, production and execution capabilities, in particular its work in hand and future commitments. It will be based upon an examination of the documentary evidence of the bidder's qualifications submitted by the bidder to the bid, as well as such other information as the Purchaser deems necessary and appropriate.

1.27.4 An affirmative determination will be a prerequisite for award of the Contract to the bidder. A negative determination will result in rejection of the bidder's bid, in which event the Purchaser will proceed to the next lowest evaluated bid to make a similar determination of that bidder's capabilities to perform satisfactorily.

1.27.5 The financial evaluation will be a two stage process. Initially the process of price bid decryption will be same as is normally executed in an e-Tender. Subsequent to this, reverse e-auction process will be executed among all the eligible bidders following the standard e-auction procedure of the portal, i.e. **https://wbtenders.gov.in.**

1.27.6 **EVALUATION:**

Bidder shall quote the Total Bid price required for setting up the 5 MW Solar PV Ground Solar Project in the financial part of bid.

All the Bidders shall be quote the following information in Bidding documents:

- Quoted Price in INR (X) excluding GST. Base price only shall be considered for evaluation.
- The Evaluation criteria for the Project cost shall be based on lowest $\mathbf{X}_{\mathbf{Revised}}$ i.e. revised Cost during reverse bidding for 5 MW AC Solar Project (in total):

$\mathbf{X}_{\mathbf{Revised}}$ (INR).

1.27.7 **Evaluation will be done on base price only quoted by the bidder during Reverse auction in their Bid where L1 will be selected on mention in cl. No. 1.19.3.2 (ii) Cl. No. 3.15.1 & Cl. No. 3.15.5.**

1.28 Not applicable.

REVERSE AUCTION

Purchaser will go for reverse bidding, after opening of price bid, only the lowest price will be displayed in the e-procurement platform without disclosing the name of the **L1** bidder. Reverse auction may hold thereafter on prescheduled date amongst technically qualified bidders. L1 price is fixed as the starting price and lowest price arrived at the end of the reverse auction will be considered as the final L 1 price.

INSTRUCTIONS TO BIDDERS FOR REVERSE AUCTION

a) This Reverse Bidding will be done through the website: https://wbtenders.gov.in/nicgep/app .

b) Tender Cum Auction is a combination of electronic Tender followed by Auction (Reverse / Forward Auction) in GePNIC. It is generally called as eRA. The Reverse (or Forward) Auction as the case may be, will be conducted after Opening of Price / Financial Bids.

c) The Tender Inviting Authority (TIA) will normally mention about conducting of eRA along with necessary instructions at the Notice Inviting Tender (NIT) stage itself. In the portal, it will be mentioned in the **Form of Contract** as 'Tender Cum Auction' against the particular tender.

d) Bidders, who are registered as privileged bidders (like MSME/ Startup / Make in India) in the portal and wish to avail the preferential treatment during financial evaluation of the tender, as per GoI policy should upload relevant

documents during bid submission to claim Preferential treatment, subject to whether the preferential treatment is permitted by TIA against that tender during the time of Publishing.

e) Bidders are advised to refer the **clause 2.31.02** in this document regarding **terminologies** being used in Tender cum Auction.

f) Generally, all bidders who are techno–commercially qualified & approved by department/organization will be eligible for participation in the Reverse (or Forward) Auction.

g) Bidder's eligibility in the Reverse (or Forward) Auction will also depends on the "Bidders Elimination Process" configured by TIA in that tender. The elimination criteria for auction is one which normally restricts one or more bidders from participation in the auction who have quoted in the tender which is exorbitantly high in case of Reverse Auction or very low in case of Forward Auction. Hence, **bidders are advised to quote reasonably in the price bid to avoid elimination from participation in the auction**. Bidders are advised to refer to NIT/ Tender documents or may contact concerned tender invitingauthority for auction elimination criteria against the tender.

h) After opening of the price (financial) bids, System will displays L1 (or H1) bidders price based on either **overall price basis** or **item wise/lot price basis** automatically.

i) The participation in the auction by an eligible bidder is voluntary. It is solely at the discretion of the bidder to participate in the auction. If a qualified bidder is not interested to participate in the auction, then price / financial bid submitted by bidder in the tender shall be treated as final price/financial bid of that bidder.

j) There will be no participation fees for the Auction. The bidders get an opportunity to change their prices by participating in the auction. In the case of Reverse Auction, the least price among the value quoted by the bidder in the auction and Financial Bid submitted will be taken as the final price quote of the bidder against the tender. In the case of Forward Auction, the highest price among the value quoted by the bidder in the auction and Financial Bid submitted will be taken as the final price quote of the bidder against the tender.

k) Using the system provided price, which would normally be considered as auction start price (but can be changed by the TIA, if required) and accordingly, will create Reverse (or Forward) Auction as the case may be and the auction will be published by the TIA.

l) The Techno-commercially qualified bidders (who are not eliminated by the system from participation in the Auction as per elimination criteria set by TIA against that tender cum auction) will receive Auction schedule intimation through e-mail. However, bidders are always advised to visit web site / portal regularly to keep them updated and to timely act upon wrt auction / other requirements of that tender.

m) The server time (which is displayed on the bidders' dash-board after login) will be considered as the standard time for referencing the deadlines for participation in live auction and other process during auction in the portal. The bidders should follow server time (Server System Clock) for all activities in the portal. The Time followed in this portal is as per Indian Standard Time (IST) which is GMT+5:30.

n) As per the configuration defined against the tender cum auction by the TIA, the system will not disclose the name of the L1 (H1) bidder, number of bids and names of the participating bidders on the portal to anybody prior to the completion of Reverse / Forward auction process.

o) **Participation in Reverse(Forward) Auction**:

i. Bidders shall login using their login ID & Password and then using DSC.

ii. Click on '**My Auctions**' button given in left side of page, to view all Auction details for which bidder is Techno-Commercially qualified.

iii. For participating in Live Auction during schedule date & time,

A. Click on **Live Auctions** Button.

B. Click on **View** button to participate in the interested Auction.

C. There is List of qualified Lots in which Bidder can participate against selected Auction. Click on **Hammer** Icon to participate in the respective lot.

D. On clicking Hammer Icon, system will show Start price, Decremental (or

Incremental) price and Current price against lot. Current Price will appear as Blank (-) in case no bidder has offered price.

E. In case of Reverse Auction : **Enter your Price** in '**My Auction Price in Rs**' in multiples of decremental value up to above Max Seal % value, and then **sign it digitally** by clicking on Sign Icon and Click on **submit** button.

F. System will then display Current Auction Price, Auction submitted Date/Time (last successfully quoted date & time), Auction scheduled date & time, Auction extended time up to (if any) etc.

G. On clicking "**Refresh**" Link in the screen, then the screen will be reloaded and will show your Latest Value / Price Quoted and system will also show Least Amount/ Rate (highest amount/ rate) which any Bidder would have quoted.

p) The live auction will be extended automatically by "Auto Extensions in minutes" if a valid and digitally signed bid has been successfully recorded in the system during the "Auction Elapse Time in minutes" before auction closing. The server time will be considered final and all bids that are received and recorded by the server before the auction close time (as per the server time) only shall be treated as valid bids. Bidder should follow the auction end/close time as displayed on the screen.

q) **During the auction the bidders are also advised to click "Refresh" link for refreshing their webpage to get the latest information about the status of the auction. The Live Auction window will remain same and also time remaining will be ticking, even in the event of disconnection of bidder computer system, Network/Internet. The bids submitted by other bidders during the time of disconnect of bidder computer system will not be displayed on your screen. The other bidder might have become L1/H1 (as the case may be) for the item during this time. To overcome this situation the bidders are also advised to click "Refresh" link for refreshing their webpage frequently.**

r) The last (latest) successful bid price quoted by bidder will be considered as valid price at any point of time during Auction.

s) The chronologically last (latest) bid submitted by the bidder till the end of the auction will be considered as the valid price bid offered by the bidder and acceptance of the same by Tender Inviting Authority will form a binding contract between Tender Inviting Authority and the bidder for entering into a contract.

t) For those bidders, who are eliminated from participating in the auction or bidders who are eligible for auction but not provided any price during auction, the rate quoted in the price/financial bid of the tender will be considered as final price.

u) Internet connectivity and other paraphernalia requirements shall have to be ensured by bidder themselves. In order to ward-off such contingent situation like internet connectivity failure, power failure etc., bidders are requested to make all the necessary arrangements / alternatives whatever required so that they are able to circumvent such situation and still be able to participate in the live auction successfully. However, the bidders are requested not to wait till the last moment to quote their bids to avoid any such complex situations.

v) Non submission of bid/quote during live auction within due date / time following due process prevalent at that time in the portal due to failure of computer system, power, network, internet connectivity or delay in performance or otherwise at Bidder's end or any other reasons for which bidder shall be held solely responsible. Neither National Informatics Centre nor concerned Tender Inviting Authority will be held responsible for the same in any manner.

w) The Tender Inviting Authority reserves the right to postpone, suspend/pause, resume and extend the Auction, if required.

x) Bidder shall not divulge their bids to any other party during auction. If a Bidder or any of its representatives are found to be involved in Price manipulation/ cartel formation of any kind, directly or indirectly by communicating with other bidders, strict action including black listing shall be taken against such bidders as per procurement guidelines/policies of the TIA.

y) After the conclusion of the online reverse (forward) auction, all bidders who have participated in Reverse(Forward) will see the overall Comparative chart ie L1(H1) price of the Auction.

z) Based on the L1(H1) price of each bidder as well as the price quoted in the tender by the non-participating bidder, comparative chart will be generated by the system and accordingly, further financial evaluation processing will be done by Tender Inviting Authority.

aa) The Tender Inviting Authority normally reserves the right to extend, reschedule or cancel the Reverse Auction process at any time, before ordering, without assigning any reason, with intimation to bidders.

bb) Bidding will be conducted only in **Indian Rupees** as indicated in the tender.

cc) Validity of bids: Price shall be valid for a period of defined number of days from the date of reverse / forward auction or as specified in the tender. These shall not be subjected to any change whatsoever.

AUCTION TERMINOLOGY

a) **Forward Auction**: A bidder shall quote over and above the Starting Price or Current Auction Price.

b) **Reverse Auction**: A bidder shall quote below the Starting Price or Current Auction Price.

c) **Auction Start Date & Time and Auction End Date &Time**:- Live auction would be conducted during this period.

d) **Auction Elapse Time in minutes**: It is the minute(s) before the 'Auction End Time' and acts as a trigger for auto extension of current auction. If a bid is received successfully within these minutes, the auction will be extended subject to number of extension is not crossed.

e) **Auction Bid Auto Extensions in minutes**: The Auction End Time will be extended by this 'Auto Extension Time in Minutes', If a bid is received successfully within 'Elapse Time' in Minutes. Process will continue till no bid is received in elapse time.

f) **Auto Extension Restriction Required**: If procuring entity (TIA) desires to restrict the number of Extensions during Live Auction then TIA will specify this parameter as 'Yes' and also specify the number of extensions required. System will accordingly, extend the Auction only for specified number of times. If unlimited Extensions is required then TIA will specify it as "No".

g) **Max Seal Percentage:** It defines maximum value a bidder can quote in multiples of incremental / decremental value in the auction.

In case of Reverse Auction, in order to displace a standing lowest bid and to become "L1", a bidder can offer a minimum bid decrement or in multiples of decremental value up to above Max Seal %.

For example:

Current price : Rs. 49,000	Decrement value : Rs. 1,000
Maximum Seal % : 50	

In this case a bidder can quote minimum decrement amount as Rs 49,000 – 1,000 = Rs. 48,000 and maximum decrement amount is 49,000 - 24,500 – 1,000 = 23,500 = 24,000* (as decrement value is in terms of 1000).

In case of Forward Auction, in order to displace a standing highest bid and to become

"H1", a bidder can offer a minimum bid increment or in multiples of incremental value up to or below Max Seal %.

Current price : Rs. 49,000	Decrement value : Rs. 1,000
Maximum Seal % : 50	

In this case a bidder can quote minimum increment amount as Rs 49,000 + 1,000 = Rs. 50,000 and maximum increment amount as 49,000 + 24,500 + 1,000 = 74,500 = 74,000* (as increment value is in terms of 1000).

Bidder Elimination process : If the procuring entity (TIA) wish to eliminate bidders who is offering high price (very low price) from participating in Reverse (Forward) Auction , then TIA will select this option as **Yes.**

In addition, if Allow Preferential Bidder Elimination is selected as **No** by **TIA**, then Preferential bidders like (MSME / Startup / Make in India) will not be eliminated even if they have quoted very high (or too low) and will be allowed to participate in Reverse (Forward) Auction. Moreover, even if **Allow Preferential Bidder Elimination** is selected as **Yes** by **TIA**, in this case during highest bidder elimination process in reverse auction, if the bidder is a privileged Bidder and their quote rate falls under the criteria L1+Tolerance percentage will not be eliminated and all others will be eliminated.

Minimum Bidder for Elimination: 4 (recommended value) - If procuring entity has received the minimum number of bids, say 4 then system will initiate the Bidder Elimination process.

Number of Bidder to Eliminate: 1(recommended value) - When the system will implement Bidder Elimination process, then system will eliminate say 1 (or mentioned number of Bidder) and start the Auction process with rest of the Bidders.

In case of Reverse Auction, H1 Bidder will be eliminated from participating in the auction even though qualified in the techno-commercial evaluation and in Forward

Auction, System will eliminate L1 Bidder.

F. AWARD OF CONTRACT

1.29 AWARD CRITERIA

1.29.1 The Purchaser will award the Contract to the successful bidder(s) whose bid has been determined to be substantially responsive and has been determined as the lowest bid provided further that the bidder is determined to be qualified to perform the Contract satisfactorily. The Purchaser shall be the sole judge in this regard.

1.29.2 For the purpose of determining the capability and capacity of the bidder to perform the Contract, the Purchaser reserves the right to verify the authenticity of the documents submitted by the bidder for meeting the qualification requirements and may undertake verification of the facilities available with the bidder.

1.30 RIGHT TO REJECT BIDS

DPL reserves the right to accept or reject any bid and to annul the bidding process and reject all bids at any time prior to award of contract, without thereby incurring any liability to affected bidder or bidders or any obligation to inform the affected bidder or bidders the reason for DPL's action.

1.31 LETTER OF AWARD

1.31.1 After approval of bid evaluation by DPL, the successful bidder may be invited for pre-award discussions. After pre-award discussions and prior to the expiry of the period of bid validity, DPL will notify the successful bidder in writing by registered letter and E-mail, that his bid has been accepted. This letter (`Letter of Award' or **LOA or Material/Service Contract**) shall mention the sum which DPL will pay to the Contractor in consideration of the execution & completion of the Works by the Contractor as prescribed under the Contract.

1.31.2 Material & Service Contract will be issued as per the BOQ Schedule submitted by L1 bidder in line with **Cl. No. 1.19.3.2 and Cl. No. 3.2**.

1.31.3 Within Seven (07) days of receipt of the LOA, the successful bidder shall sign and return one (1) original copy of the same to DPL as acknowledgment of acceptance of the same.

1.31.4 The **LOA or Material/Service Contract** will constitute the formation of the Contract as per provisions of **GCC.3.4.5**

1.32 SIGNING OF CONTRACT AGREEMENT

Within Seven (07) days from the date of acceptance of LOA, the successful Bidder submit two separate Contract Agreements for Phase I and Phase II (6 copies in original on non-judicial stamp paper of appropriate value) as per **Annexure: 1 of Section VII** in three (3) copies each incorporating all agreements between the parties duly signed by the authorized signatory of the Bidder along with the LOA.

DPL shall sign the above Contract Agreements if found in order and return one (1) copy to the Bidder.

1.33 CONTRACT PERFORMANCE GUARANTEE

1.33.1 Within fifteen (21) days of LOA from DPL, the successful bidder shall furnish to DPL Contract Performance Guarantee (CPG), as in the form of an unconditional and irrevocable Bank Guarantee equal to three percent (3%) of the Contract Prices including GST for the Contract and as per the **Annexure: 2 of Section VII.**

1.33.2 Failure of the successful bidder to submit performance security as stated herein shall constitute sufficient ground for annulment of the award and forfeiture of his bid security, in which event the Purchaser may make the award to the next lowest evaluated bidder or call for new bids.

1.33.3 **Forfeiture of Contact Performance Guarantee**

Contract Performance Guarantee shall be forfeited if,

a. The successful bidder does not execute the work within 60 days after placement of Letter of Award (LOA) and/or,
b. The successful bidder discontinue the work without prior permission of DPL and/or,
c. The successful bidder fails to install/procure the total capacity of the plant as mentioned in the Bid Document and/or,
d. The successful bidder fails to rectify/replace of the defective/damaged equipment(s)/work(s) within the Defect Liability Period.

1.33.4. **Additional Contact Performance Guarantee (ACPG)**

If L1 bidder's quoted bid is 80% or less of the estimated project cost mentioned in this tender then additional Contact Performance Guarantee (**ACPG**) as in the form of an unconditional and irrevocable Bank Guarantee (BG) equal to ten percent (10%) of the Contract Price for all the Contracts and as per the **Annexure: 2 of Section VII** have to be submitted by the bidder in the form of any scheduled commercial bank before issuance of Contract Agreement. If the bidder fails to submit the Additional Performance Security within scheduled time, his EMD will be forfeited. The Additional Performance Security shall remain valid up to the Defect Liability Period of sixty (60) calendar months and project execution period (12) months, with an additional claim period of ninety (90) days, failing which his bid security may be forfeited.

1.33.5. Forfeiture of Additional Contact Performance Guarantee (ACPG)

Additional Contact Performance Guarantee (ACPG) shall be forfeited if,

a. The successful bidder does not execute the work after (60) sixty days placement of Letter of Award (LOA) and/or,

b. The successful bidder discontinue the work without prior permission of DPL and/or,

c. The successful bidder fails to install/procure the total capacity of the plant as mentioned in the Bid Document and/or,

d. The successful bidder fails to rectify/replace of the defective/damaged equipment(s)/work(s) within the Defect Liability Period.

1.34 MISREPRESENTATION BY THE BIDDER

If the bidder conceals any material information or makes a wrong statement or misrepresents facts or makes a misleading statement in the bid, in any manner whatsoever, in order to create circumstances for the acceptance of the bid, the purchaser reserves the right to reject such bid and/or cancel the LOA/Contract , if issued.

SECTION-II

BID DATA SHEET

BID DATA SHEET (BDS)

The following bid specific data for the Equipment/ Materials / Works to be procured shall amend and/or supplement the clauses in the Instructions to Bidders (ITB). Whenever there is a conflict, the provisions herein shall prevail over those in the ITB.

Name of the Work:	**E-tender cum Reverse auction of** Design & Engineering, Manufacture/ Supply, Erection, Testing and Commissioning of 5 MW AC P-II Grid Connected Ground Mounted Solar Photovoltaic Power Plant at Durgapur Projects Limited , Durgapur, West Bengal including warrantee obligation with 05 (Five) years comprehensive Operation and Maintenance.

ITB Clause Ref., if any	**Data**
	2.1 A. SCOPE & QUALIFICATION
ITB. 1.2	Fund from Govt. of WB.
	End user of the Project The Durgapur Projects Limited
ITB.1.3	Brief Scope of Work
ITB 1.4	**QUALIFYING REQUIREMENT FOR BIDDERS:** Prescribed in NIT
ITB 1.4.1.4	Whether Joint Venture is permitted **- YES**
	Type of Bidding: National Competitive Bidding **(NCB).**

	2.2 B. THE BIDDING DOCUMENTS
ITB.1.9	Clarification or any proposed modification on bidding document may be submitted by the bidders through mail to the mail address **Email: contractcelldpl@gmail.com** as per format of **Form 9 and Form 10 of Section VI**
	Date & Time up to which request for clarifications will be received: As per NIT
	Clarifications on bidding documents may be obtained from **https://wbtenders.gov.in**
	NOTE: Late submission of queries will not be entertained.
ITB. 1.10	Pre-bid Meeting **Venue :** The Durgapur Projects Ltd. Technical Building,Durgapur-713201 **Date & Time : As per NIT**
ITB.1.10.4	The proposed modifications to the bid documents shall be sent by the bidders within the time mentioned in NIT through mail to the mail address **Email: contractcelldpl@gmail.com** as per format of **Form 10 of Section VI**
	2.3 C. PREPARATION OF BIDS
ITB.1.15.8	**Price Adjustment is not applicable.**
ITB.16.0	**Period of validity of bids: As per NIT**
ITB.1.18	**Validity of Bid Security : As per NIT**
	2.4 D. SUBMISSION OF BIDS

ITB.1.19.1 ITB.1.19.2	Bids should be submitted online through the portal **https://wbtenders.gov.in**
ITB 1.20.4	**Submission of original Bid Security (EMD) (Off line Submission) an amount :** **Rs. 50,66,000/- (Rupees Fifty Lakhs Sixty six thousand only)** **Place of Submission** **To** **The Senior Manager (Contracts) ,** The Durgapur Projects Ltd. Durgapur, PIN-713201 **Telephone: 9434713057,** **Email: contractcelldpl@gmail.com** **Date & Time : As per NIT**
ITB. 1.27	**Time to complete the Works from the Date of LOA:- As per NIT**
	Detailed Master Network for different activities **[To be submitted by successful vendor/contractor]**
	The Master Network shall include the major activities listed below showing their inter-relationship and duration so as to meet the schedule dates mentioned above: 1. Kick off Meeting 2. Start of engineering 3. Completion of engineering 4. Start of manufacturing/fabrication 5. Completion of manufacturing/fabrication 6. Commencement of supplies 7. Supplies all items 8. Completion of site delivery of spares 9. Commencement & completion of civil works (wherever

	applicable) 10. Commencement and completion of erection of equipment/materials. 11. Readiness of the system 12. PG test completion 13. Completion of Works
	The master schedule and the key milestone dates will be discussed with the successful bidder and agreed upon before the issue of LOA. Engineering Drawing and Data Submission Schedule shall also be discussed and finalised before the issue of LOA.
	After the LOA, the Contractor shall plan the sequence of work of manufacture, supply and erection to meet the above stated dates of successful completion of Works and shall ensure all work, manufacture, shop testing, inspection and shipment of the Equipment/Materials in accordance with the required erection sequence.

SECTION-III

GENERAL CONDITION OF CONTRACT (GCC)

A. CONTRACT AND INTERPRETATION

3.1. DEFINITION OF TERMS

Unless the context otherwise requires, the following terms whenever used in this document have the respective meaning:

3.1.1 **"Purchaser"** shall mean the "**The Durgapur Projects Limited(DPL)**", having its Office at Administrative Building,Durgapur-713201 and shall include its successors and assigns.

3.1.2 "**Contract**" means all the Contract Agreement(s) entered into between the Purchaser and the Contractor, together with the Contract Documents referred to therein; they shall constitute the Contract and the term Contract shall in all such documents be construed accordingly.

3.1.3 "**Contract Price**" means the sum total of contract price stated in all the Letter of Award(s) as payable to the Contractor for supply, execution and commissioning of the entire Works under the scope of Contract subject to such addition & adjustments thereto or deductions there from as may be made pursuant to the Contract(s). In cases where separate identifiable Works can be completed and taken over by the Purchaser and for which separate completion schedule is provided in the Contact, in relation to such Works, the Contract Price shall mean the price related to such Works completed and taken over by the Purchaser.

3.1.4 "**Contractor**" means the successful bidder whose bid has been accepted by the Purchaser, named as such in the Contract Agreement and included its legal successors and permitted assigns.

3.1.5 "**Bidder**" shall mean Bidding Company or a Bidding Consortium (formed through a memorandum of understanding) or any other person submitting the Bid. Any reference to the Bidder includes Bidding Company / Bidding Consortium / Member of a Bidding Consortium includes its successors, executors and permitted assigns and Lead Member of the Bidding Consortium jointly and severally, as the context may be.

3.1.6 "**Bidding Consortium**" shall mean a maximum of three(3) Bidding Companies who have signed a memorandum of understanding collectively submitted the Bid in accordance with the provisions of this RFQ cum RFP.

3.1.7 "**Project Manager**" means the person appointed by the Purchaser in the manner provided in GCC.3.19.1 hereof and named as such in the SCC to perform the duties delegated by the Purchaser.

3.1.8 "**Letter of Award**" shall mean intimation in writing by DPL placing award of contract upon the successful bidder towards execution of the contract on acceptance of the bid offered by the bidder following terms and conditions as enumerated in the tender document.

3.1.9 The '**Engineer-in-Charge'** shall mean the **Deputy** General Manager (Projects),DPL

3.1.10 The "**Controlling Officer'** shall mean the **Senior Manager (Environment)**

3.1.11 '**DPL's representative'** shall mean any person or persons or consulting firm appointed/authorized by the Company to supervise, inspect, test and examine workmanship and materials of the work under this scope

3.1.12 The '**Sub-Contractor'** shall mean any person/agency to whom any part of the contract has been sublet by the contractor with the consent in writing of the Company and will include the legal representatives, successors and permitted assigns of such persons/agency.

3.1.13 '**Equipment/materials'** shall mean and include all type of construction equipment & materials etc. required for true and satisfactory completion of the work under this contract.

3.1.14 **Workmanship'** shall mean the method/manner in which the jobs of the different items, whether included in the schedule or not but are required for true & satisfactory completion of the work under this contract, are executed.

3.1.15 **"Contractor's Equipment"** means all appliances or things of whatsoever nature required for the purposes of execution of work and which are to be provided by the Contractor but does not include any Equipment/ Materials intended to form part of Works.

3.1.16 **'Specifications'** shall mean collectively all the terms and stipulations contained in this document including the conditions of contract, technical provisions and attachments thereto and list of corrections and amendments. **Drawings' means** collectively all the accompanying general drawings as well as all detailed drawings, which may be used from time to time or desired by DPL.

3.1.17 **'Approval'** shall mean the written approval of DPL and/the statutory authorities, wherever such authorities are specified by any codes or otherwise.

3.1.18 **'Manufacturer'** shall refer to the party proposing to design/engineering and construct in complete or in part a particular job/work at their works/premises.

3.1.19 **'Labourer'** shall mean all categories of labour engaged by the Contractor, his sub- contractors and his piece workers for work in connection with the execution of the works covered by the specifications. All these labourers will be deemed to be employed primarily by the Contractor.

3.1.20 **'Plant'/'Equipment'/'Stores'** means and include plant and machineries to be provided under the contract.

3.1.21 **'Delivery of Plant'/'Delivery of Equipment'** shall be deemed to take place on delivery of the plant/equipment in accordance with the terms of the contract complete in all respect after approval by DPL.

3.1.22 **'Tests on Completion'** shall mean all such tests as are prescribed by the specification to be made by the Contractor to the satisfaction of DPL before the plant and equipment are taken over by DPL and this also includes those tests not specifically mentioned in the specification but required under various BIS codes and relevant Electricity Acts and Rules.

3.1.23 **'Commissioning'** shall mean the satisfactory, continuous and uninterrupted operation of the equipment/work as specified after all necessary initial tests, checks and adjustments required at site for a period of at least 15 (fifteen) days to the satisfaction of DPL.

3.1.24 **"Completion of Facilities"** means that all the Facilities (or a specific part thereof where specific parts are specified in the SCC) have been completed operationally and structurally as per Technical Specifications and put in a tight and clean condition and that all work in respect of Pre-commissioning of the Facilities or such specific part thereof has been completed and Commissioning has been attained as per Technical specifications.

3.1.25 **'Urgent Works'** shall mean any urgent measures, which in opinion of the Engineer–in- Charge, become necessary at the time of execution and/or during the progress of work to obviate any risk of damage to the structure, or required to accelerate the progress of work or which become necessary for security or for any other/reason DPL may deem expedient.

3.1.26 "**MNRE**" shall mean Ministry of New and Renewable Energy, Government of India;

3.1.27 "**kWp**" shall mean Kilo-Watt Peak;

3.1.28 **"Ground"** shall mean project location.

3.1.29 **Month'/'Calendar month'** means not only the period from the first of a particular month, but also any period between a date in a particular month, and the date previous to the corresponding date in subsequent month unless specifically stated otherwise.

3.1.30 **'Week'** means seven consecutive calendar days.

3.1.31 **'Writing'** shall include any manuscript, type written, printed or other statement reproduced in any visible form.

3.1.32 **"Site"** means the place or places, where Works are to be executed by the Contractor or to which Equipment machinery are to be delivered, together with so much of the area surrounding the same as the Contractor shall with the consent of the Purchaser, use in connection with the work other than merely for the purposes of access.

3.1.33 The term '**Services'** shall mean all works to be undertaken by the contractor as laid down under the head "Scope of work" or elsewhere in the specification enclosed. When the words "approved", "subject to Approval". "As directed", "Accepted", "Permitted" etc. are used, the approval, judgment, direction etc. are understood to be a function of Company.

3.1.34 **'General Conditions'** shall mean all the clauses of General conditions of the proposed contract stated hereinafter. The specification shall mean the specification annexed to or issued with the General Conditions and shall include the schedule and drawings attached thereto

3.1.35 **'Date of Contract'/'Commencement Date'** shall mean the date on which Letter of Award will be issued.

3.1.36 '**Zero Date'** will be started from the date of issuance of Letter of Award (LOA).

3.1.37 "**Program**" means the Program to be submitted by the Contractor in accordance with GCC and any approved revisions thereto.

3.1.38 "**GCC**" means the General Conditions of Contract hereof.

3.1.39 **"SCC"** means the Special Conditions of Contract.

3.1.40 **"Trial Run" means the first continuous operation of the complete Plant & Equipment including all systems & sub-systems in the of**

mode Remote operation from SCADA for fourteen (14) days with continuous daily operation, trial shall necessarily include steady operation at its evacuation capacity successfully without any unforeseen/spurious tripping of any running equipment.

3.2. CONTRACT DOCUMENTS

3.2.1. CONSTRUCTION OF CONTRACT

The Contracts to be entered into between the Purchaser and the successful bidders shall be as under:

5 MWAC Ground Mounted Solar PV Project	
First Contract: Supply, Erection and Commissioning	**"PART A"** is for ex-works supply of Equipment / Materials & Transport, transit insurance, unloading, storage, handling. **"PART B"** is for works as per Schedule of Works (Electrical & Civil), at Site, installation services (including rates and prices for all material/ labour, Contractor's Equipment, temporary works, consumables and all matters and all matters and things of whatsoever nature of such works), training of Purchaser's personnel etc and all other services specified in the Contract Documents.
"**Second Contract**" : Comprehensive O&M.	Comprehensive Operation & maintenance for five (5) years which shall be conferred after successful completion of the project.

3.2.2. The award of these Contracts shall not in any way dilute the responsibility of the Contractor for the successful completion of the Works as per Contract

Documents.

3.2.3. If the successful bidder is a Bidding Consortium then the award of this contract will be conferred on the Lead Member. But all the member of the consortium shall be severally and jointly liable and responsible for execution of the contract. In the event of failure on the part of the second member of the consortium to perform its obligation, then the Lead member of the consortium shall be solely responsible to fulfil the remaining obligations of its partner for the purpose of completion of contract including statutory obligations(refer Clasuse-3.49).

3.2.4. Subject to **Sl. no.2** of the Contract Agreement, all documents forming part of the Contract (and all parts thereof) are intended to be correlative, complementary and mutually explanatory. The Contract shall be read as a whole.

3.2.5. Subsequent to signing of the Contract Agreement, the Contractor at his own cost shall provide the Purchaser with at least **six (6) copies** of the Contract Documents within **seven (07)** days after signing of the Contract Agreement.

3.2.6. The Contractor shall provide free of cost to DPL all the engineering data, drawings and descriptive materials submitted with the bid, complete set of his bid and bidding documents, copies of all the correspondence with DPL, etc. in at least **four (4)** copies to form a part of the Contract Documents after seven (07) days of the Letter of Award (**LOA**).

3.2.7. Endorsement of Terms

The failure of either party to endorse at any time any of the provisions of the Contract or any rights in respect thereto or to an option herein provided shall in no way be construed to be a waiver of such provisions, rights or option or in any way to effect the validity of the Contract. The exercise by either party of any of his rights herein shall not preclude or prejudice either party from exercising the same or any other right it may have hereunder.

3.2.8. Effect

The Contract shall be considered to come into force on the date of issuance LOA by PURCHASER to the Contractor which may be in the form of a fax, E-MAIL or a Letter of Award. The Time for Completion shall be reckoned from that date.

3.2.9. All Contract Documents, all correspondence and communications to be given, and all other documentation to be prepared and supplied under the Contract shall be written in English, and the Contract shall be construed and interpreted in accordance with that language. If any of the Contract Documents, correspondence or communications are prepared in any language other than the governing language under this clause, the English translation of such documents, correspondence or communications shall prevail in matters of interpretation.

3.3. NOTICE

3.3.1. Notices shall be deemed to include any approvals, consents, instructions, orders and certificates to be given under the Contract. Unless otherwise stated in the Contract, all notices to be given under the Contract shall be in writing, and shall be sent by personal delivery, Registered post or e-mail followed by post confirmation to the address of the relevant party as mentioned in SCC

3.3.2. Any notice sent by registered post or speed post shall be deemed (in the absence of evidence of earlier receipt) to have been delivered **ten (10) days after dispatch**. In proving the fact of dispatch, it shall be sufficient to show that the envelope containing such notice was properly addressed, stamped and conveyed to the postal authorities for transmission by airmail or registered post.

3.3.3. Any notice delivered personally or sent by registered post shall be deemed to have been delivered if the same is properly received by the other party.

3.3.4. Either party may change its address at which notices are to be received by giving ten (10) days' notice to other party in writing.

3.4. INTERPRETATION

3.4.1. Singular and Plural

The singular shall include the plural and the plural the singular, except where the context otherwise requires.

3.4.2. Headings

The headings and marginal notes in the General Conditions of Contract are included for ease of reference, and shall neither constitute a part of the Contract nor affect its interpretation.

3.4.3. Persons

Words importing persons or parties shall include firms, corporations and government entities.

3.4.4. INCOTERMS

Unless inconsistent with any clause of the Contract, the meaning of any trade term and the rights and obligations of parties there under shall be as prescribed by the Incoterms.

Incoterms means international rules for interpreting trade terms published by the International Chamber of Commerce (latest edition), 38 Cours Albert 1er, 75008 Paris, France.

3.4.5. Entire Agreement

Subject to **GCC.3.17.3,** the Contract constitutes the entire agreement between the Purchaser and Contractor with respect to the subject matter of Contract and supersedes all communications, negotiations and agreements (whether written or oral) of parties with respect thereto made prior to the date of Contract..

3.4.6. Independent Contractor

The Contractor shall be an independent Contractor (if JV/Consortium then Lead Member) performing the Contract.

The Contractor shall be solely responsible for the manner in which the Contract is performed. All employees, representatives or Sub-contractors engaged by the Contractor in connection with the performance of the Contract subject to approval of DPL shall be under the complete control of the Contractor and shall not be deemed to be employees of the Purchaser, and nothing contained in the Contract or in any subcontract awarded by the Contractor shall be construed to create any contractual relationship between any such employees, representatives or Sub-contractors and the Purchaser.

3.4.7. Non-Waiver

Any waiver of Purchaser's/bidder's rights, powers or remedies under the Contract must be in writing, must be dated and signed by an authorized representative of the Purchaser/bidder granting such waiver, and must specify the right and the extent to which it is being waived.

Subject to above, no relaxation, forbearance, delay or indulgence by either party in enforcing any of the terms and conditions of the Contract or the granting of time by either party to the other shall prejudice, affect or restrict the rights of that party under the Contract, nor shall any waiver by either party of any breach of Contract operate as waiver of any subsequent or continuing breach of Contract.

3.4.8. Severability

If any provision or condition of the Contract is prohibited or rendered invalid or unenforceable, such prohibition, invalidity or unenforceability shall not affect the validity or enforceability of any other provisions and conditions of the Contract.

3.5. GOVERNING LAW

The Contract shall be governed by and interpreted in accordance with laws in force in India including any such Laws passed or made or coming into force during the period of the Contract. The Courts of Kolkata under the superintendence of High Court of Calcutta shall have exclusive jurisdiction in all matters arising under the Contract.

3.6. SATTLEMENT OF DISPUTE

3.6.1 If any dispute or differences of any kind whatsoever shall arise between the owner and the

Contractor, arising out of the Contract for the performance of the work whether during the progress of the Work or after its completion or whether before or after the termination, abandonment or breach of the Contract, it shall, in the first place, be referred to the settled by the Engineer, to be appointed for the purpose by Managing Director of DPL, who, within a period of thirty (30) days after being requested by either party to do so, shall give written notice of his decision to the Purchaser and the Contractor. Save as hereinafter provided, such decision in respect of every matter so referred shall be final and binding upon the parties until the completion of the Work and shall forthwith be given effect to by the Contractor who shall proceed with the Work with all due diligence, whether he or the Purchaser required arbitration as hereinafter provided or not.

If after the Engineer has given written notice of this decision to the parties, no claim to arbitration has been communicated to him by either party within thirty (30) days from the receipt of such notice, the said decision shall become final and binding on the parties.

In the event of the Purchaser falling to notify his decision as aforesaid within thirty (30) days after being requested as aforesaid, on or in the event of either the Purchaser or the Contractor being dissatisfied with any such decision, or within thirty (30) days after the expiry of the first mentioned period of thirty (30) days, as the case may be, either party may require that the matters in dispute be referred to arbitration as hereinafter provided.

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The arbitration shall be conducted by the three arbitrators, one each to be nominated by the Purchaser and the Contractor and the third to be named by the President of the Institute of Engineers , Bengal/Kolkata chapter.

The decision of the majority of the arbitrators shall be final and binding upon the parties. The expense of the arbitration shall be paid as may be determined by the arbitrators. The Expense towards Arbitration shall be bourne in equal amounts by DPL and the Bidder.The arbitrators may, from time to time, with the consent of all the parties enlarge the time of making the award. In the event of any of the aforesaid arbitrators dying, neglecting, resigning or being unable to act for any reason, it will be lawful for the party concerned to nominate another arbitrator in place of the outgoing arbitrator.

The arbitrator shall have full powers to review and/or revise any decision, opinion, directions, certification or valuation of the Purchaser in consonance with the Contract, and neither party shall be limited in the proceedings before such arbitrators to the evidence or arguments put before the Purchaser for the purpose of obtaining the said decision.

3.6.2 No decision given by the Purchaser in accordance with the foregoing provisions shall disqualify him as being called as a witness or giving evidence before the arbitrators on any matter whatsoever relevant to the dispute of difference referred to the arbitrators as aforesaid.

3.6.3 The Arbitration clause can be invoked at any time during the currency of the contract or after the expiry/ termination or closure of the contract.

3.6.4 The arbitration proceedings shall be in accordance with the prevailing Arbitration laws of India as amended or enacted time to time

3.6.5 The existence of any dispute(s) or difference(s) or the initiation or continuance of the Arbitration proceedings shall not permit the parties to postpone or delay the performance by the parties of their respective obligations pursuant to the Contract.

3.6.6 The seat of arbitration shall be Kolkata, West Bengal, India.

3.7. COMPLIANCE WITH LAWS

3.7.1 Compliance with Laws, statutes, regulations

The Contractor shall, in all matters arising in the performance of the Contract, comply with in all respects, give all notices and pay all fees required by the provisions of any national or state statute, ordinance or other law or any regulation or bye-law of any duly constituted authority.

3.7.2 Statutory Obligations

1) The agency shall have to ensure all statutory compliances and have to observe, perform and comply with all the provisions as laid down under General Terms & conditions of the NIT and specifically adhere to the provisions of (a) The Contract Labour (R&A) Act 1970, (b) The Payment of Wages Act, 1936, (c) The Employees' Provident Fund & Misc. Provisions Act, 1952 (d) The Payment of Bonus Act, 1965, (e) The Minimum Wages

Act, 1948, (f) The Employees' Compensation Act, 1923 and other law of the land as may be applicable at present and due course. However, DPL will not bear any financial liability of the workers to be deployed.

2) "The Registration under the Building and other Construction Workers (Regulation of Employment and Conditions of Service) Act 1996, has to be taken by the successful bidder. And the successful bidder have to undertake necessary compliances of the Act from their part and on the total job value of the contract quoted by them without any further reimbursement from DPL.

   DPL will not bear any financial liabilities on account of applicable **Cess** under the Building and other Construction Workers (Regulation of Employment and Conditions of Service) Act 1996".

3) The Contractor shall also be responsible and liable for such statutory compliance as stated hereinabove in respect of the Sub Contractors engaged by them. At any point of time non-compliance of the statutory provisions in respect of contract labour engaged in the job by the contractor/sub-contractors may attract penal action against Contractor from the law enforcing authorities. All liabilities arising out of the non-compliance of the Law of the land will have to be borne by the Contractor and PURCHASER will not be responsible in any manner whatsoever for the same.

3.7.3 The Contractor hall indemnify and hold harmless the Purchaser from and against any and all liabilities, damages, claims, fines, penalties and expenses of whatever nature arising or resulting from the violation of such laws by the Contractor or its personnel, including the Sub-contractors and their personnel, but without prejudice to **GCC 3.10.1** hereof.

B. SUBJECT MATTER OF CONTRACT

3.8. SCOPE OF WORKS

3.8.1 Unless otherwise expressly provided in the Technical Specifications, the Contractor's obligations cover the provision of all Equipment/ Materials including spares and the performance of all services required for the design, the manufacture (including quality assurance, construction, installation, associated civil, structural and other construction works and delivery) of the Equipment/Materials and the installation, commissioning, completion of the Works and carrying out completion tests for the Works in accordance with the plans, procedures, Specifications, drawings, codes and any other documents as specified in the Technical Specifications. Such Specifications include, but are not limited to, the provision of supervision and engineering services; the supply of labour, materials, Equipment, spare parts (as specified in **GCC 3.8.3**) and accessories; Contractor's Equipment; construction utilities and supplies; temporary materials, structures and facilities; transportation (including, without limitation, unloading and hauling to, from and at the Site), insurance and storage, except for those supplies, works and services that will be provided or performed by the Purchaser.

3.8.2 The Contractor shall, at no extra cost to the Purchaser, unless specifically excluded in the Contract, perform all such Works and/or supply all such items and materials not specifically mentioned in the Contract but that can be reasonably inferred from the Contract as being required for attaining successful completion of the Works as if such Works and Materials were expressly mentioned in the Contract.

3.8.3 The Contractor agrees to supply spare parts required for the operation and maintenance of the Works as per provision of subsequent sub clauses of **GCC 3.8.3.**

3.8.3.1. All the spares for the Equipment/material under the Contract will strictly conform to the Specification and other relevant documents and will be identical to the corresponding main Equipment/Materials supplied under the Contract and shall be fully interchangeable.

3.8.3.2. All the spares covered under the Contract shall be manufactured along with the main Equipment/ Materials as a continuous operation and the delivery of the spares will be effected along with the main Equipment/ Materials in a phased manner and the delivery would be completed by the respective dates for the various categories of Equipment/ Materials as per the agreed Program.

3.8.3.3. The Contractor will provide the Purchaser with the manufacturing drawings, catalogues, assembly drawings and any other document required by the Purchaser so as to enable the Purchaser to identify the spares. Such details will be furnished to the Purchaser during design and drawing approval.

3.8.3.4. In addition to the spares covered in the Scope of Work, if the Purchaser further identifies certain items of spares, the Contractor will submit the prices and delivery quotation for such spares within thirty (30) days of receipt of such request with a validity period of six (6) months for consideration by the Purchaser and placement of order for additional spares, if the Purchaser so desires.

3.8.3.5. The quality plan and the inspection requirement finalized for the main Equipment/ Materials will also be applicable to the corresponding spares.

3.8.3.6. The Contractor will provide the Purchaser with all the addresses and particulars of his Sub-contractors while placing the order for Equipment/ Materials covered under the Contract and will further ensure with his vendors that the Purchaser, if so desires, will have the right to place order for spares directly on them on mutually agreed terms based on offers of such vendors.

3.8.3.7. The Contractor shall guarantee the long-term availability of spares to the Purchaser for the full life of the Equipment/ Materials covered under the Contract. The Contractor shall guarantee that before going out of production of spare parts of the Equipment/ Materials, he shall give the Purchaser at least twelve (12) months advance notice so that the latter may order his bulk requirement of spares, if he so desires. The same provision will also be applicable to Sub-Contractor of any spares by the Contractor or his Sub-Contractors. Further, in case of discontinuance of manufacture of any spares by the Contractor or his Sub-Contractors, the Contractor will provide the Purchaser, two (2) years in advance, full manufacturing drawings, material specifications and technical information required by the Purchaser for the purpose of manufacture of such items..

3.8.3.8. In case the Contractor fails to supply the spares in the terms stipulated above, the Purchaser shall be entitled to purchase the same from the alternate sources at the risk and the cost of the Contractor and recover from the Contractor, the excess amount paid by the Purchaser, if any, over the rates worked on the above basis. In the event of such risk purchase by the Purchaser, the purchases will be as per the Policy and Procedures of the Purchaser prevalent at the time of such purchases and the Purchaser at his option may include a representative from the Contractor in finalizing the purchases.

3.8.3.9. It is expressly understood that the final settlement between the parties , in terms of relevant clauses of the Contract Documents shall not relieve the Contractor of any of his obligations under the provision of long term availability of spares unless otherwise discharged expressly in writing by the Purchaser.

3.8.3.10 The Contractor shall warrant that all spares supplied will be new and in accordance with the Contract Documents and will be free from defects in design, material and workmanship

3.8.4 COMPREHENSIVE OPERATION AND MAINTENANCE

Operation part consists of deputing necessary manpower necessary to operate the Solar Photovoltaic Power Plant at the full capacity. Operation procedures such as preparation to starting, running, routine operations with safety precautions, monitoring etc., shall be carried out as per the manufacturer's instructions to have trouble free operation of the complete system.

Maintenance contract shall be commence after completion of successful trial run of the plant. 05 (five) years comprehensive operation & maintenance of the plant shall also be the scope of work. The contractor needs to submit 03 (three) sets of original (from OEM) comprehensive user's manual and 02 (two) sets of original (from OEM) Operation and Maintenance manual book after commissioning of the plant.

The scope of maintenance shall include supply of spare parts, replacement of all damaged equipment and accessories with new one within the price of yearly maintenance charge.

Time for repair/ replacement of equipment or any works in case of any major failure will be granted by the Controlling Officer considering the type of failure and receiving written prayer from the contractor for the same. But in general the downtime will be 72 hours.

The period of unavailability of Grid & Force Majeure Conditions will not be considered as downtime. However bidder shall keep records of these outages including proper reasons with due signature of DPL site official.

Arrangement of security (Minimum 3 nos. of security personnel) shall be a scope of the operation and maintenance for each shift.

The contractor shall at his own expense provide all amenities to his workmen as per applicable laws and rules. The contractor shall arrange sufficient transportation arrangement (24X7) for the operation and maintenance purpose. The maintenance includes Routine and preventive, Breakdown and Capital Maintenance which shall be but not limited to the following.

3.8.4.1. **Routine and preventive maintenance:**

This shall include:

i. Regular cleaning of PV modules.
ii. Checking & tightening of all electrical connections and mechanical fittings.
iii. Earth resistance of plant as well as individual earth pit is to be measured and recorded every month. If the earth resistance is higher than the stipulated in the NIT, suitable action to be taken to bring down the same.
iv. Cleaning of Inverter and other electrical equipment.
v. Routine maintenance as recommended by the original equipment manufacturer.

The contractor shall be responsible to carry out routine and preventive maintenance and replacement of each and every damaged/faulty component/ equipment of the power plant and he shall provide all labour, material, consumables etc for routine and preventive maintenance at his own cost.

Preventive maintenance if require long stoppage of the Plant, same shall be carried out preferably during the non-sunny days or evenings. Prior information shall be provided to the Employer for such preventive maintenance prior to start.

3.8.4.2. **Breakdown maintenance:**

Breakdown maintenance shall mean the maintenance activity including repairs and replacement of any component or equipment of the power plant which is not covered by routine and preventive maintenance and which is required to be carried out as a result of sudden failure/breakdown of that particular component or equipment while the plant is running. The contractor shall be responsible to carry out breakdown maintenance of each and every component of the power plant and he shall provide the required manpower, materials, consumables, components or equipment etc. for breakdown maintenance at his own cost irrespective of the reasons of the breakdown/failure.

A maintenance record is to be maintained by the operator/ O&M-in-charge to record the regular maintenance work carried out as well as any breakdown maintenance along with the reasons for the breakdowns and steps taken to attend the breakdown, duration of the breakdown etc.

The Contractor will attend to any breakdown jobs immediately for repair/ replacement/ adjustments and complete at the earliest working round the clock. During breakdowns (not attributable to normal wear and tear) in O&M period, the Contractor shall immediately report the accidents, if any, to the Employer showing the circumstances under which it happened and the extent of damage and/or injury caused.

3.8.4.3. **Capital maintenance:**

Capital Maintenance shall mean the major overhaul of any component or equipment of the power plant which is not covered by routine, preventive and breakdown maintenance which may become necessary on account of excessive wear & tear, aging, which needs repair/replacement. The capital maintenance of power plant and all civil structures shall normally be planned to be carried out on an annual basis. For this purpose a joint inspection by the supplier and DPL shall be carried out of all the major components of the power plant, about two months in advance of the annual maintenance period, in order to ascertain as to which components of the power plant require capital maintenance. In this regard the decision of DPL will be final and binding.

However, if the condition of any plant component wants its capital maintenance at any other time, a joint inspection of DPL and supplier shall be carried out immediately on occurrence of such situation and capital maintenance shall be carried out by arranging the shutdown of the plant/part of the plant, if required, in consultation with concerned authorities. The decision of DPL shall be final and binding to the contractor.

The capital maintenance also includes painting of mechanical & civil structures etc.

The contractor shall undertake necessary maintenance /troubleshooting work of the Solar PV Power Systems. Down time shall not be more than 72 hours from time of occurrence of such faults. Adequate measures should be taken for prevention of wear and tear of the machines. Solar PV Power System is to be designed to operate with a minimum of maintenance.

The scope of support service provides preventive maintenance as & when necessary within the contract period and break down maintenance in the event of malfunctions, which prevent the operation of the power system or part of it within the stipulated time period & free replacement of spares required for maintenance.

The contractor will provide Spare parts & measuring instruments.

The contractor shall submit the detailed schedule for routine and preventive maintenance before final commissioning of the plant. The contractor shall also submit Detailed Report to DPL for any capital or breakdown maintenance mentioning the cause of breakdown, actions taken to resolve that issue and preventive measures taken to avoid failure/damage/loss of generation due to similar incidents/accidents in future etc. within 07 (seven) days from the date of recovery.

3.8.4.4. **Operation & Maintenance Report:** Maintenance register / log book must be maintained at site. However, quarterly maintenance and monthly generation report of each location as per format duly approved by DPL must be submitted in original by the contractor to DPL with certification of DPL by the contractor within 30 day of the following month. Failing of which maintenance service will be deemed to be not attended.

Daily work of the operation and maintenance in the Solar Photovoltaic Power Plant involves regular (at least once in a week) cleaning of Modules, logging the voltage, current, power factor, power and energy output of the Plant at different levels. The operator shall also note down time/ failures, interruption in supply and tripping of different relays, reason for such tripping, duration of such interruption etc.

The other task of the operators is to check battery voltage-specific gravity and

temperature. The operator shall record monthly energy output, down time, etc.

Note: During comprehensive O&M period, DPL may do any kind of performance test, analysis, audit etc. on Module, Inverter and any other equipment or overall performance of the solar plant by any reputed 3rd party / Institute like NISE from their own. O&M contractor shall assist the DPL team with manpower and tools and tackles during such test/analysis. As per recommendation of the performance test, necessary corrective action / replacement etc. should be taken without any extra cost to DPL by the successful bidder who is doing the

Comprehensive O&M job.

3.8.4.5. **Deployment of Competent Manpower:**

As the vendor is to satisfy the NMGG of the plant as per clause no. 4.10.1.

3.8.4.6. **Cost & Payment Terms for O&M Contracts:**

O&M contract value: **7.5%** of the Basic Project Cost (excluding GST) for 5(five) years.

Payment Terms:

The payment shall be made on quarterly basis and the Eligible amount will be due for payment after the certification by the Controlling Officer within 45 (forty five) days subject to satisfactory performance and submission of maintenance report in regular basis as mentioned in Clause No. 3.8.4 of GCC. The total yearly amount payable shall be subject to following table and as per certification of the Controlling Officer.

O&M Operation Year end	Amount of Payment
1st Year	1.0 % of the contract value
2nd Year	1.0 % of the contract value
3rd Year	1.5 % of the contract value
4th Year	1.5 % of the contract value
5th Year	2.5 % of the contract value

GST will be paid by DPL at actual subject to submission of proper document.

3.9. VARIATION, ADDITIONS AND OMISSIONS

The Contractor shall not modify the scope of work except under direction in writing by the DPL. The quantities provided in the schedule of works are fixed.

3.10. OBLIGATIONS OF THE CONTRACTOR

3.10.1 The Contractor shall, in accordance with the Contract, with due care and diligence, carry out the Works as necessary for successful completion of all the obligations, within the time for completion. All statutory norms has to be followed throughout the project and final No Objection Certificate has to be obtained from Site HR&A department of DPL.

3.10.2 The Contractor confirms that it has entered into this Contract on the basis of a proper examination of the conditions and circumstances at the Site affecting the Contract Price, and on the basis of information that the Contractor could have obtained from a visual inspection of the Site including existing roads and bridges and other means of access to the Site, presence of obstructions on the Site. The Contractor acknowledges that any failure to acquaint itself with all such data and information shall not relieve its responsibility for properly estimating the difficulty or cost of successfully performing the Works

3.10.3 The Contractor shall acquire in its name all permits, approvals and/or licenses from all local, state or national government authorities or public service undertakings in the country/ state where the Site is located that are necessary for the performance of the Contract.

3.10.4 The Contractor shall arrange/ construct at his own cost any storage/access, structures, bridges and approach to the work sites from public roads as may be required for execution of Works.

3.10.5 Contractor shall be responsible for all necessary statutory compliance in respect of the employees deployed by them or by the sub-contractor(s) to execute the contract. However, Form no. V for obtaining labour license under the contract labour (R&A) act , 1970 and rules framed there under shall only be issued to the Contractor.

3.11. OBLIGATIONS OF THE PURCHASER

3.11.1 The Purchaser shall provide the contractor physical possession of the Site and access thereto except where providing access is included in the scope of work of the Contractor on or after the date(s) of issuance of LOA.

3.11.2 The Purchaser shall acquire and pay for all permits, approvals and/or licenses from all local, state or national government authorities or public service undertakings where the Site is located, which such authorities or undertakings require the Purchaser to obtain in the Purchaser's name for the execution of the Contract (they include those required for the performance by both the Contractor and the Purchaser of their respective obligations under the Contract).

3.11.3 Without prejudice to the obligations of the Contractor under the Contract Agreement, if requested by the Contractor, the Purchaser shall use its best endeavours to assist the Contractor in obtaining in a timely and expeditious manner all permits, approvals and/or licenses necessary for the execution of the Contract from all local, state or national government authorities or public service undertakings that such authorities or undertakings require the Contractor or Sub-contractors or the personnel of the Contractor or Sub-contractors, as the case may be, to obtain.

3.11.4 The Bidder shall provide sufficient, properly qualified operating and maintenance personnel; shall supply, other materials and facilities and shall perform work and services of whatsoever nature for properly carry out Commissioning, Trial run and Guarantee Tests at or before the time specified in the Program furnished by the Contractor under GCC 3.23.2 hereof and in the manner thereupon specified or as otherwise agreed upon by the Bidder and make available all raw materials, utilities, lubricants, chemicals, catalysts etc. at site.

C. PAYMENT

3.12. CONTRACT PRICE

3.12.1 The Contract Price shall be specified in the Contract Agreement.

3.12.2 Subject to **GCC 3.11.1** hereof, the Contractor shall be deemed to have satisfied itself as to the correctness and sufficiency of the Contract Price, which shall, except as otherwise provided for in the Contract, cover all its obligations under the Contract.

3.12.3 No interest shall be paid by DPL for delay in making payment.

3.13. TERMS AND PROCEDURE FOR PAYMENT

The payments to the Contractor for the performance of the Contract will be made by the Purchaser as per terms and conditions specified in **Clause 4.3.3.7** No payment made by the Purchaser herein shall be deemed to constitute acceptance by the Purchaser of the Works or any part thereof. The currency of payment shall be Indian rupees**.**

3.14. SECURITIES

3.14.1 Mobilisation Advance Payment

10 % of the Contract Price (Supply & Erection contract separately for each LOA) excluding price of Mandatory spares will be paid as Mobilization advance to the contractor against submission of BG (Annexure-3) equivalent to 110 % of the Mobilization Advance. **The mobilization advance will be interest free** for the scheduled Completion period from the date of LOA of the Project against 110 % unconditional BG with a claim period of 90 days initially. The payment of Mobilisation advance shall be subject to deduction of tax at source as per law. The unadjusted mobilization advance if any after the schedule completion period, will be recovered with interest at the Rate of 8.5% p.a.

This Bank Guarantee shall be returned to the Contractor after full recovery of advance with interest if any and against request by the Contractor. The validity of Bank Guarantee would require to be extended by the Contractor, if so required by the Purchaser. The details terms and Condition of the Mobilisation Advance has been specified in **Clause No. 4.3.4.2**.

(Note:

i) Mobilization advance shall be considered excluding GST for Main Supply excluding Mandatory Spares for each LOA;

ii) Mobilization advance shall be considered including GST for Erection and services part of each LOA.)

3.14.2 Contract Performance Bank Guarantee(PBG)

3.14.2.1 The Contractor shall furnish an unconditional and irrevocable Bank Guarantee in favour of the Purchaser as per **Annexure-2** of **Section VII**, towards Performance guarantee for faithful and due fulfilment of all obligations under the Contract after placement of LOA. Bank Guarantee shall be furnished for an amount equal to three percent (3%) of the Project cost plus GST, from a Scheduled commercial Bank in India. The Bank Guarantee shall be valid for Seventy Two (72) months subject to satisfactory completion of Defect Liability Period including comprehensive O&M period (Five year) with further claim period for ninety (90) days thereafter. The value of the Bank Guarantee is not to be construed as limiting the damages under Defects Liability Period. The Purchaser reserves the right to verify the authenticity of the Bank Guarantee from the issuing bank.

3.14.2.2 The Contract Performance Bank Guarantee is liable to be invoked on demand of PURCHASER, for any breach under the Contract irrespective of any dispute or difference between PURCHASER and the Contractor, pending before any court, tribunal or any other authority,

3.14.2.3 The Performance Bank Guarantee shall be returned to the Contractor within ninety (90) days after receipt of application for release of Performance Bank Guarantee along with certification regarding completion of Defects Liability Period and O&M period. No claim shall be made against the Performance Guarantee after the issue of Defects Liability Certificate. However, no costs shall be paid for the Bank Guarantee by the Purchaser, irrespective of date of release.

3.15. TAXES, DUTIES AND OTHER LEVIES

3.15.1 *Except as otherwise specifically provided in the Contract, the Contractor shall bear and pay all the applicable taxes and duties (**GST**, **Custom related Duties, BOCW Labour welfare cess as per laws** and other Govt. Tax or Levies) and charges assessed on the Contractor, its Sub-contractors in connection with this*

Works. Excluding GST all applicable taxes, duties and levies, (eg. Customs related Duties, BOCW, Insurance etc.) where applicable and payable on Equipment/Materials, components, sub-assemblies, raw materials and any other item required for manufacture of finished Equipment/Material and services or completion of Works shall be deemed to have been included in base price.

3.15.2 The Contractor shall be solely responsible for the taxes that may be levied on the Contractor's persons or on earnings of any of his employees and shall hold the Purchaser indemnified and harmless against any claims that may be made against the Purchaser. The Purchaser does not take any responsibility whatsoever regarding taxes under Indian Income Tax Act, for the Contractor or his personnel. If it is obligatory under the provisions under the Indian Income Tax Act, deduction of Income Tax at source shall be made by the Purchaser.

3.15.3 Bidder shall submit the base price and GST in their Bid as per clause 3.15.1 & 3.15.5.

3.15.4 GST will be reimbursed at actual by the DPL to the bidder on submission of appropriate supporting document.

3.15.5 Customs related Duty (if Applicable)

3.15.5.1 The Contractor is requested to identify the value of imported components, if any and its price and accordingly the custom duty of its and should be included as base price in the price bid.

3.15.5.2 In case of any change in custom duty, entry tax. etc. during the currency of the contract, the same shall be borne by the bidder. No reimbursement shall be allowed.

3.15.5.3 All taxes and duties payable outside India shall be borne and paid by the contractor. No claim will be entertained by the PURCHASER whatsoever on this account.

3.15.6 Advance Payment

The GST payable if any in respect of advance payment may be paid to the contractor by the PURCHASER in addition to the amount of advance, subject to providing appropriate documentary evidence that GST on advance has been paid by the contractor.

3.15.7 Tax Deduction at Source (TDS) towards Income Tax/Other Taxes

Deduction of Tax at source at prevailing rate shall be effected by the PURCHASER before payment as a statutory obligation wherever applicable. Income tax and all other taxes as applicable as per statutory obligation/s enactments shall be progressively deducted from the payments released to Contractor, by the PURCHASER, for depositing with the Income tax/other Tax authorities as per Income Tax Act.

TDS on Works Contract shall be deducted at source by the PURCHASER as per statutory provisions. However, if "No Deduction at Source Certificate" is furnished from the Tax Authorities by the Contractor, deduction of TDS on Works Contract shall not be effected.

3.15.8 Personal Income Tax & Cess

Income Tax and cess, if any payable by the Contractor's Sub-contractor's employees shall be paid by the said employees directly, and the PURCHASER shall not be liable to pay the income tax & cess payable by the employee of the contractor/sub-contractor and the purchaser is not responsible for filing the tax returns of contracts employees/experts.

3.15.9 Reverse Charge Mechanism

In case the liability to discharge GST is on the employer under reverse charge mechanism, then the said fact should be clearly mentioned on the face of the invoice. Further, GST should not be charged by the vendor in such cases.

3.15.10 It shall be responsibility of the Contractor to comply with all the requirements prescribed in the GST Act and Rules as may be applicable in respect of the activities/supply made by them under the contract to enable the PURCHASER to avail entire input tax credit on timely basis. It is the responsibility of the vendor to comply with the following key compliance requirements, failing which the Contractor shall be responsible of any loss of tax credit or any other cost including interest, penalty, etc that may levied or recovered from the employer.

i. The contractor shall issue a proper tax invoice containing all the particulars as prescribed in the GST Invoice.
ii. The Contractor shall deposit the GST amount due to the Government on a timely basis.
iii. The Contractor shall file the periodic statements/returns as per the provisions of GST Law on a timely basis and include therein details of all the invoices raised during the relevant month under the present contract.
iv. The Contractor shall support DPL on a best effort and timely basis to sort out the discrepancies communicated by GSTIN, if any.

3.15.11 Under GST regulation, taxes are levied on deductions under Liquidated Damage (LD). Such taxes will be charged extra by DPL. The rate of such tax on LD would be as per laws applicable at the time of imposition of LD when a Debit Note/Invoice is raised by DPL. This is because LD is a post-delivery/performance event & is not part of initial price bid.

D. INTELLECTUAL PROPERTY

3.16. PATENT RIGHTS & ROYALTIES

3.16.1 Royalties and fees for patents covering Equipment/Materials, articles, apparatus, devices or processes used in the Works shall be deemed to have been included in the Contract Price. The Contractor shall satisfy all demands that may be made at any time for such royalties or fees and he alone shall be liable for any damages or claims for patent infringements and shall keep the Purchaser indemnified in that regard.

3.16.2 The Contractor shall, subject to the Purchaser’s compliance with GCC.16.3, indemnify and hold harmless the Purchaser , his successors or assignees ,its employees and officers from and against any and all suits, actions or administrative proceedings, claims, demands, losses, damages, costs, and expenses of whatsoever nature, including attorney's fees and expenses, which the Purchaser may suffer as a result of any infringement or alleged infringement of any patent, utility model, registered design, trademark, copyright or other intellectual property right registered or otherwise existing at the date of the Contract by reason of : (a) the installation of the Works by the Contractor or the use of the Works in the country where the Site is located; and (b) the sale of the products produced by the Works in any country. Such indemnity shall not cover any use of the Works or any part thereof other than for the purpose indicated by or to be reasonably inferred from the Contract, any infringement resulting from the use of the Works or any part thereof, or any products produced thereby in association or combination with any other Equipment/ Materials not supplied by the Contractor, pursuant to the Contract Agreement.

3.16.3 If any proceedings are brought or any claim is made against the Purchaser arising out of the matters referred to in **GCC 3.16.2**, the Purchaser shall promptly give the Contractor a notice thereof, and the Contractor may at its own expense and in the Purchaser's name conduct such proceedings or claim and any negotiations for the settlement of any such proceedings or claim. If the Contractor fails to notify the Purchaser within thirty (30) days after receipt of such notice that it intends to conduct any such proceedings or claim, then the Purchaser shall be free to conduct the same on its own behalf. Unless the Contractor has so failed to notify the Purchaser within the thirty (30) day period, the Purchaser shall make no admission that may be prejudicial to the defence of any such proceedings or claim.

3.16.4 The Purchaser shall, at the Contractor's request, afford all available assistance to the Contractor in conducting such proceedings or claim, and shall be reimbursed by the Contractor for all reasonable expenses incurred in so doing.

3.16.5 All design and drawings submitted by the Contractor will be the property of Purchaser. The Purchaser reserves the right to use the same in its future project without any further reference and additional charges to the Contractor for such use.

3.16.6 The Purchaser's Drawings, Specification and other information submitted by the Purchaser to the Contractor shall remain the property of the Purchaser. They shall not, without the consent of the Purchaser, be used, copied or communicated to a third party by the Contractor unless necessary for the purposes of the Contract. Any error in any such drawing/Specification etc. shall not absolve the Contractor of his responsibility.

3.17. CONFIDENTIAL INFORMATION

3.17.1 The Purchaser and the Contractor shall keep confidential and shall not, without the written consent of the other party hereto, divulge to any third party any documents, data or other information furnished directly or indirectly by the other party hereto in connection with the Contract, whether such information has been furnished prior to, during or following termination of the Contract. Notwithstanding the above, the Contractor may furnish to its Sub-Contractor(s) such documents, data and other information it receives from the Purchaser to the extent required for the Sub-contractor(s) to perform its Works under the Contract, in which event the Contractor shall obtain from such Sub-contractor(s) an undertaking of confidentiality similar to that imposed on the Contractor under this Clause GCC 3.17.

3.17.2 The obligation of a party under GCC 3.17.1 above, however, shall not apply to that information which

a. now or hereafter enters the public domain through no fault of that party
b. can be proven to have been possessed by that party at the time of disclosure and which was not previously obtained, directly or indirectly, from the other party hereto
c. otherwise lawfully becomes available to that party from a third party that has no obligation of confidentiality

3.17.3 The above provisions of this Clause GCC 3.17 shall not in any way modify any undertaking of confidentiality given by either of the parties hereto prior to the date of the Contract in respect of the Works or any part thereof

3.17.4 The provisions of this Clause GCC 3.17 shall survive termination, for whatever reason, of the Contract.

3.18. ADVERTISING

Any advertising stating the subject of this Contract by the Contractor in India or in foreign countries shall be subject to approval of the Purchaser prior to the publication. Publication of approved articles, photographs and other similar materials shall carry acknowledgment to the Purchaser.

E. WORK EXECUTION

3.19. PURCHASER'S REPRESENTATIVE

3.19.1 The PURCHASER shall appoint an experienced engineer designated as the **Project Manager** who shall carry out the functions and obligations of the Purchaser under the Contract.

3.19.2 The PURCHASER may from time to time appoint some other person as the Project Manager in place of the person previously so appointed, and shall give a notice of the name of such other person to the Contractor without delay. The PURCHASER shall take reasonable care to see that no such appointment is made at such a time or in such a manner as to impede the progress of Works. The Project manager shall represent and act for the Purchaser at all times during the currency of the Contract.

3.19.3 Any decision, instruction or approval given by the Project Manager to the Contractor shall have the same effect as if it had been given by the Purchaser.

3.19.4 All notices, instructions, orders, certificates, approvals and all other communications under the Contract shall be given by the Project Manager, except as herein otherwise provided.

3.19.5 The Project Manager may authorize his representative as site-in-charge for the Works. The Project Manager will also be the consignee officer for the Works.

3.20. CONTRACTOR'S REPRESENTATIVE

3.20.1 If the Contractor’s representative is not named in the Contract, then the Contractor shall appoint the Contractor’s representative and shall request the Purchaser in writing to approve the person so appointed. If the Purchaser makes no objection to the appointment within 7 days, the Contractor’s representative shall be deemed to have been approved. If the Purchaser objects to the appointment within fifteen (15) days giving the reason thereof, then the Contractor shall appoint a replacement of such objection, and the foregoing provisions of this Sub-Clause GCC 3.20.1 shall apply thereto.

3.20.2 The Contractor’s representative shall represent and act for the Contractor at all times during the currency of the Contract and shall give to the Project Manager all the Contractor’s notices, instructions, information and all other communications under the Contract.

3.20.3 Any instruction or notice which the Purchaser gives to the Contractor's representative(s) shall be deemed to have been given to the Contractor. An instruction book shall be kept at Site to record instruction of the Purchaser or his representative at the time of Site visit.

3.20.4 The Contractor shall not revoke the appointment of the Contractor’s representative without the Purchaser’s prior written consent, which shall not be unreasonably withheld. If the Purchaser consents thereto, the Contractor shall appoint some other person as the Contractor’s Representative, pursuant to the procedure set out in GCC 3.20.1

3.20.5 The Contractor’s representative may, subject to the approval of the Purchaser within 7 days, at any time delegate to any person any of the powers, functions and authorities vested in him or her. Any such delegation may be revoked at any time. Any such delegation or revocation shall be subject to a prior notice signed by the Contractor’s representative, and shall specify the powers, functions and authorities thereby delegated or revoked. No such delegation or revocation shall take effect unless and until a copy thereof has been delivered to the Purchaser and the Project Manager. Any act or exercise by any person of powers, functions and authorities so delegated to him or her in accordance with this Sub-Clause GCC 3.20.5 shall be deemed to be an act or exercise by the Contractor's representative.

3.20.6 The Contractor shall in addition to a Contractor's representative, employ one or more competent representative(s) to superintend the carrying out of the Works at Site. Such representative shall be fluent to communicate in local language for day to day work. Their names and contact addresses shall be communicated in writing to the Purchaser before commencement of Works.

3.20.7 The Purchaser may, by notice to the Contractor, object to any representative or person employed by the Contractor in the execution of the Contract ,who, in the reasonable opinion of the Purchaser, may behave inappropriately, may be incompetent or negligent, or may commit a serious breach of the Site regulations provided under GCC.3.26.6. The Purchaser shall provide evidence of the same, whereupon the Contractor shall remove such person from the Site.

3.20.8 If any representative employed by the Contractor is removed in accordance with GCC 3.20.7 the Contractor shall, where required, promptly appoint a replacement

3.21. MILESTONE OF PROJECT

The whole work must be completed within **365 (Three Sixty five)** days from the date of issuance of 'Letter of Award'. The duration of milestone will be calculated from the date of 'Letter of Award'.

Sl. No.	Description	Completion Time
1.	Approval of Detailed Design Report	45 Days
2.	Finalizations of Plant layout including area leveling & grading.	4 Months
3.	Completion of construction of Inverter Control Room	6 months

4.	Supply of transformers at site	7 Months
5.	Completion of Supply of Modules & Module Mounting structure	10 Months
6.	Completion of Modules & Electrical Equipment Erection	11 Month
7.	Complete installation of transformer and testing	11 Months
8.	Evacuation of Power	12 Months

Detailed time schedule for the site work has to be prepared based on the aforesaid milestones. Bidder has to submit the Detailed Design Report with complete time schedule within the milestones aforesaid. At the time of execution if delay occurs time extension may be allowed on the basis from request from Bidder. However all such time extension order shall be without prejudice to the terms and condition of the contract.

DPL shall have the right, without prejudice to any other clauses, to terminate contract forthwith and to take possession of the balance work/materials and have the same allotted to any other agency and the contractor shall be liable to compensate the loss that may be occasioned to the DPL on that account. Any letter in writing by the Controlling Officer shall be treated as conclusive on behalf of the Company.

3.22. SUBMISSION OF DETAILED DESIGN REPORT

The contractor shall submit 03 (three) sets of the Detailed Design Report along with editable soft copy in a compact disk for approval from DPL.

The contractor shall submit 05 (five) sets of the finalized Detailed Design Report along with editable soft copy in a compact disk to the General Manager (Project), DPL within 45 (Forty Five) days from the date of issuance of 'Letter of Award' to carry out further course of action.

3.23. PROGRAMME

3.23.1 The Contractor shall supply to the Purchaser and the Project Manager a chart showing the proposed organization to be established by the Contractor for carrying out the Works. The chart shall include the identities of the key personnel together with the curricula vitae of such key personnel to be employed after placement of LOA. The Contractor shall promptly inform the Purchaser and the Project Manager in writing of any revision or alteration of such an organization chart.

3.23.2 The Contractor shall submit to the Purchaser for his approval the Program schedule in the form of **MS Project Network**, within fifteen(15) days of placement of LOA, with respect to Contract ,where such programme schedule is required, which shall, interalia, contain the following:

a. the order in which the Contractor proposes to carry out the Works (including but not limited to design, engineering, manufacture, supply, finalization of Sub-contractors , Quality plans, transport, delivery to Site, assemble, erection, testing and commissioning),
b. the date(s) by which the Contractor reasonably requires that the Purchaser shall have fulfilled its obligations under the Contract so as to enable the Contractor to execute the Contract in accordance with the Program and to achieve Completion and conductance of guarantee test of the Works in accordance with the Contract
c. the times of submission and approval of the Contractor's Drawings

3.23.2.1. The Program so submitted by the Contractor shall be in accordance with the Time Schedule mentioned above and any other dates and periods specified in the Contract. The Contractor shall update and revise the Program as and when appropriate or when required by the Project Manager, but without modification in the Time for Completion of the milestone and any extension granted in accordance with GCC 3.46 and shall submit all such revisions to the Project Manager.

3.23.2.2. This Program shall show clearly all activities and its duration along with earliest and latest dates and the first and last dates of submission of the drawings and each date of shop inspection by the Purchaser and critical path for the Works. The Program approved by the Purchaser shall form part of the Contract.

3.23.2.3. The approval by the Purchaser of the Program shall not relieve the Contractor from any obligation under the Contract towards timely completion of Works.

3.23.2.4. Once the programme schedule has been finalized, no revision shall normally be permitted as long as the scope of work remains unchanged. However, in cases of increase in quantities, while executing the work as per original scope; suitable adjustments may be made without affecting the time for completion. The revision in programme schedule, for aforesaid reasons, shall be done with the approval the Purchaser.

3.23.2.5. If the scope of work undergoes changes during execution stage resulting into additional scope over that originally provided, for which the Contractor insists extension in time for completion, such extension shall be granted while ordering additional scope of work. Contractor shall submit revised programme schedule for approval of the Purchaser.

3.23.2.6. In case the scope of work does not change but the time for completion is extended because of delayed commencement of the work on account of non-fulfilment of obligations by Purchaser or because of any other reasons not attributable to Contractor, programme schedule shall be suitably revised as per the extended time for completion. Once the time for completion has been extended with the approval of Purchaser, Contractor shall submit revised programme schedule for the approval of Purchaser.

3.23.3 Progress Report

The Contractor shall monitor progress of all the activities specified in the Program referred to in **GCC 3.23.2,** and supply a progress report to the Project Manager every month, with a copy to officials as mentioned in the SCC. Guarantee

2.23.3.1. The progress report shall be in a form acceptable to the Project Manager and shall also indicate: (a) percentage completion achieved compared with the planned percentage completion for each activity; and (b) where any activity is behind the Program, giving comments and likely consequences and stating the corrective action being taken.

3.23.4 Progress of Performance

If at any time the Contractor's actual progress falls behind the Program referred to in GCC 3.23.2, or it becomes apparent that it will so fall behind, the Contractor shall, at the request of the Purchaser or the Project Manager, prepare and submit to the Project Manager a revised Program, taking into account the prevailing circumstances, and shall notify the Project Manager of the steps being taken to expedite progress so as to attain Completion of the Works within the Time for Completion as stated in SCC, or within such extended time to which the Contractor shall be entitled under GCC 3.46.2

3.24. DESIGN AND ENGINEERING

3.24.1 The Contractor shall submit to the Purchaser for approval:

Within the time given in the Contract or in the Program such documents including drawings, samples, models or information as may be called for therein, and in the numbers therein required, in a sequential order of execution and during the progress of the Works, such documents of the general arrangement and details of the Works as specified in the Contract. The Purchaser shall signify his approval or disapproval as detailed in the schedule and procedure of documents approval indicated below.

3.24.2 The Contractor shall prepare (or cause its Sub-contractors to prepare) and furnish to the Project Manager the documents, including Manufacturing Quality Plan and Field Quality Plan wherever required or review as specified and as in accordance with the requirements of GCC 3.23.2.

3.24.3 Any part of the Works covered by or related to the documents to be approved by the Purchasers Representative shall be executed only after the Project

Manager's approval thereof

3.24.4 The Contractor shall supply additional copies of approved documents in the format and numbers stated in the Contract.

3.24.5 The Schedule and Procedure of Documents Approval shall be finalised at the **Kick off** meeting and also Contract Coordinate Procedure (**CCP**) shall be prepared.

3.24.6 If any dispute or difference occurs between the Purchaser and the Contractor in connection with or arising out of the disapproval by the Project Manager of any document and/or any modification(s) thereto that cannot be settled between the parties within a reasonable period, then such dispute or difference may be settled in accordance with **GCC 3.6.1** thereof.

3.24.7 The Project Manager's approval, with or without modification of the document furnished by the Contractor, shall not relieve the Contractor of any responsibility or liability imposed upon it by any provisions of the Contract except to the extent that any subsequent failure results from modifications required by the Project Manager.

3.24.8 Approval of any documents by the Purchaser shall not relieve the Contractor of his responsibility for the accuracy thereof or modification required during actual execution or for any deviation in scheme from Technical Specification with accepted deviations if there be any

3.24.9 Approval of Design and Drawing:

The contractor shall have to prepare and submit the designs and drawings associated with civil, mechanical, electrical and other work which includes design of foundation, structure cable sizing, fabrication work, layout design, wiring diagram etc. and obtain approval prior to the execution of work and for this purpose the contractor shall submit Detailed Design Report for obtaining approval from DPL. The contents of the Detailed Design Report shall be as mentioned in the scope of work (Clause no. GCC 3.24.10).

Materials: Contractor shall obtain prior approval for the materials deliverable under the project from DPL as mentioned in the technical specification.

3.24.10 Detailed Design Report (DDR)

Contractor shall prepare and submit the Detailed Design Report to DPL which must contain site meteorological data considered with necessary supporting documents, calculations for annual energy generation, schedule of site works (L1 & L2 network), Design of modules layout with MMS and associated calculations for selection of different major equipment for the plant based on the site location and relevant code of practice. **Design calculation of Solar Plant shall be submitted in latest version of PVsyst software with maximum accurate data like latest version of Solargis.**

The Detailed Design Report shall contain detailed Billing Breakup (BBU) for supply as well as Erection of Plant.

3.25. PROCUREMENT

3.25.1 The Contractor shall manufacture or procure and transport all the Equipment/Materials in an expeditious and orderly manner to the Site.

3.25.2 Defective Material

If in the opinion of the Engineer, any of the machineries/ equipment/ materials

etc. brought to the site for use are not of the quality or kind specified in the contract and/or are unfit for the work, he shall be at liberty to order the removal of the said items and the contractor shall remove the same within twenty four (24) hours after notice has been given to him and if he fails to remove them within the time the engineer may cause them to be removed anywhere at the risk of the Contractor and any cost incurred in so doing shall be deducted from the dues to the contractor under the contract. In such case, items as prescribed by the Controlling Officer or his representative are to be substituted immediately.

3.25.1 Transportation.

3.25.3.1. The Contractor shall at its own risk and expense transport all the Equipment/Materials and the Contractor's Equipment to the Site by the mode of transport that the Contractor judges most suitable under all the circumstances

3.25.3.2. Unless otherwise provided in the Contract, the Contractor shall be entitled to select any safe mode of transport operated by any person to carry the Equipment/Materials and the Contractor's Equipment.

3.25.3.3. Upon dispatch of each shipment of the Equipment/ and the Contractor's Equipment, the Contractor shall notify the Purchaser by courier, email, post or by fax followed by post confirmation of the description of the Equipment/Materials and of the Contractor's Equipment, the point and means of dispatch, and the estimated time and point of arrival at the Site. The Contractor shall furnish the Purchaser with relevant shipping documents to be agreed upon between the parties

3.25.3.4. The Contractor shall be responsible for obtaining, if necessary, approvals from the authorities for transportation of the Equipment/ Materials and the Contractor's Equipment to the Site. The Purchaser shall use its best endeavors in a timely and expeditious manner to assist the Contractor in obtaining such approvals, if requested by the Contractor. The Contractor shall indemnify and hold harmless the Purchaser from and against any claim for damage to roads, bridges or any other traffic facilities that may be caused by the transport of the Equipment/ Materials and the Contractor's Equipment to the Site.

3.25.3.5. **Transit Damages:** In the event of receipt of goods in damaged condition or having found them so upon opening of packages at site, Contractor shall make good of all such damages within a reasonable time from such intimation by DPL.

3.26. CONTRACTOR'S CONSTRUCTION MANAGEMENT:

3.26.1 Setting Out

The Contractor shall be responsible for the true and proper setting-out of the Works in relation together, reference marks and lines provided to it in writing by or on behalf of the Purchaser.

The Contractor shall set out the Works in relation to original points, lines and levels of reference given by the Purchaser in writing and provide all necessary instruments, appliances and labour for such purposes. If at any time during the execution of Works, any error appears in the positions, levels, dimensions or alignment of the Works, the Contractor shall rectify the error at his cost. The checking of any setting-out by the Purchaser shall not relieve the Contractor of his responsibility for the accuracy thereof.

3.26.2 Contractor's Supervision

The Contractor shall give or provide all necessary superintendence during the installation of the Works, and the Contractor’s representative(s) shall be constantly on the Site to provide full-time superintendence of the installation. The Contractor shall provide and employ only technical personnel who are skilled and experienced in their respective callings and supervisory staff who are competent to adequately supervise the Works

3.26.3 Labour

3.26.3.1 The Contractor shall provide and employ on the Site in the installation of the Works such skilled, semi-skilled and unskilled labour as is necessary for the proper and timely execution of the Contract. The Contractor is encouraged to use local labour that has the necessary skills

3.26.3.2 Unless otherwise provided in the Contract, the Contractor shall be responsible for the recruitment, transportation, accommodation and catering of all labour, local or expatriate, required for the execution of the Contract and for all payments in connection therewith.

3.26.3.3 The Contractor shall be responsible for obtaining all necessary permit(s) and/or visa(s), if required, from the appropriate authorities for the entry of all labour and personnel to be employed on the Site.

3.26.3.4 The Contractor shall at its own expense provide the means of repatriation to all of its and its Sub-contractor's personnel, employed on the Contract, at the Site to their various home countries. It shall also provide suitable temporary maintenance of all such persons from the cessation of their employment on the Contract to the date programmed for their departure.

3.26.3.5 The Contractor shall at all times during the progress of the Contract use its best endeavours to prevent any unlawful, riotous or disorderly conduct or behaviour by or amongst its employees and the labour of its Sub-contractors.

3.26.3.6 The Contractor shall, in all dealings with its labour and the labour of its Sub-contractors currently employed on or connected with the Contract, pay due regard to all recognized festivals, official holidays, religious or other customs and all local laws and regulations pertaining to the employment of labour.

3.26.4 Contractor's Equipment

3.26.4.1 The Contractor shall provide all erection Equipment haulage & power if necessary to complete the Works as per Time for completion, including transport at his own cost. The Contractor shall provide additional manpower as well as haulage and other erection equipment as necessary for maintaining the Time schedule of completion.

3.26.4.2 Contractor's all equipment shall, when brought to the Site, be deemed to be exclusively intended for the execution of Contract. The Purchaser shall have lien on all such Equipment brought to Site for the purpose of erection, testing and commissioning of the Equipment/Materials.

3.26.4.3 The Contractor shall not remove from the Site any such Equipment, except when it is no longer required for the completion of the Works, or when the Purchaser has given his consent.

3.26.5 Purchaser's Equipment

The Contractor shall pay hire charges as may reasonably be levied for the Purchaser's equipment, if used, and also provide the transport, haulage, power etc. thereof at his own cost.

3.26.6 Site Regulations and Safety

The Purchaser and the Contractor shall establish Site regulations setting out the rules to be observed in the execution of the Contract at the Site and shall comply therewith. The Contractor shall prepare and submit to the Purchaser, with a copy to the Project Manager, proposed Site regulations for the Purchaser's approval, for which approval shall not be unreasonably withheld.

Such Site regulations shall include, but shall not be limited to, rules in respect of security, safety of the Works, gate control, sanitation, medical care, and fire prevention. Details are mentioned in **SCC clause no.4.8**.

3.26.7 Environment & Social Policy and Procedures (ESPP) of Purchaser

The Contractor shall make himself aware of the ESPP of the Purchaser and shall execute the scope of work under the Contract in compliance with the said provisions.

3.26.8 Watching and Lighting

The Contractor shall provide and maintain at its own expense all lighting, fencing, and watching when and where necessary for the proper execution and the protection of the Works, or for the safety of the Purchasers and occupiers of adjacent property and for the safety of the public.

3.26.9 Clearance of Site

The Contractor shall, from time to time during the progress of the Works clear away and remove all surplus/ rejected materials and debris from Site. On completion of the Works, the Contractor shall remove all Contractors' Equipment and leave the whole of the Site clean and in a workmanlike condition to the satisfaction of the Purchaser.

3.26.10 Communication

The Contractor may require the Purchaser to confirm in writing any decision or instruction of the Purchaser which is not in writing. The Contractor shall promptly notify the Purchaser of such requirement.

3.26.11 Authority of Access

No persons other than the employees of the Contractor or his authorised representative shall be allowed at the Site. Purchaser or his representative shall have access to the work sites at any time.

3.26.12 Emergency work

3.26.12.1 If, by reason of an emergency arising in connection with and during the execution of the Contract, any protective or remedial work is necessary as a matter of urgency to prevent damage to the Works, the Contractor shall immediately carry out such work.

3.26.12.2 If the Contractor is unable or unwilling to do such work immediately, the Purchaser may do or cause such work to be done, as the Purchaser may determine it necessary in order to prevent damage to the Works. In such event the Purchaser shall, as soon as practicable after the occurrence of any such emergency, notify the Contractor in writing of such emergency, the work done and the reasons thereof. If the work done or caused to be done by the Purchaser is such that the Contractor was liable to do at its own expense under the Contract, the reasonable costs incurred by the Purchaser in connection therewith shall be paid by the Contractor to the Purchaser.

3.27. INSPECTION & TESTING

3.27.1 Inspecting Agency

The Purchaser may delegate inspection and testing to an outside agency in place of personnel of PURCHASER with due notice, not less than Twelve (12) days, to the Contractor. Any such outside agency shall be considered as a **Project Manager.**

3.27.2 Inspection and Testing During Manufacture

3.27.2.1 The Purchaser or his designated representative shall be entitled during manufacture to inspect, examine and test the materials and workmanship and check the progress of manufacture of all Equipment to be supplied under the Contract. This shall take place on the Contractor's premises during working hours.

3.27.2.2 No such inspection, examination or testing shall relieve the Contractor of his obligation under the Contract regarding quality of material and soundness of manufacture.

3.27.2.3 No inspection call will be valid before drawings are approved under approval category without comments.

3.27.3 Dates for Inspection and Testing

After getting the related drawings approved under approval category, the Contractor shall give the Purchaser notice of inspection along with factory test results in writing of the date and the place at which any Equipment/Materials will be ready for testing as provided in the Contract. The Purchaser shall attend at the place so named within twelve (12) days of the date which the Contractor has stated in his notice. The Purchaser shall give the Contractor notice, in writing, of his intention to attend the tests. The above notices shall be given at first by the quickest possible means and confirmed later in writing. The Contractor shall render all possible assistance in carrying out inspection in time

3.27.4 Facilities for Testing

3.27.4.1 Where the Contract provides for tests on the premises of the Contractor or Sub-contractor, the Contractor shall provide such assistance, labour, materials, electricity, fuel, stores, apparatus and instruments as may be necessary to carry out the tests efficiently without any extra charges. If the facilities are inadequate to carry out tests as per standard, the Contractor shall have to arrange suitable testing place having all such required facilities and the cost towards this will be on Contractor's account.

3.27.4.2 The Contractor shall be responsible for proper execution of the quality plans. The Works beyond Purchaser's hold points will progress only with Purchaser's prior written consent. The Purchaser may also undertake quality surveillance and quality audit of the systems and procedures and quality control activities. Any change in the Quality Plans shall be made only with Purchaser's prior written approval.

3.27.4.3 The Contractor shall provide the Purchaser with the necessary facilities for carrying out quality audit and quality surveillance of the Contractor's and its Sub-Contractors' Quality Assurance System and manufacturing activities

3.27.5 Routine and Acceptance Tests

All routine tests and acceptance tests shall be carried out at manufacturer's works or test site of the Contractor/Sub-contractor/test laboratory as per stipulation of relevant Indian Standard and relevant IEC in presence of Project Manager. All tests shall be carried out on every lot offered for inspection as per relevant I.S. and IEC.

3.27.6 Type Test

3.27.6.1 The successful bidder shall submit complete test reports of all tests (including type tests) as stipulated in the relevant I.S. and IEC and carried out in a Govt. recognized Test House or laboratory/NABL accredited laboratory on Equipment/ Materials of identical design conforming to our Technical Specification, along with submission of drawing during detailed engineering stage

3.27.6.2 PURCHASER may also undertake Proto checking and quality approval of structural items (wherever applicable) before erection. Each type test report shall provide the following information with test results:

Complete identification, date and Serial No.

Method of application where applied, duration and interpretation of each test.

3.27.7 Repetition of Tests

If any of the type tests, routine or acceptance tests fails to pass, the Contractor shall arrange for repetition of the tests, after rectification or replacement, at his own cost and expenses. If, however, the tests fail for the 2nd time, the related Sub-contractor shall be rejected immediately and the Contractor will be required to furnish the name of another Sub-contractor immediately either from the already approved list of Sub-contractor for that particular Equipment/Materials, or any new Sub-contractor along with submission of all relevant documents in support, towards approval of the new Sub-contractor as stated in this tender document.

3.27.8 Reports of Inspection and Tests

After the factory tests have been completed at the Contractor's or Sub-contractor's works, the Contractor shall submit **three (3) copies** of Test Reports for approval of Purchaser. The Purchaser in turn will approve the same on being satisfied. The Contractor shall provide the Purchaser with **four (04) copies** of Approved Reports of all inspection and tests.

3.27.9 If the Purchaser or his designated representatives fails to attend the test and/or inspection or if it is agreed between the parties that such persons shall not do so, then the Purchaser may advice the Contractor in writing to proceed with the test and/or inspection in the absence of such persons. The Contractor should provide the Purchaser with a certified report of the results thereof.

3.27.10 Covering Up

3.27.10.1 The Contractor shall give the Purchaser full opportunity to examine, measure and test any work on Site which is about to be covered up or put out of view.

3.27.10.2 The Contractor shall give due notice to the Purchaser whenever such work is ready for examination, measurement or testing.

3.27.10.3 The Purchaser shall then notify the Contractor within **seven (07) days** that the Purchaser shall carry out the examination, measurement or testing. Unless it is notified specifically the Contractor may proceed with the work as per Programme submitted.

3.27.11 Uncovering Works

If so instructed by the Purchaser, the Contractor shall expose any parts of the Works. The Contractor shall reinstate and make good such parts to the Purchaser's satisfaction at the risk, cost and responsibility of the Contractor

3.28. TESTS ON COMPLETION.

3.28.1 Notice for Tests on Completion

The Contractor shall give to the Purchaser seven (07) days' notice of the date after which he will be ready to conduct the Tests on Completion. Unless otherwise agreed upon, the Tests shall be carried out as per agreed schedule.

3.28.2 Delayed Tests

If the tests are being unduly delayed by the Contractor, the Purchaser may, by notice, require the Contractor to make the tests within twenty one (21) days after the receipt of such notice. If the Contractor fails to make the tests **Fifteen (15) days** of such notice, the Purchaser may himself proceed with the tests. All tests so made by the Purchaser shall be at the risk and cost of the Contractor and cost thereof shall be deducted from the Contract Price. The tests shall also be deemed to have been made in the presence of the Contractor and shall be accepted as accurate and no claim whatsoever in this respect of the Contractor shall be entertained.

3.28.3 Facilities for Tests on Completion

The Contractor, except where otherwise specified, shall arrange such labour, material, fuel, water, stores and testing apparatus as may be reasonably required to carry out such tests efficiently, without any extra charge

3.28.4 Re-testing

If the work or any portion thereof fails to pass the Tests, the Purchaser or the Contractor may require such tests to be repeated on the same terms and conditions. All costs of such retesting will be borne by the Contractor.

3.28.5 Consequences of Failure to Pass Tests on Completion

If the Works or any portion thereof fails to pass the tests or the repetition thereof under GCC.28, the Purchaser, after due consultation with the Contractor, shall be entitled to

Order one further repetition of the Tests under the conditions of GCC 3.26, or

Reject the Works or portion thereof in which event the Purchaser shall have the same remedies against the Contractor as are provided under GCC 3.27 or

Issue a Taking-Over Certificate, if the Purchaser so wishes, notwithstanding that the Works are not complete. The Contract Price shall then be reduced by such amount as may be agreed by the Purchaser and the Contractor or, failing agreement, as may be determined under GCC 3.6. As soon as the work or any portion thereof has passed the tests, the Purchaser shall issue a Completion certificate to the Contractor to that effect.

3.29. REJECTION

Purchaser may not accord approval to test results if those results are not in conformity with Guaranteed Technical Particulars with given tolerable limits as per relevant standard or the results and procedure followed are found not in line with standard. The results may be rejected even if the Project Manager had witnessed the test. On approval of Test results only, Material Inspection Clearance Certificate will be issued by the Purchaser. Approval of Test results will not relieve the Contractor of its obligation as regards quality, standard and suitability of the Equipment/ Materials.

3.30. PERMISSION TO DELIVER

3.30.1 The Contractor shall apply in writing to the Purchaser for permission to deliver any Equipment / Materials to the Site. No Equipment/ Materials shall be delivered to the Site without the Purchaser's written permission

3.30.2 The Contractor shall make arrangement for receipt of all Equipment/ Materials delivered to Site under the scope of Contract besides all other Equipment/Materials required for the purpose of execution. Upon arrival at Site, the Contractor shall give a notice to the Purchaser when and where materials has arrived and been stored.

3.31. COMPLETION OF WORKS

3.31.1 As soon as execution of the Works or any part for which a separate completion schedule is provided in the Contract has, in the opinion of the Contractor, been completed operationally and structurally and put in tight and clean condition as specified in the Technical Specifications, excluding minor items not materially affecting the operation or safety of the Works, the Contractor shall so notify the Purchaser in writing within **seven (7) days** of the date of completion.

3.31.2 If, for reasons not attributable to the Contractor, the Works cannot be completed in next six (6) months, the Purchaser at, its discretion, may take up the inspection of the portion of the Works already completed, the balance payment due to the Contractor can be released against Bank Guarantee of equivalent amount. The Bank Guarantee validity shall be initially for a period of twelve (12) months or until three (3) months after expected date of commissioning, whichever is earlier. If the completion and thereafter commissioning does not take place within the validity period of the Bank Guarantee, the validity shall be extended from time to time up from the date from which the concerned work was held up on aforesaid account. The Contractor shall also be required to extend the validity of the Contract Performance Guarantee.

3.31.3 For ‘Works’ not involving Commissioning

3.31.3.1 Within **fifteen (15) days** of intimation from the Contractor regarding completion of Works, the Project Manager shall cause to inspect the Works to verify the completion status, in presence of the Contractor’s representative.

3.31.3.2 If the Works are found to be completed and acceptable in all respects (except for minor defects and deficiencies, if any), Completion Certificate (**Annexure-6**) / Taking over Certificate (TOC) shall be issued by the Purchaser after the Contractor’s intimation. The Completion certificate shall generally contain the following details: (a) Date of completion; (b) Defects to be rectified; (c) Items not conforming to Specification but can be accepted at a reduced rate; (d) items not acceptable at all and need to be re-done

3.31.3.3 If, on inspection, Works are not found to be completed or rectification of major nature is required, the Purchaser shall, within twenty-one (21) days of Contractor’s intimation, inform the incomplete works/ defects & deficiencies to the Contractor in writing advising him to take necessary action and to inform PURCHASER after completion/ rectification. The Purchaser shall give reasonable time to the Contractor for remedying the defects/ deficiencies. However, if the Contract specifies separate completion period for different parts of works for the purpose of taking over also, Completion certificate/ TOC shall be issued in respect of portion of works that are completed and are acceptable.

3.31.3.4 The provisions contained in GCC 3.31.3.1 to GCC 3.31.3.3 shall also be applicable in relation to a part of the Works for which separate schedule of completion has been provided in the Contract and such part of Works can be taken over independently

3.32. TAKING OVER

3.32.1 The Works shall be taken over by the Purchaser after completion, either in full or in part (where for part completion, separate completion schedule has been provided in the Contract), upon successful erection, testing and commissioning of Works at Site by the Contractor in accordance with provisions of Contract.

3.32.2 On successful completion of Works or any part thereof as provided in GCC 3.32.1 and upon request of the Contractor for taking over the Works and issuance of TOC, the Purchaser shall, within forty-five (45) days after the receipt of the Contractor's application, or within fifteen (15) days from the date of actual handing over of relevant Works, either issue the TOC or reject the application giving his reasons and specifying the work required to be done by the Contractor to enable the TOC to be issued

3.32.3 Notwithstanding the above mentioned provisions, the issuance of TOC shall not be held up for delay in completion/ rectification of works of minor nature that do not affect the performance/ use of the building/installation/ equipment/sub-system/system at rated voltage. In such a case the Contractor shall however be required to give an undertaking that in case he fails to complete/rectify within a mutually agreed period, the Purchaser shall be at liberty to carry out the work at his risk and cost, and deduct an amount as may be considered appropriate by the Purchase.

3.32.4 Issuance of such certificates shall not relieve the Contractor of any of his obligations which otherwise comes under the terms and conditions of the Contract.

3.32.5 Issuance of TOC for any part of the Works is only for the purpose of facilitating the Contractor to receive the payment for part of the Works completed and for determination of liquidated damages in respect thereof and shall not relieve the Contractor of his responsibilities under the Contract towards other parts of the Works.

3.32.6 TOC is issued to the Contractor on stating the date on which the Works or any part thereof were complete and ready for taking over, after ascertaining the following:

a. The Works under the Contract have been satisfactorily completed by the Contractor as per the provisions of Contract.

b. Submission of required number of reproducible of approved as-built drawings (hard copies & soft copy in CDs), design documents duly authenticated by Purchaser, O&M manuals, data sheets, test reports, pamphlets and manuals of spares, maintenance and testing equipment by the Contractor.

c. The Contractor has cleared the Site of all the surplus materials, removed all scaffoldings, shuttering materials, labour huts/ sheds, cleaned the dirt from part of building, sanitary arrangement, water supply connection and all electrical gadgets/ equipment/ switches, wiring, any wood work or any such item, to the satisfaction of the Project Manager, except those required for carrying out rectification works.

d. All the defects have been rectified to the complete satisfaction of the Project Manager

3.32.7 Notwithstanding the above mentioned provisions, the issuance of TOC shall not be held up for delay in completion/ rectification of works of minor nature that do not affect the performance/ use of the building/installation/ equipment/sub-system/system at rated voltage. In such a case the Contractor shall however be required to give an undertaking that in case he fails to complete/rectify within a mutually agreed period, the Purchaser shall be at liberty to carry out the work at his risk and cost, and deduct an amount as may be considered appropriate by the Purchaser.

F. GUARANTEES AND LIABILITIES

3.33. NET MINIMUM GUARANTEED GENERATION (NMGG)

Contractor shall have to ensure Net Minimum Guaranteed **Generation @ 1.60 MU/MW AC for the first year after final commissioning and at a reduced rate of 0.7% per year for subsequent years**. Initially, the above Guarantee shall be required for the 5 years i.e. in the O&M period. The same guarantee shall continue for extended O & M period, if agreed on mutual terms & conditions. The Contractor shall design their plant to achieve the Net Minimum Guaranteed Generation.

The Point of Measurement of Generated energy and net send out will be measured at the Outgoing feeder(from 6.6 KV Solar Switchyard).

- If contractor is not responsible for non-availability of Grid hours subject to submission of proper documentary evidence with due signature of DPL official then said hours may be considered for finalization of NMGG calculation if it satisfies following condition:

a) Grid outage where contractor is not responsible.

b) If outages occur only during the daytime (daytime shall be considered as per data available in WMS).

c) MW reduction calculation for grid outage for above two reasons:

   i) **If grid outage for few hours :**
   The average generation of that particular day (MWHr/generation Hr) X number of hours of Grid outage (supporting document)

   ii) **If Grid outage for days:**
   Average generation of that week (MWHr/days generation) X number of days grid outage (supporting document)

d) All the above records shall need to be accepted and signed by the DPL site authority.

3.34. LIQUIDATED DAMAGES

3.34.1 **LD for Time Delay:**

Compensation of loss on account of late delivery/completion (notionally/actually) where loss is pre-estimated and mutually agreed to is termed as LD. Law allows recovery of pre-estimated loss provided such a term is included in the contract and there is no need to establish actual loss due to late supply/execution of work. Time schedule of delivery/completion shall be essence of the order. If the contractor fails to execute the order in full or part thereof within the fixed period or any time repudiate the contract before the expiry of such period, the Corporation may, without prejudice to any other right or remedy available, recover damages for breach of contract and to safe-guard its interest.

'In the event of any delay in the supplies of ordered materials/execution of works beyond the stipulated delivery/completion schedule including any extension permitted in writing, the Corporation reserves the right to recover from the contractor a sum equivalent to 0.5% of the value of delayed supply/work for each week of delay and part thereof subject to a maximum of 10% of the Project cost plus GST.'

3.34.2 **LD for Milestone Delay**

Delay in attaining the milestones by the contractor shall lead to imposition of intermediary Liquidated damages **upto the maximum extent of 5(Five) Percent of the Contract price.**

Intermediary Liquidated damages shall be calculated based on the following Milestones.

Sl. No.	Description	Completion Time from date of LOA	% of LD
1.	Approval of Detailed Design Report	45 Days	0.25

2.	Finalizations of Plant layout including area leveling & grading.	4 Months	0.75
3.	Completion of construction of Inverter Control Room	6 months	0.25
4.	Supply of transformers& switchgears at site	7 Months	0.75
5.	Completion of Supply of Modules & Module Mounting Structure	10 Months	1.0
6.	Completion of Modules & Electrical Equipment Erection	11 Month	1.5
7.	Complete installation of transformer and testing	11 Months	0.25
8.	Evacuation of Power	12 Months	0.25

3.34.3

The total/aggregate LD (LD for Time Delay + LD for Milestone Delay) leviable for this contract shall not exceed 10 (Ten) Percent of the Contract price.

However no LD for milestone delay will be charged if the project is completed within the stipulated time provided such milestone delay does not hamper scheduled execution of any other related projects/activities thereby causing loss or damage to the Purchaser

3.35. DEFECTS LIABILITY

3.35.1 The Contractor warrants that the Works or any part thereof shall be free from defects in the design, engineering, materials and workmanship of the Equipment/Materials supplied and of the work executed.

3.35.2 The Defect Liability Period shall be as specified in the SCC. Where any part of the Works is taken over separately, the Defects Liability Period for that part shall commence on the date it was taken over.

3.35.3 If during the Defect Liability Period any defect should be found in the design, engineering, materials and workmanship of the Equipment/Materials supplied or of the work executed by the Contractor, the Contractor shall promptly, in consultation and agreement with the Purchaser regarding appropriate remedying of the defects, and at its cost, repair, replace or otherwise make good (as the Contractor shall, at its discretion, determine) such defect as well as any damage to the Works caused by such defect.

3.35.4 The Contractor shall not be responsible for the repair, replacement or making good of any defect or of any damage to the Works arising out of or resulting from any of the following causes:

a. improper operation or maintenance of the Works by the Purchaser
b. operation of the Works outside Specifications provided in the Contract
c. normal wear and tear

3.35.5 The Purchaser shall give the Contractor a notice stating the nature of any such defect together with all available evidence thereof, promptly following the discovery thereof. The Purchaser shall afford all reasonable opportunity for the

3.35.6 The Purchaser shall afford the Contractor all necessary access to the Works and the Site to enable the Contractor to perform its obligations under this Clause GCC 3.35. The Contractor may, with the consent of the Purchaser, remove from the Site any Equipment/Materials or any part of the Works that are defective, if the nature of the defect and/or any damage to the Works caused by the defect is such that repairs cannot be expeditiously carried out at the Site

3.35.7 If the repair, replacement or making good is of such a nature that it may affect the efficiency of the Works or any part thereof, the Purchaser may give to the Contractor a notice requiring that tests of the defective part of the Works shall be made by the Contractor immediately upon completion of such remedial work, whereupon the Contractor shall carry out such tests.

If such part fails the tests, the Contractor shall carry out further repair, replacement or making good (as the case may be) until that part of the Works passes such tests. The tests, in character, shall in any case be not inferior to what has already been agreed upon by the Purchaser and the Contractor for the original part of the Works.

3.35.8 If the Contractor fails to commence the work necessary to remedy such defect or any damage to the Works caused by such defect within a reasonable time (which shall in no event be considered to be less than fifteen (15) days), the Purchaser may, following notice to the Contractor, proceed to do such work, and the reasonable costs incurred by the Purchaser in connection therewith shall be paid to the Purchaser by the Contractor or may be deducted by the Purchaser from any money due to the Contractor or claimed under the Performance Securities.

3.35.9 If the Works or any part thereof cannot be used by reason of such defect and/or making good of such defect, the Defect Liability Period of the Works or such part, as the case may be, shall be extended by a period equal to the period during which the Works or such part cannot be used by the Purchaser because of any of the aforesaid reasons.

At the end of the Defect Liability Period, the Contractor's liability ceases except for latent defects. The Contractor 's liability for latent defects warranty for the Equipment/Materials, including spares, shall be limited to a period of five (5) years from the end of Defect Liability Period of the respective Equipment/Materials, including spares. For the purpose of this clause, the latent defects shall be the defects inherently lying within the material or arising out of design deficiency which do not manifest themselves during the Defect Liability Period as defined in this Clause GCC 3.35 but later.

3.35.10 Except as provided in Clauses GCC 3.35 and GCC 3.40, the Contractor shall be under no liability whatsoever and howsoever arising, and whether under the Contract or at law, in respect of defects in the Works or any part thereof, the

Equipment/Materials, design or engineering or work executed that appear after Completion of the Works or any part thereof, except where such defects are the result of the gross negligence, fraud, criminal or wilful action of the Contractor.

3.35.11 In addition, the Contractor shall also provide an extended warranty for any such component of the Works and for the period of time as may be specified in the SCC. Such obligation shall be in addition to the defect liability specified under GCC 3.35.2.

The Contractor shall transfer the warranty/ guarantees of the equipment as such from the OEM/ supplier in the name of the Owner. The period of the warranty / guarantee for each equipment shall be as per the "Technical Specifications".

During the Comprehensive Operation & Maintenance period, the Contractor shall be responsible for any defects in the work due to faulty workmanship or due to use of sub-standard materials in the work. Any defects in the work during the guarantee period shall therefore, be rectified by the Contractor without any extra cost to the Owner within a reasonable time as may be considered from the date of receipt of such intimation from the Owner failing which the Owner reserves the right to take up rectification work at the risk and cost of the Contractor.

3.35.12 Defects Liability Certificate and No-Claim Certificate

When the Defects Liability Period for the Works or any part thereof has expired and the Contractor has fulfilled all his obligations under the Contract for defects in the Works or the part, the Purchaser shall issue to the Contractor a Defects Liability Certificate to that effect within next twenty-eight (28) days. A No-Claim Certificate should be furnished by the Contractor for contract closure before the issuance of the final Defects Liability Certificate by the Purchaser. Such No-claim certificate shall be furnished by the Contractor as per Annexure of Section VIII. The Contractor is expected to complete all formalities for closure of Contract including their final claims relating to the Contract. All claims will be deemed to be settled and no further claims of the Contractor will be entertained after the furnishing of the No-Claim Certificate by the Contractor.

3.36. LIMITATIONS OF LIABILITY

3.36.1 Liability after Expiry of Defects Liability Period

The Contractor shall have no liability to the Purchaser for any loss of or damage to the Purchaser's physical property unless caused by Gross Misconduct or negligence of the Contractor provided that this exclusion shall not apply to any obligation of the Contractor to pay Liquidated Damage to the Purchaser.

The aggregated liability of the Contractor to the Purchaser under the Contract shall not exceed the Contract Price provided that this limitation shall not apply to any obligation of the Contractor to the cost of repairing or replacing the defective Equipment/ Materials or to indemnify the Purchaser with respect to patent infringement.

G. RISK DISTRIBUTION

3.37. TRANSFER OF PURCHASERSHIP

3.37.1 Purchasership of the Equipment/Materials (including spare parts) procured from within/outside the country shall be transferred to the

Purchaser when the Equipment/Materials (including spare parts) are loaded on to the mode of transport to be used to convey the Equipment/Materials (including spare parts) from the works to the Site and upon endorsement of the dispatch documents in favour of the Purchaser.

3.37.2 Purchasership of the Contractor's Equipment used by the Contractor and its Sub-Contractors in connection with the Contract shall remain with the Contractor or its Sub-contractors.

3.37.3 Notwithstanding the transfer of Purchasership of the Equipment/Materials, the responsibility for care and custody thereof together with the risk of loss or damage thereto shall remain with the Contractor pursuant to GCC 3.39 hereof until Completion of the Works or the part thereof in which such Equipment/Materials are incorporated

3.37.4 **Indemnity by Contractors:** As this is a turnkey project and the contractor is fully responsible for handling, erection and commissioning of the equipment and materials so for executing the work, the Contractor shall execute an Indemnity Bond in favour of the Purchaser for keeping the equipment in safe custody and to utilize the same exclusively for the purpose of the said Contract. The Indemnity Bond shall be furnished as per proforma in **Annexure-5** of Section VII. The Purchaser shall also issue a separate Authorization Letter to the Contractor to enable him to take physical delivery of Equipment/ Materials from the Purchaser as per proforma enclosed in **Annexure-11** of Section VII.

3.37.5 After material reached at site contractor shall apply for gate pass to enter the material as per the **Annexure-10B** in the Purchser's site. The Materials reached at site will be in the custodian of the Contractor.

3.37.6 The Purchaser will issue a Materials Receipt Certificate (MRC) as per **Annexure-12** after receiving the materials and equipment at site.

3.38. RISK AND RESPONSIBILITY

3.38.1 Allocation of Risk and Responsibility

The Risks of loss of damage to physical property and of death and personal injury which arise in consequence of the performance of the Contract shall be allocated between the Purchaser and the Contractor as follows

a. the Purchaser: the Purchaser's Risks as specified in GCC 3.38.2
b. the Contractor's Risks as specified in GCC 3.38.3

3.38.2 Purchaser's Risks

a. War and hostilities (whether war be declared or not), invasion, act of foreign enemies,
b. revolution, insurrection, military or usurped power or civil war,

c. use or occupation of the Works or any part thereof by the Purchaser,
d. the use or occupation of the Site or any part thereof, for the purposes of the Contract, or interference, whether temporary or permanent with any right-of-way, any easement, way leave or right of a similar nature which is inevitable result of the construction of the Works in accordance with the Contract.
e. the right of the Purchaser to construct the Works or any part thereof on, over, under, in or through any land,
f. damage (other than that resulting from the Contractor's method of construction) which is the inevitable result of the construction of the Works in accordance with the Contract,
g. the act, neglect or commission or breach of Contract or of statutory duty of the Purchaser

3.38.3 Contractor's Risks

The Contractor's Risks are all risks other than those identified as the Purchaser's Risks.

3.39. CARE OF WORKS

3.39.1 Contractor's Responsibility for the care of Works, man and Materials/Equipment

The Contractor shall be responsible for the care of Works, man and materials/equipment from the Commencement Date until the Risk Transfer Date applicable thereto under GCC 3.39.2

3.39.2 Risk Transfer Date

The Risk Transfer Date in relation to the Works is the date of occurrence of any of the following

a. the date of issue of the TOC, or
b. the date of expiry of the notice of termination when the Contract is terminated by the Purchaser or the Contractor in accordance with these Conditions

The risk of loss or damage to the Works shall pass from the Contractor to the Purchaser on the Risk Transfer date applicable thereto.

3.39.3 Making Goods Damage

3.39.3.1 **After risk transfer date**

To making good forthwith loss or damage caused by the Contractor prior to the completion of the Defects Liability period

3.39.4 Till such time the system is not commissioned or taken over by PURCHASER, its custody and watch and ward shall remain with Contractor who shall accordingly be required to maintain a skeleton establishment at Site. Charges towards insurance cover for Contractor supplied Equipment /Material shall be paid by the Contractor till completion of the defect liability period.

3.40. ACCEDENT OF INJURY

Damage to property and injury to persons, accident or injury to workers

3.40.1 Contractor's liability

Except as provided under GCC 3.43 the Contractor shall be liable for and shall indemnify the Purchaser against all losses, expenses and claims in respect of any loss of or damage to physical property (other than Works), death or personal injury to the extent caused by :

a. defective design, material or workmanship of the Contractor, or
b. negligence or breach of statutory duty of the Contractor, his Sub-contractors or their respective employees and agents

3.40.2 Accidents

The Contractor shall be liable for and shall indemnify the Purchaser against all losses, expenses or claim arising in connection with the death of or injury to any person employed by the Contractor or his Sub-contractors for the purposes of the Works.

The Contractor/Sub-contractor shall obtain necessary insurance coverage under the Employees compensation Act-1923 to cover the risk of payment of compensation in case of injury/death arising in course and out of employment to any employee.

3.41. INSURNCE

3.41.1 The Contractor at his own cost shall arrange, secure and maintain all insurances as may be pertinent to the Works and obligatory in terms of law to protect his interest and interests of the Purchaser against all perils detailed herein in the type and up to the limit of such insurance as defined herein together with the underwriter in each case shall be acceptable to the Purchaser. The identity of insurers and the form of policies shall be subject to the approval of Purchaser which shall not be unreasonably withheld. However, irrespective of such acceptance, the responsibility to maintain adequate insurance coverage at all times during the period of Contract including the Comprehensive O&M period and extended period of Contract shall be of Contractor alone. Contractor shall maintain seamless insurance cover during construction and O&M phases. Copy of policies shall be given to DPL.

3.41.2 Any loss or damage to the Equipment/ Materials during transportation, handling, storage, erection, putting into satisfactory operation and all activities to be performed till the successful completion of commissioning of the Equipment and handed over to the Purchaser shall be to the account of the Contractor. The Contractor shall be responsible for preference of all claims and make good the damages or loss by way of repairs and/or replacement of the Equipment/ Materials, damaged or lost. The transfer of title shall not in any way relieve the Contractor of the above responsibilities during the period of Contract. The Contractor shall provide the Purchaser with copy of all insurance policies and documents taken out by him in pursuance of the Contract. Such copies of documents shall be submitted to the Purchaser immediately after such insurance coverage. The Contractor shall also inform the Purchaser in writing at least sixty (60) days in advance regarding the expiry/cancellation and/or change in any of such documents and ensure revitalization, renewal etc. as may be necessary well in time at his cost, risk and responsibility.

3.41.3 The perils required to be covered under the insurance shall include, but not be limited to fire and allied risks, miscellaneous accidents (erection risks), workman compensation risks, loss or damage in transit, theft, pilferage, riot and strikes and malicious damages, civil commotion, weather conditions, storms, accidents of all kinds, terrorist attacks, war risks etc or combination of the said incidents.

All other insurance like – transit insurance (Marine/ Cargo/ others as applicable), Construction All Risk, Erection All Risk, workmen compensation, fire, third party liability, insurance against theft, Contractor's Equipments, machinery breakdown policy, business interruption insurance, Property damage Insurance & Environmental risk insurance as required during the Construction and O&M period of the Plant shall be in the contractor's scope & shall borne by the Contractor.

The scope of such insurance shall be adequate to cover the replacement/ reinstatement cost of the Equipment/Materials for all risks up to and including delivery of goods on ex-works basis and shall also cover transportation and other costs till the Equipment/ Materials are delivered, erected and installed upto successful completion of O&M period and also for the extension of time without prejudice to the terms of the contract. Notwithstanding the extent of insurance cover and the amount of claim available from the underwriters, the Contractor shall be liable to make good the full replacement/ rectification value of all Equipment/Materials and to ensure their availability as per project requirements at its cost.

Any FIR required to be lodged to local Police Station shall be the responsibility of the Contractor. The Contractor shall arrange to supply/ rectify/ recover the materials without waiting for settlement of the insurance claim and even if the claim is unsettled for timely completion of the project. The final financial settlement with the insurance company shall rest upon the Contractor.

3.41.4 The Contractor shall ensure that for all activities to be performed under the Contract viz. transportation, storage, erection, testing, commissioning etc. till the Works are handed over to the Purchaser; the insurance cover shall only be taken from Indian Government Insurance Companies.

The Contractor shall be responsible to take suitable insurance(s) and claim management during and till the completion of the O&M contract and indemnify the Owner from all associated risks whatsoever.

3.42. CHANGE IN LAWS AND REGULATIONS

If, after the date seven (7) days prior to the last date of bid submission, in the country where the Site is located, any law, regulation, ordinance, order or bye-law having the force of law is enacted, promulgated, abrogated or changed (which shall be deemed to include any change in interpretation or application by the competent authorities) that subsequently affects the costs and expenses of the Contractor and/or the Time for Completion, the Contract Price shall be correspondingly increased or decreased, and/or the Time for Completion shall be reasonably adjusted to the extent that the Contractor has thereby been affected in the performance of any of its obligations under the Contract. However, these adjustments would be restricted to direct transactions between the Purchaser and the Contractor and shall also not be applicable on the bought out items despatched directly by Sub-contractor(s) to Site.

Notwithstanding the foregoing, such additional or reduced costs shall not be separately paid or credited if the same has already been accounted for in the price adjustment provisions where applicable.

3.43. FORCE MAJEURE

3.43.1 “Force Majeure" shall mean any event beyond the reasonable control of the Employer or of the Contractor, as the case may be, and which is unavoidable notwithstanding the reasonable care of the party affected.

3.43.2 In the event of either party being rendered unable by Force Majeure to perform any obligation required to be performed by them under this Contract, relative obligation of the party affected by such Force Majeure shall be treated as suspended during which the Force Majeure clause last.

3.43.3 The term "Force Majeure" shall herein mean riots (other than among the Contractors employee), Civil commotion, War (whether declared or not), invasion, act of foreign enemies hostilities, civil war, rebellion, revolution, insurrection, military coup, damage from aircraft, embargoes, quarantines, acts of god such as earthquake, floods, fires not caused by Contractor's negligence and other causes which the Contractor has no control. Normal climatic conditions such as rainy season, monsoons, storms, etc., are not to be considered as Force Majeure.

3.43.4 Upon occurrence of such causes, the party claiming that it has been rendered unable as aforesaid, thereby, shall notify the other party in writing by registered notice within 10 (Ten) days of the beginning of the event thereof giving full particulars and satisfactory evidence in support of its claim.

3.43.5 The burden of proof as to occurrence of the event of Force Majeure and its effect shall be upon the party claiming the Force majeure event and such claim shall be supported by documentary evidence in the form of a Certificate Issued by a recognized Chamber of Commerce or any other local, state or national authority.

3.43.6 Time for Performance of the relative obligation suspended by the event of force majeure shall stand extended by the period for which such clause lasts.

3.43.7 If works are suspended by Force Majeure conditions lasting for more than two months leading to prolonged force majeure, the parties shall hold consultation to find a solution/resolve the problem satisfactorily. Provided, The Employer shall reserve the right to cancel the Order/Contract, wholly or partly, in order to meet the overall project schedule and make alternative arrangement for completion of delivery and other schedules.

3.43.8 The Contractor shall not claim any compensation for Force Majeure conditions and shall take appropriate steps to insure men and materials utilized by it under the Contract well in advance.

3.44. WAR RISKS

3.44.1 War risks shall mean any of the following events occurring or existing in or near India:

a. war, hostilities or warlike operations (whether a state of war is declared or not), invasion, act of foreign enemy and civil war
b. rebellion, revolution, insurrection, mutiny, usurpation of civil or military government, conspiracy, riot, civil commotion and terrorist acts, and
c. any explosion or impact of any mine, bomb, shell, grenade or other projectile, missile, munitions or explosive of war

3.44.2 Notwithstanding anything contained in the Contract, each party to the Contract shall bear its own costs for any loss or damages as may be incurred on accounts of war risks with respect to:

a. destruction of or damage to Works or any part thereof to the extent not taken over by the Purchaser shall be the Contractor's risk and for those taken over by the Purchaser, it shall be the Purchaser's risk.
b. injury or loss of life of its personnel

H. CHANGE IN CONTRACT ELEMENTS

3.45. VARIATIONS

3.45.1 Changes originating from Purchaser

The Purchaser may, by variation order to the Contractor, at any time before the Works are taken over, instruct the Contractor to alter, amend, omit, add to or otherwise vary any part of the Works. The Contractor shall not vary or alter any of the Works, except in accordance with a variation order from the Purchaser. The Contractor may, however, at any time propose variations of the Works to the Purchaser.

The Contractor shall not request for extension of time for completion in case of variation of quantity during execution for an increase as specified in SCC.

3.45.2 Variation order procedure

Prior to any variation order under **GCC. 3.44.1** the Purchaser shall notify the Contractor of the nature and form of such variation. As soon as possible after having received such notice, the Contractor shall submit to the Purchaser:

a. A description of Works, if any, to be performed, its anticipated quantity, the proposed rate in case of a new item and total adjustment to the Contract Price. In case of items for which the rates are available in the Contract the extra quantities shall be executed by the Contractor at the same rates upto the variation limit provided in the Contract.
b. The Contractor's proposals for any necessary modifications to the Program according to **GCC 3.23** or to any of the Contractor's obligations under the Contract.

3.45.3 Following the receipt of the Contractor's submission, the Purchaser shall, after due consultation with the Contractor, decide whether or not the variation shall be carried out.

3.45.3.1 If the Purchaser decides that the variation shall be carried out, he shall issue a variation order clearly identified as such in accordance with the Contractor's submission or as modified by mutual agreement.

3.45.3.2 Pending issue of variation order, the Purchaser may require the Contractor to proceed ahead with the Works to avoid delay in the progress of Works. In such situations, subject to physical verification, payment shall be made up to sixty percent (60%) of rates as provided in the Contract, for items for which separate rates are available beyond prescribed limit of quantity variation as per the Contract.

3.45.3.3 In case of new items/ substituted items, up to forty (40%) of amount estimated by the Purchaser shall be paid to the Contractor subject to reasonableness of the claim. If the Purchaser and the Contractor are unable to agree to the adjustment of the Contract Price, the provisions of **GCC 3.45.4** shall apply.

3.45.4 Disagreement on adjustment of the Contract Price

3.45.4.1 If the Contractor and the Purchaser are unable to agree on the adjustment of the Contract price, the adjustment shall be determined in accordance with the rates specified in the approved Price Schedules(BBU), subject to ceiling in Contract Price variation as specified in SCC.

3.45.4.2 If the rates contained in the Price Schedules are not directly applicable to the specific Works in question, suitable rates shall be established by the Purchaser reflecting the level of pricing in the Price Schedules.

3.45.4.3 Where rates are not contained in the said Price Schedule, for the sake of reaching a reasonable rate in respect of any sort of erection work, the number of skilled, semi-skilled and unskilled labour and minimum wage rate declared by the Govt. of West Bengal and/or the rates specified on the latest PWD/CPWD Schedule, overhead, profit and consumables shall be the basis for determination of reasonable rate.

3.45.4.4 For any supply item, reasonable rates shall be reached based on current purchase rate of identical equipment purchased by PURCHASER. The

Contractor shall also be entitled to be paid:

the cost of any partial execution of the Works rendered useless by any such variation, and

The cost of making necessary alterations to Equipment already manufactured or in the course of manufacture or of any work done that has to be altered in consequence of such a variation.

The Purchaser shall on this basis determine the rates or prices to enable on account payment to be included in certificates of payment.

3.45.5 Contractor to Proceed

On receipt of a variation order, the Contractor shall forthwith proceed to carry out the variation and be bound to these conditions in so doing as if such variation was stated in the Contract. The Works shall not be delayed pending the granting of an extension of the Time for Completion or an adjustment to the Contract Price under **GCC 3.45.4**.

3.45.6 Records of costs

In any case where the Contractor is instructed to proceed with a variation prior to the determination of the adjustment to the Contract Price, keeping in mind that the adjustment of Contract Price due to this variation shall be guided by **GCC 3.45.4** the Contractor shall keep the necessary records of the cost of undertaking the variation and of time expended thereon. Such records shall be open to inspection/ verification by the Purchaser at all reasonable times.

3.45.7 Quantity variation

PURCHASER, during execution of the Contract, reserves the right to increase or decrease the quantity of goods and services for the items included in the Contract with variation of the total Contract Price as specified in SCC but without any change in unit price or other terms and conditions. The quantity of the individual items of goods and services may however vary up to any extent within the overall ceiling limit of the Contract Price.

3.46. EXTENSION OF TIME FOR COMPLETION

3.46.1 Delivery and installation of Equipment/Materials as per requirement of work Program shall be made by the Contractor in accordance with Time Schedule pursuant to the SCC or within such extended time to which the Contractor shall be entitled under **GCC 3.46.2**

3.46.2 **<u>Reasons for Extension of Time for Completion</u>**

The Contractor may seek an extension of the Time for Completion if he is or will be delayed in completing the Works by any of the following reasons:

a. extra or additional work ordered in writing under GCC 3.45.
b. the delay in completion of Works caused for no fault on the part of the Contractor due to orders/instructions issued by the Purchaser
c. Force Majeure as per GCC 3.43.
d. any changes in laws and regulations as provided in GCC 3.42.
e. any other matter specifically mentioned in the Contract

3.46.3 The Contractor shall give notice to the Purchaser of his intention to make a claim for an extension of time within fifteen (15) days of the occurrence of any of the above cause(s). The notice shall be followed as soon as possible by the claim with full supporting details.

3.46.4 The Contractor shall demonstrate to the Purchaser's satisfaction that it has used its best endeavour to avoid or overcome such causes for delay and the parties will mutually agree upon remedies to mitigate or overcome causes for such delays.

3.46.5 Notwithstanding the provisions of clause **GCC 3.46.4**. the Contractor shall not be entitled to an extension of time for completion, unless the Contractor, at the time when circumstances specified in **GCC 3.46.2** arises, has immediately notified the Purchaser in writing that it may claim such extension as caused by such circumstances. The Purchaser on receipt of such notice may agree to extend the Contract completion period as may be reasonable and mutually agreed but without prejudice to other terms and conditions of the Contract.

3.46.6 **Earlier Completion:**

The Purchaser may require completion of the Works or part thereof earlier than the Time for Completion, as mutually agreed between the Purchaser and the Contractor. The earlier completion date so agreed, if not achieved, shall not be considered for the purpose of levy of Liquidated damages.

3.47. TERMINATION

3.47.1 **Termination for Purchaser's Convenience**

3.47.1.1 The Purchaser may at any time terminate the Contract for any reason by giving the Contractor a notice of termination that refers to this sub-clause **GCC 3.47.1**.

3.47.1.2 Upon receipt of the notice of termination under **GCC 3.47.1.1**, the Contractor shall either immediately or upon the date specified in the notice of termination

a. cease all further work, except for such work as the Purchaser may specify in the notice of termination for the sole purpose of protecting that part of the Works already completed, or any work required to leave the Site in a clean and safe condition.

b. terminate all subcontracts, except those to be assigned to the Purchaser pursuant to paragraph (d)(ii) of sub-clause GCC 3.47.2.

c. remove all Contractor's Equipment from the Site, repatriate the Contractor's and its Sub-contractors' personnel from the Site, remove from the Site any wreckage, rubbish and debris of any kind, and leave the whole of the Site in a clean and safe condition.

d. In addition, the Contractor, subject to the payment specified in **GCC 3.47.3**, shall

   i deliver to the Purchaser the parts of the Works executed by the Contractor up to the date of termination.

   ii to the extent legally possible, assign to the Purchaser all right, title and benefit of the Contractor to the Works as at the date of termination, and, as may be required by the Purchaser, in any subcontracts concluded between the Contractor and its Sub-contractors.

   iii deliver to the Purchaser all non-proprietary drawings, Specifications and other documents prepared by the Contractor or its Sub-contractors as at the date of termination in connection with the Works.

3.47.1.3 **Risk Purchase Clause:** In case of failure of supplier, DPLL at its discretion may make purchase of the materials / services NOT supplied / rendered in time at the RISK & COST of the supplier. Under such situation, the supplier who fails to supply the goods in time shall be wholly liable to make good to DPL any loss due to risk purchase.

In case of items demanding services at site like erection and commissioning, vendor should send his servicemen /representatives within 7 days from the service call. In case a vendor fails to attend to the service call, DPL at its discretion may also make arrangements to attend such service by other parties at the RISK & COST of the supplier. Under such situation the supplier who fails to attend the service shall be wholly liable to make good to DPL any loss due to risk purchase / service including additional handling charges due to the change.

3.47.1.4 In the event of termination of the Contract under **GCC 3.47.1.1**, the Purchaser shall pay to the Contractor the following amounts:

a. the Contract Price, properly attributable to the parts of the Works executed by the Contractor as of the date of termination
b. the costs reasonably incurred by the Contractor in the removal of the Contractor's Equipment from the Site and in the repatriation of the Contractor's and its Sub-contractors' personnel
c. any amounts to be paid by the Contractor to its Sub-contractors in connection with the termination of any sub-contracts, including any cancellation charges
d. costs incurred by the Contractor in protecting the Works and leaving the Site in a clean and safe condition pursuant to paragraph (a) of **GCC. 3.47.1.2**

the cost of satisfying all other obligations, commitments and claims that the Contractor may in good faith have undertaken with third parties in connection with the Contract and that are not covered by paragraphs (a) through (d) above

3.47.2 **Termination for Contractor's default**

3.47.2.1 The Purchaser, without prejudice to any other rights or remedies it may possess, may terminate the Contract forthwith in the following circumstances by giving a notice of termination and its reasons thereof to the Contractor, referring to this **GCC 3.47**:

a. if the Contractor becomes bankrupt or insolvent, has a receiving order issued against it, compounds with its creditors, or, if the Contractor is a corporation, a resolution is passed or order is made for its winding up (other than a voluntary liquidation for the purposes of amalgamation or reconstruction), a receiver is appointed over any part of its undertaking or assets, or if the Contractor takes or suffers any other analogous action in consequence of debt.

b. if the Contractor assigns or transfers the Contract or any right or interest therein in violation of **GCC. 3.48** if the Contractor, in the judgment of the Purchaser has engaged in corrupt or fraudulent practices in competing for or in executing the Contract.

For the purpose of this Clause:

"**corrupt practice**" means the offering, giving, receiving or soliciting of anything of value to influence the action of a public official in the process or in Contract execution.

"**fraudulent practice**" means a misrepresentation of facts in order to influence a process or the execution of a Contract to the detriment of the Purchaser and includes collusive practice among bidders (prior to or after bid submission) designed to establish bid prices at artificial non-competitive levels and to deprive the Purchaser of the benefits of free and open competition.

3.47.2.2 **If the Contractor:**

a. has abandoned or repudiated the Contract
b. has without valid reason failed to commence Works promptly
c. persistently fails to execute the Contract in accordance with the Contract or persistently neglects to carry out its obligations under the

Contract without just cause

d. refuses or is unable to provide sufficient materials, services or labour to execute and complete the Works in the manner specified in the Program furnished under GCC.20 (at rates of progress that give reasonable assurance to the Purchaser that the Contractor can attain Completion of the Works by the Time for Completion as extended.

Then the Purchaser may, without prejudice to any other rights it may possess under the Contract, give a notice to the Contractor stating the nature of the default and requiring the Contractor to remedy the same. If the Contractor fails to remedy or to take steps to remedy the same within fifteen (15) days of its receipt of such notice, then the Purchaser may terminate the Contract forthwith by giving a notice of termination to the Contractor that refers to this clause GCC 3.47.2.

3.47.2.3 Upon receipt of the notice of termination under **GCC 3.47.1** or **GCC 3.47.2** the Contractor shall, either immediately or upon such date as is specified in the notice of termination

a. cease all further work, except for such work as the Purchaser may specify in the notice of termination for the sole purpose of protecting that part of the Works already executed, or any work required to leave the Site in a clean and safe condition
b. terminate all subcontracts, except those to be assigned to the Purchaser pursuant to paragraph (d) of GCC 3.47.2.3
c. deliver to the Purchaser the parts of the Works executed by the Contractor up to the date of termination

d. to the extent legally possible, assign to the Purchaser all right, title and benefit of the Contractor to the Works and to the Equipment/Materials as at the date of termination, and, as may be required by the Purchaser, in any subcontracts concluded between the Contractor and its Sub-contractors

e. deliver to the Purchaser all drawings, Specifications and other documents prepared by the Contractor or its Sub-contractors as at the date of termination in connection with the Works

3.47.2.4 The Purchaser may enter upon the Site, expel the Contractor, and complete the Works itself or by employing any third party. The Purchaser may, to the exclusion of any right of the Contractor over the same, take over and use with the payment of a fair rental rate to the Contractor, with all the maintenance costs to the account of the Purchaser and with an indemnification by the Purchaser for all liability including damage or injury to persons arising out of the Purchaser's use of such Equipment/ Materials, any Contractor's Equipment owned by the Contractor and on the Site in connection with the Works for such reasonable period as the Purchaser considers expedient for the supply and installation of the Works.

3.47.2.5 Upon completion of the Works or at such earlier date as the Purchaser thinks appropriate, the Purchaser shall give notice to the Contractor that such Contractor's Equipment will be returned to the Contractor at or near the Site and shall return such Contractor's Equipment to the Contractor in accordance with such notice. The Contractor shall thereafter without delay and at its cost remove or arrange removal of the same from the Site.

3.47.2.6 The Purchaser shall not be liable to make any further payments to the Contractor until the costs of execution and all other expenses incurred by the Purchaser in completing the Work or its turnkey work Package, as the case may be, have been ascertained.

If the Cost of Completion when added to the total amounts already paid to the Contractor as at the date of termination exceeds the total amount which would have been payable to the Contractor for the execution of the Work or work

Package, as the case may be, the Contractor shall upon demand, pay to the purchase the amount of such excess. Any such excess shall be deemed a debt due by the Contractor to the Purchaser shall be recoverable accordingly. If there is no such excess the Contractor shall be entitled to be paid the difference (if any) between the value of the Work or its work package and the total of all payment received by the Contractor as a the date of termination.

3.47.2.7 If the Purchaser completes the Works, the cost of completing the Works by the Purchaser shall be determined.

If the sum that the Contractor is entitled to be paid, pursuant to **GCC 3.47.2.6** , plus the reasonable costs incurred by the Purchaser in completing the Works, exceeds the Contract Price, the Contractor shall be liable for such excess.

The Purchaser and the Contractor shall agree, in writing, on the computation described above and the manner in which any sums shall be paid.

3.47.2.8 No account shall be taken of any increased cost which results from the Contractor's default or negligence.

3.47.3 In this clause **GCC 3.47**, the expression “Works executed” shall include all work executed, Installation Services provided, any and Equipment/ Material acquired (or subject to a legally binding obligation to purchase) by the Contractor and used or intended to be used for the purpose of the Works, up to and including the date of termination.

3.48. ASSIGNMENT & SUB-CONTRACTING

3.48.1 The whole of the works included in the Contract shall be executed by the contractor and the contractor shall not directly or indirectly transfer, assign or underlet the contract or any part, share or interest therein without the written consent of the Purchaser.

3.48.2 No sub-contracting shall relieve the Contractor from the full and entire responsibility of the Contract or from the active superintendence of the work during their progress.

3.48.3 The contractor has to engage specialized agencies / personnel depending upon the nature and complexity of the work with the prior approval of the Purchaser.

To this regard, the contractor has to submit the completion certificates / required documents of similar type of works executed by the subcontractor/ specialized agencies to establish the sub-contractor's/specialized agencies' workmanship. Also the contractor has to submit drawings done by the specialized agency for approval of Purchaser before procuring and installing the item. This does not in any way relieve the contractor of his obligations to get the quality work and architectural design as desired by the Purchaser.

3.49. JOINT AND SEVERAL LIABILITY

3.49.1 If the Contractor has formed a Consortium of not more than 3 (three) persons for implementing this Solar Power Project:

(a) These persons shall, without prejudice to the provisions of this Agreement, be deemed to be jointly and severally liable to the DPL for the performance of the Agreement; and

(b) The Contractor shall ensure that no change in the composition of the Consortium is effected without the prior consent of the DPL

3.49.2 In case of Consortium, without prejudice to the joint and several liability of all the members of the Consortium, the Lead Member shall represent all the members of the Consortium and shall at all times be liable and responsible for discharging the functions and obligations of the Contractor. The Contractor shall ensure that each member of the Consortium shall be bound by any decision, communication, notice, action or inaction of the Lead Member on any matter related to this Agreement and the DPL shall be entitled to rely upon any such action, decision or communication of the Lead Member only. The DPL shall have the right to release payments solely to the Lead Member and shall not in any manner be responsible or liable for the inter se allocation of payments among members of the Consortium.

If member of the consortium fails to perform satisfactorily then the Lead Member will be liable and responsible for completion of the project.

3.49.3 Issue of LOA and Contract agreement will be conferred with the Lead Member only.

SECTION –IV

SPECIAL CONDITION OF CONTRACT (SCC)

The following Special Conditions of Contract are supplementary, to the General Conditions of Contract. Whenever there is a conflict, the provisions herein shall prevail over those in the General Conditions of Contract.

4.1 DEFINITION

4.1.1. Purchaser means The Durgapur Projects Limited (DPL).

4.1.2. Project as well as details of the site is shown below:

AC Capacity of the Plant	5 MW
Minimum DC Capacity	5.75 MWp (Minimum)
PV Technology	Si Mono Crystalline
Design life of the Plant	25 years
Selected Location	DPL, Durgapur , Paschim Bardhaman, West Bengal.
Latitude:	23.55°N
Longitude:	87.32°E
Altitude:	65 meters
Nearest Major Towns	Durgapur
Nearest police station:	Coke Oven PS (2 km from site)
Nearest hospital:	Bidhannager S D Hospital (6 km from site)
Seismic Zone	Zone-III as per IS 1893-1984.

Access by Road	National Highway NH-2
Nearest Rail Station	Durgapur.
Access by Sea	Nearest port is Kolkata.
Access by Air	Andal, Durgapur

4.1.3. Engineer In Charge

The General Manager (O&M), DPL, Durgapur or his authorised Engineer will be Engineer in Charge.

4.1.4. Controlling Officer

The Deputy General Manager (Projects), DPL or his authorized Engineer will be Controlling Officer.

4.1.5. Project Manager

Project Manager will be selected and will be finalized in the kick off meeting at the Durgapur office of DPL.

4.1.6. Paying Officer

The General Manager (F&A), DPL , Durgapur

4.2 CONTRACT PERFORMANCE GUARANTEE

4.2.1. Within fifteen days (15) from date of LOA, the Contractor shall furnish Performance Guarantee to the Purchaser amounting to **10%** of the accepted Project Cost Plus GST in the form of Bank guarantee from scheduled bank. The (Earnest Money Deposit) EMD shall be returned on submission of the Performance Guarantee of success full bidders.

4.2.2. **Release of Contract Performance Bank Guarantee (PBG):** 100% after the successful completion of defect liability period i.e. 5 years after successful commissioning of the Project.

4.2.3. Contract agreement will be executed after submission of PBG, deposited EMD shall be returned.

4.3 PAYMENTS

Following terms of payment shall be applicable –

4.3.1. Performance Security:

5% of contract value will be retained and it will be released in five equal installments every year in the defect liability period subject to compliance of following conditions by the contractor:

Integrated project performance of **Net Minimum Guaranteed** Generation (NMGG) solar energy at rate of **1.60 MU/MW AC** annually **with degradation of 0.7%** for any reason, from second year onwards. If generated units fall short, then Rs.4.00 per unit of short fall will be deducted from DPL payments every year up to 5th year. Maximum cap for NMGG shall be 1% per year i.e. totality 5% of Project Cost plus GST for 5 years.

4.3.2. Total Integrated System Warranty/Contract Performance Guarantee

a. After completion of the project the contract performance bank guarantee (CPBG) will be the total integrated System warrantee.

b. CPBG will be returned to the contractor after successful completion of the defect liability period including O&M period.

4.3.3.PAYMENT TERMS

4.3.3.1 After receiving an application for payment, duly complete in all respects, the Purchaser shall pay the amount certified after issue of each certificate of payment to the Contractor at his principal place of business. However, delay in making payment by the owner to the contractor shall not attract any interest on the due and payable amount.

4.3.3.2 Mobilization Advance:

i. 10 % of the Contract Price (Supply & Erection contract) will be paid as interest Free Mobilization advance to the contractor against submission of Mobilisation Advance Bank Guarantee (**ABG-Annexure-3**) equivalent to **100 %** of the Mobilisation Advance, subject to the conditions mentioned below.

ii. Unconditional acceptance of the LOA.

iii. Taking over of site from DPL

iv. Execution of Contract Agreement

v. Submission of Performance Bank Guarantee

vi. Submission of duly authenticated Activity Schedule in MS Project showing the entire execution of work.

vii. The Bank Guarantee shall be valid for a total period of 12 (twelve) months plus a claim period of 3 (three) months from date of LOA.

viii. Mobilization advance, so paid shall be recovered as mentioned in the **Payment Schedule (clause SCC 4.3.3.7)**.

ix. Bank Guarantee submitted for mobilization advance will be released after full recovery of the mobilization advance and on receipt of written request of the contractor for release of the same.

The mobilization advance will be interest free for the scheduled Completion period from the date of LOA of the Project against 100 % unconditional BG with a claim period of 90 days initially. However, in case of there is delay in completion of the facilities covered under the package, the validity of this Bank Guarantee shall be extended by the period of such delay along with additional 90 (ninety) days' claim period.

4.3.3.3 Not Applicable

4.3.3.4 Supply of Materials:

Payment for supply of material at site will be given as per the **Payment Schedule (clause SCC 4.3.3.7)** given below after satisfactory acceptance by Purchaser of the supplied materials along with the relevant following documents mentioned:

I. **Deduction:-**

a) Statutory deductions like TDS, GST if any shall be deducted.

b) Adjustment of any excess / short payment made in the earlier bills, at the time of making payments.

c) Electricity charges will be recovered as per tender terms.

d) Value of chargeable materials if any issued by the Purchaser.

e) Any other recovery if due as per tender terms & conditions.

II. **Documents:-**

a) Computerized detailed measurements, Running Accounts Bills and Final Bill will be Prepared by contractor and submitted to DPL for verification and approval.

b) Document for claiming subsidy from MNRE should be submitted to DPL for release of first RA bill.

c) **DPL gate entry documents/MRC(M**aterial Receive Certificate)

d) Relevant test, type test, joint inspection reports warranty and guarantee Certificate for the items supply as per quality criteria mentioned in tender document.

e) Copy of Performance guarantee for contract execution from Scheduled bank valid till completion of work for 3% of purchase order value.

f) Labor License (as per statutory requirements).

g) EPF Code Registration number with RPFC.

h) Insurance – Contractor's All Risk (CAR) Policy.

i) Workmen compensation policy, Proof for PF deduction and remittance.

j) GST or Other tax invoice, Service Tax registration number.

k) Indemnity Bond (ANNEXURE-5) in standard pro forma to indemnify the Purchaser against all risks arising during the performance of the contract.

l) Proof of deployment of project engineers as specified in SCC.

m) Challans / receipts of taxes paid to statutory authorities i.e., GST, Custom duties etc.

n) A Certificate towards the effect that minimum Technical and Safety man power was employed for the work execution Certified by the DPL representative as per the Tender Clauses.

o) Undertaking for compliance of all labor laws.

4.3.3.5 Payment of Bills for installation, erection, testing, integration, successful commissioning of integrated system in total, PG test and ready for handing over to DPL

Payment for the Bill amount for the materials quoted will be paid as per the **Payment Schedule (clause SCC 4.3.3.7)** after joint inspection / measurements by the Purchaser for installation, erection, testing, integration, successful commissioning of integrated system in total and ready for handing over to DPL by the contractor and submission of following document :

I. Deduction:-

a) Deduction of the Value of chargeable materials if any issued by the Purchaser.

b) Statutory deductions like TDS, GST, Custom duties if any.

c) Charges for Electricity.

d) Any other recovery if due as per tender terms & conditions.

e) Adjustment towards any excess / short payment made in the earlier bills.

II. Document:-

a) Relevant test/ type test/ joint inspection reports warranty and guarantee Certificate for the items installed, integrated & commissioned as per quality criteria mentioned tender document.

b) DPL gate entry documents if any.

c) Proof of deployment of project engineers as specified in SCC.

d) Challans / receipts of taxes paid to statutory authorities i.e., GST, Customs Duties etc

e) A Certificate towards the effect that minimum Technical and Safety man power was employed for the work execution Certified by the DPL representative as per the Tender Clauses.

f) Undertaking for compliance of all labor laws.

g) Valid Indemnity Bond in standard pro forma indemnifying the Purchaser against all risks arising during the performance of the contract.

h) Valid Workmen compensation policy, Proof for PF deduction and remittance.

i) Valid Insurance – Contractor's All Risk (CAR) Policy.

j) Valid Labor License (as per statutory requirements).

k) MNRE–Govt of India format project complication certificates / documents to DPL.

l) Computerized detailed measurements, Running Accounts Bills and Final Bill will be prepared by contractor and submitted to DPL for verification and approval.

4.3.3.6 Payment of Bill – Performance testing of total integrated system – final Payment / bill:

Payment for materials bill shall be paid as per the **Payment Schedule as per clause SCC 4.3.3.7** after performance testing of total integrated system for two months in all respect. The final bill complete in all respect shall be submitted by the contractor accompanied with the following documents.

I. Deduction:-

a) Statutory deductions like GST and other tax and duties if any shall be deducted DPL reserves the right to adjust any excess / short payment made in the earlier bills, at the time of making payments.

b) Adjustment of any excess / short payment made in the earlier bills, at the time of making payments.

c) Electricity charges will be recovered as per tender terms.

d) Value of chargeable materials if any issued by the Purchaser.

e) LD if any and other recovery if due as per tender terms & conditions.

II. Document:-

a) Computerized detailed measurements, Running Accounts Bills and Final Bill will be Prepared by contractor and submitted to DPL for verification and approval.

b) Validated of Total Integrated System Warranty PBG/Contract PBG valid for 72 months with additional 90 days claim period for the value equivalent to 10% of the total final project value from scheduled bank.

c) Proof of project completion and relevant documents as per MNRE format for release of subsidy by MNRE, Govt of India to DPL.

d) Job completion certificate by DPL (**Annexure-6**).

e) No claim certificate on Purchaser's prescribed pro forma - if any deduction is to be made for short fall, Purchaser shall record the same in this document.

f) Site clearance certificate by DPL.

g) Indemnity certificate towards labor payment and all statutory payments.

h) Copy of the insurance policy. (Workmen compensation act and contractors all risk policy).

i) Operation and Maintenance manuals and testing and commissioning reports with Guarantee and Warranty certificates.

j) Statutory Compliance certificate towards payment of insurance, GST and other taxes if any applicable.

k) A compliance Certificate should be submitted towards deployment of Technical & Safety man power as per the relevant Tender Clauses.

l) Proof of deployment of annual maintenance & operation manpower for operation of the plant.

m) Valid Workmen compensation policy,

n) Valid Insurance – Contractor's All Risk (CAR) Policy for AMC personnel.

o) Valid Labor License (as per statutory requirements) for AMC personnel.

p) Relevant test/ type test/ joint inspection reports warranty and guarantee Certificate for the items installed, integrated & commissioned as per quality criteria mentioned tender document.

q) DPL gate entry documents if any.

r) Undertaking for compliance of all labour laws.

s) Valid Indemnity Bond (ANNEXURE-5) in standard pro forma indemnifies the Purchaser against all risks arising during the performance of the contract.

t) Final acceptance certificate issued by DPL /Purchaser.

4.3.3.7 Payment Schedule:

Sl. No.	Work Head	Pattern of Release of Billing Amount
1	**Mobilization Advance**	**10%** of the total contract price of Supply and Erection & Commission (excluding quoted price of Mandatory spare) will be paid as Mobilization advance to the contractor against submission of Mobilisation Advance Bank Guarantee (**ABG**) equivalent to 100 % of the Mobilization Advance, subject to the conditions mentioned in the Clause No. SCC **4.3.3.2** above. The mobilization advance will be recovered as per the sub clause mentioned in Sl. No. 2 (i) & 3(i) below. NOTE:- (1) For **supply of material**, submission of Mobilisation Advance Bank Guarantee (**ABG**) should be equivalent to 100 % of the Mobilization Advance **excluding GST**. (2) For **Erection & Commissioning**, submission of Mobilisation Advance Bank Guarantee (**ABG**) should be equivalent to 100 % of the Mobilization Advance **excluding GST**. However, after completion of scheduled completion period,balance Mobilization Advance shall be recovered. In case of time extension without prejudice to the original contract, said BG shall have to be extended by the contractor upto the given extension period with an additional 90 days of claim period.
2	**Contract for Supply of Material**	i. 70% of the contract value will be given on prorate basis after material received at site (Mobilization Advance paid as per Clause 4.3.3.4 will be recovered proportionately from bills under this phase of payment).

Sl. No.	Work Head	Pattern of Release of Billing Amount
		ii. 15% of the contract value will be paid prorata basis after completion of the erection of the material
		iii. 5% of the contract value on prorate basis after successful completion of the commissioning and testing.
		iv. 5% of the contract value after successful completion of PG test.
		v. 5% of the Contract value will be retained for Performance warranty as per clause no. SCC 4.3.4. However, 5% of contract amount shall be released against submission of 5 numbers unconditional BG of equivalent amount i.e. 1% each of Basic project cost plus GST after completion of Project and before start of comprehensive O&M contract. 5 nos. BG shall be submitted with different validity period upto successful completion of O&M contract with a claim period 90 days. Penalty due to NMGG if any shall be recovered from these BG during O&M period. Details of BG validity is mentioned under clause no. SCC 4.3.4.
3	**Contract for Erection & Commissioning**	i. 65% of the contract value will be given prorata basis after erection of the material (Mobilization Advance paid as per Clause 4.3.3.5 will be recovered proportionately from bills under this phase of payment).

Sl. No.	Work Head	Pattern of Release of Billing Amount
		ii. 20% of the contract value will be paid prorata basis after completion of the Testing and Commissioning of the material
		iii. 5% of the contract value after successful completion of PG TEST.
		iv. 5% of the contract value after successful completion of all facilities and Final Acceptance Certificate of project.
		v. 5% of the Contract value will be retained for Performance warranty as per clause no. SCC 4.3.4. However, 5% of contract amount shall be released against submission of 5 numbers unconditional BG of equivalent amount i.e. 1% each of Basic project cost plus GST after completion of Project and start of comprehensive O&M contract. 5 nos. BG shall be submitted with different validity period upto successful completion of O&M contract with a claim period 90 days. Penalty due to NMGG if any shall be recovered from these BG during O&M period. Details of BG validity is mentioned under clause no. SCC 4.3.4.
4	**Supply of Mandatory spares**	i) 70% of the Supply price of Mandatory spare will be given on prorata basis on production of invoices, satisfactory evidence of shipment including Insurance and Material Dispatch Clearance Certificate (MDCC) issued by DPL.

Sl. No.	Work Head	Pattern of Release of Billing Amount
		ii) 30% of the Supply price of Mandatory spare will be given on prorata basis on receipt and storage at site and on physical verifications by DPL Site Official of having received and stored the spares at site.
5	**Additional advance (Other than Mobilization advance)**	NA

4.3.4. Performance Penalty:

a) Integrated project performance of minimum solar energy to be generated 1.60 MU/MW annually with degradation of 0.7% for any reason, from second year onwards. If generated units fall short, then Rs.4.00 per unit of short fall plus applicable taxes (if any) will be deducted from DPL payments every year up to 5th year upto 1% maximum of Project cost plus GST per annum.

b) Penalty will be deducted from retained amount against Performance warranty maximum 5% of Project cost plus GST.

c) DPL reserves the right to adjust any excess / short payment made in the earlier bills, at the time of making payments.

d) However retention amount i.e. 5% of Project cost plus GST per annum shall be released against submission of 5 numbers unconditional BG of equivalent amount i.e. 1% each of project cost plus GST after completion of Project and start of comprehensive O&M contract. 5 nos. BG shall be submitted with different validity period upto successful completion of O&M contract with a claim period 90 days. Penalty due to NMGG if any shall be recovered from these BG during O&M period.

BG validity:

- 1% of Project cost plus GST: validity period of One year from O&M start date plus additional claim period of 90 days.
- 1% of Project cost plus GST: validity period of Two years from O&M start date plus additional claim period of 90 days.
- 1% of Project cost plus GST: validity period of Three years from O&M start date plus additional claim period of 90 days.
- 1% of Project cost plus GST: validity period of four years from O&M start date plus additional claim period of 90 days.
- 1% of Project cost plus GST: validity period of five years from O&M start date plus additional claim period of 90 days.

4.3.5. Defects Liability Period:

Defect liability period is 60 calendar months from the date of Issue of Final completion certificate/Acceptance of work by DPL.

4.3.6. INCREASE IN COST:

Quoted prices are firm and no escalation charges on any account are allowed in this work. No claim will be entertained on this account in future.

4.3.7. DISALLOWANCE OF PAYMENT:

If payment has been made for any item but later on some defect is noticed, Purchaser is authorized to disallow payment of the subsequent bill till rectification /replacement of the item.

4.3.8. ESCALATION

No Escalation shall be paid on any account.

4.3.9. LIQUIDATED DAMAGES

a) **Time Delay:** 0.5 % of the Project cost plus GST for per week delay or part there of subject to a maximum of 10% of the Project cost plus GST.

b) **Milestone Delay:** Delay in attaining the milestones by the contractor shall lead to imposition of intermediary Liquidated damages of delay (as per GCC Cl. No. 3.34) upto the maximum extent of 5%(Five) Percent of the Project cost plus GST. However, if bidder shall complete the dependent subsequent milestone within the scheduled time, earlier penalty due to milestone delay if any shall be waived off.

However this condition shall not relief the bidder from the other stipulations mentioned in the NIT.

c) **Total LD value** (LD for Time Delay + LD for Milestone Delay) shall **not exceed 10%** of total Project cost plus GST.

4.6 NOTICES

For the purpose of all notices, the following shall be the address of the Purchaser and the Contractor.

PURCHASER:

The GM (O&M)
The Durgapur Projects Ltd.

Durgapur -713201

CONTRACTOR:

____

____

(To be filled in at the time of Signing of the Contract)

4.7 LABOUR

4.7.1. The Contractor shall make his own arrangements for the engagement of all staff and labor, local or other, and for their payment, housing, food, transport etc. No labor to stay at site.

4.7.2. The Contractor shall, if required by the Purchaser deliver to the Purchaser a return in detail, in such form and at such intervals as the Purchaser may prescribe, showing the staff and the numbers of the several classes of labor from time to time employed by the Contractor on the Site and such other information as the Purchaser may require.

4.7.3. Compliance with labor regulation: During continuance of the contract, the Contractor shall abide at all times by all existing labor enactments and rules made there under, regulations, notifications and bye-laws of the State or Central Government or local authority and any other labor law (including rules), regulation by laws that may be passed or notifications that may be issued under any labor law in future either by the State or the Central Government or the local Authority. The Contractor shall keep the Purchaser indemnified in case any action is taken against the Purchaser by the competent authority on account of contravention of any of the provisions of any Acts or rules made there under, regulation or notifications including amendments. If the Purchaser is caused to pay or reimburse, such amount as may be necessary to cause or observe, or for non-observance of the provisions stipulated in the notifications/bye laws/Acts/Rules /regulations including amendments, if any, on the part of the Contractor, the Purchaser shall have the right to deduct from any money due to the Contractor including his amount of performance security. The Purchaser shall also have right to recover from the Contractor any sum required or estimated to be required for making good the loss or damage suffered by the Purchaser

4.7.4. The employees of the Contractor and the Sub-Contractor in no case shall be treated as the employees of the Purchaser at any point of time.

4.7.5. No labor shall stay at site. Temporary storage space provision should be made by contractor.

4.7.6. The rates shall be complete in all respects i.e. inclusive of all taxes, local taxes, work contract tax, Insurance charges nothing on any account shall be paid over the approved rate.

4.7.7. All specialized and specific jobs shall be carried out by approved agencies/vendors only.

4.7.8. The Contractor shall arrange temporary drinking water and sanitation facilities for his workmen.

4.7.9. Minimum Wage as applicable

i. The contractor shall pay not less than fair wages to laborers engaged by him on the work.

ii. “Fair” wages means wages whether for time or piecework notified by the Government from time to time for the locality of work.

iii. The contractor shall not-with-standing the revisions of any contract to the contrary cause to be paid to the labor directly engaged on the work including any labor engaged by the sub-contractor in connection with the said work, as if the laborers had been directly employed by him.

iv. In respect of labor directly or indirectly employed in the works for the purpose of the contractors part of the agreement the contractor shall comply with the rules and regulations on the maintenance of suitable records prescribed for this purpose from time to time by the Government. He shall maintain his accounts and vouchers on the payment of wages to the laborers to the satisfaction of the Purchaser.

v. The Purchaser shall have the right to call for such record as required to satisfy himself on the payment of fair wages to the laborers and shall have the right to deduct from the contract amount a suitable amount for making good the loss suffered if any by the worker or workers by reason of the “fair wages” clause to the workers.

vi. The contractor shall be primarily liable for all payments to be made and for the observance of the regulations framed by the Govt., from time to time without prejudice to his right to claim indemnity from his sub-contractors.

vii. As per contract labor (Regulation and abolition) Act.1970 the contractor has to produce the license obtained from the licensing officers of the labor department along with the tender viii. Any violation of the conditions above shall be deemed to be a breach of his contract.

viii. Equal wages are to be paid for both men and women if the nature of work is same and similar.

ix. The contractor shall arrange for the recruitment of skilled and unskilled labor local and imported to the extent necessary to complete the work within the agreed period as directed by the Purchaser.

x. The Contractor/Sub-contractors to be engaged by the contractor subsequently, must have its own PF & ESI Code.

4.8 SAFETY MEASURES & SECURITY

The contractor shall take necessary precautions for safety of the workers and preserving their health while working in such jobs, which require special protection and precautions. The Purchaser has standard safety guidelines for the Contractor. The Contractor is to be followed the above guidelines which will be provided with placement of the Award of Contract.

4.8.1 SECURITY SERVICES:

a) The contractor has to arrange proper security system round the clock including deputation of security personnel at his own cost for the check vigil for the Solar Power Plant for both phases for the complete scope of works including comprehensive O&M period.

b) The security staff may be organized to work on suitable shift system; proper checking & recording of all incoming & outgoing materials vehicles shall be maintained. Any occurrence of unlawful activities shall be informed to Employer immediately. A monthly report shall be sent to DPL on the security aspects.

4.9 FIRE FIGHTING MEASURES

i. The contractor shall provide and maintain adequate firefighting equipment and take adequate fire precaution measures for the safety of all personnel and temporary and permanent works and shall take action to prevent damage and destruction by fire of trees shrubs and grasses.

ii. Separate payment will not be made for the provision of fire prevention measures.

4.10 DEPLOYMENT OF MANPOWER DURING O&M

The contractor shall deploy the following minimum man power at site to take instructions from DPL Staff & Report the site activities on day to day basis execution of quality work and maintain all statutory records as per Govt. norms/as directed by the Purchaser.

i) Engineer with 2 years' experience in Solar Plants and SCADA operation - 1no.

ii) ITI grade in electrical with 2 years' experience – 1 nos.

iii) Other skilled manpower as per the requirement of the contractor to meet NMGG requirement.

iv) All relevant statutory compliances also to be fulfilled during O&M period.

v) No objection certificate for HR obligation from STPS HR&A department for the billed months.

vi) For day to day operation the contractor shall furnish the name, bio data, experience of the personal intended to be posted at site at the start of work. Site Engineers are to be deployed irrespective of the Contractor or the Director / Manager of the contracting company being an Engineer himself. Attendance register should be kept at the office of DPL designated engineer.

vii) The Contractor shall maintain supervisor staff to Labour ratio as per standard government / CPWD norms.

viii) The Contractor may employ more number of site Engineers other than above mentioned Engineers to get quality workmanship and maintain all statutory.

ix) However, the site-incharge must be on payroll of the EPC vendor during entire tenure of Project and comprehensive O&M period upto completion of defect liability.

4.11 QUANTITY OF WORK

The quantities of work shall be mentioned in the LOA.

4.12 PROJECT MANAGEMENT

4.12.1. CONTRACTOR'S ORGANISATION

The Contractor shall supply to the Purchaser and the Project Manager a chart showing the proposed organization to be established by the Contractor for carrying out work on the Facilities within twenty-one (21) days of the Effective Date. The chart shall include the identities of the key personnel and the curricula vitae of such key personnel to be employed shall be supplied together with the chart. The Contractor shall promptly inform the Purchaser and the Project Manager in writing of any revision or alteration of such an organization chart.

4.12.2. PROGRESS REPORT

Monthly progress report of the previous month along with photographs of work progress shall be submitted to the Purchaser before the 7th day of every month. Format for the monthly report will be finalized in the Kick Off meeting.

4.12.3. PROJECT REVIEW MEETING

Progress of the project will be evaluated in the Project Review Meeting (PRM) in every month. Date of the PRM will be informed by the Purchaser to the entire contractor.

4.12.4. SPECIAL PROJECT REVIEW MEETING

Special Project Review Meeting (SPRM) will be held after completion of the each milestone mentioned in the data sheet (approved network by the Purchaser).

In this meeting the total time taken for complete the milestone will be finalized/evaluated and if any delay occurs then delay contribution will be calculated and minuted by both Contractor as well as Purchaser.

4.13 ACCIDENT OR INJURY TO WORKMEN

4.13.1. DPL shall not be responsible for any injury or loss of life of any worker of the contractor that may take place while on work. Any compensation or expenditure towards treatment for such loss of life or injury shall be the sole responsibility of the contractor.

4.13.2. The contractor is solely responsible for any damage injury or accident that may occur to any of his personnel working under this contract. He will not claim any compensation from DPL.

4.14 REGULATIONS & STANDARDS

The installation shall confirm in all respects to the Indian Standard Code of Practice for Electrical Installations. It shall also be in conformity with the current Indian Electricity Rules and Regulations and requirements of the local Electric Supply Authority in so far as these become applicable to the installation. Wherever the specifications of this tender document calls for higher standard of material and/or workmanship than those required by any of the above codes and regulations then this specifications shall take precedence over the said regulations and standards.

4.15 INSPECTION AND TESTING

4.15.1. Required Inspection charges at the Factory are under scope of Contractor.

4.15.2. Inspection & Testing for Imported items like Inverters, Transformer, Battery and charger, SCADA, 33 kV HT switchgear, RMU etc may be done at Manufacturer local testing facilities / at the works of the contractor in India as desired by the contractor. In such case the contractor shall provide full set up for inspection and routine test of the item.

4.16 DESIGN AND ENGINEERING

4.16.1. The optimum system design and sizing of the equipment of the PV power plants shall be done by using design software PV- Syst or any other modern tools by using the accurate meteorological data base like latest version of Solargis approved by DPL.

4.16.2. The contractor shall develop the general layout drawing of Array Yard, Inverter, AJB, Grid Interfacing panel, Single line diagram and other drawing as may be required. All designs & drawings are to be developed based on specification given in the tender, relevant BIS unless otherwise specified.

4.16.3. The Power Plants shall have to be designed considering optimal usage of space without compromising the effect of shadow, cooling, accessibility, losses, protection, security and safety etc.

4.16.4. Document to be submitted during approval of the Design and Drawing:

During approval of drawing and design of the PV Power Plant the documents have to be submitted by the contractor which shall be includes but not limited as follows:

1) Power plant design document
2) PV Array and other component layout drawing of the PV Power Plant
3) Drawing of different equipment of PV power Plant
4) Design and drawing of PV Module mounting along with the MMS arrangement of PV array as per technical specification.
5) List of Equipment and Component and its capacity and manufacturer name to be used in the PV Power Plants
6) Type test report of Transformer, Inverters, PV Module, Cables, 220 kV Switchyard equipment like CB, isolator, CT etc, 33 HT Switchgear, SCADA, Battery and charger, LT Panel, etc
7) Original Technical Manual of the Equipment and Component.

4.17 COMPLETION OF WORK

The Contractor shall assist the beneficiaries and liaison with the DISCOMs if necessary for installation of Net Meter by the DISCOM and complete the formalities for availing net metering benefit to make the power plant is operational from the date of commissioning.

4.18 TRAINING OF OWNER'S ENGINEER

Contractor will arrange for training on Manufacturing, Erection, Operation and Maintenance of owner's engineer at STPS site on following equipment:

i) Inverter;
ii) Solar Module;
iii) SCADA;
iv) RMU;

Bidder shall provide minimum Thirty-Two man days (eight man for four days) training at site along with study material.

4.19 TAKING OVER:

4.19.1 OPERATIONAL ACCEPTANCE FOR EACH PHASE:

i) All As-Built Drawings & Design of the power plant in Six (06) sets and two sets soft copy in two separate Pen Drives.
ii) Detailed Engineering Document with detailed specification, schematic drawing, Design and test results, manuals for all deliverable major items, Operation, Maintenance & Safety Instruction Manual and other information about the project in Six (06) sets and two sets soft copy in two separate Pen Drives.
iii) Commissioning & completion of trial run certificate;.

The project completion certificate issued by DPL upon successful completion of all the facilities including successful PG test pertaining to the scope of work. Contractor shall approach DPL in writing for "final completion of project work" of the plant. On receipt of such request, DPL shall issue to the contractor a Project completion certificate as a proof of the of the system other than successful completion of defect liability period including comprehensive O&M. Such certificate shall not relieve the Contractor of any of his obligations which otherwise survive, by the terms and conditions of the Contract after issuance of such certificate.

4.19.2 HAND-OVER OF PROJECT:

During handing over of the complete project work, the contractor will submit the followings for considering final payment.

a) Bill of materials
b) Site wise documentation as per MNRE GoI Format.
c) Performance Guarantee Certificate of PV modules from the original manufacturer (**Cl. No. 4.22 & 4.23**).
d) Handover of all Mandatory spares at DPL store.
e) Completion certificate as per prescribed format provided by DPL
f) Approved PG Test reports and yearly NMGG reports.
g) Site maintenance logbook.
h) Handover of Insurance policies.

4.20 COMPREHENSIVE MAINTENANCE

Comprehensive Maintenance during defect liability period

4.20.1. All the equipment to be installed for commissioning of each of the grid connected solar PV power plant and the power plant in whole shall be under Comprehensive Maintenance Contract within the scope of the tender for 5(five) years from the date of commissioning. The equipment or components, or any part thereof, so found defective during Comprehensive Maintenance Contract period will be forthwith repaired or replaced within the scope of guarantee obligation to the satisfaction of the Purchaser.

The maintenance of grid connected solar PV power plant include routine & periodic maintenance, overhauling, breakdown maintenance, and repairing or replacement of defective PV modules, invertors, and other components, providing of consumables. The Down-Time of PV system should not be more than 72 hours (03 days). Details of the maintenance scope are mentioned in the clause no. **GCC 3.8.4**

4.20.2. Routine maintenance:

In order to carry out routine maintenance of the power plant, the contractor will provide all labour, material, consumables etc. within the scope of maintenance service. Recommended tusks under the scope of routine maintenance will include but not limited to the followings:

Sl No.	Type of Routine Maintenance
01	Checking and tightening of all electrical connections
02	Checking and tightening of mechanical fittings
03	Checking and restoring of earthing system
04	Dusting and cleaning of Inverter and other electrical equipment
05	Routine maintenance as recommended by the Original Equipment Manufacturer (OEM)

4.20.3. Rental and Other Periodic Charges

The contractor shall pay the rental charge of the SIM / Telephone Bill web connectivity in order to transfer of data related to Web Based Remote Monitoring System. The contractor shall also pay the rental charges for server of the web based remote monitoring system for storing and access the data till the defect liability period is over. The Contractor shall also pay necessary charges periodically for refilling of the Fire Extinguisher till defect liability period is over.

4.20.4. Comprehensive maintenance shall be the integral part of the contract. 5% of the total order value (Supply and service) shall be retained and shall be released after successful completion of project and submission of 5 nos. of unconditional BG. This is apart from the 3% Contract Performance Guarantee (CPG). The Comprehensive maintenance shall be included the rental charges mentioned under **SCC.4.20.3**

4.20.5. The payment towards Operation & Maintenance shall be made on yearly basis on submission of bills in triplicate to DPL along with a copy of the maintenance report during the claimed period which will already to be submitted on quarterly subject to satisfactory performance as per Clause no: **SCC.4.3.2** and its sub clauses on submission of maintenance report on regular basis.

4.21 WARRANTY OF EQUIPMENT

The contractor shall warrant that all equipment, hardware and accessories are new, unused, most recent or current models and incorporate all recent improvements in design and in accordance with the contract documents and free from defects in material and workmanship. The contractor shall also guarantee for defect free operation of the materials supplied and workmanship towards erection for a period of **sixty (60) calendar months** commencing immediately upon date of TOC.

4.22 MANUFACTURER WARRANTY CERTIFICATE:

The manufacturer should warrant the supplied equipment, hardware and accessories free from the defects and/or failures specified below from the official start date of comprehensive O&M activity.

i. Defects and/or failures due to manufacturing
ii. Defects and/or failures due to quality of materials
iii. Non conformity to specifications due to faulty manufacturing and/or inspection processes.

If any equipment fails to conform to this warranty, the manufacturer will replace the equipment, at the Purchaser's sole option.

The contractor has to submit the Warrantee Certificate issued by the Equipment Manufacturer to DPL against delivery of each lot. The Warrantee Certificate issued by the contractor must comprise order no. of DPL, name of the project and name of DPL.

The contractor should submit the warrantee certificate issued by the original Equipment manufacturer to DPL after delivery of every lot. The warrantee shall include but not limited to the following:

a. Capacity and model of the Equipment,
b. Equipment serial no.
c. Warrantee period:
d. Name of DPL (as principal purchaser)
e. Order reference of DPL
f. Order reference & name of the of vendor

4.23 PERFORMANCE GUARANTEE OF PV MODULE:

i) The manufacturer should warrant the output of Solar Module(s) for at least 90% of its rated power after initial 10 years & 80% of its rated power after 25 years from the official start date of comprehensive O&M activity. The contractor shall collect the Warranty Certificate for performance of the modules from the manufacturer and submit the same to DPL prior to delivery of the products to the respective sites.

ii) The modules shall be warranted, against all material/ manufacturing defects and workmanship for minimum of 5 years from the start date of Comprehensive O&M activity.

If, Module(s) fail(s) to exhibit such power output in prescribed time span, the Contractor will bound to either deliver additional PV Module(s) to compensate the shortfall power output with no change in area of site used with no extra cost claimed.

The contractor has to transfer Performance Guarantee Certificate of the module from the original manufacturer to the Purchaser for subsequent arrangement after completion of the guarantee period i.e. successful completion of Defect liability period.

4.24 KICK OFF MEETING

Project Kick Off meeting to be held within 15 days of the placement of LOA and venue of the kick off meeting will be the Corporate Office of DPL.

4.25 CLEAN-UP AND WASTE DISPOSAL

Contractor shall keep the power project site reasonably clean and otherwise free from accumulation of waste materials, rubbish and other debris resulting from performance of the EPC Contractor's obligations. Contractor shall be maintained project site at hygienic standards and shall be kept reasonably free from debris, litter and malodour on or before Final Performance Acceptance. The EPC Contractor shall remove from the power plant site area all petroleum, waste materials, rubbish and other debris, as well as all tools, construction equipment, machinery and surplus material which the Client does not hold title, and shall make the power plant area in a neat, clean and usable condition. The EPC Contractor shall remove, transport and dispose-off hazardous material transported into the power plant site or any subcontractor or created, used or handled as part of contractor's or any subcontractor's construction activities at the power plant site.

The EPC Contractor shall notify Purchaser immediately upon the discovery of presence of any hazardous material on, or the release of hazardous material on or from, the power plant site. All clean up and disposal activities of contractor (including, without limitation, the transportation and disposal of any hazardous materials taken from the power plant site) shall be conducted in accordance with all Applicable Laws and Applicable Permits. All these shall be applicable during the O&M period also in mutatis mutandis.

4.26 CO-ORDINATION WITH STATUTORY BODIES AND OUTSIDE AGENCIES AND COMPLIANCE OF STATUTORY REQUIREMENT.

The Bidder shall be fully responsible for carrying out all co- ordination and liaison work with Electrical Inspectors, Factory Inspector, and other statutory bodies for implementation of the work including all the licence fees, statutory fees etc. Applications on behalf of the Owner, for submission to the Electrical Inspector and other statutory bodies along with necessary drawings complete in all respects shall be prepared by the Bidder. Approved drawings and certificates shall be submitted to the Owner/Consultant well ahead of schedule so that the actual commissioning of equipment does not get delayed for want of inspection and approval by the Inspector and other statutory bodies. The actual inspection work by the Electrical Inspector shall be arranged by the Bidder and necessary coordination and liaison work in this regard shall be the responsibility of the Bidder.

Bidder has to comply with all the latest statutory requirements including CEA grid connectivity standards, technical standards, plant and equipment safety standards, MNRE guidelines/OM/Advisory/Clarifications. All equipment, materials and services whether explicitly stated in Technical Specifications or otherwise and that are necessary for the successful commissioning of Solar Plant as per latest statutory regulations/procedures issued by bodies like MNRE, SERC, SLDC, Orders of Govt. of WB and other Ministry etc. shall be deemed to be included in the scope of work of the Contractor.

4.27 PERFORMANCE GUARANTEE (PG) TEST

4.27.1 The final acceptance test as to prove the Performance Guarantee shall be conducted at Site by the Contractor in presence of the Employer. The PG test shall be conducted on the basis of PG test procedure to be submitted by the contractor and approved by DPL. **Maximum LD for P G Test Failure is 5% of the Total Contract Price plus GST**

4.27.2 This test shall be binding on both the parties of the Contract to determine compliance of the equipment with the functional guarantee. Any special equipment, instrumentation tools and tackles and manpower, required for the successful completion of the Performance Guarantee Test shall be provided by the Contractor free of cost. The accuracy class of the instrumentation shall be as per the relevant clause of documents.

4.27.3 The procedure for PG demonstration test shall be as follow: Any consecutive three months period for the purpose of conducting performance guarantee test shall be chosen on the discretion of DPL.

1. “Target Generation” for the month and corresponding Global horizontal insolation of the site for is given below table.

Table -A : "Target Generation" for the month and corresponding Global horizontal insolation of the site		
Month	GHI (Monthly) kWhr / m^2	Target Generation for the month (kWh)
January	129.3	553599
February	141.2	604549
March	187.3	801926
April	193.5	828472
May	198.2	848595
June	166.8	714156
July	151.7	649505
August	159.1	681188
September	146.1	625529
October	145.8	624244
November	127.8	547176
December	121.7	521061
Total yearly Generation	1868.5	8000000

The value of the target generation and corresponding Global Horizontal Insolation shall be on pro-rata basis in case the PG test does not start from the first day of the month.

2. In addition to the two pyranometers to be supplied under the scope of work, the contractor shall install calibrated pyranometers at horizontal plane at locations mutually agreed by Contractor and DPL. The additional pyranometer shall be free of cost on returnable basis.

3. Contractor shall also install data logger to store all the pyranometers data during test period. A valid test reports for the installed pyranometers shall be

submitted by the Contractor for approval to DPL. The output of both pyranometers mounted on horizontal plane shall be made available at SCADA during the complete PG test duration i.e. three month period.

4. During the PG test period, the module tilt shall be kept as per approved design.

5. Actual energy exported from the plant shall be recorded for three consecutive month period. For this purpose, the net energy exported at the metering point and pyranometers reading shall be noted at agreed frequency on daily basis for entire PG test period.

6. This measured value of energy at step-5 shall be compared with "Target Generation" for the PG test given in Table-"Target Generation".

Following factors shall be considered for computing the "target Generation"

a) Effect of any meteorological parameters shall not be considered except of solar radiation.

b) Variation of PG on account of Generation loss due to grid outage (or power evacuation system which is not in the scope of the Bidder): The measured global solar radiation of the period of the outage of the power evacuation system shall be excluded to calculate the cumulative global Insolation for the month. Under such situation, the radiation corresponding to the warm-up time of inverter as per data sheet shall also be adjusted to arrive at the cumulative global insolation for the month.

7. If the difference of reading between the two horizontally mounted pyranometers exceeds more than 2%, the test shall be halted and resumed only after rectification of errors which has led to mismatch. The data of that particular day(s) shall be discarded and test period shall be extended by same numbers of day(s).

The test shall be repeated in case of outage of following equipment for more than 7 days.

1. Inverter transformer

2. Power Conditioning Unit

3. SCADA and data logger combined

4. Both pyranometers.

8. If bidder is not able to demonstrate PG test during these three (03) months he will be given one more chance within 15 (Fifteen) days to demonstrate the PG test after incorporation of suitable corrective measures. In that case the steps for PG test shall be repeated again as above after carrying out necessary modification / rectification.

9. The test shall be repeated in case of outage of following equipment for more than 7 days.

i) Converter transformer

ii) Power Conditioning Unit

iii) SCADA and data logger combined

iv) Both pyranometers.

10. Further, if the plant is not able to achieve the target generation as per the PG procedure during the test period, then an amount equivalent to the loss of generation based on sample calculation @ **37.03** shall be deducted.

Shortfall LD will be levied @ **37.03** shortfall.

LD limit for Performance shortfall in PG shall be 5% of the Total Project cost plus GST.

A sample calculation for shortfall in energy generation for period from 16th February to 17th May (90 days), LD calculation for the site is given in the next page.

11. The following Sample calculation of PG test and O&M penalty shall be considered :

Table - B : Sample calculation for 5 MW Ground Solar Total Shortfall in Energy during P G Test

Month	Global Solar Insolation of the month (kWhr/ m^2 x day) (Data provided by DPL)	Target Generation (KWhr) (Data provided by DPL)	No of test days of the month	Reference Solar Insolation	Modified Target Generation of the month kWhr	Measured Global Horizontal Solar Insolation (kWhr/ m^2)	Corrected Target Generation(kWhr)	Measured Generation at 6.6 kV outgoing switchgear panel (kWhr)	Shortfall in energy for PG test (kWhr)
	(a)	(b)	(c)	(d)=[(a) X (c)]/(Ndm *)	(e)	(f)	(g)=(e) x (f)/(d)	(h)	(g-h)#
February	141.7	893,658	12	60.7	382,996	71.7	452,402.43	472,667.73	-20265.3
March	187.2	1,144,116	31	187.2	1,144,116	180.2	1,101,333.60	1,073,345.90	27987.7
April	197.8	1,182,396	30	197.8	1,182,396	190.6	1,139,356.21	1,110,666.41	28689.8
May	190.7	1,144,578	17	104.6	627,672	114.4	686,478.35	693,543.05	-7064.7

*** Ndm= Nos of days in the month**

**** Test is assumed to start from 16th February till 17th May**

 # -ve value denotes excess generation

Total Short fall in Energy for the test period	**(ΔGTP)**	(-20265 + 27987.7 + 28689.8 - 7064.7) KwHr = **29347.5 KwHr**
Total Modified Target generation for the test period	**(GTP)**	3379571 Kwhr
Target yearly generation	**(GY)**	11,200,000 KwHr
Yearly shortfall in generation	**(ΔGY): GY x ΔGTP / GTP**	11200000 X (29347 /3379571) = 97258 KwHr
LD for per unit shortfall in generation	**N x P = Rs.37.03/kWhr**	
P = Penalty per unit= Rs. 4.00. N=Applicable future value=9.2575		
Yearly loss of Revenue and applicable LD	**: ΔG_Y x P x N**	97258 X 37.03 = **Rs. 36.01 Lakhs**

4.27.1 Calculation of LD during O&M for the project :

Methodology for Calculation of penalty during O&M:

- Target Generation: G1 (Target Generation Table A).
- Reference Global Horizontal Solar intensity (kWhr/sq.m): H1 (Table A).
- Measured Generation during O&M period= G2.
- Measured Global Horizontal Solar intensity (kWhr/sq.m) during O&M period: H2.
- Modified target generation during O&M period: $\mathbf{G_{modified}}$.
- Energy charges during O&M period (P): Rs. 4 / Unit.
- Maximum penalty 1% of Project cost plus GST per year.

$$\mathbf{G_{modified} = (H2/H1) \times G1 \times MPDF \times 0.98125}$$

Where, MPDF: Module performance degradation factor per year: (1-Year of operation completed x 0.007),

MPDF is 1 for the 1st year of O&M as year of operation completed is 0. For 2nd year MPDF shall be 0.993 as year of operation completed is 1 and so on.

Shortfall in generation in kWhr: $\Delta \mathbf{G} = (\mathbf{G_{modified}} - \mathbf{G2})$

Therefore, penalty due to shortfall in generation in INR:

($\mathbf{G_{modified}}$ - **G2**) x (**Energy charges**)

= Δ**G x P** + GST in INR

If $G_{modified}$ < or = G2, then no penalty for the corresponding O&M period.

In case, the GHI is not available because of instrumentation or SCADA problem, the corresponding solar intensity and generation shall be excluded from the time block for estimation of loss of generation.

Generation loss due to the grid outage not attributed to the bidder shall also be excluded for arriving loss of generation.

One day shall be equally divided into 96 blocks of 15 minutes each starting from 00.00 Hrs. i.e. 66th time block shall be from 16.15 – 16.30 Hrs.

Sample Calculation for 10 MW O&M penalty due to shortfall in generation:

- Base Value of the Project cost= Say Rs. 50 Cr.
- Maximum yearly Penalty deductible from O&M contract = 1 % of Contract value plus GST =Rs. (50 / 100) = Rs. 50 Lakh + GST

  Sample Procedure for determining penalty for shortfall in generation during O&M period as follow:
- G1 = 10 MU (say)
- O&M Period being considered 2nd Year i.e. MCF=0.993
- Reference Global Horizontal Insolation (H1) = 1600 kWh/m2-year (say)
- Measured Generation by the Bidder (G2 in Million Units) = 9 MU (say)
- Measured Global Horizontal Insolation during the O&M period(H2)=1700

kWh/m2-year (say)

- Modified target Generation during the 2nd year of the O&M period (**$G_{modified}$**)

=10 x (H2/H1) x MPDF x 0.98125

= 10 x (1700/1600) x 0.993 x 0.98125 = 10.3528 MU

ΔG = $\mathbf{G_{modified}}$ - G2 = 10.35 – 9.0 MU = 1.3528 MU

Since $\mathbf{G_{modified}}$ > G2, LD applicability =Yes

Value of yearly penalty base value in INR = Shortfall in Generation (unit) x Tariff

= INR ΔG x R

= INR13,52,800 x 4

= Rs. 54,11,203/-

As maximum yearly base amount of penalty is Rs. 50 Lakh, so applicable penalty value for the said project shall be Rs. 50 lakh plus applicable GST.

(Only philosophy of sample calculation shall be followed.)

Section V

Technical Specification

A. Lead Specification

Any item/equipment not mentioned in the Technical Specification, but required for successful completion of the project shall be deemed to be a part of the scope of the work and the same shall be included by the bidder in their Billing Break Up (BBU).

5.01.00 INTENT OF SPECIFICATION

This specification is intended to installation of 5 MW AC Solar PV Power Plantat DPL Power Plant area.

The scope shall include design, engineering, manufacture, inspection and testing at manufacturer's works, packing, forwarding, shipment and delivery at site. In addition, the Bidder's scope shall also include all necessary civil / structural / architectural works, erection / installation including unloading storage and handling at site, site testing, commissioning, trial run, performance and guarantee tests and other services including supply co-ordination, engineering and project management related to the equipment/systems as specified hereinafter and in accordance with the requirements, conditions, drawings, etc. stated in Section-V which shall be considered as a part of this Section as completely as if bound herewith.

This specification shall be read and construed in conjunction with the drawings to determine the scope of work. The quantities shown on drawings are indicative only. Any variation arising during finalization of detailed engineering and erection & services shall be duly taken care of by the Contractor without any extra cost and time to the owner.

The Bidder shall be responsible for providing all material, equipment and services, specified or otherwise which are required to meet the intent of this specification, ensuring high degree of reliability and ease of operation and maintenance. The equipment and system/sub-systems shall conform in all aspects to high standards of engineering, design and workmanship and shall be capable of performing in continuous operation, in a manner acceptable to the Owner and shall also be in line with the prevailing good engineering practice for reliable and efficient functioning of the plant.

Owner shall interpret the meaning of the specification, drawings, requirement of operation, maintenance, redundancy etc., and shall have a right to reject or accept any work or material which in his assessment is not technically complete to meet the requirements of this specification and/or applicable National and International standards.

Bidder is required to carefully examine and understand the specifications and seek clarifications, if required, to ensure that he has understood the specifications as intended by the Owner. In the absence of any specific clarifications made by the Owner during bidding stage, the interpretation of Owner shall be final. The Bidder's offer should not carry any sections like clarifications, interpretations and/or assumptions. All such points are required to be clarified during bidding stage.

In the event of conflict between requirements of any two clauses of this specification/documents or requirements of different codes/standards, specified, the more stringent requirement as per the interpretation of the Owner shall apply. Owner shall not bear any extra cost for such interpretation.

The Bidder may also make alternate offer provided such offers are superior in his opinion, in which case, adequate technical information and justifications, operating feedback data etc. shall be enclosed with the offer, to enable the Owner to assess the superiority and reliability of the alternative offered without the need to seek any further clarification or explanation. In case of each alternative offer, its implications on the performance, guaranteed efficiency, auxiliary power consumptions and definite advantages the Owner can positively derive shall all be clearly brought out along with commercial implications, if any for the Owner to make an overall assessment. In any case, the base offer shall necessarily be in line with this specification. Under no circumstances shall the specified equipment/plant be brought out as an alternative offer.

In case all the above requirements are not complied with, the offer may be considered as incomplete and non-responsive and may be liable for rejection.

Whenever a material or article is specified or described by the name of a particular brand, manufacturer or vendor, the specific items mentioned shall be understood as establishing type, function and quality desired. Other manufacturer's products may be considered provided sufficient information is furnished so as to enable the Owner to determine without the need for any additional clarification/data, that the products proposed are proven and comparable or superior to those named. However, it is Owner’s sole prerogative to accept or reject such similar product from other manufacturer.

5.02.00. SCOPE OF SUPPLY

5.02.01.SUPPLY OF SYSTEMS AND EQUIPMENT

The scope of supply & works as mentioned hereunder for 5 MW Ground mounted Solar PV Project shall be contracted on a Single Package Turnkey basis. Scope of supply & works shall be read and construed in conjunction with the detailed technical specification, drawings, data sheets, annexure etc. as per the relevant sections as specified in the respective Section of this Tender Specification.

The supply to be rendered by the Bidder shall include but shall not be limited to the following:

#	Supply Item
I	**Civil Construction Material for**
1.	Road, Drain
2.	Water Tank
3.	Main Control Room
4.	Inverter cum Switchgear Room-PEB
5.	Dry Transformer Shed
6.	Module Mounting Structure
7.	Watch Cabin/Security Cabin
8.	Module Washing System
II	**Electrical Supply-DC**
1.	Solar Module
2.	MC4 Connector
3.	Solar Cable
4.	DC Battery, Battery Charger, DCDB
5.	Grid Connected Inverter
6.	String Combiner Box

7.	Communication Cable
8.	Earthing Material
III	**Electrical Supply-AC**
1.	Inverter Duty Transformer
2.	Station Auxiliary Transformer
3.	LT Switchgear
4.	HT Switchgear
5.	ACDB
6.	LT Cable, HT Cable
7.	SCADA
IV	MISCELLNEOUS
1.	Fire Protection System
2.	Weather Monitoring System
3.	CCTV
4.	Ventilation System
5.	Air Conditioning System

Supply of all mandatory spares, commissioning and recommended spares, special tools & tackle including required after-sales services during and after the warranty period till the life cycle of the Plant is also the scope of supply of the Contractor.

5.03.00. SCOPE OF SERVICES

5.03.01.The services to be rendered by the Bidder shall include but shall not be limited to the following:

i. Services for complete engineering, co-ordination and project management as detailed in Section-V of this specification volume.

ii. Land Leveling and grading job

iii. Construction of Water Tank

iv. Construction of Paving of the Plant area

v. Service Construction of Main Control Room (MCR), Inverter Room, Transformer Shed.

vi. Installation and Commissioning of all Solar plant equipment to complete the Solar PV System.

vii. Power evacuation system

viii. 5Yrs Comprehensive Operation & maintenance after hand over the plant

ix. Services for construction, fabrication, equipment erection, testing as well as trial run & commissioning of various equipment and accessories
under the contract. The details of such services are indicated in Section-V of this specification volume.

x. Services for shop tests and quality assurance, etc. as detailed elsewhere in this specification.

5.04.00. System layout

a	**Capacity of the Plant**	**5 MW(AC)**
b	**Minimum DC Capacity**	5.75 MWp (MIN) Plant capacity is cumulative capacity of Inverters and which not be less than 5 MW. Inverter Transformer capacity shall not be less than the sum of the connected Inverter capacity
c	**Land area**	**17.6 Acres**

MODULE

Module** for this plant should be monofacial/bifacial and mono- crystalline capacity **should not be less than 500Wp

INVERTER

i. Capacity:

The continuous combined rating of all Inverters shall not be less than plant capacity at (i) unit power factor at ambient temperature of 50°C and (ii) 0.95 power factor at 40°C.

ii. DC Overloading

Minimum solar inverter DC over loading shall be 10%. Bidder needs to submit all the relevant technical document/test report from Inverter manufacturer (OEM) during details engineering stage in support of declared solar inverter design DC overloading capacity.

6.6 kV INDOOR SWITCHGEAR

i. **Bus Bar Rating of HT Switchgear:** As per Single Line Diagram (attached)

ii. **System Fault Current Rating:** As per Single Line Diagram

iii. **Dynamic withstand Current rating** – 2.5 times of system fault current

iv. Spare 6.6 kV breaker panels with VCB, relay and all other accessories shall be provided, as per Single Line Diagram

v. DC supply shall be used for control and protection system of switchgear. In case UPS AC supply are considered for auxiliary control and protection supply for switchgear, then suitable rated AC/DC converter/power pack shall be used to meet the DC control supply requirement of switchgear panels.

vi. The 6.6kV Switchgear shall have an internal Arc Classification of IAC FLR 25kA, 1sec.

vii. Remote Control Desk shall be provided for newly installed Breakers Panels.

viii. Control Supply should be DC with Charger & Battery. AC Control by UPS is not accepted.

GRID CONNECTIVITY

Power is to be evacuated from a new 6.6 kV indoor switchgear which to be placed at in the new constructed 5 MWAC Solar Plant Main Control Room. Power shall be ultimately evacuated by means of two numbers of 3 Core 300 square mm XLPE cables layed undergroung for a distance of approximately 2.2 kms to 7 MWAC Solar 6.6 KV BUS (beside Administrative building).

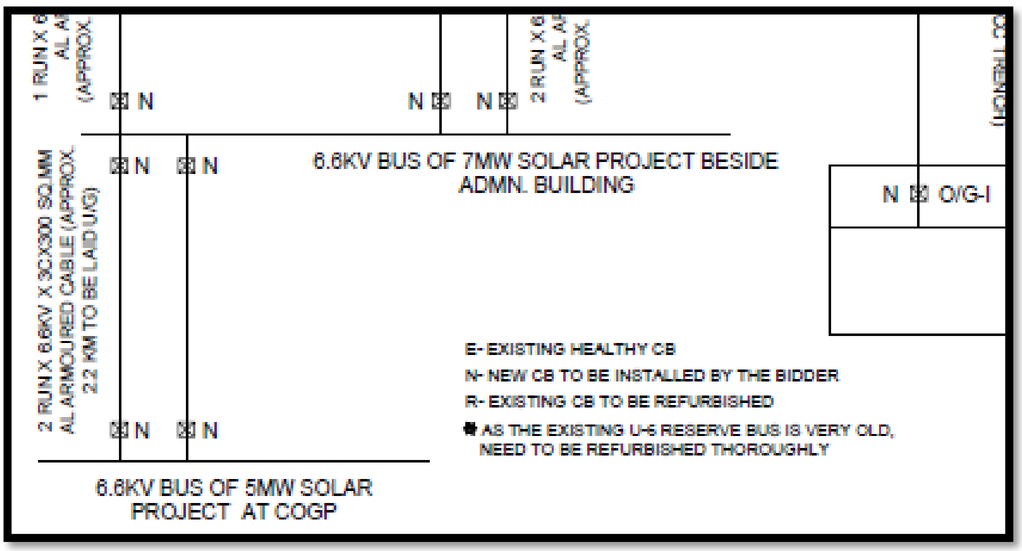


Figure: Indicative Power Evacuation

INVERTER TRANSFORMER

Dry type ANAN step up Transformer (0.4kV/6.6kV) for this Ground Solar shall be connected to the outputs of one inverters as input and the 6.6kV HV output sides will be connected to the Unit No.6, 6.6kV Reserve Switchgear bus.

STATION AUXILIARY TRANSFORMER (SAT)

One numbers 6.6kV/415V outdoor dry type Station transformer having a minimum capacity of 250 kVA has to be considered for light and other auxiliary purposes. **Another LT source shall be available within 100**

metres of site, however, supply & cabling etc. shall be in the scope of the bidder.

HT side of the transformer shall be connected to the 6.6kV Bus through a VCB and LT side shall be terminated to the Station Service Board (SSB)/ 415 V LT Switchgear. Successful bidder shall consider entire LT load under this Aux. Transformers during detail engineering.

LT SWITCHGEAR:

One number indoor type 415V LT switchgear has been considered for supply of 415 V (3 phases, 1 neutral and single phase for lighting etc). The said LT bus has to be made with proper protection system with two Incomers and one Bus- coupler philosophy. Incomer and Bus-coupler shall be Vacuum Circuit Breaker of similar rating. Two out going feeders and One spare feeder of capacity 200A MCCB also to be considered under this LT Board from both bus-sections. Control Voltage 220 V DC.

Bidder shall consider following Vacuum Circuit Breaker feeders:

Sl. No	Description	Quantity
1.	Incomer feeder VCB	Two (02 nos.)
2.	Bus coupler VCB	One (01 no.)
3.	Outgoing Feeder	Two(2)
4.	Spare MCCB	200 A capacity each, two (01 nos.)

B. GENERAL INFORMATION

5.05.00 INTRODUCTION

5.05.01 The Durgapur Projects Limited, (hereinafter called "DPL" or "Purchaser"),wishes to install 5 MW AC Grid connected Ground Mounted Solar PV Plant inside its Plant premises. The 5 MW AC solar power project shall be implemented in a single EPC package under Domestic Competitive bidding based on Open e-tendering mode.

5.05.02 The Durgapur Projects Limited site is West of Durgapur town by the side of the Kolkata-Delhi High Road Paschim Burdwan District, West Bengal and around 200 KM from Kolkata, India. Nearest broad gauge Railway station is Durgapur and nearest Airport is Andal.

5.06.00 MODE OF EXECUTION

The entire work shall be executed on Lump sum turnkey basis. Any item(s) not included in the specification / schedule but required for completion of the work shall have to be carried out/supplied without any extra cost. Such works, not listed in the schedule of works but elaborately described to perform or to facilitate particular operation(s) required for completion of the project shall be deemed to have been included in the scope of this work and the bidder shall supply, install the same without any extra cost to DPL.

The work shall be executed in conformity with the relevant standard of Bureau of Indian Specification (or equivalent International Standard), Indian Electricity Rules, 1956 (as amended up to date), Indian Electricity Act 2003(as amended up to date), BARC/DAE rules, Explosive Act 1948, Petroleum Act 1934, National Building Code and relevant Rules in vogue at the time of execution including operation & maintenance period.

The bidder shall comply with all applicable laws or ordinances, codes, approved standards, rules, and regulations and shall procure all necessary Panchayat / Municipal and Government permits & licenses etc at his own cost.

All sub systems /components such as cables, connectors, Junction boxes, surge protection devices, etc., shall conform to the relevant international and national standards for electrical safety besides that for Quality required for ensuring Expected service life and Weather resistance.

5.07.00 SITE INSPECTION

The bidder is advised to visit and examine project site and its surroundings and obtain for himself, on his own responsibility, all information that may be necessary for entering into contract. The bidder will assess and satisfy himself as to the adequacy of the local conditions such as approach roads to the site, adequacy of existing culverts/bridges/roads for the expected traffic, water and power supply, nature of ground and sub soil condition, water table level, accommodations required during the contract, climatic conditions, local terrain, availability of labour and construction materials, details of taxes, duties and levies as applicable and any other information required. The cost of visiting the site shall be at the bidder's own expenses.

5.08.00 FACILITIES BY THE OWNER

i. Land, free of charge as may be available for the construction of:

a) Residential accommodation to be built by the Contractor at his own expense for his employees near site. Land required for construction of labour hutment shall be provided.

b) Site office, pre-assembly/storage yard to be constructed at Contractor's own expense within plant site.

ii. Construction water at One (1) points at Power Plant Area shall be made available free of cost. No separate drinking water supply shall be given.

iii. Construction Power at one (1) point at 415V AC ± 10%, 3 Ph, 50 Hz. ± 5% near the erection site within the Power Plant premises shall be made available with meter; cost of electricity will be charged at standard the then prevailing rate of WBSEDCL.

5.09.00. Project Start-Up and Functional Requirements

The project start-up and functional requirements shall essentially meet all the supply, installation, testing and commissioning by Bidders as required to complete the job. Following are the non-exhaustive requirements that shall be offered by Bidders.

5.09.1 Topographical Survey and Geotechnical Investigations

Bidders are advised to understand the site conditions with an obligatory visit to the project site before submitting their bids.

The site preparation shall include all the work as required for installation of a utility-scale Ground mounted solar PV plant as per international practices.

The Modules so mounted and the associated structure should be such that the system is able to withstand Wind velocity of 180 kmph.

Only the area required for building the SPVPP and necessary infrastructure (e.g. internal roads, buildings, water evacuation systems, etc.) shall be levelled. This shall essentially include but not be limited to:

Earthwork

Following the site visit prior to bidding and based on previous experience, Bidders are expected to estimate the volume of earthwork required.

Site preparation may require:

a. Clearing of weeds, chopping down of small bushes and trees
b. Levelling of land with excavation and back filling of soil
c. Movement of soil for levelling within the site or by bringing soil from outside
d. Grading the soil with slope
e. Preparing land and water flows to prevent erosion of roads, infrastructure, PV structure foundation, etc.
f. Compaction of minimum 95% shall be required for finish

Internal Roads and Pathways

a. The contractor will be responsible for the construction of the approach road from peripheral boundary road to the project site and all internal roads, after necessary consultation with the Employer and after getting due approval for the same. IRC: 15-2002 (Third Revision) shall be followed.

b. Provision of future road expansion should be kept in general layout.

c. As a part of site preparation, all roads shall be designed and constructed to allow smooth onsite transportation and delivery of all project components to installation location and heavy equipment to their respective work locations.

d. Main approach and internal roads shall be designed for a life of 25 years.

e. At all cable crossings, Hume pipes of appropriate sizes shall be embedded within the road.

f. Roads shall be constructed at minimum 250 mm above ground with required cutting of soil, grading to the given slopes, backfilling and due compaction.

g. The main approach road shall be of WBM as per relevant Indian standards.

h. Minimum width of main approach road and internal roads shall be 4 m and 3.0 m respectively excluding road sides.

i. All buildings, including inverter and transformer prefab units, will be connected by internal roads.

j. Taken into account the final slopes and water flows an appropriate drainage system shall be in place to protect the roads from erosion. Maximum longitudinal slope shall be 7%.

Site Office and Material Storage

The Employer shall not be liable for any on site infrastructure. The Contractor shall have its own onsite project management office and site warehouse for material storage with required security. Maintaining the hygiene, the Contractor shall have adequate mobile toilets at multiple locations within the site.

Weather Station

Not later than three months after Contract signing, the Contractor shall provide and install on site a weather station to provide adequate meteorological data to evaluate system performance. The station shall essentially include sensors for the following parameters:

a. Global irradiation on tilted plane
b. Global horizontal irradiation
c. Air temperature and humidity
d. Wind direction and speed
e. Atmospheric pressure
f. Precipitation
g. Evaporation from water surface
h. Atmospheric effects including the most hazardous (storms, whirl, thunderstorm activity, dust storms and etc.).

Dedicated pyranometers shall be used for measurement of global irradiation on tilted plane and global horizontal irradiation.

The weather station along with data logger shall be located at a strategic point and shall be capable of collecting the data points and sample frequency. The weather station shall have capability of recording and storing environmental data without AC power for at least 12 months.

Both the weather station and the data logger will be tamper proof. This weather station will be integrated in the control system before commissioning. The Contractor will provide a remote secure connection through the external internet line for data export to the main offices of the Employer at its Kolkata Office or any other entity that the Employer might designate. Format of the data transmission and interfaces will be defined in agreement with the Employer.

Lightning Protection System

a. The entire PV plant area including all buildings will be protected from lightning.

b. The protection system will be based on early streamer emission lightning conductor air terminals.

c. The air terminals shall provide an umbrella protection against direct lightning strike covering a radial distance of maximum 60 m.

d. The air terminal will be capable of handling multiple strikes of lightning current and should be maintenance free after installation.

e. The earthing stations for the lighting discharges shall be provided with test links of phosphorus bronze and located at 150 mm above ground level in an easily accessible position for testing.

Illumination System

a. All the main and internal roads shall be lit with external lighting system strategizing site security and maintenance requirements; utmost care should be taken for avoiding any shading effect due to the poles.

b. The light fittings shall be highly efficient having longer life. LED-based system shall be used.

c. Bollards of height up to 1 meter shall be installed along the main approach road towards modular plot.

d. Additionally, flood lights may be mounted at strategic locations along the boundary of the PV plant. All the light shall be fed from power supply system available at the plant.

Water Supply, Water Drainage and Water Treatment

1 General Requirements

a. The water supply and internal drainage system must be suitable for operation under the extreme climatic conditions prevailing at the site. The respective technical measures planned by the Contractor shall be described in detail in the Detailed Design Report.

b. Water Supply and water tanks are under responsibility of the Contractor.

c. The Contractor will describe the state-of the art technology he will use to save and re-use water during plant construction and operation, the source of water he will use.

d. The Contractor is expected to study the site topography and plan the surface drains accordingly. Utmost care shall be taken to maintain slopes and to prevent water clogging at the site and to avoid erosion of any infrastructure foundation or road.

e. Proper water drain channels shall be designed wherever necessary. The water drains shall have an outlet at a suitable location on the plot periphery to connect with the natural water flow.

f. Water waste from buildings will be conducted to a septic tank. All other water waste will be conducted to an evaporation pond.

2 Water Quality Requirements and Water Flow

Three following systems of water supply shall be provided in the SPVPP:

a. Drinking water supply providing water to meet the needs of the buildings and facilities on the site
b. Water supply for solar panels cleaning
c. Firefighting water supply

The Contractor shall provide a connection for drinking water and firefighting water for the substation as needed.

3 Water Drainage System

The following facilities shall be designed on the SPVPP site:

1. Household sewage system for waste diversion
2. Rain/flood water drainage
3. Oil emergency drains
4. Septic tanks
5. Ponds

5.10.01.Scope of Supply of Plant and Services by the Contractor

[1] The Project concerns the construction of a solar photovoltaic (PV) power plant with an installed capacity of at least 5 MW (AC) as detailed below. The overall objective is to ensure the successful and timely implementation and operation of the Project in accordance with international best practices and industry standards.

[2] Scope of work shall include site preparation, designing, planning, engineering, procurement (manufacturing & supply), construction / erection, testing, commissioning, five (5) years of operation and maintenance of the

utility-scale solar PV plant as a turnkey contract according to terms and conditions set out herein.

[3] The project region is vulnerable to natural hazards. The environmental conditions need to be reflected in the design of the PV power plant and the respective water evacuation systems from the site.

[4] The Bidder will be responsible for the optimal design of the Project. Flexibility is allowed as long as the Technical Proposal and the Project complies with all the Technical Requirements and Standards set out herein.

[5] The following specifications attempts to outline the Technical Requirements and Standards in suitable detail but can by no means be considered as complete and comprehensive. It is rather the Bidder's responsibility to critically verify the Technical Requirements, Standards and Services indicated and to extend, reduce or amend it, wherever he deems necessary according to his own professional judgment and the knowledge he will acquire during the preparation of his proposal.

[6] Bidders are requested to provide their own calculations and experience for the preparation of the technical proposal and are fully responsible for this.

5.10.02.Document Approval Philosophy

All documents and drawings shall be submitted to the Employer both in soft as well as hard copies (3 nos.) for review and approval. Every drawing shall also be submitted in '*.dwg' format. In case of design calculations done in spread sheet, editable (working) soft copy of thespread sheet shall also be submitted along with 'pdf' copies during every submission. The Employer shall return, as suitable, either soft or hard copies to the Contractor with category of approval marked thereon. The drawings/documents shall be approved in any one of the ollowing categories based on nature of the comments/ type of drawing or document.

Category-I: Approved

Category-II: Approved subject to incorporation of comments. Re-submit for approval after incorporation of comments

Category-III: Not approved. Re-submit for approval after incorporation of comments

Category-IV: Kept for record/ reference

Category-IVR: Re-submit for record/ reference after incorporation of comments

(Note: Approval of document neither relieves the Vendor/ Contractor of his contractual obligations and responsibilities for correctness of design, drawings, dimensions, quality &specifications of materials, weights, quantities, assembly fits, systems/ performance requirement and conformity of supplies with Technical Specifications, Indian statutory laws as may be applicable, nor does it limit the Employer/ Purchaser's rights under the contract)

After LOA, the Contractor shall submit complete Master Document & Drawing list (MDL) to the Employer within 1 week. The MDL shall list all the Drawings & Documents envisaged for submission/ approval from the Employer and shall also have all the required information like drawing no , title, scheduled date of submission, actual date of submission and approval. The category of approval shall be decided mutually between Contractor and the Employer at the time of finalization of the MDL which shall be the basis for drawing & document approval process during project execution.

The construction shall be done only as per drawings approved under Category – I, II & IV.

The specifications provided with this bid document are functional ones; any design provided in this document is only meant as an example. The

Contractor must submit a detailed design philosophy document for the project to meet the functional requirements based upon their own design in-line with the above. The bidders are advised to visit the site and satisfy themselves before bidding.

C. Civil Specification

5.11.00. GENERAL

All civil works shall conform to the latest Indian standards, codes etc as where applicable or to equivalent applicable international standards approved by the engineer-in-charge. Civil works include but not limited to the following items whichever necessary for implementation of the Solar PV Project. All civil works should be carried out as per the relevant standards and guidelines.

5.11.01. SCOPE OF WORK-Civil

The scope of civil work comprises all necessary investigations, survey, foundations, Earthworks and Infrastructure required for the complete 5 MW Ground mounted Solar PV plant.

The work under this Section consists of Civil and Structural works, but not limited to items mentioned below.

a. Topographical Survey of the proposed solar plant site.

b. Geotechnical investigation of the Soil of the plant site.

c. Site leveling and grading

d. Design of Module Mounting Structures

e. Civil Structural work like construction of Main Control Room, Switchgear, Inverter Room, Transformer Shed.

f. Construction of PEB

g. Construction of Plant Drainage System

h. Paving of the Plant area.

i. Adequate protective measures are to be taken for existing underground water pipelines so as to prevent damages to the pipelines and also ensure that repair works of the pipelines can be carried out when required.

j. Entire area to be fenced off.

5.11.2. CODES AND STANDARDS

Following is a general listing of Codes and Standards to be used in the design of the Plant. Specific applicable codes and standards will be identified in System Design Descriptions/Technical Specifications as appropriate. The latest editions/revision of following codes and standards along with addendums/amendments, if any, shall be followed:

i. GENERAL

a) Internationally accepted design Codes and Standards which are equivalent or more stringent than corresponding Indian Standards.

b) National Building Code of India.

c) "Accepted Standards" and "Good Practice" listed in the appendix to National Building Code of India. IS-1200: Method of measurement of Building and Civil Engineering Works.

d) IS-1256: Code of Practice for Building Byelaws.

ii. EARTHWORK

IS-1498	:	Classification and identification of soils for General Engineering purposes
IS-3764	:	Safety Code for excavation work.
IS-7293	:	Safety Code for working with construction machinery.

iii. CONCRETE

IS-269	:	Ordinary and low heat Portland cement.
IS-383	:	Coarse and fine aggregate from natural sources for concrete.
IS-432	:	Mild Steel and medium tensile steel bars and hard drawn steel wire for concrete reinforcement.
IS-455	:	Portland Slag Cement.
IS-456	:	Code of Practice for Plain and reinforced concrete
IS-460	:	Test Sieves (all parts).
IS-516	:	Methods of test for strength of concrete.
IS-1199	:	Methods of sampling and analysis of concrete.
IS-1489	:	Portland Pozzolana Cement.
IS-1566	:	Hard drawn steel wire fabric for concrete Reinforcement.
IS-1786	:	High strength deformed steel bars and wires for concrete reinforcement.
IS-1834	:	Hot applied sealing compounds for joints in concrete.
IS-2386	:	Methods of test for aggregates for concrete (all parts).
IS-2502	:	Code of practice for bending and fixing of bars for concrete reinforcement.
IS-3370	:	Code of practice for concrete structures for storage of liquids (all parts).
IS-3414	:	Code of practice for design and installation of joints in buildings.
IS-4948	:	Welded steel wire fabrics for general use.
IS-6452	:	High Alumina Cement for Structural use.
IS-7320	:	Concrete slump test apparatus.
IS-7861	:	Code of practice for extreme weather concreting (all parts).
IS-8041	:	Rapid Hardening Portland Cement.
IS-8112	:	High strength ordinary Portland Cement.
IS-10262	:	Recommended guidelines for concrete mix design

iv. MASONRY

IS-712	:	Building limes.
IS-1077	:	Common Burnt Clay Building Bricks.
IS-1127	:	Recommendations for dimensions and workmanship of
		natural building stones for masonry work.
IS-1597	:	Code of practice for construction of stone masonry (all parts)
IS-2212	:	Code of practice for brickwork
IS-2116	:	Sand for masonry mortars
IS-2185	:	Concrete masonry units
IS-2250	:	Code of practice for preparation and use of masonry mortars.
IS-2572	:	Code of practice for construction of hollow concrete block masonry
IS-2691	:	Burnt clay facing bricks.
IS- 12894	:	Fly Ash Lime Bricks
IS-1948	:	Aluminium doors windows and ventilators.
IS-1949	:	Aluminium windows for industrial building.
IS-2191	:	Wooden flush door shutters (Cellular and hollow core type).
IS-2202	:	Wooden flush door shutters (solid core type).
IS-3103	:	Code of practice for Industrial ventilation.
IS-3548	:	Code of practice for glazing in buildings.
IS-3614	:	Fire check doors.
IS-4021	:	Timber door, windows and ventilator frames.
IS-4351	:	Steel door frames.
IS-6248	:	Metal rolling shutters and rolling grills.

v. ROOF AND FLOORING

a) IS-2204 : Code of practice for construction of reinforced concrete shell roof.

b) IS-3201 : Criteria for the design and construction of precast concrete trusses.

c) IS-2210 : Criteria for Design of R.C. shell structures and folded lates.

d) IS-809 : Rubber flooring materials for general purposes.

e) IS-1195 : Bitumen mastic for flooring.

f) IS-1196 : Code of practice for laying bitumen mastic flooring.

g) IS-1198 : Code of practice for laying, fixing and maintenance of oleumfloors.

h) IS-1237 : Cement concrete flooring tiles.

i) IS-1443 : Code of practice for laying and finishing of cement concrete flooringtiles.

j) IS-2114 : Code of practice for laying in situ terrazzo floor finish.

k) IS-2571 : Code of practice for laying in situ cement concrete flooring.

l) IS-5491 : Code of practice for laying in situ granolithic concrete floor topping.

m) IS-5766 : Code of practice for laying burnt clay brick flooring.

vi. SOIL ENGINEERING

a) IS-1498 : Classification and identification of soils for general engineeringpurposes.

b) IS-1892 : Code of practice for sub-surface investigation for foundations.

c) IS-2131 : Method for standard penetration test for soils.

d) IS-2720 : Methods of test for soils (all parts).

e) IS-2911 : Code of Practice for Design and Construction of pile foundations.

vii. WATER SUPPLY, DRAINAGE AND SEWERAGE

a) IS-404 : Lead pipes

b) IS-458 : Concrete pipes

c) IS-651 : Salt glazed stoneware pipes and fittings.

d) IS-771 : Glazed fire-clay sanitary appliances (all parts).

e) IS-774 : Flushing cisterns for water closets and urinals other than plasticcisterns.

f) IS-783 : Code of practice for laying of concrete pipes.

g) IS-1172 : Code of basic requirements for water supply, drainage and sanitation.

h) IS-1626 : Asbestos cement building pipes, gutters and fittings (all parts).

i) IS-1742 : Code of practice for building drainage.

j) IS-2064 : Code of practice for selection, installation and maintenance ofsanitary appliances.

k) IS-2065 : Code of practice for water supply in buildings.

l) IS-2470 : Code of practice for installation of septic tanks (all parts).

m) IS-3114 : Code of practice for laying of Cast Iron pipes.

n) IS-4127 : Code of practice for laying of glazed stoneware pipes.

o) IS-12251 : Code of practice for Drainage of Building Basement.

viii. PAVING AND ROAD WORKS

a) IS-73 : Paving bitumen

b) IS-702 : Industrial Bitumen

c) IS-1201 : Method of testing tar and bituminous materials. thru' 1220

ix. PAINTING

e) IS-348 : Specification for French Polish.

f) IS-427 : Specification for Distemper, dry colour as required.

g) IS-428 : Specification for Distemper, oil emulsion, colour as required.

h) IS-1477 : Code of practice for painting of ferrous metal [I & II] in buildings.

i) IS-2338 : Code of practice for finishing of wood and [I & II] wood-basedmaterials.

j) IS-2339 : Specification for Aluminium Paints for general purposes in dualcontainers.

k) IS-2395 : Code of practice for painting concrete, masonry and plaster surface.

l) IS-2932 : Specification for enamel, synthetic, exterior - a) undercoating, b)finishing.

m) IS-2933 : Specification for enamel, exterior - a) undercoating, b) finishing.

n) IS-5410 : Specification for cement paint.

x. STEEL WORKS:

i. IS 800: General construction in Steel structures

5.11.3. TOPOGRAPHY SURVEY

Bidder shall conduct the Topographical Survey for the allocated plot in the proposed solar project. The scope of work and technical specification for the same is as below:
The Bidder shall carry out the Topographical Survey and preparation of Plans (Maps) and report of the entire area for locating the Solar PV Power plant and its other systems.

Carrying out the Bench Mark (GTS) to site(s) under survey by parallel levelling, establishing and constructing benchmark, grid and reference pillars in the field and spot level survey of the entire area at specified intervals and development of the contours. Bidder can also use DGPS for establishing the coordinates.

Carrying out cross-section of river/canal taking spot levels at on an average 20 meters intervals or less depending upon the site conditions.

Furnishing all field data & drawings with Longitude and Latitude of all corners and strategic points. Furnishing of the survey report as described in details in the succeeding paragraphs is also included in the scope of this work.

The work shall include construction of two permanent Bench Marks and reference pillars which shall be shown on the survey drawings.

Latitude and Longitude: The work shall be carried out in UTM grids system.

At least 50 meter width of the adjoining solar plots and area shall also be covered in the survey for correlation with adjoining plots. Presence of any well and/or tube well in the site or adjoining areas and water level in them shall be marked in the documents

Topographical Survey and Mapping

Positions, both in plan and elevation, of all natural and artificial features of the area like waterways, railway tracks, trees, cultivation, houses, fences, pucca and kutcha roads including culverts and crossings, foot tracks, other permanent objects like telephone posts and transmission towers etc. are to be established and subsequently shown on survey maps by means of conventional symbols (preferably, symbols of survey of India Maps), all hills and valleys within the area/areas is to be surveyed and plotted on maps by contours.

Method of the survey, contour intervals etc. shall be decided by an owner on site in case of steep slopes and dense jungle etc. where grading is not possible. Any unusual condition or formations on the ground, locations of rock outcrops (if visible on the surface) and spring/falls, possible aggregate deposits etc. shall also be noted and plotted on the maps.

The field work shall be done with Total Station Equipment in the following steps:

Establishing horizontal and vertical controls and locating reference grids and benchmark in the area. Surveying for establishing spot levels and plotting contours. Surveying for locating the natural and manmade details as described earlier.

The grids for the survey work shall be established in N-S & E-W direction (Corresponding to Magnetic North).

Contouring

Bidder shall carry out spot level surveying at an interval of on an average 5.00 meters for contouring the area. Levels shall also be taken on all traverse stations and on salient points located at random over the area (ground points). Contours are to be interpolated at 0.5 M intervals after the above points are plotted.

Preparation & Submission of Survey Maps and Documents

The Contractor shall submit survey maps of the site in 1:10,000 scale indicating grid lines and contour lines, demarcating all permanent features like roads, railways, waterways, buildings, power lines, natural streams, trees etc. All the maps should be prepared in digitised forms using computer software like AutoCAD – Release 2016 or latest.

5.11.4. GEOTECHNICAL INVESTIGATION

i) The Contractor is advised to and is solely responsible to carry out detailed Geotechnical investigation to ascertain soil parameters of the proposed site for the use of planning / designing / construction / providing guarantee / warranty of all civil work including but not limited to foundations / piling for module mounting structures, HT lines, etc. The Contractor shall carry out soil investigation through any Govt. approved / certified soil consultant. The Bidder shall submit the credentials of the proposed agency along with relevant certificates in support thereof for verification/ approval of the Investigation Agency by the Engineer.

ii) The scope of work includes execution of complete soil exploration including boring and drilling with rotary drilling rig, standard penetration test (SPT), collecting disturbed (DS) and undisturbed samples (UDS), collecting ground water samples, trial pits, electrical resistivity tests (ERT), field & laboratory CBR tests, conducting laboratory tests on collected samples of soil & ground water and preparation and submission of report. The starting depth of SPT shall be 0.5m from ground level. UDS shall be collected at every 1.5m interval or at change of strata. The min. size of trial pit shall be 2.0m x 2.0m x 2.5m deep.

iii) The field investigations shall mainly include drilling of min. 5m deep BHs (50% of total No. of boreholes shall be 10m deep), conducting SPT and collecting Disturbed (DS) and Undisturbed samples (UDS), conducting in-situ CBR test for approach road to the plant, internal roads & peripheral road; Trial pits (TP) and Electrical resistivity tests (ERT). Number and location of BHs, California bearing ratio (CBR) tests, ERTs and TPs shall be decided as per the project layout, site topography and soil conditions in consultation with the Employer. The proposed locations shall fairly represent the total project site to get the complete required geotechnical information. The BH near MCR and ICR shall be 10m deep. There shall be minimum 1 no. of BH per 5 acres of the area (However, total number of bore holes shall not be less than 5), 3 nos. of Trial pits, 5 nos. of CBR test & ERT, 5 nos. of Ground water samples for laboratory investigations. The soil/ rock samples for laboratory investigations shall be collected from each bore hole and trial pit in sufficient nos. (Note- In case the project plot is divided into number of discrete blocks separated from each other, min. 3 nos. of bore holes, 2 trial pits, 2 ERT and 2 CBR tests shall be taken per such block with at least 1 No. of BHs per 5 acres as specified above).

iv) The proposed Geotechnical investigation plan indicating proposed locations of TPs, BHs, water sample collection points, CBR test & ERT shall be submitted to the Employer for review and approval before start of work.

v) Laboratory tests shall be conducted on DS & UDS samples and ground water samples in sufficient no. & shall include, Soil classification, Grain size analysis including Hydrometer analysis, determination of Bulk and dry density, Specific gravity, Natural moisture content, Atterberg limits, Tri-axial shear tests (Unconsolidated Un drained –UU) on UDS, Un drained shear test, Consolidation tests, Unconfined compression tests(UCS), Free swell index, chemical analysis of soil and water samples to determine the nitrates, pH, Organic matter and any other chemicals harmful to concrete and reinforcement/ steel. Laboratory tests on rock samples shall be carried out for Hardness, Specific Gravity, Unit Weight, Uni-axial Compressive Strength (in-situ & saturated), Slake Durability etc. Laboratory CBR test on soaked samples shall also be conducted on min. 5 no. of soil samples to ascertain the suitability of soil for sub-grade and requirement of any treatment of sub grade soil in case of CBR<2% as per IRC requirements.

vi) After completion of field and laboratory work, the Bidder shall submit a Geotechnical Investigation Report for Engineer's approval. All bore log details and lab test results shall be presented in the report as per provisions of relevant BIS standards indicating BH coordinates, Existing GL, Depth of water table, Method of drilling etc. The report shall include a Map showing the locations of various field tests including coordinates, calculations and recommendations for foundation type and safe bearing capacity (SBC) for various Plant buildings (ICR, MCR etc.) and Open installations, Switch Yard structures & Sub-Station (as applicable), Transformer foundation, HT lines (as applicable), MMS foundation etc. corresponding to settlement of 25mm.

vii) The report shall include the study for "Liquefaction potential assessment of the ground and suggestions for any ground improvement measures" as required.

viii) The report shall also include ground water analysis (water sample collected from bore well) to ascertain its suitability for construction purposes, recommendations for type of cement, grade of concrete & minimum cement content as per prevalent soil characteristics with respect to presence of aggressive chemicals and environment exposure conditions as per relevant BIS specifications.

ix) In case the Bidder wishes to adopt concrete pile foundation for MMS supports the Geo-tech. report shall also include the calculations, based on soil properties, for safe pile capacity under direct compression, lateral load and pull out as per IS:2911. For single pile, Lateral load capacity shall be min. of the values obtained as per IS:2911 & Brom's method corresponding to free pile head. The report shall also include recommendations about type of pile, its depth and dia. to be used.

a) In coastal areas and in marshy or swelling type soil, under reamed or driven precast concrete pile shall be used. In case Bidder wishes to use helical piles the design, fabrication and installation shall conform to IBC (International building code).

b) The Bidder shall carry out field trials for initial load test on pile to verify the pile capacity under direct compression, Lateral load and Pull out. The min. of the two values (design value as per soil characteristics & field test results) shall be adopted.

c) The nos. of piles to be tested under each category shall be finalized corresponding to geotechnical characteristics at site, plot area etc. However, minimum 5 nos. of piles shall be tested {min. 3 nos. in each block (block size < 25 acre) and min. 5 nos. in each block (block size.25 acres) if the plant site is divided in discrete blocks separated from each other} under each category of load.

d) The locations of test piles shall be distributed over the plant site and to be finalized in consultation with Engineer. In case the MMS column is fixed using base plate-anchor bolt assembly, the adequacy of provided pile reinforcement in job (working)pile corresponding to the set of test loads shall be reviewed by the Bidder for any additional requirement of reinforcement and the same shall be provided in the pile to be cast for initial load test.

e) In case the Bidder proposes to embed the Column leg in the pile for fixing, the test pile shall be provided with embedded column leg as per approved design and any dowels as required for application of test load. The drawing for the Test pile shall be submitted to Engineer for his approval before casting the test pile. The load test on pile shall be conducted after min. of 28 days from the date of casting. In case the Bidder desires to conduct the test earlier than 28 days, he may use suitable higher-grade concrete or if there is substantial evidence from earlier cube test results on design grade concrete to demonstrate the early gain of required compressive strength prior to application of the test load.

f) However, under no circumstances the test shall be conducted before 15 days of the date of casting the pile. All the dial gauges and hydraulic jack assembly shall be properly calibrated as per the requirements of relevant BIS standards and valid calibration certificate to this effect from Govt. / NABL accredited Test agency shall be submitted to the Engineer before use.

g) The Bidder shall submit detailed methodology for conducting the tests in line with IS: 2911 (Part 4) for Engineer’s approval before commencement of any test. After completion of these tests the Bidder shall compile the test results and submit the report in a proper format as specified in the BIS standard with recommendations/conclusions for Engineer’s approval. The pile work shall start only after approval of the final pile design duly verified/ confirmed with initial load test results.

These reports shall be furnished to the Employer prior to commencing work. All RCC works shall be provided of required grade of concrete as per relevant IS specifications as well as soil data considering appropriate earthquake seismic zone, wind velocity, whether effect, soil characteristics etc.

5.11.5. OTHER INVESTIGATIONS

Successful Bidder shall obtain and study earthquake and wind velocity data for design of module mounting structure, and considering all parameters related to the weathers conditions like Temperature, humidity, flood, rainfall, ambient air etc. The Successful Bidder shall carry out Shadow Analysis at the site and accordingly design strings and arrays layout considering optimal use of space, material and man-power and submit all the details / design to Employer for its review / suggestions / approval.

5.11.6. AREA GRADING LAND DEVELOPMENT FOR SITE ACTIVITIES

The turnkey contractor is responsible for making the site ready and easily approachable by clearing of bushes, felling of trees (if required with appropriate approval from concerned authority), leveling of ground (wherever required) etc. for commencing the project. It is to ensure that land must be graded and leveled properly for the flow of water. It is advisable to follow the natural flow of water at the ground. If the land pocket needs any filling of sand, it is to ensure that the filled earth must be well compacted as per the relevant IS standards. In case the filled earth is brought out from outside the plant, the contractor shall provide the necessary challans. On the other hand, additional earth, if any, must be disposed of properly. Bidder shall take reasonable care to ensure that the plant is aesthetically designed.

i) The Finished Grade Level (FGL) of the proposed plant shall be fixed with reference to the highest flood level (HFL) and surrounding ground profile at proposed site to avoid flooding of plant site. The data regarding HFL at proposed site shall be obtained from the metrological department by the Bidder. In case of absence of this data, the Bidder shall assess the required information through local site reconnaissance. The area at and around (up to 25m beyond external wall/ area including access road &parking whichever is minimum) all buildings/ plinth for open installations (ICR, MCR etc.), transformer yard and switch-yard shall be uniformly levelled at suitable RL (i.e.

FGL) to be finalized considering topography and HFL at site. The minimum plinth level of all buildings/ open installations shall be 450mm above FGL. Module mounting structure foundation/ Pile cap or any other pedestal shall be min. 200mm above FGL. Top of transformer foundation pedestal shall be min, 500mm above the FGL.

i) A detailed drawing for site levelling and grading (if necessary) shall be submitted by the Bidder before commencement of construction of all buildings, plinth for open installation and transformer/switchyard works. The estimated volume of cutting and filling shall also be marked on the Grading drawings for reference. The final grade levels to be adopted for different blocks shall be clearly marked on the Plant Layout/ Array Layout drawing.

ii) Based on the spot level, contour survey done and meeting above requirements, bidder can propose different site grade levels. The site levelling may be carried in patches/blocks. Bidder may also propose the site leveling and grading matching with the **natural topography** of the land considering the optimized use of the land, however bidder shall ensure to meet the desired power generation capacity in the allotted plot area. Bidder shall also ensure that no water ponding and flooding occurs in the low lying areas & effective drainage is provided in the whole plot area, in all kind of site levelling and grading or plant at natural topography schemes, bidders has to provide proper and effective drainage system in line with “Drainage System” chapter. After performing the optimization of levels from the detailed site survey by the Bidder, the final formation level of the plot in various areas shall be finalized. The area shall be suitably cut and filled to suit the layout requirement. The site levelling and grading scheme incorporating the above aspects shall be submitted to DPL for prior approval.

All buildings & switchyard area/sub-station area shall be constructed in levelled area. No foundation shall be allowed on back filled soil and in that case the depth of foundations shall reach up to NGL. Final Level will be approved in detail engineering.

The slope protection measure shall be provided in case inter levelled patches level difference is more than 2.0m. Random rubble/boulder/stone pitching/concrete blocks etc. shall be provided for the slope protection for road side slope, storm water ditches/drainage, embankment slopes, inter levelled patches slopes etc. as per design requirements.

iii) The Bidder is responsible for making the site ready and easily approachable by clearing bushes cutting, filling with selected excavated earth. Except in exceptional cases (with approval of the Engineer), filling shall be made up of cohesive non-swelling material. The filling for levelling/reclaiming the ground/area shall be done in layers not more than 150mm of compacted thickness compaction up to 95% (proctor density). It shall be ensured that the land-grading and levelling is done properly to ensure for free flow of surface run-off and the grade levels shall be fixed with respect to high flood level at site, drainage pattern and system requirements. It shall be ensured that the land is used optimally to have maximum solar power generation considering full utilization of the plot areas. It is advisable to follow the natural flow of water at the ground as far as possible for drainage design.

iv) Suitable sand erosion control measure shall be provided in case any sand dune falls inside the plot area. The same may be made with Random rubble/boulder/stone pitching/concrete blocks etc. Bidder shall also provide sufficient grass/bush/tree covers on these dune.

5.11.7. SURFACE/ AREA DRAINAGE

i) The Bidder shall design and construct storm water drainage network for smooth disposal of storm water from the plant to the nearest available drainage outlet.

ii) The storm water drainage system shall be designed and planned to ensure no water stagnation in the plant. Average annual rain-fall shall be considered as 1400 mm and design rainfall intensity for storm water drainage shall be 70 mm/hr.

iii)The plant drainage system shall be designed for maximum hourly rainfall intensity and relevant time of concentration.

iv) The design shall conform to the provisions of IRC SP 42 and best Industry practices.

v) The coefficient of run-off for estimation of design discharge shall be considered as per catchment characteristics, however it shall not be less than 0.6.

vi) The drainage scheme shall be designed considering the plant plot area and the surrounding catchment area contributing to the plant area drainage as per the topography.

vii) The storm water drainage system shall be a network of open surface drains (with rectangular or trapezoidal cross section) and shall generally be designed to follow the natural flow of water and ground contours.

viii) Suitable size plant peripheral drain as per design (min. 500mm wide x 500mm deep) along inside of plant boundary wall/ fence shall be provided for smooth channelization of outside storm water and to avoid flooding in the plant. The size of all internal and road side drains shall not be less than 450mm (bottom width) x 500mm (depth).

ix)All trapezoidal drains shall have side slopes not steeper than 1:1 and shall be lined with brick. The min. Thickness of the lining shall be 115 mm for brick masonry, The lining shall be in CM(1:4) and the joints shall be raked and pointed with CM (1:3).

x) In case of rectangular drain, the thickness of the wall shall be checked against structural stability under action of the design loads as specified in Cl. No. 10.0 'Design Loads'. However, Min. thickness shall be 230mm for brick masonry, 300mm for RR masonry and 125mm for RCC work, except for garland drain around buildings where the min. wall thickness can be 115 mm, brick masonry.

xi) The structural design of drains shall be as per provisions of relevant BIS standards and good industry practice.

xii) The drain outfall shall be connected to the nearest existing natural drain(s)/ water body outside plant premises and it shall be ensured that the drainage water shall not re-enter the plant nor encroach/ flood in the adjacent property/ plot.

xiii) The proposed drainage scheme along with design calculations and drawings shall be submitted to the Engineer for review/approval before start of construction. 'The Bidder explore for providing rain water harvesting system for water conservation by constructing suitable collection wells along the drains or through provision of detention ponds or percolation/recharge pit etc. The scheme for rain water harvesting along with design calculations shall be submitted for approval.

5.11.8. MODULE MOUNTING STRUCTURE

SCOPE

This section covers activities related to design, Manufacturing, testing, supply, insurance, transportation, delivery at project site, storage, erection, testing of module mounting structure as detailed hereunder.

Adequate number of mounting structure shall be provided for installation of the required number of PV module. The contractor shall provide the details of this item in the detailed design report.

STANDARD:

The applicable codes and standards are as mentioned below.

1	IS 875: Part 1 & 2	Code of practice for the design loads for buildings and structures-
2	IS 875: Part 3	Code of practice for the design loads for buildings and structures-Wind Loads
3	IS 800 : 2007	Code of practice for use of structural steel in general building construction
4	IS-4759	Hot-dip zinc coatings on structural steel and other allied products
5	IS 1868	Anodic Coatings on Aluminium and its Alloys
6	IS: 4736	Hot-dip zinc coatings on mild steel tubes

TECHNICAL REQUIREMENTS

- The Module structure design shall be appropriate and innovative. It must follow the existing land profile. Modules shall be mounted on non-corrosive support structures.
- The structure shall be designed to allow easy replacement of any module and shall be in line with the site requirement. All the panels shall have provision to adjust it at three optimum angular positions. The optimum angle shall be proposed by the bidder and finalized during detailed engineering.
- Design drawings with material selected and their standards shall be submitted for prior approval.
- The bidder shall design the structure height considering highest flood level at the site. The minimum clearance between the lower edge of the module and the ground shall be the higher of (i) above highest flood level at the site and ii) 800 mm min.

- There must be sufficient gaps between the rows of the panels and all precautions to be taken to avoid any shading on the panels.
- The module alignment and tilt angle shall be calculated to provide the maximum annual energy output. This shall be decided based on the location of array installation.
- The mounting structure shall be designed for simple mechanical and electrical installation. It shall support SPV modules at a given orientation, absorb and transfer the mechanical loads to the base properly. Minimum thickness of MMS column post shall be 2.00 mm and the minimum thickness of light gauge members shall be 2 mm. All materials shall be fabricated in shop. Welding of structures at site shall not be allowed.
- The array structure shall be so designed that it will occupy minimum space without sacrificing the output from SPV panels at the same time.
- In the ground mounting structure system with seasonal tilt arrangement, the column post and rafter/beam at seasonal tilt point of rotation shall be hinge plate and bolt system.
- Nut & bolt, supporting structures including module mounting structures shall have to be adequately protected from atmosphere and weather prevailing in the area and should be Stainless steel/ MS chrome plated (GI).
- The Mounting structure shall be grounded properly using maintenance free earthing kit.
- The support structure & foundation shall be designed with reference to the Soil test report in order to withstand wind speed of the location as given in relevant Indian code for wind (IS 875 Part III, latest edition where the minimum value of these factors shall be considered as K1 = 1.0, K2 = 1.0 & K3 = 1.0 for the design) and seismic load (IS 1893, latest edition). However the basic wind speed of the site shall be taken as 180 km/hr.
- The Ground mounting structure system which constitute a photovoltaic array(s) shall be designed to withstand the extreme fair wind (positive pressure) and adverse wind (negative pressure) on design tilt angle of solar photovoltaic array(s).
- The module mounting structure shall be designed as per prevailing IS codes and shall be safe against wind and seismic force. Detailed design including STAAD output file shall be submitted for final approval. An increase in allowable stresses of structural materials should not be considered during

design analysis. The members are to be designed taking into consideration the humid nature of the area.

- Wind pressure for following loads shall be considered as follows:

  i) Dead Load of steel with all members, fittings & panels.

  ii) Load due to fair wind direction on design tilt angles of solar mounting structural members.

  iii) Load due to adverse wind direction on design tilt angles of solar mounting structural members.

  iv) Load on side face of mounting structural members.

  Wind pressure coefficient, load and load combination shall be as per Indian standards (latest revision) such as IS: 875, IS: 800. The load (4) shall be considered with load (2) and load (3) in relevant load combination.

  Design analysis and the forces on MMS (Compressive force, uplift force, shear and moment) shall be used for design of foundation system.

- The array structure shall be made hot dipped galvanized steel of suitable size. The specification of steel should be as per relevant IS standards. The thickness of galvanization should be as per the relevant standards for galvanization but minimum 70 microns. It is to ensure that before galvanization the steel surface shall be thoroughly cleaned of any paint, grease, rust, scale, acid or alkali or such foreign material as are likely to interfere with the galvanization process. The bidder should ensure that inner side should also be galvanized. All galvanized materials shall withstand tests as per IS 2633.

- Foundation is to be provided with piling if necessary. Foundation should be of minimum M25 grade of concrete (with provision of cube test as per relevant IS code) and minimum Fe500 reinforcing steel (conforming to IS 1786, latest edition). Design should be such that the foundation should be safe against the Soil Load Bearing Capacity as obtained at site. The work includes necessary excavation, concreting, curing, back filling, shoring & shuttering etc.

- For multiple module mounting structures located in a single row, the alignment of all modules shall be within an error limit of maximum 10 mm.

- The bidder/manufacturer shall specify installation details of the PV modules and the support structures with appropriate diagram and drawings.

- Bidder must submit all the quality test documents and test certificates complying with the requirement of the structure.

Design Parameters

A. MMS design & analysis to be done on computer software (preferably STAAD) and the Bidder shall submit a write-up on the computer program used and its input (soft format) and output data for review and approval.

B. An increase in allowable stresses of structural materials should not be considered during design and analysis.

C. Wind pressure for following loads shall be considered as follows:
(1) Dead Load of steel with all members, fittings & panels.

(2) Load due to fair wind direction on design tilt angles of solar mounting structural members.

(3) Load due to adverse wind direction on design tilt angles of solar mounting structural members.

(4) Load on the side face of mounting structural members.

D. Wind pressure coefficient, load and load combination shall be as per Indian standards (latest revision) such as IS: 875, IS: 800, IS 801.

E. Design analysis and the forces on MMS (Compressive force, uplift force, shear and moment) shall be used for the design of foundation system.

Vertical Deflection and Horizontal Sway Limits:

Limiting Deflection: The limiting permissible vertical deflection for structural steel members shall be as per: Maximum vertical deflection in purlin = Span/180, Maximum vertical deflection in rafter (cantilever span) = Span/180 and Maximum lat eral deflection in column/vertical post = Height/240.

Materials Specification & Coating for Structural Steel Works:

A. Hot-rolled/Cold-formed steel sections:				
Members	**Reference code**	**Yield strength, min, MPa**	**Coating, Referencecode**	**Min Thickness** (mm)
Column/ Vertical Post	IS 2062 /	250		**2.0**

<table>
<tr><td>Bracing/Rafter/ Beam/Purlin</td><td>IS 1079</td><td></td><td rowspan="4">70 micron
(IS 4759)
(minimum)</td><td>2.0</td></tr>
<tr><td>Steel Tubes in all sections</td><td>IS 1161</td><td>240</td><td>2.0</td></tr>
<tr><td>Hollow Steel in all sections</td><td>IS 4923</td><td>240</td><td>2.0</td></tr>
<tr><td>Coupler/Plate/Cleat Splice/SagAngle</td><td>IS 2062</td><td>250</td><td>2.0</td></tr>
<tr><td></td><td></td><td>Yield strength, MPa</td><td>Coating Class Designation</td><td></td></tr>
<tr><td>Rafter/ Beam/ Purlin
(Pre-Galvanized steel sections)</td><td>ASTM A653M/ IS 1079</td><td>255-380</td><td>Z600 (ASTM A653M/ IS277)</td><td>1.6</td></tr>
<tr><td colspan="5">1. Minimum elongation % shall be as per relevant Standard and Code.
2. Materials shall be fabricated in the shop.
3. Minimum coating requirement mentioned above in the table.
4. All structural calculations of cold formed steel section for checking the adequacy for strength and deflection criteria is to be done taking into consideration the maximum permissible negative tolerance over specified BMT i.e, the lower limitof BMT is to be considered.
5. The tolerance on Base Metal Thickness (BMT) thickness of steel shall be as given in IS 1852.</td></tr>
<tr><td colspan="5">B. Hot-dip Aluminium-Zinc alloy metallic coated sheet steel strip andsheet sections:</td></tr>
<tr><td>Members</td><td>Reference code</td><td>Yield strength, MPa</td><td>Coating Class Designation</td><td></td></tr>
<tr><td>Rafter/ Beam</td><td>ASTM A792M/ IS 15961</td><td>250 - 450</td><td>AZM150 (ASTM A792M) / (IS 15961)</td><td></td></tr>
<tr><td>Purlin</td><td>ASTM A792M/ IS 15961</td><td>250 - 450</td><td>AZM150 (ASTM A792M) / (IS 15961)</td><td></td></tr>
<tr><td colspan="5">1. Minimum elongation % shall be as per relevant Standard and Code.
2. Materials shall be fabricated in the shop.
3. Minimum coating requirement mentioned above in the table.
4. All structural calculations of cold formed steel section for checking the adequacy for strength and deflection criteria is to be taking into consideration the maximum permissible negative tolerance over specified BMT i.e. lower limit of BMT is to be considered.
5. The tolerance on Base Metal Thickness (BMT) thickness of steel sheets and coils shall be as given in IS/ISO 16163</td></tr>
</table>

Painting of Steel Surfaces embedded in Concrete: For the portion of Steel surfaces completely embedded in Concrete as reinforcement for foundation systems, the surface shall be prepared by Manual Cleaning and provided with Primer Coat of Chlorinated Rubber based Zinc Phosphate Primer of Minimum 50 Micron Dry Film Thickness (DFT).

Bidder shall also use principles governing design that shall prevent or reduce the risks of corrosion as per IS 9172 and relevant IS codes.

Connections :

Sl No	Connection	Grade
1	Solar PV module to purlin/structure connection.	SS304, A2-70
2	Bolts required loosing and tightening seasonally for seasonal tilting in the module mounting structure.	
3	Other structural fixed connections.	HDG 5.6 & 8.8
4	Foundation Anchoring.	HDG 4.6

Note: Fastener shall conform to IS 1367

A. SS304 Fasteners (nuts, bolts, washers and U-bolts) shall be of corrosion-resistant austenitic steel. SS 304 Fasteners shall have a good anti-seize finish with proper wax coating for better durability and firm resistance to all types of failure including seasonal removal and re-fixing of bolts.

B. All fasteners shall be provided according to the connection design requirement. All bolts shall be tightened with designed torque mechanically immediately after the erection of MMS to avoid any possible damage due to any incidental storm during the erection stage.

C. One set of fastener shall consists of one hexagonal head nut, one hexagon shape bolt, and two washers. The bots and nuts with inbuilt washers may also be provided.

D. In the ground mounting structure system with seasonal tilt arrangement, the column post and rafter/beam at seasonal tilt point of rotation shall be preferably hinged plate and bolt system.

APPROVAL

- Proposed layout of the Module mounting structure fitted with equipment & other ancillaries as required over the proposed developed land profile shall be submitted with Detailed Design Report for approval.

- Design, drawings, specifications of all components with material selected & installation details shall also be included with Detailed Design Report.
- Approval of the Engineer in charge should be taken before execution of the work at site.
- The contractor shall deliver the product to the site only after receipt of such approval against their prayer in writing from DPL.
- Design, drawings, specifications of all components with material selected & installation details shall also be included with Detailed Design Report.
- Joint inspections and testing will be done by DPL and the authorized representatives of the contractor at the manufacturer's workshop on regular basis for quality assurance and testing. Acceptance Tests as per relevant Standard shall be carried out at the module Mounting Structure's workshop.
- Approval of the Engineer in charge should be taken before execution of the work at site.

The contractor shall deliver the product to the site only after receipt of such approval against their prayer in writing from DPL.

FOUNDATION SYSTEM

Top of concrete/ height of collar for MMS foundation shall be minimum 150 mm above Finish ground level. The proposed foundation system for MMS shall be based on findings/results of the approved geo technical investigation report. Following kind of foundation may be provided:

a) Short pile foundation (Min. 300mm dia.)

b) Rock anchor with concrete collar (Min. 700 sq.cm.)

c) Isolated, strip or raft foundation

d) Concrete ballast foundation

List of applicable Indian standards

IS 2062 - Hot Rolled Medium and High Tensile Structural Steel. IS 811 - Cold Formed Light Gauge Structural Steel Sections.

IS 1161- Steel Tubes for Structural Purposes.

IS 4923 - Hollow steel sections for structural use.

IS 4759 - Hot-dip zinc coatings on structural steel and other allied products IS 4736 - Hot-dip zinc coatings on mild steel tubes

IS 1868 - Anodic coatings on aluminium and its alloys.

IS 2629 - Recommended practice for hot-dip galvanizing of iron and steel.

IS 15961 - Hot dip aluminium-zinc alloy metallic coated steel strip and sheet (plain)

IS 9172 -Recommended design practice for corrosion prevention of steel structures .

5.11.9 ROADS

a) Suitable approach road (as applicable) from nearest public road up to plant Main gate, Access road from Main gate to Main control cum office room (MCR), Internal roads connecting MCR and other facilities/ buildings/ open installations like Local control room(s) (LCR)/ Inverter control room(s) (ICR), Sub-station & Switch yard (as applicable) etc. and peripheral road along inside of the boundary fence/ wall shall be provided for safe and easy transportation of men, material and equipment during construction and maintenance.

b) The Approach road connecting nearest public road and the Main gate shall be of 4.0m wide carriage way with 0.5m wide shoulders on either side. The access road connecting Main gate and MCR and internal access road(s) connecting MCR to various facilities/buildings/open Installations shall be of 3.75m wide carriage way with 0.5 m wide shoulders on either side while the peripheral road shall be of 3m wide carriage way with 0.5m shoulders on either side. The top of road (TOR) elevation shall be minimum150 mm above FGL to avoid flooding of roads during rains. The roads shall be provided with alongside drains as per design requirements of drainage system for effective disposal of storm water and to avoid cross flow of storm water over the road. The roads shall be designed as per IRC SP-72 corresponding to traffic category T3 and critical field CBR value of the sub grade. Shoulder shall be of min. 150mm thickness.

c) However, following minimum road section details shall be followed:

i) Base course WBM (CBR>100%) conforming to Cl. 405 of MORD specs: 75mm compacted thick, Grade III

ii) Base course WBM (CBR>100%) conforming to Cl. 405 of MORD specs: 75 mm compacted thick, Grade II

iii) Granular/ gravel sub-base course (CBR>20%), conforming to Cl. 401 of MORD, specs: 175 mm compacted thick, compacted to 100% of max dry density

iv) Compacted sub grade: 300mm thick below sub-base (non-expansive soil with max.dry density > 1.65 kN/m3) conforming to Cl 303 of MORD specs, compacted up to98% of standard proctor density in layers of 150mm thickness. In case of expansive soils like black cotton soil suitable treatment as per Cl. 403 of MORD specs shall be provided before laying sub-base course.

v) Shoulders conforming to Cl 407 of MORD specs: 150mm compacted thick, compacted to 100 % of max. dry density

d) The construction of road shall conform to MORD specifications for Rural roads published by IRC.

e) Drain, cable or any other crossing shall be provided with RCC box or precast concrete pipe culvert. The culvert design shall conform to relevant IRC standard. The pipes for road culverts shall be of minimum class NP3 conforming to IS 458 with min. soil cover of 750mm above the pipe. In case of soil cushion less than 750mm the pipe shall be provided with 100 mm thick M20 reinforced concrete encasement with 10 dia. reinforcement rods @ 150mm c/c both ways. However, the water supply pipe for module cleaning and service/ drinking water shall be routed through Medium class GI steel pipe of required dia. conforming to IS: 1161.

f) Minimum dia. of casing pipe to be used at any facility like electric cable, water pipe line etc. shall be 150mm.

g) Maintenance pathways of min. 1200 width shall be provided between SPV arrays for easy movement of maintenance staff, tools, equipment and machinery, washing of modules etc. Finished surface with ground topography/ grade to be maintained to avoid accumulation of water in the region and allowing its free flow to keep the area devoid of mud/ sludge.

h) However, about 2.5m wide corridor shall be left along inside of the plant boundary suitably maintained clean of any vegetation and shall be provided with adequate illumination for movement of security personnel. Any undulations shall be made good with locally available coarse grained material to have fairly level passage way.

i) All pathways from central road to Inverter room road shall be Concrete road. Also, all cable crossings and other crossings shall be provided with GI/ Hume pipes considering electrical standard and safety. Concrete road upto each entrance and exists of the Inverter cum control room building shall be provided.

j) Bidder shall have to construct suitable approach from main gate for each phase with 3mtrs wide concrete road to inverter cum control room. Approach road between Phase 1 and Phase II inverter room shall be provided with similar 3 mtrs. wide concrete road.

k) The design and drawings for approach road, all internal roads and culverts shall be submitted to the Engineer for approval before execution.

ROADS DURING CONSTRUCTION:

Bidder, if necessary, shall build other temporary access roads to the actual site of construction for his own work at his own cost. The Bidder shall be required to permit the use of the roads so constructed by him for vehicles of any other parties who may be engaged on the project site. The Bidder shall also facilitate the construction of any permanent roads should the construction there of starts while he is engaged on this work. He shall make allowance in his Tender for any inconvenience he anticipates on such account. Non-availability of access roads, railway siding and railway wagons for the use of the Bidder shall in no case condone any delay in the execution of work nor be the cause for any claim for compensation against the Employer/ Owner.

PAVING OF THE AREA:

- The area of the proposed plant both at Phase II (area which actually comes after detailed engineering for the installed solar panel) has to properly leveled and graded with slope to have a collection of surface runoff and drainage water obtained from the cleaning of the panels. Entire area has to be developed with CLSM (refer below).
- The road will of rigid pavement. (Concrete road with reinforcement).
- All road crossing for DC cable will be on reinforced concrete box culvert with sufficient spacing from electrical and serviceability considerations. However for HT cable, road boaring shall be used with GI pipes with a minimum thickness of 5mm & depth of 1 mtrs. from the top of Road.
- If, any of the existing structure is damaged or required to be dismantled for convenience of the erection, the same has to mend good as per the original.
- Existing drainage system must have to be maintained. If, any cable crossing is required to cross over the existing toe drain of the reservoir, the same has to be properly blocked with RCC wall and drainage to be restored to the original.

CLSM (CONTROLLED LOW STRENGTH MATERIAL):

1.0 A flowable lean concrete mix consisting of a mixture of cement, fly ash, densely graded mineral aggregates, water, and admixtures. CLSM shall be capable of freely flowing to fill all voids etc. with a minimum 100 mm thickness over entire areas of both Phase I and Phase 2 apart from the roads without compaction or other additional effort, shall be of uniform density, shall be of low permeability to prevent migration of adjacent fines into the set mix, and shall excavate easily at any later time with minimal risk of damage to buried services.

Bidder shall do the preparation & consolidation of sub-grade with power roller/ hand roller and providing & laying of CLSM mix with coarse sand with proper sloping arrangement to remove the surface water through drains. Surface dressing of ground including removing vegetation and inequalities not exceeding 15 cm deep and disposal of rubbish after proper levelling as per instruction of site engineer.

2.0 Fly ash shall be issued free of cost by DPL, however the contractor has to transport the same to site from the location shown by DPL is in scope of the contractor. Transportation of fly ash shall be done by the contractor properly to avoid spillage of ash. The contractor has to clean the spilled ash at his own cost. The contractor has to make trial mix with different water and cementations material ratio.

3.0 Note:-

a) The contractor has to mobilize all the resources as per requirement of the work, so that the job is completed as per instruction of DPL site official.

b) The contractor has to make their own arrangement from nearest source of power & water supply.

c) Works will be carried out as per latest CPWD specification/IS codes.

d) The contractor has to take all necessary measures for safety of personnel and all the erected equipment and cabling system.

e) Providing and laying Controlled low strength material having compressive strength not less than 150 psi by using fly-ash, cement (PPC cement shall be approx 50kg per cu.m, which may vary as per mix design), fine aggregate (i.e. coarse sand), water etc. as mentioned above in required proportion including mechanical mixing as per approved drawing. The contractor has to make trial mix with different water and cementations material ratio to get the desired strength. Necessary Cube testing to be done by the contractor from reputed govt. approved test lab to reach at the final design mixture of compressive strength 150 psi. Compressive strength testing shall be carried out as per procedure adopted during mix design or ASTM standard.

5.11.10. MCR, INVERTER ROOMS, SECURITY ROOM & STORE ROOM

The Control Room Inverter Room , Transformer room etc should be such that it can accommodate facilities of both the Phase II projects namely 5 MWAC located opposite Administrative building and **8MWAC** behind 220 KV switchyard .

The following structures shall be designed and provided by the bidder:

i. CONTROL ROOM:

For the operation and maintenance of SPV Plant one Main Control Room (MCR) with Switchgear room shall be provided. The MCR building shall consist of the following:

i. Air-conditioned SCADA Room.

ii. Inverter, Battery Room, ACDB and Switchgear Room.

iii. Store Room.

iv. Air-conditioned Main Control Room.

v. Toilets (Male and female).

vi. Pantry.

Bidder to adhere to the Tender Drawing and equipment's manufacturers requirements for development of layout. The MCR shall be RCC framed structure with fly ash bricks/concrete blocks masonry walls. The MCR shall have entry lobby and portico with a roof for vehicle stoppage.

Parking shed to accommodate at least 2 cars and 5 bikes shall be provided near the MCR building. The parking shed shall be made of structural steel conforming to IS 1079/2062 with permanently color coated roof sheets. The minimum size & requirements of the MCR Building & all items shall be done by the Contractor as per the equipments layout and requirement.

ii. INVERTER ROOMS

Inverter rooms consist of PCU's, LT panels, batteries, etc. shall be provided based on manufacturer recommendation, easy passage of O&M persons and cable trench layout required.

The inverter rooms shall be made of as mentioned below:

a. Pre-Engineered Building in line with PEB Tender drawing & technical specification (Inverter R PEB shall be provided only in non-coastal area),

b. Steel Containerized solutions.

c. The battery and its associated equipment shall be suitably segregated inside the Inverter room with proper ventilation arrangement.

d. The equipment inside the inverter room shall be placed so as to provide sufficient space for their maintenance.

iii. SECURITY ROOM

Prefabricated security room or fly ash brick/stone masonry with RCC slab near the entry of the main gate. The toilet room shall be made of fly ash brick/stone masonry with water facility

iv. TRANSFORMER ROOM

Five nos. 4-Inverter Transformer and 1-Auxiliary Dry type transformer will be placed in the PEB room.

v. STORE ROOM

One store shall be constructed near MCR Room for storage of Mandatory Spares during O&M Period by bidders and later on for DPL after O&M period. The buildings and allied works shall be designed to meet NATIONAL BUILDING CODE (SP: 07 2016) requirements. Finish floor level of all building/rooms shall be minimum 450 mm above from Finish graded level.

SPECIFICATION FOR PRE-ENGINEERED BUILDINGS (PEB)

The Power Conditioning Unit (PCU)/Inverter and Dry Type Transformer Rooms shall be made of Pre-Engineered Buildings (PEB). The architectural and civil works drawing of Pre Engineered - Inverter Rooms are provided in the technical specification, tender drawings. Bidder shall prepare the detailed fabrication and civil construction drawings based on tender drawing title: "**Pre Engineered Building"** and submit to DPL for approval before the start of work. PEB shall be manufactured, supplied and erected by the bidder/PEB agency. The PEB shall be made of structural steel construction with double skinned metal roofing and wall cladding of approved profile. PEB shall be complete with painting, metal facia, metal gutter, rainwater down comers, sun-shades, openings, etc., along with associated structural steel, cladding and roofing work insulation, Trims & Flashings. Each item of PEB like panels, masonry, plastering, flooring, foundation, fittings etc. shall be suitable for the complete life of the solar plant. The construction methodology for PEB shall also be submitted for DPL approval before the start of works.

The layout of the Inverter room shall be designed so as to divert the heat generated from each inverter outside the room. The inverter room shall be designed for a life of 25 years. The PEB shall have a robust water tightness at all joints and connections. The building shall have a high class durability and performance during the adverse weather conditions. The PEB supplied shall be complete in all respect meeting all the requirements of tender drawings and other technical and functional requirements like lighting, ventilation system etc. to ensure effective functioning.

PEB length can be determined based on actual requirement, however, the grid spacing shall be maintained as per tender drawing title: "**Pre Engineered Building"**.

A. The PEB inverter room structural members shall meet the requirements of Tender Drawings. All hot rolled primary structural members and

Rod/Angle/Pipe bracing etc. shall conform to IS: 2062, minimum Grade E250 Quality A. Secondary members for Purlins and Girts shall conform to the specification of IS 811 or ASTM A1003-12 made from steel sheets conforming to ASTM A1011-12b Grade 50 having a minimum yield strength of 345 MPa. The minimum thickness of secondary members shall be 3mm. All other miscellaneous secondary members shall have the minimum yield strength of 250 MPa.

B. Insulated wall cladding or roofing shall consist of double skin metal cladding with Poly Urethane Foam (PUF). PUF must be made of continuous method PU foam and must be CFC free, self-extinguishing, fire retardant type with density 40 +/-2 kg/m3 and thermal conductivity 0.019-0.023 W/(m.K) at 10°C. The PUF panels shall be a factory made item ready for installation at site.

Inside room/enclosure temperature shall not be more than design temperature of the equipment.

C. Fasteners & Connections:

Special coated self-drilling screws/fastener shall be used conforming to class 3 as per AS 3566.1 and AS 3566.2. Steel bolts, nuts and washers complying with AS 1112:2000. High Strength Bolts for Primary Connections IS 1367 (Part III) Gr.

8.8 / ASTM A325. Bolts for Secondary Connection IS 1367 (Part III) Gr. 4.6 / ASTM A307. Anchor/foundation Bolts shall conform to IS 5624 and relevant IS code.

D. Roof and Wall cladding:

PUF panels shall be made of troughed permanently colour coated metal sheets of steel for roofing and side cladding (internal and external) shall conform to the requirements of Table-1 and IS: 513 for Hot-dip Zinc coated or Al/Zn coated sheets. The insulation material thickness and details shall be as specified at the relevant para in the specification.

PUF insulated panels Metal Sheet for Roofing and side cladding consist of an external sheet as troughed permanently colour coated sheet & internal sheet as plain permanently colour coated sheet.

The chemical composition of Troughed permanently colour metal sheet for roofing and side cladding shall conform to the provisions of same reference code to which the mechanical properties conform to.

Plain permanently colour coated steel metal sheet for ridge and hips, flashing, trimming, closure for vertical and horizontal joints, capping etc. shall conform to the same requirements as those of troughed permanently colour coated metal sheet for roof and side cladding.

The maximum spacing of the fastener shall be 390 mm c/c along the length of purlins/runners. However exact spacing shall be as per the design was done by the bidder of the fastener considering the wind load, self-load and other associated load. The minimum diameter of the fastener shall be 5.5 mm and at- lest 3 nos. of fastener shall be used per sheet.

Fillers blocks as a trough filler shall be used to seal cavities formed between the profiled sheet and the support or flashing. The fillers blocks shall be manufactured from black synthetic rubber or any other material approved by the engineer.

E. Roof Insulation and type

Both metal sheets shall have an under insulation of minimum 40 mm thick PUF with density 40 +/- kg/m3 and thermal conductivity 0.019-0.023 W/(m.K) at 10°C with gutters and down take pipes along with Flashing & Top cap of the required size and colour complete with all necessary hardware complete. The roof shall be projected at-least 300 mm from the wall.

Stiffening ribs / subtle fluting for effective water shedding and special male / female ends with full return legs on side laps for purlin support and anti-capillary flute inside lap.

Both upper and lower sheets shall be separated through spacers and fastened through zinc /zinc-tin coated self-drilling screws. The fastener size shall be calculated as per the design or manufacturers recommendations.

F. Wall Insulation:

All voids of external and internal metalled walls shall have an under insulation of minimum 40 mm thick PUF with density 40 +/- kg/m3 and thermal conductivity 0.019-0.023 W/(m.K) at 10°C with proper supports etc. as approved.

Both the walls should be separated by spacers system made up of cold-formed steel bars and fastened through zinc /zinc-tin coated self-drilling screws.

The external wall of Inverter room facing the transformer area shall be as per IS: 1646 - Code of practice for fire safety of buildings (general): electrical installations.

G. Doors Frames:

Door frames shall be of the iron frame of mild steel sections. All doors shall be provided necessary fittings like hinges, handles, mortice locks, tower bolts, stopper, hydraulic door closer, etc. of CP brass complete fixed to Pre-Engineered structure including necessary filling up of gaps at junctions with required PVC/neoprene felt etc. including hinges / pivots and double action hydraulic floor spring of approved brand and manufacture IS: 6315 marked, lock, handle and all necessary fittings as detailed in tender drawing or submitted by bidder in shop drawing and approved by DPL.

The door entrance shall include Mild Steel single leaf door. The structural steel shall conform to IS 7452 and IS 2062. The holdfasts shall be made from steel flats (50 mm and 5 mm thick). The fixtures, fastenings and door latch are to be made with same materials.

Bidder can also proposed uPVC extruded casement/ sliding doors, with complete fitting, accessories and panels as per items mentioned in DSR 2016.

H. Windows Frame

Aluminum black powder coated section, frame shall be of 92x31 mm, minimum 16G thick as per approved design. Tinted glass and aluminum grill shall be provided.

The Bidder can also propose uPVC extruded casement/ sliding windows with complete fitting and accessories as per items mentioned in DSR 2016.

I. Ventilators:

Aluminum black powder coated frame of minimum size 62x25 mm and 16G thick as per approved design. Ventilators/duct shall be provided with bird guard. Size of opening at the wall for ducts shall be as per PCU manufacture and min 18 gauge GI sheet. Ducts shall be supported with suitable means, as approved during detail engineering.

All accessible ventilators and windows of all buildings shall be provided with min. 4mm thick float glass, tinted for preventing solar radiations. Suitable sunshades made out of approved colour sheet will be provided to all external windows and door. The minimum projection for the sunshades will be 450 mm and 300mm wider than the width of the opening.

J. Rolling shutter:

Rolling shutter (Hand operated) shall be fabricated from 18 gauge steel and machine rolled with 75 mm rolling Shutter with effective bridge depth of 12 mm lath sections, interlocked with each other and ends locked with malleable cast iron clips to IS:2108 and shall be designed to withstand a wind load without excessive deflection. Metal rolling shutters and rolling grills as IS: 6248.

K. Plinth Protection:

500 mm wide plinth protection minimum with 75 mm thick of cement concrete 1:3:6 (1cement : 3 coarse sand : 6 graded stone aggregate 20 mm nominal size) over 75 mm bed of dry brick ballast 40 mm nominal size well rammed and consolidated and grouted with fine sand including finishing the top smooth, shall be provided around the Pre-Engineered Building.

L. Floor Finish:

Flooring, including preparation of the surface, cleaning etc. shall be of cement concrete flooring as per IS: 2571 with ironite hardener. The inverter room floor shall be at least 450 mm above the ground level.

M. Paint and Coating:

Metal sheet shall be colour coated with total coating thickness of 25 microns (nominal) dry film thickness (DFT) comprising of silicon modified polyester (SMP with silicon content of 30% to 50 %) paint or Super Durable Polyester (XRW) paint of 20 microns (nominal) on one side (exposed face) on 5 micron (nominal) primer coat and 10 microns (nominal) SMP or Super Durable Polyester paint over 5 micron (nominal) primer coat on other side. SMP and polyester paints system shall conform to Product type 4 as per AS/ANZ 2728.

The structural steel shall be hot-dipped galvanized, conform to IS: 4759 or relevant Indian standard

N. Lighting

The inverter room shall be provided with electric light to achieve an average illumination level of 150 Lux. However, room should be designed to utilize maximum natural light during the day.

O. Descriptions of PEB Structures:

Primary Members: Primary structural framing shall include the transverse rigid frames, columns, corner columns, end wall wind columns, beams, truss member, base pate.

Secondary Members: Secondary structural framing shall include the purlins, girts, eave struts, bracing, flange bracing, base angles, clips, flashings and other miscellaneous structural parts. Suitable wind bracings sag rods to be reckoned while designing the structure.

Sealant: Sealant used for cladding shall be butyl based, two parts polysulphide or equivalent approved, non-staining material and be flexible enough not to interface with fit of the sheets.

Closures: Solid or closed cell closures matching the profiles of the panel shall be installed along the eaves, rake and other locations

Flashing and Trim: Flashing and / or trim shall be furnished at the rake, corners, eaves, and framed openings and wherever necessary to provide weather tightness and finished appearance. Colour shall be matching with the colour of the wall. The material shall be 26 gauge thick conforming to the physical specifications of sheeting.

Gutters and Down Comers: Gutters shall be fabricated out of same metal sheet. Material shall be same as that of sheeting. Down comers shall be of galvanized steel pipes or PVC designed to ensure proper roof drainage system.

SPECIFICATION FOR RCC BUILDING (MCR) AND OTHER RCC/MASONRY STRUCTURE.

The MCR building shall be made of RCC framed structure with fly ash bricks/concrete blocks masonry walls. The thickness of outer masonry walls shall be minimum 230mm in case of bricks and minimum 200mm thick in case of concrete blocks. The roof shall be designed for a minimum superimposed load to 150 kg/m2. The bidder shall also provide rainwater harvesting system for MCR building roof.

A. REINFORCED CONCRETE STRUCTURE

a. All RCC works shall be designed mix as per IS 456 (2000). For structural concrete items, Ordinary Portland cement (43 Grade) conforming to IS: 8112 and Fly ash based Portland pozzolana cement conforming to IS: 1489 (Part-1) shall be used for superstructure. Type of cement for sub-structures shall be decided based on the final Soil Investigation report.

b. Coarse aggregate for concrete shall be crushed stones chemically inert, hard, strong, durable against weathering of limited porosity and free from deleterious materials. It shall be properly graded. It shall meet the requirements of IS: 383.

c. Sand shall be hard, durable, clean and free from adherent coatings of organic matter and clay balls or pellets. Sand, when used as fine aggregate in concrete shall conform to IS: 383. For plaster, it shall conform to IS: 1542 and for masonry work to IS: 2116.

d. Reinforcement steel: Non-coastal area

Reinforcement steel shall be of high strength deformed TMT steel bars of grade minimum Fe-500 and shall conform to IS: 1786. Ductile detailing in accordance with IS: 13920 shall be adopted for superstructure and substructure of all RCC buildings/structures

e. Concrete :

The following minimum grades of concrete for design mix and nominal mix shall be adopted for the type of structures noted against each unless not specified elsewhere

Grade as per IS 456	**Non-coastal area**
M25	All RCC structural elements above and below ground level, precast concrete, MMS foundation, cable trench, oil pit, Grade Slab, Paving, culverts & road.
M20 (Equivalent nominal Mix of	Fencing work

1:1.5:3)*	
M15 (Equivalent Nominal Mix of 1:2:4)	Base slab of drains.
M10 (Equivalent Nominal Mix of 1:3:6)	Plain Concrete Cement.

The bidder shall carry out the design mix of M-25 and M-20 grade concrete on priority. The design mix shall be approved from DPL before the start of work.

* The use of nominal mix for M-20 grade may be accepted only in exceptional cases subject to approval of DPL Engineer-In-Charge. The same shall be the adopted subject to approval from DPL for specific work.

f. In case Geotechnical investigations require any special kind of cement or higher grade of concrete, the same shall be provided. The foundation system shall be made which transfer loads safely to the soil for the module mounting structures, depending on soil conditions, geographical condition, regional wind speed, bearing capacity, slope stability etc. All foundation system and foundation depth shall be decided based on the approved geotechnical investigation report. No foundation allowed on back filled soil and the foundation depth to reach upto natural ground level (NGL).

g. All loads shall be considered in line with IS: 875. Seismic loads for design shall be in accordance with IS: 1893 and relevant Standards.

h. IS: 2502 Code of Practice for Bending and Fixing of Bars for concrete Reinforcement must complied for reinforcements. IS 5525 and SP 34 shall be followed for reinforcement detailing.

i. A minimum 75 mm thick PCC shall be provided below RCC wherever RCC structure is laid over the ground. Proper and sufficient formwork/shuttering shall be provided for the required period as per IS 456.

B. Masonry Work

a) Brickworks shall be fly ash brick of minimum compressive strength 7.5 N/mm2 of approved quality as per IS: 1077, IS: 2212 and IS: 3495. Concrete blocks shall be of a minimum compressive strength of 7.5 N/mm2 and shall be of Grade-A as per IS: 2185. Stone masonry work with hard stone in building works, foundation, plinth and drains shall be Coursed Rubble or Random Rubble masonry work with the stone of good quality and durability. The masonry surface shall be plastered with minimum 18mm plaster in case of CMCS walls. The stone masonry work shall be in line with IS: 1597, IS: 1122 and IS: 1126.

b) The cement mortar for all kind of masonry work shall be in the ratio 1 cement and 4 sand by weight.

c) Bricks/blocks required for masonry work shall be thoroughly soaked in the clean water tank for approximately two hours. Brick shall be laid in English bond style. Green masonry work shall be protected from rain. All masonry work shall be kept moist on all the faces for a period of seven days.

d) Bricks of class designation 5.0 N/mm2 and 3.5 N/mm2 may be permitted to have slight distorted & rounded edges provided no difficulty shall arise on this account in laying of uniform courses in non-load bearing structures and shall be subject to the approval of DPL. Tolerances on dimensions up to +/- 8% shall be permitted. Dimension test to be carried out as per IS code.

e) The external wall for the building shall be 230 mm thick walls and internal wall 230/115 thick as per requirements. The external wall of CMCS facing the transformer area shall be as per IS: 1646 - Code of practice for fire safety of buildings (general): electrical installations.

f) The suitable damp proof course shall be provided the proportion of cement, sand & aggregate shall be 1:2:4 using 6 mm down stone chips with a waterproofing admixtures. The thickness of the damp-proof course shall be minimum 40 mm.

C. ROOFING

The roof of the building shall be insulated and waterproofing shall be done as per relevant IS standard.

D. PLASTERING

All external surfaces shall have 18 mm cement plaster in two coats, under layer 12 mm thick cement plaster 1:5 and finished with a top layer 6 mm thick cement plaster 1:6 (DSR 2013-13.11). White cement primer shall be used as per the manufacturer’s recommendation.

At least one coat of plaster shall be applied to interior walls by hand or mechanically, to a total thickness of 12 mm using 1:6, 1 cement and 6 sand. Plastering shall conform to IS 1542, IS 1661, IS 1630. Oil bound washable distemper on smooth surface applied with minimum 2 mm thick Plaster of Paris putty for the control room. Plaster of Paris (Gypsum Anhydrous) conforming to IS: 2547 shall be used for plaster of Paris punning.

E. PLINTH PROTECTION

Plinth protection 1000 mm wide shall be provided around all the buildings.

F. WHITE WASHING & COLOUR WASHING

White washing and colour washing work shall be conforming to IS 6278.

- Internal walls - Acrylic distempering as per IS 427.
- External walls – Heat reflective synthetic enamel as per IS 428.

For cement painting IS 5410 shall be followed.

For painting of steel doors, ventilators IS 2338, IS 1477 (Part I & II) shall be followed.

G. FLOORING

Flooring shall be of cement flooring in concrete mix 1:2:4 using 10 mm aggregates as per IS 2571. Flooring for control room, shall be of vitrified tiles 10 mm. For toilet area, the floor shall be of non-skid vitrified tiles of 8 mm thicknesses. The floor finishing must include skirting up to a suitable height. The wall tiles, if proposed, shall be glazed tiles of 6 mm thickness and provided up to lintel level. Flooring before the Indoor HT switchgear should be matched with the panel base in such a way that all the VCB trucks shall be racked outside smoothly

H. DOORS & WINDOWS

Steel framed doors, Windows and ventilators shall conform to IS – 1081 with necessary glass panels including of all fixtures and painting etc. complete. Doors and windows shall be made of aluminum sections. All sections shall be 20 microns anodized. Sections of door frame and window frame shall be adopted as per industrial standards. Door shutters shall be made of aluminum sections and combination of compact sheet and clear float/ wired glass. The control room shall require a number of windows/ louvers to provide ventilation/ fresh air circulations.

I. WATER SUPPLY

GI pipes of Medium quality conforming to IS 1239 (Part I-1990) or CPVC pipes conforming to IS 15778 shall be used for all portable hot and cold water distribution supply and plumbing works.

The Syntax or equivalent make PVC storage water storage tank conforming to IS: 12701 shall be provided over the roof of the CMCS with adequate capacity for 10 No person and 24 hour requirement, complete with all fittings including float valve, stopcock etc. The capacity of the tank shall be minimum 1000 litres.

J. TOILET

Toilet shall be designed for 10 persons; and constructed with following finish

- Floor: Vitrified tiles/ ceramic tiles
- Door window: made out of aluminium sections, 6mm float glass
- Ventilators: Mechanical exhaust facility
- Plumbing fixtures: Repute make
- Sanitary ware: Repute make
- EWC: 390 mm high with health facet, toilet paper roll holder and all fittings
- Urinal (430 x 260 x 350 mm size) with all fittings.
- Wash basin (550 x 400 mm) with all fittings.
- Bathroom mirror (600 x 450 x 6 mm thick) hard board backing
- CP brass towel rail (600 x 20 mm) with C.P. brass brackets
- Soap holder and liquid soap dispenser.
- GI pipes (B class) of reputed makes
- Overhead water tank of minimum 1000 litre capacity

K. OVERHEAD WATER TANK

The Syntax or equivalent make PVC storage water storage tank industrial grade conforming to IS: 12701 shall be provided over the roof of the Control Room building with adequate capacity for 10 No person and 24 hr requirement, complete with all fitting including float valve, stop cock etc.

L. DRAINAGE FOR TOILETS

Drainage pipes shall be of PVC (6 kg/cm2) Supreme, Prince or equivalent make. Gully trap, inspection chambers, septic tank for 10 person and soak well to be constructed for abovementioned requirement.

M. ELECTRIFICATION OF BUILDING

Electrification of all building shall be carried out as per IS 732-1989, IS: 4648-1968 and other relevant standards.

5.11.11. Structural Steel

Structural steel design shall be carried out as per IS 800 and IS 801. Structural steel shall conform IS 2062 / IS 1079 or equivalent, Pipe shall be as per medium/high grade of IS 1161, Chequered plates shall conform to IS 3502 and Hollow steel sections for structural use shall conform to IS 4923.

Structural Steel/Steel Sheet Painting

All non-hot dip galvanised structural steel (excluding Module Mounting & SCB structure)/ Outdoor metal containers/ Enclosure/ Rolling shutter items shall be provided with paint designed for a minimum maintenance-free life of fifteen (15) years (high durability) as per ISO 12944 and IS 800. For finishing coat suitable colour pigment shall be added. All paints including primer shall be of the reputed brand/manufacturer and as approved by the Engineer-In-charge. The method of application shall be as per the recommendations of the manufacturer. For corrosive category of refer appendix of site specific data.

Grouting

Cement mortar (1:2) grout with non-shrink additives shall be used for grouting below base plate of a column. The grout shall be high strength grout having a minimum characteristic compressive strength of min 30 N/mm2 at 28 days.

5.11.12. PIPE /CABLE RACKS & TRENCHES

a. The conventional methods of cables laying and installation shall conform to IS 1255 for laying direct in ground, drawing in ducts, laying on racks in air, laying on racks inside a cable tunnel and Laying along buildings or structures, etc.

b. Outdoor RCC Cable Trenches: RCC outdoor cable trenches in switchyard area shall be provided with pre-cast RCC removable covers with lifting arrangement. The top of outdoor trenches shall be kept at least 100 mm above the gravel level so that rainwater does not enter the trench.

c. Indoor RCC Cable Trenches: RCC indoor cable trenches shall be provided with 50X50X5 mm angles grouted on the top edge of the trench wall for holding minimum 6 mm thick mild steel checkered plate covers conform to IS: 3502 with lifting arrangement.

d. RCC cable trenches shall be constructed with wall thickness minimum 100 mm.

e. Trench Drainage: The trench bed shall have a slope of approx. 1/500 along the run & 1/250 perpendicular to the run. In case straight length exceeds 30 m, suitable expansion joint shall be provided at appropriate distances. The expansion joint shall run through vertical wall and base of the trench. All expansion joints shall be provided with approved quality PVC water stops. Suitable drainage at the lowest point of the trench shall be provided.

5.11.13. WATCHMEN / SECURITY CABIN

Contractor shall provide prefabricated Watchman's portable cabin at each corner of the Plant at least 4 locations surrounding of the plant including Main gate. The Minimum size of watchmen's (Security Cabin) cabin is 1.2 metre x 1.8 metre size and height of 2.4m with appropriate roof at the top. Location of the watch Cabin (Security Cabin) will be as directed by the Purchaser. The Prefabricated Security Cabin of size 3 metre x 3 metre at the main entrance gate shall be designed and constructed by the Successful Bidder keeping in view the safety and security of the power plant. All Watchman / Security cabins shall have separate LED light, fan and 5 pin plug with individual switches.

5.11.14. PERIPHERAL BOUNDARY WALL/FENCE

- The plant peripheral boundary shall be provided with Chain link fencing.
- The boundary fence shall be provided along the Solar PV plant boundary to demarcate the plant boundary and to keep away the unauthorized access to the plant. The fence/ wall shall be provided with Main entry gate. The fencing/ wall shall be with 2.5m height above grade level including 400mm dia. GI concertina wire along with 3no. of barbed wires on either arm to be fixed on Y shape angle brackets. The main gate shall be min. 6.0m wide (clear) (4.5 m carriage way + 1.5m wicket gate).

CHAIN LINK FENCING

a) The fencing shall be of Chain link (poly coat GI as specified) mesh fabric with internal, corner and stay posts of RCC (min 200mm x 200mm size, M30 grade), along with 230 thick brick/toe wall, with 100mm thick M15 PCC foundation (min. width450mm and min. depth 450 mm below GL).

b) The toe wall shall project 150mm above GL. The intermediate, corner and stay posts shall be supported by angle struts that shall have the same foundation as that of the main posts.

c) The brick masonry toe wall shall be plastered with 15 thick CM (1:4) plaster on both faces and shall have min. 50 thick PCC (1:2:4) coping finished smooth and projecting 35 mm on either side of the wall with top sloping inwards.

d) Spacing of intermediate posts shall not be more than 2.5m. Every 10th intermediate post shall be provided with a stay post while every corner post shall be provided with two stay posts on either side.

e) Joints in RR masonry shall be properly raked and pointed with CM (1:3).

f) In case of pond/ drain crossing the fence, RCC beam of adequate size supported on RCC columns on either side and suitable grill of MS square rods (vertical spacing not more than 150mm) of min. Size 25x25 mm and min. 3 no. horizontal 20 SQ MS rods or 50 mm x 8 mm thick flats secured to RCC beam and columns; shall be provided in place of toe wall for smooth flow of water.

g) The GI chain link mesh fabric (40 x 40 mm with min. wire gauge 3.15mm, both ends twisted) and fencing shall conform to IS: 2721. Poly coat GI chain link mesh (50 x 50 mm) shall conform to ASTM 668 and fencing shall conform to ASTM 668.

h) Each fence panel, in lieu of tie wire, shall be provided with 35x35x3mm GI edge angle at top and bottom with mesh fabric firmly secured to them and to intermediate support angles.

i) All MS sections shall be painted with 2 coats of epoxy paint of approved make and shade over 2 coats of suitable primer.

MAIN GATE

1 The Main entry gate of size as specified elsewhere (2.5m height) shall be of rugged design with solid MS steel sections (25x25mm). The spacing of vertical members shall not be more than 150 mm. for each phase.

2 The gate shall be complete with MS flat guide track, castor wheel(s), GI fittings & fixtures like hinges, aldrop, locking arrangement, posts etc.

3 The main gate shall be of 2.5m height and shall have 4.5m wide Gate for vehicular movement and an adjacent 1.5m wide wicket gate for pedestrian movement.

4 Area near the main gate extending from 500 mm (min) outside the gate to 2700 mm(min) inside the gate, shall be brought to Top of Road elevation with respect to the approach road at main gate for full width of the gate. This shall be achieved by providing 200 mm thick PCC (1:2:4) over 100 mm thick PCC (1:4:8) further underlain with 300 mm thick well compacted boulder soling with interstices filled with sand, resting over well compacted sub-grade.

5 The gate shall be provided with the Project name plate (2.5mx 1m, 3mm thick MS plate). The gate shall be painted with 2 coats of epoxy paint over 2 coats of suitable primer.

6 The column posts of the gate shall be supported through RCC pedestal and footing. Min. depth of foundation shall be 1200mm below NGL.

7 All design and drawings for peripheral boundary fence/ Wall and Main gate shall be submitted for Engineer's approval before execution.

Details of main gate, wall/ fencing, material details along with design report and drawings are to be submitted for approval.

5.11.15. MODULE WASHING SYSTEM & WATER STORAGE:

a) The Contractor shall design and install the effective module cleaning system.

b) The Bidder shall do the cleaning the photovoltaic modules at least once in every week in order to operate the plant at its guaranteed plant performance i.e. NMGG stipulation. All necessary arrangement for wet cleaning of the solar panels shall be in the scope of the bidders and accordingly the agency has to provide suitable portable manual operated cleaning equipment with all the necessary accessories, tool & tackles, electrical pumps, tankers, brush and small water storage, piping arrangement etc. which as may be required for the same.

c) PV module cleaning purpose shall be of potable quality and fit for cleaning the modules with TDS generally not more than 75 PPM. However, water with TDS more than 200 PPM shall not be used directly for module cleaning without suitable treatment to control the TDS within acceptable limits. The water must be free from any grit and any physical contaminants that could damage the panel surface. Module cleaning procedure and pressure requirement at discharge point shall be as per the recommendation of PV module manufacturer. However, discharge pressure at outlet shall not be less than 50 kg/cm^2 (5 MPa).

d) The Contractor is advised to ascertain the availability of good quality ground water at site for construction, drinking and module cleaning purpose. The Contractor shall estimate the water requirements for cleaning the photovoltaic modules at least once in a week or at closer frequency at site, in order to operate the plant at its guaranteed plant performance. Also, the contractor is required to plan the water storage accordingly with provision of a tank of suitable capacity for this purpose. However, min. consumption of 2 Ltr / Sqm of surface area of SPV module shall be considered in estimation of required quantity of water storage.

e) The module cleaning system shall include construction of RCC tank of required storage capacity, pumps (including 1 No. standby pump), water supply mains and flexible hose pipes, taps, valves (NRV, Butterfly valve, Ball valve, Gate valve, PRV, scour valve etc.), Water hammer arrester(s), pressure gauge, flow meter etc. as per the planning & design. The opening pipes for fixing the movable/Hose pipes for spraying water on module shall be made of GI pipe.

Low lying area below HFL (High flood level) to be converted into natural pond (Water Reservoirs) with garlanding for beautification purpose. This water is not to be used for Module Cleaning.

f) Bidder shall provide the piping and the instrumentation diagram (P&ID) of water washing arrangement including the physical sequence of branches, reducers, valves, pressure gauge, cleaning points with location of pump(s) and water storage tanks to DPL for approval during detailed engineering.

g) In case of over ground water storage tank, the contractor shall check its effect on plant performance through shadow analysis. The PVC storage tank shall conform to IS: 12701. The valves shall conform to IS: 778. A suitable metal sheet canopy for protection from direct sunlight shall be provided over the tank area.

h) The water supply mains could be either of GI, uPVC or HDPE, however, the vertical pipe connecting supply main to the discharge point shall be of GI.

i) Masonry chamber shall be provided for Main gate valve at pump end. Whereas, as per requirements, at other locations either a masonry or GI/ HDPE pipe chamber may be provided.

j) Module cleaning procedure and pressure requirement at discharge point shall be as per the recommendation of PV module manufacturer. However, discharge pressure at outlet shall not be less than 5 kgf/cm2 (0.5 MPa).

k) All the pipes thus laid shall be buried in ground at least 150mm below FGL or laid above ground clamping on suitable concrete support blocks. In case of above ground piping only GI pipes shall be used.

l) Separate storage tank shall be provided for Phase I.

WATER STORAGE TANK

a. The Contractor is advised to ascertain the availability of good quality ground water at site for construction, drinking and module cleaning purpose.

b. Construction of Bore Well is not permitted in Plant premises.

c. The Contractor shall arrange the water to clean the photovoltaic modules at least once in a week or at closer frequency, in order to operate the plant at its guaranteed plant performance. Bidder may apply for DPL's water coneection for this purpose which might be provided within 100 metre from the propsed plant area. Vendor shall install flow meter for measurement of water consumption which will be chargeable as per DPL's water charge rate which may vary time to time.

d. The module cleaning system shall include construction of RCC tank or supply and installation of Ground mounted PVC tank (s) of required storage capacity for Backup purpose and laying network of GI pipe. System shall also include booster pumps, valves (NRV, Butterfly valve, Ball valve, Gate valve, PRV, scour valve etc.), Water hammer arrester(s), pressure gauge, flow meter, GI pipelines, bends/ joints/ couplers, tap assemblies (at delivery points), nozzles, hose pipes etc. Opening from the GI pipe with manual isolating valves should be provided at regular intervals.

Water level indicator shall be provided for automatic stopping of water pumping. Pressure gauge shall be installed at every pump end. Water hammer arrester(s), pressure gauge, flow meter etc. as per the planning & design.

e. In case the vendor proposes to provide ground mounted PVC Tanks then vendor to locate the PVC storage tank in the Array layout so as to minimize it’s shadow effect on the Array layout. The PVC storage tank shall conform to IS:12701. The valves shall conform to IS: 778. A suitable metal sheet canopy for protection from direct sunlight shall be provided over the tank area.

f. Drainage system to be made in such a way so that all rain water and module cleaning water should collect in the Natural Pond lying below HFL.

g. Capacity of the Pump should in such a way so that the Pump can supply water at least 1/3 portion of the Solar Plant.

h. The water supply mains could be either of GI, UPVC or HDPE, However, the vertical pipe connecting supply main to the discharge point shall be of GI. Masonry chamber shall be provided for Main gate valve at pump end. Whereas, as per requirements, at other locations either a masonry or GI/ HDPE pipe chamber may be provided.

i. All the pipes thus laid shall be buried in ground at least 150mm below FGL or laid above ground clamping on suitable concrete support blocks. In case of above ground piping only GI pipes shall be used.

Water supply pipe for module cleaning crossing the road shall be laid through Medium class GI steel pipe conforming to IS: 1161 of dia minimum 150 mm.

Suitable RCC pedestals shall be constructed for mounting the booster pumps that shall be housed within suitable shed/shelter arrangement.

Specifications for Underground Water Tank.

The top of the UG tank shall be 250 mm above FGL. The tank shall have clear free board of 300mm above MWL. The tank bottom shall have a slope of 1:100 towards drainage sump (500x500x500 mm deep). The slope shall be provided either in structural slab or in screed concrete (1:2:4) trawl finished. 1000x1000 mm size Manhole in roof slab and 20 mm MS rung ladder shall be provided for easy access to the storage tank and silting chamber for periodic cleaning. The manhole shall be covered with RCC precast cover. 50x50x6 mm MS angle with lugs shall be rovided around precast cover and tank slab opening for edge protection. Rungs shall be painted with 2 coats of epoxy paint over 2 coats of primer. The underground RCC tank shall be designed for following load conditions:

External earth pressure + hydrostatic pressure due to ground water table (to be considered at FGL for design purposes) + Surcharge of 20 kN/ Sqm and Tank Empty.

Tank full up to MWL and no external loads

The design shall conform to IS: 3370 with maximum crack width of 0.1mm for wall,bottom slab and roof slab.

Min. grade of concrete shall be M30 (M35 in coastal areas, marshy and saturated soils) conforming to IS: 456. Suitable construction joints shall be provided as per provisions of IS: 3370 (Part 1). Water proofing admixture conforming to relevant IS standard and of approved make shall be added to concrete as per manufacturer's recommendations. The underground water tank shall be tested for water tightness as per the provisions of IS 3370 (Part-4). In case any leakage is noticed the same shall be repaired by injection of cement grout installing suitable nozzles around affected areas. Outside face of water tank in contact with water and soil and underside of roof slab shall be painted with 2 coats of epoxy paint.

D.Electrical System

5.12.1
Solar Power generated from each unit will be converted to 415V AC and synchronized with the grid. 415V AC will be stepped up at 6.6KV through 2MVA/3 MVA (for 5 MW AC Solar Plant , Bidder should provide atleast 2 numbers of 6.6 KV Dry Transformers), 415V/415V/6.6KV transformer which will be connected to 6.6KV line through 6.6KV outdoor VCB and 6.6KV Isolator. Control panel for operation of 6.6KV VCB along with protection, metering, indication etc will be installed at local control room. There will be a Main Control Room (MCR) consisting of Control & Relay Panel, DC System (Battery Bank with Battery Charger, DCDB etc.), ACDB, 6.6KV Switch Gear415 V Switchgear, Protection and Metering System, Monitoring System etc. The exact position and layout of control Room and Switchyard will be finalized after detailed engineering. The 6.6 kV Switchyard will be located adjacent to the Main ControlRoom.

The *MCR* will contain:

- Control & Relay Panel.
- AC Distribution Board.
- ***DC System (220V):*** DC storage Battery Bank with Charger along with DCDB for the operation of CBs, Protection System, Annunciation, Emergency Light etc. However, separate DC systems may be used for local control rooms considering the performance of the DC Network based on site condition and length of

control cable.

- *Trenching work* for cabling.
- *PC* for local monitoring and security system etc.
- Office room , conference room, security cabin, pantry etc.

The 6.6 kV Switchyard will contain:

- 6.6 kV incoming feeder from the SPV Plant, 6.6 kV outgoing feeder to the nearby 6.6 kV Sub-station, one (01) no. bay for Auxiliary Power System, One no. Bus Coupler bay.
- One No. 0.4-0.4/6.6 kV Three Winding transformer.
- *50 kVA, 6.6 kV/433V, Dyn11 Distribution Transformer:* one (01) no. for the auxiliary AC power supply.
- *CT, PT, 6.6 kV indoor Type VCB, 6.6 kV Isolator* etc.
- 7.2 kV *LA* for each feeder
- *Earthing System:* Earth pit, earthing rods etc. (As per IS 3043 / IEEE 80-2000).
- *Mechanical structure*.

Solar Power generated from this Solar PV plant is proposed to be fed to the 6.6 kV bus of the Main Control Room which is nearest to the site.

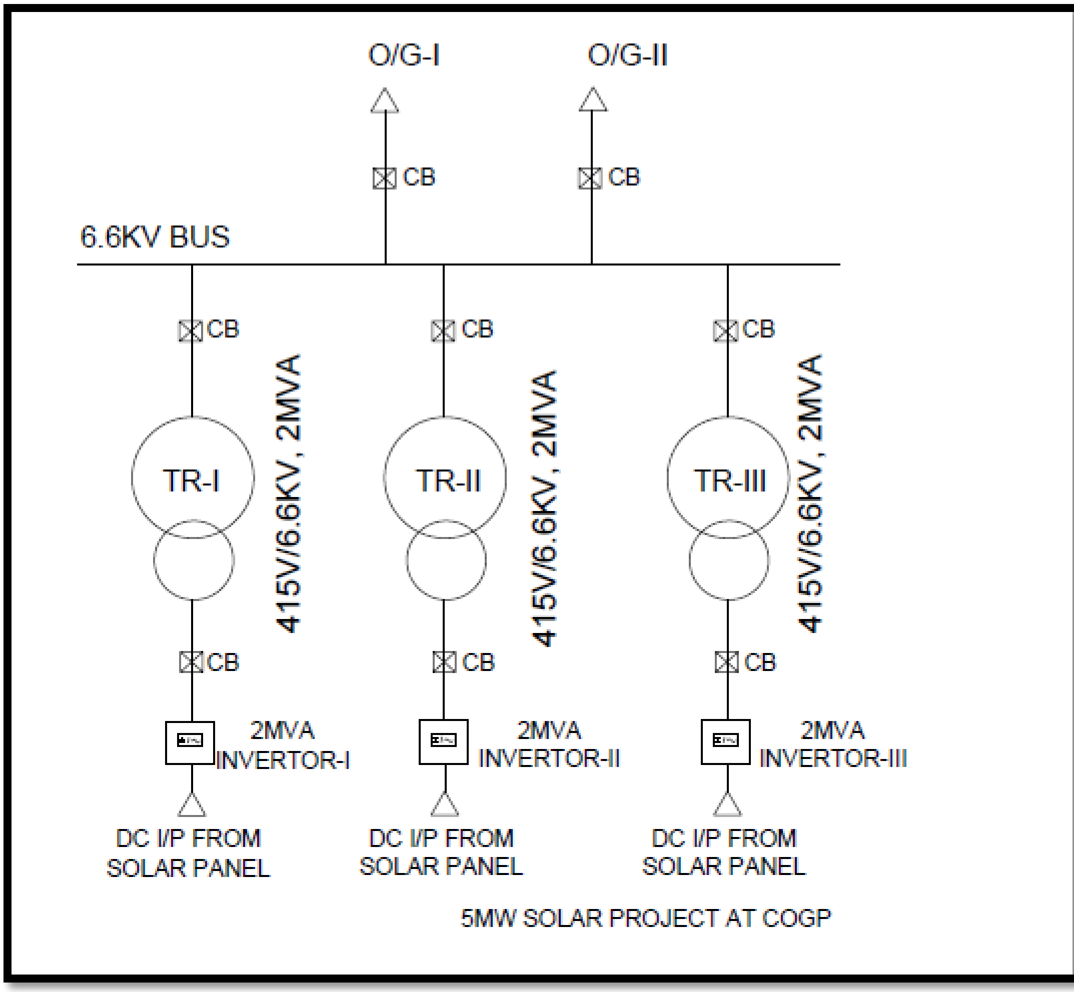


Figure 1: Indicative Drawing of 6.6 KV Evacuation network

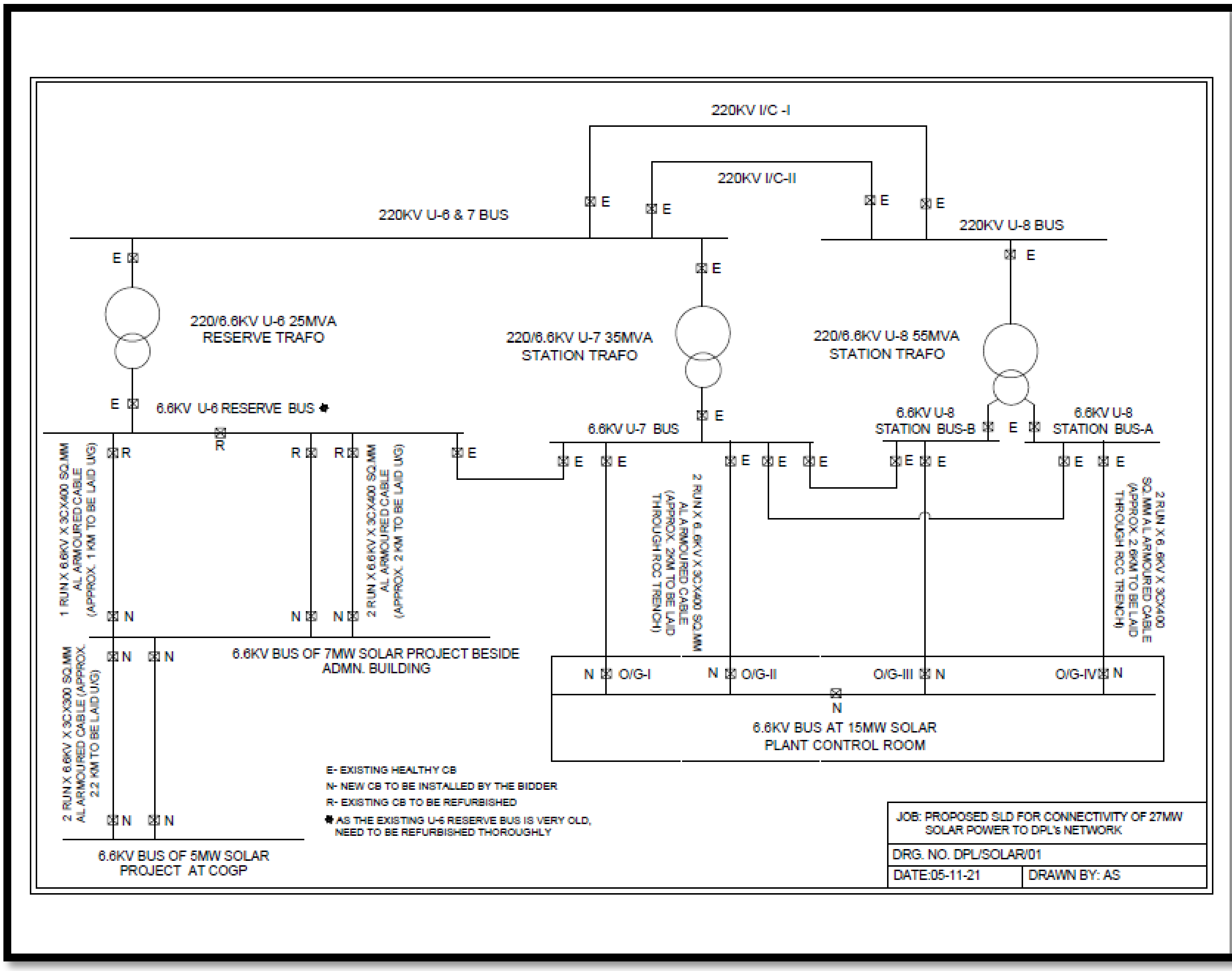


Figure – 2: Electrical Evacuation Network (Indicative)

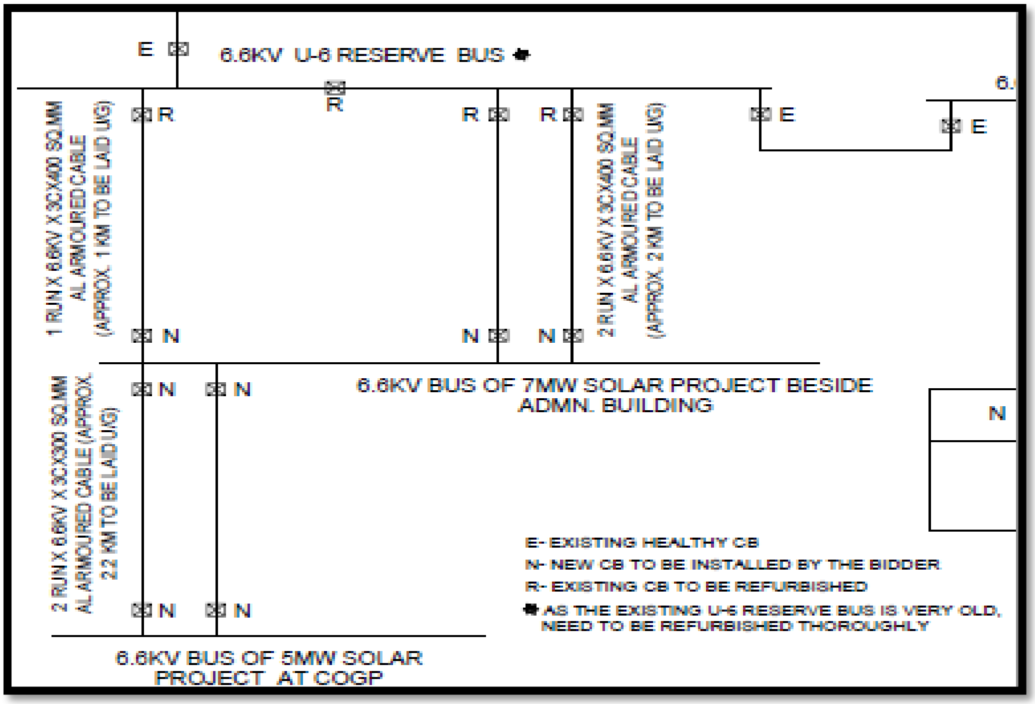


Figure 3: Evacuation network,Indicative

The Outgoing feeders from Control Room of 5 MW Solar will terminate at 6.6 KV 7 MW (beside Administrative building) AC Solar Bus as per Figure 3 (above).
Armoured cable to be laid through trench .
Approximate length is 2.2 KM (each cable)

5.12.2. Photovoltaic Modules

This section covers activities related to design, manufacturing, testing at works, supply, insurance, transportation and delivery at Project site, storage, erection,testing, commissioning of solar modules as detailed hereunder.

Solar Mono crystalline modules having capacity minimum 500 Wp shall be used for the project for 5 MWSolar PV Power Plant .
The make, capacity, technical specification and the model no. of each module connected with an individual inverter should be the same

Total capacity of the Solar PV modules shall be designed to ensure 5 MW**AC with minimum 15% overloading on DC capacity** and the net minimum guaranteed generation

5.12.2.1 Standards & Codes

Photovoltaic Modules shall comply with the specified edition of the following standards and codes.

Sl. No.	Latest Standards	Description
1	IEC: 61215/IS: 14286	Terrestrial Photovoltaic (PV) Modules – Design qualification and type approval Part 1 – Test Requirements
		Terrestrial Photovoltaic (PV) Modules – Design qualification and type approval Part 1 – Test Requirements Section 1- Special requirements for testing of crystalline silicon photovoltaic (PV) modules
		Crystalline silicon terrestrial photovoltaic modules – Design qualification and type approval.
2	IEC: 61730 – Part 1	Photovoltaic (PV) module safety qualification – Requirements for construction.
3	IEC: 61730 – Part 2	Photovoltaic (PV) module safety qualification – Requirements for testing.
4	IEC: 61701/IS: 61701	Salt Mist Corrosion Testing of the module.
5	IEC: 62804	Test method for detection of Potential Induced Degradation of photovoltaic (PV) modules.

The proposed PV Module must be enlisted in the latest ALMM (Approved List of Module Manufacturer) of MNRE, Government of India.

The PV Module shall have the Test Certificate issued from accredited test laboratories of Ministry of New and Renewable Energy, Government of India.

The manufacturers should get their samples tested as per the new format/ procedure/Official memorandum i.e. latest ALMM list from MNRE, Govt. of India shall be followed.

5.12.2.2 Technical Requirements

SI. No.	Item	Description
1	**Type**	**Crystalline Silicon – Mono Park**
2	Efficiency of module	Minimum 19 % at STC
3	**Cell efficiency** (Type of Cell : Only Class A)	Minimum 20 % at STC
4	**Fill Factor**	**Minimum 78 %** at STC
5	No. of cells per module	at least 72
6	Module Frame	Non-corrosive and electrolytically compatible with the structural material, preferably anodized Aluminium (not less than 15 micron).
7	Termination box	Thermo-plastic, IP 67, UV resistant
8	Blocking diodes	Schottky type
9	Bypass Diode	Yes, as required
10	**Power Rating**	**The nominal power of a single PV module shall be minimum 500 Wp**
11	Power tolerance	upto +5 %
12	Temperature co-efficient of power	Less than - 0.40% / °C
13	Glass	High transmittance glass with Anti Reflective Coating (ARC)
14	RF Identification tag for each solar module	Shall be provided inside or outside the module and must be able to withstand environmental conditions and last the lifetime of the solar module as per MNRE Norms.

- Modules should be Mono Park crystalline type having capacity of **minimum 500 Wp.** Higher capacity Solar PV modules will be preferred.

- **The PV Module must be enlisted in the latest ALMM (Approved List of Module Manufacturer) of MNRE at the time of delivery.**

- All the modules in the PV plant should be arranged in a way so as to minimize the mismatch losses.

- Module shall be uniformly laminated without any lamination defects.

- The module frame shall be made of aluminium or corrosion resistant material (minimum thickness 15 micron), which shall be electrically compatible with the structural material used for mounting the modules. Grounding / Earthing provision shall be provided.

- Solar module shall be laminated using lamination technology using established polymer (EVA).

- The back sheet used in the crystalline silicon based modules shall be of 3 layered structures. Outer layer of fluoropolymer, middle layer of Polyester (PET) based and Inner layer of fluoropolymer or UV resistant polymer. Back sheet with additional layer of Aluminium also will be considered. The thickness of back sheet should be of minimum 300 microns with water vapour transmission rate less than $3g/m^2/day$. The Back sheet shall have voltage tolerance of more than 1000 V.

- The EVA used for the modules should be of UV resistant in nature. No yellowing of the back sheet with prolonged exposure shall occur.

- The sealant used for edge sealing of PV modules shall have excellent moisture ingress protection with good electrical insulation (Break down voltage >15 kV/mm) and with good adhesion strength.

- The solar modules shall have suitable encapsulation and sealing arrangements to protect the silicon cells from the environment. The arrangement and the material of encapsulation shall be compatible with the thermal expansion properties of the Silicon cells and the module framing arrangement/material. The encapsulation arrangement shall ensure complete moisture proofing during the whole life of the solar modules.

- The Module shall be made of high transmittance glass front surface giving high encapsulation gain. The glass used to make the crystalline silicon modules shall be toughened low iron glass with minimum thickness of **4 mm**. The solar cell shall have surface anti-reflective coating to help to absorb more sunlight in all weather conditions. The glass used shall have transmittance of above 90% and with bending less than 0.3% to meet the specifications.
- Module rating is considered under standard test conditions, however Solar Modules shall be designed to operate and perform as per installation site condition.
- The peak-power point voltage and the peak-power point current of any supplied module and/or any module string (series connected modules) shall not vary by more than 2 % (two percent) from the respective arithmetic means for all modules and/or for all module strings, as the case may be.
- All materials used shall be having a proven history of reliable, light weight and stable operation in external outdoor applications and **shall have service life of 25 years.**
- **The modules should be 100% PID (Potential Induced Degradation) tolerant** and should comply with IEC 62804.
- Solar PV Module design shall conform to following requirement:
    - Weather proof DC rated MC connector and a lead cable coming out as a part of the module, making connections easier and secure, not allowing for any loose connections.
    - Resistant of water, abrasion, hail impact, humidity & other environment factor for the worst situation at site.
    - The PV Junction Box shall confirm IP 65 and shall have sufficient bypass diodes to avoid shadowing effects.
- Modules shall perform satisfactorily in relative humidity up to 95% and temperature between -10^{0}C and 85^{0}C (module temperature).

- The developer shall arrange for the details of the materials along with specifications sheets of from the manufacturers of the various components used in solar modules along with those used in the modules sent for certification. The Bill of Material (BOM) used for modules shall not differ in any case from the ones submitted for certification of modules.
- The I-V characteristics of all modules as per specifications to be used in the systems are required to be submitted at the time of supply.
- SPV module shall have module safety class as per MNRE standard and should be highly reliable, light weight and must have a service life of more than 25 years.

5.12.2.3 **IDENTIFICATION AND TRACEABILITY**

Each PV module must use a RF identification tag (RFID), which must contain the following information:

i. Name of the manufacturer of PV Module
ii. Name of the Manufacturer of Solar cells
iii. Type of Cell (Mono)
iv. Month and year of the manufacture (separately for solar cells and module)
v. DCR (separately for solar cells and module)
vi. I-V curve for the module
vii. Peak Wattage, I_m, V_m and Fill Factor for the module
viii. Unique Serial No and Model No of the module
ix. Date and year of obtaining DCR PV module qualification certificate
x. Name of the test lab issuing DCR certificate
xi. Other relevant information on traceability of solar cells and module as per ISO 9001 series.

RFID for each solar module shall be provided inside of the module and must be able to withstand environmental conditions and last the lifetime of the solar module as per latest MNRE norms.

5.12.2.4 AUTHORIZED TESTING LABORATORIES/ CENTERS

PV modules must qualify (test reports/ certificate from IEC/NABL accredited laboratory should be enclosed) as per the relevant IEC standard. Additionally the performance of PV modules at STC conditions must be tested and approved by one of the IEC / NABL Accredited Testing Laboratories including National Institute of Solar Energy (NISE).

5.12.2.5 PERFORMANCE WARRANTY

A. **Material Warranty:** The manufacturer should warrant the Solar Module(s) to be free from the defects and/or failures specified below for a period not less than five (05) years from the date of full commissioning of each phase i,e, Phase I and Phase II separately to the original customer (i.e. EPC Contractor).

   i. Defects and/or failures due to manufacturing
   ii. Defects and/or failures due to quality of materials
   iii. Non conformity to specifications due to faulty manufacturing and/or inspection processes.

   Bidder will replace the solar module(s)with new one of same approved parameters. The contractor shall be responsible to contact with the module manufacturer if any of the above mentioned cases occurred within defect liability period.

B. **Performance Warranty:** The manufacturer should warrant the output of Solar Module(s) for at least 90% of its rated power upto initial 10 years & 80% of its rated power upto 25 years from the completion of trial run at site/date of final commissioning. The contractor shall collect the Warranty Certificate for performance of the modules from the manufacturer and submit the same to DPL prior to delivery of the products to the respective sites.

   If, Module(s) fail(s) to exhibit such power output in prescribed time span, the Contractor will bound to either deliver additional PV Module(s) to replace the missing power output with no change in area of site used or replace the PV Module(s) with no extra cost claimed at Purchaser 's sole option.

Manufacturer of proposed PV modules must have the ISO 9001:2008 or ISO 14001 Certification for their manufacturing unit for their said manufacturing item.

Note: Only indigenously manufactured PV modules & cell (i.e. Domestic Content Requirement of manufacturer of module & cell) should be used in Grid Connected Ground Mounted Solar PV Power Plants under this scheme. However, other imported components can be used, subject to adequate disclosure and compliance to specified quality norms and standards and approval of the Purchaser**.**

5.12.2.6 **Approval**

2.6.1 The Contractor shall provide commercial datasheet and Guaranteed Technical Particular datasheet

2.6.2 The Contractor shall provide the Bill of Materials (BOM) of the module that is submitted for approval along with the datasheets of each component. The component datasheet shall contain all the information to substantiate the compliance for component specifications mentioned above. The Contractor shall also provide complete test reports and certifications for the module proposed as per above. The BOM proposed shall be the subset of Constructional Data Form (CDF)'s of all the test reports.

5.12.2.6.3 The Contractor shall obtain the approval of the proposed module make & model prior to manufacturing/ inspection call.

5.12.2.7 APPROVAL, TESTING & INSPECTION

- The Detailed Design Report Submitted by the contractor to DPL must contain but not limited to the following details of the solar modules:
  - Detailed specification
  - Necessary Drawings
  - Type Test Report and Necessary MNRE Certificates (for DCR) etc.

PRE-DISPATCH INSPECTION PROCEDURE

5.12.2.7.1. OBJECTIVE:

The objective of this document is to establish General inspection protocol with objectivity for verification of Quality Parameters of Solar Modules by the customer (or its authorized inspection agency) prior to dispatch. The decision rules and procedure specified herein seek to uphold quality standards based on industry best practices and technical specifications laid out in tender documents as well as to control risks associated with item procurement.

5.12.2.7.2. STANDARDS AND CODES:

1. Sampling for determining Acceptance Quality Level (AQL) shall follow ISO 2859-1: 1999.

2. IEC TS 60904-1-2:2019 - Photovoltaic devices - Part 1-2: Measurement of current voltage characteristics of bifacial photovoltaic (PV) devices

5.12.2.7.3. DEFINITIONS:

1. Lot: All products/items manufactured in one batch.

Notwithstanding the aforementioned definition, the customer or authorized inspection agency can lay down alternate/additional criteria for determining a lot at appropriate time.

2. Major Defect: A defect that reduces the usability or causes the product to fail to fulfil its nominal characteristic function.

3. Minor Defect: A defect that does not reduce the usability of the product, but does not meet the quality standard.

5.12.2.7.4. INSPECTION SCHEDULE:

Customer representative shall propose the schedule for Pre-despatch Inspection of Finished Goods to the Customer well in advance, and in no case less than 3 working days prior to commencement of Inspection at a location within India and 7 days in case of a foreign country.

5.12.2.7.5. SCOPE OF INSPECTION:

Supplier representative will accompany the Inspector while doing the inspection which shall typically consist of 2 steps for clearance of each Lot: BOM verification: To be conducted prior to the commencement of production.

The details of materials used will be verified from the ERP/Manufacturing data and corroborated with the Construction Data Form (CDF). This shall include verification of following:

Item	Method of Verification
Shelf life of the following BOM items: • EVA • PV Module Back sheet • Sealant and potting material (Silicone)	Verify the expiry date/shelf life and storage conditions *The PV Module manufacturer shall submit all required information to prove that materials being used are within their shelf life.*

Note : Contractor shall provide the necessary documents for approval of BOM as per IEC standards and tender Technical Specifications.

Witness Tests:

Manufacturer shall assist the Inspecting agency to witness following checks, the details of which are provided elsewhere in this document:

I. Flash test- As per sampling Plan

For Bifacial Modules, Measurement of current-voltage characteristics shall be done as per IEC TS 60904-1-2:2019 - Photovoltaic devices - Part 1-2

II. Visual Inspection- As per sampling Plan

III. EL Inspection-As per Sampling Plan

IV. Electrical Characteristics (Other than Flash Test)- As per Sampling Plan

Note: The Supplier shall furnish soft and hard copy of the Production Quality Plan prior to commencement of the Inspection.

5.12.2.7.6. SAMPLING PROCESS:

a. Supplier shall provide the list of modules in a lot ready for despatch, along with flash test data (Measured Electrical Data, Pmax) prior to commencement of Inspection tests.

Note: Smallest lot size for Inspection: 20% of the capacity as per the PO.

b. Supplier will arrange to move the PV Modules from FG to Inspection area.

c. Same samples shall be used for all Witness Tests stated at 5.2 above.

d. Inspector shall commence Inspection process by randomly selecting samples from the list of serial nos. (pallet-wise) provided by Supplier as per ISO 2859: Single Sampling Plan for Normal Inspection, General Inspection plan level-I. However, the Inspector shall reserve the right to switch to tightened or reduced level of Inspection as per the lot quality.

5.12.2.7.7. DECISION RULES FOR ACCEPTANCE/REJECTION

Following is a summary of Decision Rules for Acceptance/Rejection of a given Sample in a lot offered for Inspection:

Table 1: AQL Levels

Defect Type	**AQL (%)**
Major (Ma)	2.5
Minor (Mi)	4

Table 2: Inspection Levels

Inspection steps	**Inspection item**	**Inspection level**
1	Flash Test	General inspection level I
2	Visual	General inspection level I
3	EL	General inspection level I

4	EC (Other than Flash Test)	10 Nos. per lot

5.12.2.7.8. INSPECTION PROCESS

a. Electrical Inspection – Flash Tests

For Electrical inspection following preparation will be done:

• Module Temp Stabilization: Modules will be kept in controlled environmental condition till it reaches 25 ±2ºC

• Calibration of Sun-simulator: Sun-simulator will be calibrated as per Calibration Reference. Reference should calibrated against Calibration Reference tested from reputed testing lab TUV / Fraunhofer etc. Testing of modules will be done at STC condition, AM=1.5

Note:

(i) All modules selected for sampling inspection will be re-tested in the sunsimulator.

A Pmax retest (repeatability test) variation of ± 2 % on actual flash

Pmax value will be acceptable. (ii) The Supplier shall provide a valid calibration certificate of the apparatus used.

b. Visual Inspection:

• Customer representative will verify the module visual characteristics as per the Visual Acceptance norms.

• The Visual Inspection shall be carried out in a well-lit room. It shall be the responsibility of the Supplier to ensure adequate brightness in the room.

c. Electroluminescence (EL) Inspection:

• The EL image shall have sufficient resolution for analysis of defects.

• Hi-pot test shall be done as per IEC procedure. The Supplier shall provide a valid calibration certificate of the apparatus used.

5.12.2.7.9. RE-INSPECTION AND REVIEW

In case of minor non-conformities like cleaning issues, label mismatch, etc. which can be easily reworked, Supplier shall rework/replace the modules and offer them for re-inspection to Inspector.

5.12.2.7.10. INSPECTION SUMMARY:

Once the inspection is completed Customer Representative will compile his Inspection Summary Report and share with Supplier and give necessary recommendation on despatch depending upon the audit findings based on the observations made. This report shall be provided within same day of inspection (Format Attached).

5.12.2.7.11. DISCLAIMER:

Inspection by DPL does not absolve the responsibility of the Supplier/vendor to ensure quality during production of the material and its transport to site. Any damages during transport/ handling shall be replaced before erection at site as directed by Engineer-in-charge without any extra cost to the purchaser.

5.12.2.8 ACCEPTANCE CRITERIA:

- 25 no modules to be carried out all the type tests as per IS from NISE, Delhi or MNRE accredited test centers.
- For the above (i.e.25 no. modules) special packing if required may be done to shift the panels from manufactory site to testing lab as directed by DPL.
- Sufficient competent man power to be deployed at test centre for sufficient days to unpack and repack the modules after the test.
- Contractor has to coordinate with DPL for all the testing activity.

5.12.3 DC CABLES

5.12. 3.1. CODE & STANDARDS

	Standards	Description
1.	EN 50618:2014	Cables for PV System, Low smoke halogen- free, flexible, single-core power cables withcross-linked insulation and sheath.
2.	IS: 7098	Cross linked Polyethylene Insulated Thermoplastics Sheathed Cables for voltage 3.3kV to 33kV
3.	EN50395	Voltage Withstand capacity
4.	IEC 60228 Class 5.	conductor flexibility
5.	EN 60216-2	Thermal Endurance
6.	EN 60811-3-1	High Temperature Pressure
7.	EN 60068-2-78	Damp – Heat Resistance Test
8.	EN60811	Tensile Strength
9.	EN60811	Elongation of Insulation
10.	EN 60811-1-3	Shrinkage
11.	EN 60811-2-1	Mineral Oil Resistance
12.	EN 50396 part 8.1.3 Method B	Ozone Resistance
13.	HD 605/A1 or DIN 53367	UV Resistance
14.	IEC 60332-1-2	Flame retardant
15.	ASTM D-2843/ IEC 61034-2/ EN 50268-2	Low smoke emission
16.	IEC 60754-2	Conductor conductivity
17.	IEC 60332-1-2	Fire Performance
18.	IEC 60811-1	Oil & Chemical resistance
19.	IEC 60754-1	Halogen Free

5.12.3.02 The DC Cables in a solar PV plant are used in the following areas

i. Interconnecting SPV modules
ii. From SPV Modules upto String Combiner Box(SCB)
iii. From SCB upto the Inverter.

5.12.3.3. DC CABLES

(Interconnecting SPV MODULES and from SPV Modules to SCB) Each module, provided by Employer, shall have two 4 sq.mm stranded UV resistant cables and terminated with DC plug-in connector directly. The positive (+) terminal shall have a male connector while the negative (-) terminal a female connector. All the modules in the PV plant shall be arranged in a way so as to minimize the mismatch losses. Cables used shall be of min. 1.5 kV (DC) grade. In case bidder offers 1500V DC system, 3.3 kV (E) grade cables shall be provided.

Cables used between SCB and Inverter

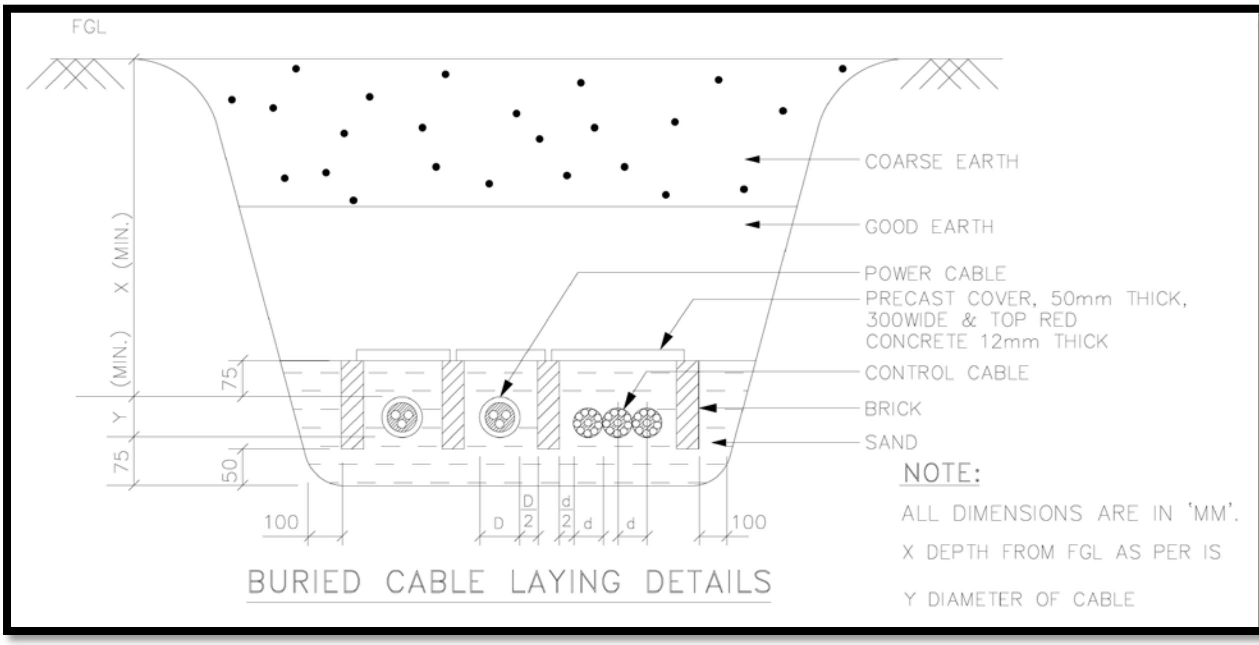
Cables used between SCBs and Inverters shall be of 3.3kV (E) grade. These
Power cables shall have compacted Aluminium/copper conductor, XLPE insulated, PVC inner-sheathed (as applicable), Armoured/ Unarmoured, FRLS PVC outer sheathed conforming to IS: 7098 (Part-II). These cables shallconfirm to the requirements of the standards & codes specified in the relevant chapter.
DC cable voltage drop criteria: From Module to Inverter Transformer shallbe limited to 3% of rated voltage. These cables shall also meet the fire resistance requirement as per the abovestandard and shall be electron beam cured.
All cables except module cable used for (+) ve and (–)ve shall have distinct color identification.
In addition to manufacturer's identification on cables as per EN50618,
following marking shall also be provided over outer sheath.
i. Cable size and voltage grade
ii. Word 'FRNC' at every 5 metre
iii. Sequential marking of length of the cable in metres at every one
metre

5.12.3.4 CABLE LAYING METHOD

i. Cable shall be laid without intermediate joints throughout the lengths from one termination to the other.
ii. Cables shall be handled carefully during installation so that no any damage
to outer sheath, inner insulation or conductor should take place Cables.
Protective pipes shall be used as and when required as directed.
iii. Before laying, Insulation resistance value of all the cables shall be measured with the help of 2.5KV Megger for phase to phase and from phaseto ground (Armour) with

2.5 kV Megger in presence of Engineer in charge and record shall be maintained and same shall be measured after layingalso and has to be compared with the original measured values so that defective cables can be shorted out before laying or after laying and suchcable shall be rejected.

iv. The cable shall be laid in an underground trench. The cable trench shall be prepared by the prospective consumer with following drawing.



5.12.4. String Monitoring Box/Array Junction Box (SMB)

5.12.4.1. SCOPE

This section covers activities related to design, manufacturing, testing at works, supply, insurance, transportation and delivery at Project site, storage, erection, testing, commissioning of array junction box as detailed hereunder.

a. Adequate number of String Monitoring Boxes shall be provided for termination of array string with inverter.

b. The number and specification of PV String Monitoring Box will be as per plant configuration.

The String Monitoring Boxes shall be suitable for interfacing with SCADA system and all necessary transducers shall be included in the scope of supply.

5.12.4.2. STANDARDS

The String Monitoring Boxes shall conform to the latest edition of following Standards except where specified otherwise in this specification:

Sl. No.	Standards	Description
1	IS 13703: Part 1	Low voltage fuses for voltage not exceeding 1000V AC or 1500V DC: General Requirements
2	IEC 60269: Part 4 /IS 13703: Part 4	Low-voltage fuses: Supplementary requirements for fuse-links for the protection of semiconductor devices
3	IEC 60269-4: Part 6	Low-voltage fuses: Supplementary requirements for fuse-links for the protection of solar photovoltaic energy systems
4	VDE 0636	Low-voltage fuses

Equipment meeting with other authoritative standards which ensure an equal or better quality is also acceptable. Where the equipment conforms to any other authoritative standard, the salient points of difference between the standard adopted and IS/IEC shall be clearly brought out in the tender. Complete set of documents and standards in English shall be supplied by the bidder without any extra charge. It shall, however, be ensured that equipment offered comply with one consistent set of standards except in so far as they are modified by the requirement of this specification.

5.12.4.3 TECHNICAL REQUIREMENTS

- The junction Boxes shall have suitable arrangement for the followings (typical):
    - Combine groups of modules into independent charging sub-arrays that will be wired into the controller.
    - Provide arrangement for disconnection for each of the groups.
    - Provide a test point for each sub-group for quick fault location finding.
    - To provide group array isolation.

- The string monitoring box shall be dust, vermin, and waterproof and made of Polycarbonate Plastic.
- The string monitoring box shall be of IP 65 or better.
- The terminal will be connected to bus-bar arrangement of proper size. The junction boxes shall have suitable cable entry points fitted with cable glands of appropriate sizes for both incoming and outgoing cables.
- Suitable markings shall be provided on the bus-bars for easy identification and Cables shall be fitted at the cable termination points as per appropriate polarity.
- Each String shall be terminated through Fuses of required current rating
- The string monitoring box shall be provided with suitable Surge Protection Device (SPD).

5.12.4.4 APPROVAL

The Detailed Design Report Submitted by the contractor to DPL must contain but not limited to the following details of the String Monitoring Boxes:

- Detailed specification
- Necessary drawings etc.

Prior to the delivery of the product, the contractor shall submit but not limited to the following documents:

- Guarantees
- Instructions for installation and operation, manuals
- Necessary test certificates

The contractor shall deliver the product to the site only after receiving such approval against their prayer in writing from DPL.

5.12.5. SOLAR INVERTERS/POWER CONDITIONING UNIT(PCU)

5.12.5.1 SCOPE

This section covers the activities related to design, manufacturing, testing at works, supply, insurance, transportation and delivery at project site, storage, erection, testing, commissioning of **Solar Inverters/Power Conditioning Unit (PCU)** as detailed hereunder.

a. Adequate number of Indoor Solar Central Inverter of **minimum capacity 500 kW** having high quality, high efficiency and reliable operation. Total inverter capacity of the plant should not be less than 8000 kW (AC).

b. **The scope of supply shall also include necessary spares**, if any, required for normal or any breakdown maintenance for at least 05 (five) years and special tools & plants required for erection & maintenance. Corresponding parts of all the equipment & spares shall be of the same specification & workmanship and shall be interchangeable.

All the material & workmanship shall be of reputed make as have proven successful in their respective uses in similar services & under similar condition.

The solar inverter/power conditioning unit shall be suitable for **interfacing with SCADA system** and all necessary transducers shall be included under the scope of supply.

5.12.5.2 STANDARDS

The equipment and materials covered by this specification shall conform to the latest edition of following Indian Standards or equivalent IEC standards except where specified otherwise in this specification:

Sl. No.	Standards	Description
1	IEC/IS: 61683	Photovoltaic systems – Power Conditioners – Procedure for measuring efficiency
2	IEC 62093	Balance-of-system components for photovoltaic systems – Design qualification natural environments
3	IEC 60068-2-1:2007	IEC 60068-2-1:2007 Environmental testing - Part 2-1: Tests - Test A: Cold
4	IEC 60068-2-2:2007	Environmental testing - Part 2-2: Tests - Test B: Dry heat
5	IEC 60068-2-14:2009	Environmental testing - Part 2-14: Tests - Test N: Change of temperature

6	IEC 60068-2-30:2005	Environmental testing - Part 2-30: Tests - Test Db: Damp heat, cyclic (12 h + 12 h cycle)
7	IEC 62116 / IEEE 1547/UL 1741/ equivalent IS standard	Islanding Prevention Measurement
8	IEC 61727 Relevant CEA/ CERC regulation and grid code (amended up to date)	Interfacing with utility grid
9	IEC 61000-6-2 Ed. 2	Electromagnetic compatibility (EMC) - Part 6-2: Generic standards - Immunity standard for industrial environments
10	IEC 61000-6-4 Ed. 2.1	Electromagnetic compatibility (EMC) - Part 6-4: Generic standards - Emission standard for industrial environments
11	IEC 62109 (1&2), EN 50178 or equivalent	Safety of power converters for use in photovoltaic power systems - Part 1: General requirements
12	IEC 62109 (2), EN 50178 or equivalent	Safety of power converters for use in photovoltaic power systems - Part 2: Particular requirements for inverters
13	IEC 62093 or equivalent	Reliability test standard
CEA Technical Standards for Connectivity to the Grid Regulations 2007 with 2013 Amendment		
As per the Solar Photovoltaics, Systems, Devices and Components Goods (Requirements for Compulsory Registration) Order, 2017, Inverters used in the grid connected solar power projects shall be registered with BIS and bear the Standard Mark as notified by the Bureau of Indian Standards.		

Solar Inverters should have certificate and approval from VDE, IEC etc. The inverters should have CE conformity according to LVD (Low Voltage Directive) and EMC (Electro Magnetic Compatibility) Directive for safety purpose.

Type test certificate issuing authorities should be any NABL/IEC Accredited Testing Laboratories or MNRE approved test centers.

Equipment meeting with other authoritative standards which ensure an equal or better quality is also acceptable. Where the equipment conforms to any other authoritative standard, the salient points of difference between the standard adopted and IS/IEC shall be clearly brought out by the contractor.

5.12.5.3 TECHNICAL REQUIREMENTS

- The inverter should be 3-Φ static solid state type power conditioning unit.
- Inverter/PCU shall be centralized grid tied in nature, shall consist of MPPT controller. Inverter shall be selected based on array design. Associated control and protection devices shall be an integrated part of the PCU.
- Degree of protection of the Outdoor Inverters shall confirm at least IP-65. For PEB it shall be IP-55.
- The inverters shall be built in with data logging system for remote monitoring of the plant performance through external PC. (PC shall be provided as a part of the Solar PV Plant).
- The dimension, weight, foundation details etc. of the PCU shall be clearly indicated in the technical specification to be submitted with the detailed design report.
- The PCU shall be capable of complete automatic operation, including wake-up, synchronization & shut down independently & automatically. Inverters / PCU shall operate in sleeping mode when there will no power connected.
- The Inverter shall have internal protection arrangement against any sustained fault in output line and lightning in the grid. AC protection boxes shall be provided at the inverter output which shall include over current, under voltage protection etc.
- Both AC & DC lines shall have suitable fuses & surge arrestors and contactors to allow safe start up and shut down of the system (Type-I & II DC side and Type-II AC side).

- PCU shall be capable to synchronize independently & automatically with grid power line frequency to attain synchronization and export power generated by solar plant to grid.
- In case of grid failure, the PCU shall be re-synchronized with grid after revival of power supply. Bidder to furnish the time taken by PCU to be re-synchronized after restoration of grid supply during detailed engineering.
- Inverters shall have the features like Low Voltage Ride through (LVRT), High Voltage Ride through (HVRT) etc. for grid support and connection.
- Inverters should run in synchronized manner, effect of one inverter should not be reflected to the others. The PCU shall be capable of operating in parallel with the grid utility service and shall be capable of interrupting fault line currents, line to ground fault currents and short circuit currents.
- The PCU shall be able to withstand an unbalanced load conforming to related IEC standard (+/- 5% voltage). The PCU shall include appropriate self-protective and self-diagnostic features to protect itself and the PV array from damage in the event of PCU component failure or from parameters – beyond the PCU's safe operating range due to internal or external causes. The self-protective features shall not allow signals from the PCU front panel to cause the PCU to be operated in a manner which may be unsafe or damaging. Faults due to malfunctioning within the PCU, including commutation feature, shall be cleared by the PCU protective devices and not by the existing site utility grid service circuit breaker.
- Operation outside the limits of power quality as described in the technical data sheet should cause the power conditioner to disconnect the grid. Additional parameters requiring automatic disconnection are over current, earth fault, short circuit and reverse power.
- The inverter itself shall consist of one circuit breaker for isolation from the circuit during any fault or maintenance purpose.
- PCU shall have active power limit control, reactive power and power factor control feature. Plant operator shall be able to provide (manual intervention) Active power, reactive power and power factor control/limit

set point through SCADA HMI and local control display unit (or Laptop computer). PCU shall be provided with remote start and stop facility from SCADA HMI. All required hardware and software required for this purpose shall be provided by Bidder.

- PCU shall have necessary limiters in build in the controller so as to ensure safe operation of the PCU within the designed operational parameters.
- The Contractor shall ensure by carrying out all necessary studies that the PCU will not excite any resonant conditions in the system that may result in the islanded operation of PV plant and loss of generation. In case there is excitation of any resonant condition in the system during PV plant operation that may result in the islanding/tripping of the PV plant and affect the power transfer, it shall be the responsibility of contractor to rectify the design and carryout required modification in the equipment of his supply.
- PCU shall be provided with Mobile user interface facility for monitoring of inverter by plant O&M personal for better O&M and highest yield from the PV plant. In case PCU does not have this facility, then Bidder can provide the same facility through plant SCADA system.
- PCU shall have AC and DC side monitoring capability and reporting to SCADA system (measured analog and digital value measured within PCU). Any special software if required for this purposes shall be provided for local and remote monitoring and report generation.
- All-important alarm and trip signals shall be configured in the PCU and their corresponding modbus address shall be provided for SCADA configuration. Signal shall necessarily be included such as LVRT/HVRT in action & trip operated, islanding protection operated, over current operated, Inverter cabinet temperature high alarm and all other important signals.
- DC Overloading:- Maximum PCU DC overload loading shall be limited to its design PV Array Power to PCU nominal AC power ratio. Bidder needs to submit all the relevant technical document/test report from PCU manufacturer (OEM) during details engineering stage in support of declared PCU design DC overloading capacity.

- The inverters shall operate satisfactorily within the operating ambient temperature range of -15°C to +60°C. The contractor shall assure that the inverter should not de-rate upto 50°C ambient temperature. Humidity: 95 % non-condensing.
- To take care of PID (Potential Induced Degradation), the inverters should have active negative grounding kit.
- **EARTHING OF INVERTERS-** The PCU shall be earthed as per manufacturer recommendation (OEM). The Bidder needs to submit the details earthing arrangement of PCU and system earth pit requirement during detail engineering stage. The detail specification for panel earthing for safety has been mentioned elsewhere in this specification

5.12.5.3.1 OPERATING MODES OF PCU

- **Low Power Mode:-** The PCU shall be able to wake-up automatically when PV array open circuit voltage value is equal/more than preset value in the PCU program. Once its start generation the PCU shall automatically enter maximum power mode.
- **Maximum Power Point Tracking (MPPT):-** In order to maximized the energy collection from solar PV array, the PCU shall have inbuilt maximum power point tracker (MPPT) controller and MPPT shall be able operate the PV array at its maximum power point by adjusting output voltage of PV array system according to atmospheric condition. PCU MPPT controller shall ensure that it operate the PV array system at its global maximum power point and it shall not trapped into PV array local maximum power point during cloudy atmospheric condition.
- **Sleep Mode :-** PCU shall automatically go into sleep mode when the output voltage of PV array and/or output power of the inverter falls below a specified limit. During sleep mode the inverter shall disconnect from grid. Inverter shall continuously monitor the output of the PV array and automatically start when the DC voltage rises above a pre-defined level.

- **Standby Mode:-** In standby mode the PCU DC & AC contactor are open, inverter is powered on condition and waiting for start command.
- String Monitoring facility:- PCU shall be provided with current monitoring transducer at incoming DC cables from each string combiner box (SCB) for PV array zone monitoring purpose. The current transducers used for this purpose shall have accuracy of 1.0 class or better.
- The PCU should be designed for parallel operation through galvanic isolation. Solid state electronic devices shall be protected to ensure smooth functioning as well as ensure long life of the inverter. Parallel operated PCU system are also accepted subjected to recommendation of PCU manufacturer. In such case, PCU design shall also ensure that no abnormal interaction shall take place among the PCU unit during any grid operating condition which may result in outages.
- Bidder may offer liquid cooling system subject to DPL approval. In case Liquid cooled inverters are offered, Bidder to ensure that coolant is used in closed cycle. Complete inverter along with cooling system shall be of proven design.
- The Inverter shall have suitable arrangement for negative grounding of solar PV array system and the ground current shall be limited to safe limit. Ground current shall be measured continuously and alarm shall be generated in case ground current reaches to predefined set value. Inverter shall trip in case ground current more than safe operating limit.
- Inverter shall have emergency stop push button for tripping of inverter with complete DC & AC electric isolation.
- Following protections shall be provided with the inverter.
    - Over voltage and under voltage both at input & output
    - Over current both at input & output
    - Over/under grid frequency
    - Heat sink over temperature
    - Cooling system failure protection
    - PV array ground fault monitoring & detection
    - PV array insulation monitoring

- LVRT protection
- Anti-islanding protection
- Over temperature protection.
- Short circuit
- Protection against lightning
- Surge arrestors to protect against Surge voltage induced at output due to external source
- Direct earth fault protection and body earthing
- Set point pre-selection for VAR control
- Synchronization loss protection
- Grid monitoring
- Any other protections required

- Inverters should have user friendly LED/LCD or touch display for programming and view on line parameters in LED/LCD as well as in SCADA. These parameters are:
  - Inverter per phase Voltage, current, kW, kVA, frequency and power factor.
  - 15 minute, Daily, monthly & Annual energy generated by the solar system(kWh)
  - Solar system temperature
  - Ambient temperature
  - Grid Voltage, frequency and power factor
  - AC and DC side voltage and current
  - Power factor on AC side
  - DC injection into the grid
  - Inverter Import export kWh summation
  - Solar kWh summation
  - Inverter ON/OFF
  - Grid ON/OFF
  - Inverter under voltage/over voltage
  - Inverter over load
  - Inverter over temperature etc.
  - Total Current Harmonics distortion in the AC side

- Total Voltage Harmonic distortion in AC side
- Efficiency of the inverter
- Solar system efficiency
- Display of I-V curve of the solar system
- Fault details with time when occur.
- Any other parameter considered necessary by supplier of the solar PV system based on prudent practice.

5.12.5.4 SPECIFICATION

SI. No.	Operating Parameter	Desired specification
1	Input (DC)	
	PV array connectivity capacity	As per site requirement
	MPPT Voltage range	Compatible with the array voltage
	Number of MPPT Channel	Number of MPPT channel shall be minimum one. One spare MPPT channel shall be provided.
2	Output (AC)	
	Nominal AC Power output	500 kW (minimum)
	Number of Grid Phase	3
	Adjustable AC voltage range	+/- 10%
	Nominal output voltage frequency	50Hz
	Continuous operating frequency range	47.5 Hz to 52 Hz
	Frequency range	+/- 5%
	Operating power factor range	Operating power factor (adjustable) shall be 0.8 Lead to 0.8 Lag.
	AC wave form	Sine wave
	THD	Less than 3%
	Switching	H.F. transformer/transformer less
3	General Electrical data	
	Efficiency	97.5 % (minimum)
	VAR Control	The PCU shall be capable of supplying reactive power as per grid requirement. PCU shall have night SVG function (Q at night).

SI. No.	Operating Parameter	Desired specification
	No load loss	< 1% of rated power
	Maximum loss in sleep mode	< 0.05% of rated power
4	Protection	
	DC Side	As mentioned in the Technical Requirement
	AC side	As mentioned in the Technical Requirement
	Isolation Switch	PV array Isolation switch (DC)
	Ground fault detection device (RCD)	To be provided
5	Display	
	Display type	LED/LCD or touch display
	Display parameter	
	DC	As mentioned in the Technical Requirement
	On grid connected mode	As mentioned in the Technical Requirement
9	Interface (Communication protocol)	Suitable port must be provided in the inverter for i. On site upgrade of Software ii. On site dumping data from the memory iii. Plant based remote monitoring system
10	Storage of Data	At least for 1 year. Separate data logger may be provided to meet the criteria.
11	Monitoring	Matched with the monitoring and data logging system (SCADA)
12	Mechanical Data	
	Protection Class	As mentioned in the Technical Requirement
	DC Switch	Integrated
	Operating ambient temperature	-15° C to 60° C
	Relative Humidity	15 to 95 %
	Noise Emission	Less than 75 dB (A) @ 1 meter (indoor)
	Cooling	Forced cooling

5.12.5.5 Communication interface

The project envisages a communication interface which shall be able to support:

- Real time data logging

- Event logging
- Supervisory control
- Operational modes
- Set point editing

5.12.5.6 COMMUNICATION SYSTEM

Communication System shall be an integral part of inverter. All current values, previous values up to 40 days and the average values of major parameters shall be available on the digital bus.

5.12.5.7 DATA LOGGER SYSTEM

Data logger system (Hard ware) and the software for study of effect of various environmental & grid parameters on energy generated by the solar system and various analyses would be required to be provided.

The communication interface shall be suitable to be connected to local computer and also remotely via the Web using either a standard modem or a GSM / WIFI modem.

PCU shall have suitable communication card (Modbus TCP/IP) for networking and SCADA integration and same shall support dual master communication. PCU shall include all important measured & internal calculated analog values and alarm & trip signals for remote monitoring, storing and report generation purpose in SCADA system. Details list of above such parameters shall be provided along with their Modbus address during detail engineering stage

5.12.5.8 APPROVAL

The Detailed Design Report Submitted by the contractor to DPL must contain but not limited to the following details of the Solar Inverter/Power conditioning Unit:

- Detailed technical description of the complete unit
- Necessary Drawings
- Type Test Report etc.

Joint inspections and testing will be done by DPL and the authorized representatives of the contractor at the manufacturer's workshop on regular basis for quality assurance and testing. Acceptance Tests as per relevant Standard shall be carried out at the module manufacturer's workshop. Following tests shall be carried out on certain number of Inverters from a lot (decided by DPL) as acceptance tests of Solar Inverters:

- Visual Inspection
- Performance Test and measurement of AC & DC parameters
- DC reverse polarity protection
- Islanding Protection
- Over Voltage & Under Voltage withstand
- Over Frequency & Under Frequency withstand
- Night consumption
- Any other test as desired by DPL

Arrangements for the aforesaid testing and inspection at manufacturer's end are to be provided by the contractor.

Prior to the delivery of the product, the contractor shall submit but not limited to the following documents:

- Guarantees
- Instructions for installation and operation, manual
- Safety precautions
- Test reports for routine tests and acceptance tests etc.

The contractor can deliver the product to the site only after receiving such approval against their prayer in writing from DPL.

5.12.5.9 Acceptance:

Type test:

During detailed engineering, the contractor shall submit all the type test reports including temperature rise test and surge withstand test carried out within last five years from the date of LOA of this project for Owner's approval. These reports should be for the test conducted on the equipment similar to those proposed to be supplied under this contract and the test(s) should have been either conducted at an independent laboratory or should have been witnessed by a client.

However if the contractor is not able to submit report of the type test(s) conducted within last ten years from the date of techno-commercial bid opening, or in the case of type test report(s) are not found to be meeting the specification requirements, the contractor shall conduct all such tests under this contract at no additional cost to the owner either at third party lab or in presence of client/owners representative and submit the reports for approval.

Routine tests:

Routine tests to be conducted as per IEC norms in the presences of DPL staff along with warranty & guarantees certificates.

5.12.6.0 INVERTER TRANSFORMER

5.12.6.1 SCOPE

This section covers the activities related to design, manufacturing, testing at works, supply, insurance, transportation and delivery at Project site, storage, erection, testing, commissioning of step up transformers and associated equipments as detailed hereunder.

a. Adequate number of 3-Φ, , **DRY** type transformers with suitable capacity shall be provided to step up voltage from 3-Φ, Grid tied Solar Inverter output to **6.6 kV** voltage level for feeding the generated power to the 6.6 kV Bus.

b. The LV winding of the transformer will be connected to the outputs of inverters and the HV sides will be connected to the 6.6 KV Bus/line through VCB, Isolator etc. The Dry type transformer will be placed inside each local control room.

The scope of supply shall also include necessary spares required for normal operation & maintenance of transformers for a period of 5 (five) years & special tools & plants required for erection & maintenance. Corresponding parts of all the equipments & spares shall be of the same material & dimensions, workmanship & finish and shall be interchangeable. All the material & workmanship shall be of suitable commercial quality as have proven successful in their respective uses in similar services & under similar condition.

The transformers and associated equipment shall be suitable for interfacing with SCADA system and all necessary transducers shall be included in the scope of supply.

5.12.6.2 STANDARDS

The equipments and materials covered by this specification shall conform to the latest edition of following Indian Standards or equivalent IEC standards except where specified otherwise in this specification:

Sl. No.	Standards	Description
1	IS: 2026 (Part I to IV), IS:6600, IEC: 60076	Power Transformer
2	IS: 2099, IEC:60137, IS:3347, IS 12676	Transformers bushings
3	IS: 2705, IEC 60185	Current transformers
4	IEC 60296	Transformer oil
5	IS: 3637	Gas and oil operated relay
6	IS: 5120	Fittings and accessories for power transformers
7	IS: 6088	Dimensions for porcelain transformer bushings
8	IS: 3347	Loading guide for oil-immersed transformers
9	CBIP No. 295	CBIP Manual on Transformers Publication

Equipment meeting with other authoritative standards which ensure an equal or better quality is also acceptable. Where the equipment conforms to any other authoritative standard, the salient points of difference between the standard adopted and IS/IEC shall be clearly brought out in the tender. Complete set of documents and standards in English shall be supplied by the contractor without any extra charge. It shall, however, be ensured that equipment offered comply with one consistent set of standards except in so far as they are modified by the requirement of this specification.

5.12.6.3 TECHNICAL REQUIREMENTS

The transformers shall be three winding, Dry type , 3-Φ, Step Up transformers

Sl. No.	Standards	Description
1	Type	Three Winding
2	No. of phases	Three
3	Duty, Service, Application & Installation	Continuous Solar Inverter application and converter duty (Outdoor), Outdoor on rails/channel
4	**Rated continuous MVA at**	**As required according to Solar Inverter**

Sl. No.	Standards	Description
	maximum ambient temperature of 50ºC	**capacity/Transformer designing**
5	% Impedance at 75°C, rated current & frequency	As per system requirement & as per Inverter manufacturer recommendation.
6	Type of cooling	AN/AF
7	Winding material	Copper
8	Connection	
	HV	Delta
	LV	Star-Star
9	Vector group	As per system requirement
10	Voltage	
	HV	6.6 kV
	LV	As per Solar Inverter Output Voltage
11	Rated Frequency	50 Hz
12	Type of Bushing	
	HV Winding	Porcelain/ XLPE bushing
	LV Winding	Porcelain bushing
13	Insulation level (impulse withstand)	
	HV	60 KVp(Peak)
	LV	NA
14	Insulation level (Power freq. withstand)	
	HV	20 kV (rms)
	LV	3 kV (rms)
15	Tapping	OCTC
	Range	+5% to -5% @ 2.5%
16	Temperature rise of oil/ winding over design ambient temperature of 50°C (irrespective of taps)	50ºC / 55ºC
17	Hot spot temperature over a maximum yearly weighted average ambient temperature of 32 °C	105°C
18	Short circuit current for 1 sec. on HV side	25 kA
19	Short circuit withstand time	2 sec
20	Insulation	

Sl. No.	Standards	Description
	HV winding	Class B (Winding insulation shall be able to withstand 6.6 kV continuously)
	LV winding	Class B (Uniformly insulated)
21	Voltage withstand capacity during sudden disconnection of load	1.4 times the rated voltage for 5 sec. 1.25 times the rated voltage for 1 min. 1.1 times the rated voltage for continuous operation.
22	Noise level	< 90 dB As per NEMA TR-1 standard
23	Cooling medium	Mineral oil (as per IS 335)
24	Earthing	LV neutrals solidly earthed through neutral CT, HV side should also be earthed.
25	Minimum efficiency	98%

5.12.6.5 DESIGN CRITERIA

The rating of the Transformers shall be sufficient to evacuate generated power from the Solar Inverter under full load conditions. The Transformers shall be able to evacuate generated power under all conditions of ambient temperature, frequency and voltage variations

The transformers will have Off Circuit Tap Changer (OCTC) with tap ranging +5% to -5% in steps of 2.5 % at HV side. The transformers will operate without injurious heating at the rated capacity at any voltage within +/-10% of the rated voltage of that particular tap. The transformer will be designed to deliver rated MVA continuously even at the lowest tap without exceeding specified temperature rise.

HV line terminals shall be brought out through 6.6 kV class weather proof, shaded porcelain bushing.

Ambient air temperature for the transformer

- Maximum ambient air temperature: 50^0 C
- Maximum daily average ambient air temperature: 40^0 C
- Maximum yearly weighted average ambient air temperature: 32^0 C
- Minimum ambient air temperature: - 5^0 C

The transformers shall be designed to withstand short circuit current of 25 KA for 2 seconds without any damage. This capability shall be demonstrated by type test report.

The transformers will be capable of being loaded in accordance with IS 3347 - loading guide for oil immersed power transformers. The transformers shall also be designed for operation at unbalanced loading conditions.

The transformers shall be suitable for co-ordination and integration with SCADA System and necessary contacts and/or ports for the purpose shall be provided.

Earthing arrangement of the transformers shall be provided as per the relevant Indian Standard.

Necessary protection arrangement should be provided in the transformer.

Construction of different parts of the transformer shall conform to the latest edition of IS 2026.

Fittings and accessories as per relevant Indian Standard shall be provided within the scope of the work.

5.12.6.6 APPROVAL

The Detailed Design Report Submitted by the contractor to DPL must contain but not limited to the following details of the transformers:

- Detailed specification including Fittings and Accessories
- Necessary Drawings shall contain but not limited to the following:

  - Outline dimension drawings of transformers and fittings/accessories
  - Assembly drawings and weight of main components.
  - Transport drawings, showing main dimensions and weight of each package.
  - Foundation details
  - Tap-changing equipment
  - Name-plate diagrams

- Necessary test certificates and type test reports.

A joint inspection and testing will be done by DPL and the authorized representatives of the contractor at the manufacturer’s workshop. Testing and inspection of the transformers will be carried out as per relevant Indian Standard. Arrangements for the aforesaid testing and inspection at manufacturer’s end are to be provided by the contractor.

Prior to the delivery of the product, the contractor shall submit but not limited to the following documents:

- Guarantees
- Instructions for installation and operation, manual
- Safety precautions
- Test reports for routine tests and acceptance tests etc
- Detailed schematics of all power instrumentation and control equipment and subsystems along with their interconnection diagrams. Schematics shall indicate wiring diagrams, their numbers and quantities, type and ratings of all components and subsystems etc.

The contractor can deliver the product to the site only after receiving such approval against their prayer in writing from DPL

5.12.7.0
6.6 kV INDOOR SWITCHGEAR & POWER EVACUATION

5.12.7. 1 SCOPE

This specification covers the design, manufacture, testing at manufacturer's works before dispatch, supply, delivery at site, transit insurance, storage at site, erection, testing & commissioning of 6.6 kV Indoor Switchgear & Power Evacuation System with all necessary equipments and accessories required for satisfactory operation of 5 MWSolar PV Power Plant in DPL, Durgapur, West Bengal. The contractor shall make the arrangement for feeding the generated power to the 6.6 kV Switchyard of the Solar PV Power Plant. **Cabling from site 6.6 KV 5 MWAC switchyard to 6.6 KV 7 MWAC P-I Switchyard building, 2200 m away, along with jobs mentioned elsewhere in the contract in the scope of the bidder.**

The scope of supply shall also include necessary spares required for normal operation & maintenance of switchgear equipments & Power Evacuation System for a period of 5 (five) years & special tools & plants required for erection & maintenance.
Corresponding parts of all the equipments & spares shall be of the same material & dimensions, workmanship & finish and shall be interchangeable. All the material & workmanship shall be of suitable commercial quality as have proven successful in their respective uses in similar services & under similar condition.

5.12.7.2 STANDARDS

The equipments covered under this chapter shall comply with the requirement of latest edition of following IS/BS/IEC specifications as amended up to date except where specified otherwise.

Sl.	Standards	Description
1	IS: 5	Colors for ready mixed paints & enamels
2	IEC-62271-100,200; IEC-600298 / 600694; IS-3427	AC metal enclosed switchgear & control gear for rated voltages above 1 kV & up to & including 52 kV.
3	IS: 13947/ IEC 60529	Degree of protection provided by enclosures for switchgear.
4	IS: 1901	Specification for visual indication lamps
5	IEC-60056 / IS	High Voltage Alternating current Circuit

Sl.	Standards	Description
	13118 /IEC	Breakers
6	IS: 2705 - (Part I - IV)/ IEC 60185	Current Transformers
7	IS: 3156 - (Part I - IV)/ IEC 60186	Voltage Transformers
8	IEC: 60694	Common clauses for high voltage switchgear & control gear
9	IS: 1248	Indicating Electrical measuring instruments
10	IS: 8084	Inter connecting Bus bars for AC voltage between above 1 kV up to and including 36 kV
11	IS-3231 & 3842 / IEC 60255	Electrical relays for Power Systems
12	IEC: 62271-102/ IEC 60129	Alternating current disconnectors and earthing switches
13	IEC-99-4	Metal oxide surge arresters without gates for A.C. systems

Equipment meeting with other authoritative standards which ensure an equal or better quality is also acceptable. Where the equipment conforms to any other authoritative standard, the salient points of difference between the standard adopted and IS/IEC shall be clearly brought out in the tender. Complete set of documents and standards in English shall be supplied by the contractor without any extra charge. It shall, however, be ensured that equipment offered comply with one consistent set of standards except in so far as they are modified by the requirement of this specification.

5.12.7.1SPECIFIC TECHNICAL REQUIREMENTS/PARAMETERS:

All indoor switchgear panels shall have minimum technical parameters for design consideration as mentioned hereunder:

BUS BARS/CUBICLE

SI. No.	Description	Requirement
1	Nominal/Highest System Voltage	6.6/6.9 KV
2	Type of Installation	Indoor
3	Max. Ambient Temp. and Temp. Rise	As per IS/IEC
4	Min. Clearances in air (Phase to Phase and Phase to Earth)	As per IS/IEC

SI. No.	Description	Requirement
5	Degree of protection	IP 52
6	Continuous current rating	As per system design which is capable of handling 5 MWpower
7	Short Time Current Rating for 1 sec	25 kA
8	Rated Power Frequency withstand voltage	20 kV (rms)
9	Rated Lightning Impulse Withstand Voltage	60 kV (peak)
10	Cable Entry	From bottom

CIRCUIT BREAKERS:

SI. No.	Description	Requirement
1	Type	Vaccum
2	No. of Poles	3
3	Nominal/Highest System Voltage	6.6 kV/6.9 kV
4	Type of Installation	Indoor
5	Duty Cycle	O- 0.3 sec-CO-3 min-CO
6	Operating cycles	Min. 10000
7	Control Voltage	220 V DC (10% to -15%)
8	Short Time Current Rating for 3 sec	25 kA
9	Continuous current rating	As per system design
10	Symmetrical Breaking Current Capacity	25 kA (rms)
11	Short Circuit Making Current	62.5 kA
12	Degree of protection	IP 55
13	Operating mechanism	Spring Charged

CURRENT TRANSFORMER

SI. No.	Description	Requirement
1	Type	Cast Resin Type
2	Nominal/Highest System Voltage	6.6 kV/6.9 kV
3	Short Time Current Rating for 1 sec	25 kA
4	No. of Phases	Single

SI. No.	Description	Requirement
5	Insulation Class	Class B or better
6	Rated Power Frequency withstand voltage (Primary/secondary)	20 kV (rms)/3 kV (rms)
7	Rated Lightning Impulse Withstand Voltage	60 kV (peak)
8	Protection Class	5P20
9	Diff./REF Protection Class	PS
10	Metering Class	Class 0.2 and ISF <=5

POTENTIAL TRANSFORMER

SI. No.	Description	Requirement
1	Type	Cast Resin Type
2	Nominal/Highest System Voltage	6.6 kV/6.9 kV
3	Short Time Current Rating for 1 sec	25 kA
4	No. of Phases	Single
5	Insulation Class	Class B or better
6	Rated Power Frequency withstand voltage (Primary/secondary)	20 kV (rms)/3 kV (rms)
7	Rated Lightning Impulse Withstand Voltage	60 kV (peak)
8	Accuracy Class	0.2 for metering and of 0.5 for other purposes.

ISOLATOR/DISCONNECTING SWITCH

SI. No.	Description	Requirement
1	Type	Cast Resin Type
2	Nominal/Highest System Voltage	6.6 kV/6.9 kV
3	Short Time Current Rating for 1 sec	25 kA
4	No. of Poles	3
5	Continuous current rating	As per system design which is capable of handling 20MW power
6	Short Time Current Rating for 3 sec	25 kA
7	Rated Power Frequency withstand voltage	

SI. No.	Description	Requirement
	a. To earth & between poles	20 kV (rms)
	b. Across isolating distance	28 kV (rms)
8	Rated Lightning Impulse Withstand Voltage	
	a. To earth & between poles	60 kV (peak)
	b. Across isolating distance	75 kV (peak)

SURGE ARRESTOR

SI. No.	Description	Requirement
1	Type	Metal Oxide Gapless
2	Rated Voltage	6 kV
3	Nominal Discharge Current	As per IS
4	Installation	Indoor
5	Rated Power Frequency withstand voltage	20 kV (rms)
6	Rated Lightning Impulse Withstand Voltage	60 kV (peak)

5.12.7.2 GENERAL REQUIREMENTS

The 6.6 kV Indoor Switchgear shall be designed considering the minimum general requirements as mentioned hereunder:
The switchgear shall be indoor, free standing, sheet metal clad, draw out type and shall be fully compartmentalized.

The Switchgear enclosures shall be totally enclosed design, dust tight and vermin proof.

Switchgear panels shall be made of cold rolled sheet steel of not less than 2.5 mm thickness suitably reinforced to provide rigid flat level surfaces resistant to vibration during transportation and installation having smooth finish and elegant appearances.

The Switchgear shall confirm IP 52 or higher degree of protection.

Each panel shall be equipped with space heaters to prevent moisture condensation within the enclosure and shall be complete with MCB, thermostats and auxiliary relay (if required).

Switchgear design shall comprise of fully compartmentalized execution having separate vertical sections for each circuit.

Structure, buses and control wiring shall be designed and arranged in such a manner so that future extension of the switchboard would readily be feasible.

All corresponding components of the circuit-breakers and switchgear of same rating shall be fully interchangeable.

Bus-bar of high conductivity Aluminium alloy of adequate uniform cross-section shall be provided.

From 6.6 kV Switchgear at each local control room, the power shall be fed to the 6.6 kV Swichyard under Master Control room building through 6.6 kV XLPE Cable.

The circuit breaker shall be suitable for remote electrical and local electrical/manual operation. The closing coil shall operate correctly between 85% to 110% of its rated voltage and shunt trip shall operate correctly under all operating conditions of the circuit breaker between 70% to 110% of its rated voltage.

The disconnecting switches shall be provided with local electrical/manual control from the panel. The disconnecting switches shall be fitted with earthing link wherever required. The disconnecting switch shall be connected between the transformer and circuit breaker for the power incoming from solar PV and for synchronization between the bus bar and transmission line through breaker, at 6.6 kV.

The supplier shall ensure that the current transformers shall have adequate VA output for the type of protection & metering offered. The supplier shall also ensure that the current transformers quoted by him have adequate output for prescribed accuracy class and accuracy limit factor for the type of relays and instruments connected in their circuits. PS class CTs shall have low secondary resistance and high knee point voltage so as to avoid any possibility of CT saturation under through fault conditions.

Three single phase voltage transformers shall be suitable for connecting in a bank of three phase voltage transformers for protection and measurement purpose for each incomer and outgoing feeders. Separate and dedicated voltage transformers shall be provided for synchronization.

The lightning arrester & voltage transformer (LAVT) cubicles for 6.6 kV shall comprise of lightning arresters and capacitors (for surge protection) and voltage transformers. Degree of protection for all indoor cubicles shall be IP-52. The LAVT & VT cubicles shall be dust tight, vermin-proof.

Each cubicle shall be equipped with space heaters, thermostats, illumination lamps & 240 V AC, 5A receptacle.

Suitable single compression type, heavy duty brass cable glands with check nuts, rubber sealing ring and brass washers mounted on a removable gland plate shall be supplied with the switchgear to support all power and control cables entering the switchgear.

Cables for each equipment must be tagged with permanent metal tag of impregnated cable number as per drawings at MCC/switchgear end and equipment terminal end as well as in the mid portion of the cables at certain distances as instructed by the owner or his authorized representative.

The switchgear units shall have safety interlock features.

The accuracy class of indicating instruments shall be 1 or better as per IS. The accuracy class of meters for commercial metering shall be 0.2 All instruments shall have means for calibration, testing and adjustment at site.

Three phase watt hour meters conforming to latest issue of relevant Indian standard shall be provided with test link for CTs & PTs. Meters shall be compensated for temperature errors and factory calibrated to directly read the primary quantities.

Following equipments at 6.6 kV switchgear shall be monitored from SCADA:
1) Circuit breaker - On/Off status & Control
2) Transformer - Winding temperature & Oil temperature Alarm status
3) Energy meters
4) Numerical Relays
5) Voltmeters
6) Ammeters
Suitable earthing arrangement must be designed for the system.

Finishing work like painting etc. for switchgears should be as per relevant IS.

7.3 **APPROVAL**

The Detailed Design Report submitted by the contractor to DPL must contain but not limited to the following details of the Ring Main Unit/Switchgear:

- Detailed specification of all the items.
- Necessary Drawings
- All necessary test certificates and approvals etc.

The successful bidder required to produce all necessary test certificates and approvals of the product as per relevant standard with the Detailed Design Report.

Prior to the delivery of the product, the contractor shall submit but not limited to the following documents:

- Guarantees
- Instructions for installation and operation, manual
- Electrical diagrams
- Safety precautions
- Detailed schematics of all power instrumentation and control equipment and subsystems along with their interconnection diagrams. Schematics shall indicate wiring diagrams, their numbers and quantities, type and ratings of all components and subsystems etc.

The contractor can deliver the product to the site only after receiving such approval against their prayer in writing from DPL.

5.12.8.0 STATION AUXILIARY TRANSFORMER

5.12.8.1 SCOPE

This section covers the activities related to design, manufacturing, testing at works, supply, insurance, transportation and delivery at Project site, storage, erection, testing, commissioning of 6.6 KV / 0.415 kV dry type station auxiliary transformers and associated equipments as detailed hereunder.

The scope of supply shall also include necessary spares required for normal operation & maintenance of transformers for a period of 5 (five) years & special tools & plants required for erection & maintenance. Corresponding parts of all the equipments & spares shall be of the same material & dimensions, workmanship & finish and shall be interchangeable. All the material & workmanship shall be of suitable commercial quality as have proven successful in their respective uses in similar services & under similar condition.
The transformers and associated equipment shall be suitable for interfacing with SCADA system and all necessary transducers shall be included in the scope of supply.

5.12.8**.2 STANDARDS**

The equipments and materials covered by this specification shall conform to the latest edition of following Indian Standards or equivalent IEC standards except where specified otherwise in this specification:

Sl. No.	Standards	Description
1	IS: 2026 (Part I to IV) /IEC 76	Power Transformer
2	IS: 2099/IEC 137	Transformers bushings
3	IS: 2705/IEC 185	Current transformers
4	IS: 1180	Three phase distribution transformer
5	IS: 6088	Dimensions for porcelain transformer bushings
6	IS: 3347	Loading guide for oil-immersed transformers
7	IS: 335	Transformer oil
8	CBIP No. 295	CBIP Manual on Transformers Publication

5.12.8.3 **TECHNICAL REQUIREMENTS**

Capacity of the Station Auxiliary Transformer (SAT) shall be calculated based on the total auxiliary load of the plant.

HT side of the transformer shall be connected to the 6.6 kV Bus through a VCB and LT side shall be terminated to the Station Service Board (SSB)/ 415 V LT Switchgear.

SAT shall be installed at the switchyard of the Solar plant.

The transformer shall be able to perform satisfactorily under voltage variation limit of +/- 10 % and frequency variation limit of +/- 5 %.

Vector group of the transformer shall be Dyn11

Off Circuit Tap changer (-10% to +10% @1.25% steps) shall be provided with the transformer. Owner will take final decision regarding this based on the proposal submitted by the contractor.

% Impedance, type of bushing, class of insulation, temperature rise etc. shall be as per relevant Indian Standard.

The transformers shall be suitable for co-ordination and integration with SCADA System and necessary contacts and/or ports for the purpose shall be provided.

Earthing arrangement of the transformers shall be provided as per the relevant Indian Standard.

Necessary protection arrangement like should be provided in the transformer.

Construction of different parts of the transformer shall conform to the latest edition of relevant Indian Standard.

Fittings and accessories shall be provided as per relevant Indian Standard code.

Transformer oil shall conform to latest edition of IS 335.

5.12.8.4 APPROVAL

The Detailed Design Report Submitted by the contractor to DPL must contain but not limited to the following details of the transformers:

- Detailed specification
- Fittings and Accessories
- Necessary Drawings shall contain but not limited to the following:
  - Outline dimension drawings of transformers and fittings/accessories
  - Assembly drawings and weight of main components.
  - Transport drawings, showing main dimensions and weight of each package.
  - Foundation details
  - Tap-changing equipment
  - Name-plate diagrams
  - Schematic control and wiring diagrams for all aux. equipment etc.
- Tools and spare parts etc.
- Type Test Reports and certificates etc.

A joint inspection and testing will be done by owner and the authorized representatives of the contractor at the manufacturer’s workshop, if desired so by the owner. Testing will be done as per relevant IS Code.

Prior to the delivery of the product, the contractor shall submit but not limited to the following documents:

- Guarantees
- Instructions for installation and operation, manual

- Test Reports for routine and acceptance tests etc.

The contractor can deliver the product to the site only after receiving such approval against their prayer in writing from DPL.

5.12.9.0 LT SWITCHGEAR

5.12.9.1 SCOPE

The scope of work under this specification covers the design, manufacture, assembly, testing at manufacturer's works, transportation, transit insurance, delivery at site, storage, installation, testing, and commissioning of indoor type following 415V LT Switchgear complete with all accessories and spares.

The Scope shall include supply of 415 V (3 phase, 1 neutral and single phase for lighting etc.) transmission line (all Al conductor) for the entire area from the LT switchgear at both location with necessary breaker, switch fuse unit as and when required, Boards as above along with gland plates for all power and control cables, base frames, special tools i.e. operating handles, trolley necessary for removing the circuit breakers for maintenance etc. Isolators should be provided in the line to connect or isolate the connection from both the station auxiliary transformer.

The scope shall include all associated devices, components, relays, contactors, switches etc. required for satisfactory operation of the switch boards as per the proposed logic control scheme. The scope of supply shall also include necessary spares required for operation & maintenance of switchgear equipments for a period of 5 (five) years & special tools & plants required for erection & maintenance.

Corresponding parts of all the equipments & spares shall be of the same material & dimensions, workmanship & finish and shall be interchangeable. All the material & workmanship shall be of suitable commercial quality as have proven successful in their respective uses in similar services & under similar condition.

5.12.9.2 **STANDARDS**

The equipments covered under this chapter shall comply with the requirement of latest edition of following IS/BS/IEC specifications as amended up to date except where specified otherwise.

Sl. No.	Standards	Description
1	IS: 13947 (Part 1 to 5)	Specification for Low-Voltage Switchgear and Control gear.
2	IS: 10118 (Part 1 to 4)	Code of practice for selection, installation and maintenance of switchgear & control gear.
3	IS: 1248	Specifications for Electrical Indicating Instruments
4	IS: 2633	Hot dip Galvanizing
5	IS: 2705	Current Transformers
6	IS: 3156	Voltage Transformers
7	IS: 3231	Electrical Relays for Power System Protection
8	IS: 5082	Wrought Aluminium and Aluminium Alloy bars, tubes and sections for electrical purposes.
9	IS: 8623	General requirement for factory built

		assemblies up to 1000V.
10	IS: 8828	Circuit breakers for over current protection for household and similar installations
11	IS: 13703	Low Voltage fuses for voltages not exceeding 1000V AC
12	IS: 11353	Guide for uniform system of marking and identification of conductors and apparatus terminals.

5.12.9.3 TECHNICAL REQUIREMENTS

- Main Incoming A.C. circuits on Station service Board shall be controlled through microprocessor based numerical relay with draw out type Air Circuit Breaker. Type and capacity of the breakers shall be proposed by the bidders in their bid considering the total auxiliary load of the plant.
- The LT switchgears shall be suitable for indoor installation in the control room.
- **A)** LT switchgears shall be placed in each site control room of Phase I and shall be connected to each other by means of 415 V (3 phase, neutral) transmission line along the whole area. LT switchgear at main control room shall be connected with Station Auxiliary Transformer . This LT switchgear shall feed all auxiliary supply of Plant.

  B)

- The Station Service Board shall be sectionalized in two parts through sectionalizing breakers of equivalent capacity and protection of incomers on the bus to ensure continuity of supply to the auxiliaries in case of failure/fault on one section. This Switchboard shall be considered as two Incomers and one Bus-coupler method with a castle key interlock.
- For interconnection with various boards and all outgoing feeder circuits, 50 kA, 3 pole draw-out type MCCBs with adjustable current setting shall be provided.
- The Air Circuit Breakers, Boards etc. shall have at least the following ratings:
  - No. of phases : Three
  - Rated voltage : 1.1 kV
  - Service voltage : 415 V ± 10%

- Frequency : 50 Hz. ± 5%
- Rated short time current rating switchgear : 50 kA for 1 sec. for bus & switchgear
- Normal control voltage : 220V DC
- Degree of Protection : IP 42 or higher

➢ The following equipment at LT switchgear shall be monitored from SCADA /DCS.

1) All Air Circuit breaker – On & Off status, test, service, spring charged, TCS healthy, 86 operated, DC fail etc.

2) All ACB - On and Off command from DCS /SCADA

3) Energy meters

4) Voltmeters-from transducer

5) Ammeters-from transducer

6) Numerical Relays should also be integrated with SCADA.

➢ The 415V switchboards shall be metal-enclosed draw out type, free standing, self-supporting, floor mounted, indoor type, totally enclosed and compartmentalized to house the switchgear. Circuit breakers and other switchgear components shall be arranged in compartments, vertically in a multi-tier formation. All metering and protection equipment associated with a particular circuit shall be housed in separate and independent compartment earmarked for particular circuit and in the fixed portion of the vertical panel in case of breaker panels.

➢ Construction of all the switchboards and equipment shall conform to the latest edition of relevant IS codes.

➢ All cable glands and aluminum crimping type cable lugs for all power and control cables shall be in the bidder’s scope of supply. Panels shall be suitable for bottom entry of cable unless otherwise specified.

➢ The bidder shall indicate clearly the de-rating factors, if any, employed for each component and furnish the basis for arriving at these de-rating factors duly considering the specified current ratings, ambient temperature etc.

- The equipment shall comply with all safety requirements during erection and operation as per relevant standards.
- The neutral of the incoming transformer secondary shall be connected to the neutral bus of the auxiliary boards. The neutral shall be connected to the common earthing system of the switchyard/control room.
- All auxiliary devices for control, indication, measurement and protection such as push buttons, control and selector switches, indicating lamps, Power monitors, kWh meters and protective relays shall be mounted on the front side of the respective compartment. The design shall be such that unless required for maintenance / inspection purposes, all power ON/OFF or START / STOP and relay reset operations shall be performed without opening the panel door.
- The switchboard panels shall be provided with thermostatically controlled space heaters to prevent moisture condensation.
- Tube light / CFL lamp fittings along with necessary isolating switches shall be provided for illumination inside the panels. Each panel shall be provided with an industrial grade power socket as well.
- The 415V bus shall be of suitable cross-section so as to be able to carry the required continuous and short circuit currents within the limits of temperature rise for the site conditions.
- Control and selector switches shall be rotary type with escutcheon plates clearly marked to show the function and positions. The switches shall be of sturdy construction suitable for mounting on panel front.
- AC Distribution Board is to be provided in the main switchgear room and in the particular local control room having auxiliary transformer as per requirements.
- Instrument transformers shall be provided and shall conform to the relevant standard.
- The relay for the switchgear units shall have all the normal design features/ as specified elsewhere in the contract .

- All relays shown in the drawing and others required for operation of the system as per the specification shall be included in the scope of supply. The relays shall be of electromagnetic/ static/numerical type/ microprocessor based conforming to the requirements of IS: 8686 or IEC: 255.
- All instruments and meters shall be suitable for operation under the climatic conditions prevailing at site. The instrument cases shall be dust-proof, water tight, vermin proof, specially constructed to adequately protect the instruments against damage or deterioration due to high ambient temperature and humidity.
- The VA burden of instrument coils/elements shall be as low as possible, consistent with the best modern design.
- Watt hour meter shall be suitable for 3-Phase, 4-wire unbalanced system and shall comply generally with the requirements of relevant IS code and shall be of first grade for the purpose of accuracy classification. Watt hour meters shall be provided in each LT switchgears as well as each 6.6 kV switchgears.
- Panels shall be supplied completely wired internally to equipment and terminal blocks for connection to external cables entering the panel from the bottom. Terminal blocks shall be complete and provided with necessary terminal accessories for cable ends.
- Engraved PVC labels shall be provided on incoming and all outgoing breaker compartments, the exact details of legend to be engraved shall be furnished later to the contractor.
- All vertical cubicles shall be connected to earth bus bar running throughout the length of the switchboard. All doors and movable parts shall be connected to the earth-bus with flexible copper connections. Provision shall be made to connect the earthing bus bar to the main earthing grid at two ends. All non-current carrying metallic parts of the mounted equipment shall be earthed. Earthing bolts shall be provided to ground cable armours.
- Finishing work like painting etc. for switchgears should be as per relevant IS.

5.12.9.4 APPROVAL

The Detailed Design Report submitted by the contractor to DPL must contain but not limited to the following details of the LT Switchgear:

- Detailed specification of all the items.
- All necessary drawings
- All necessary test certificates and approvals etc.

Prior to the delivery of the product, the contractor shall submit but not limited to the following documents:

- Guarantees
- Instructions for installation and operation, manual
- Electrical diagrams
- Safety precautions
- Detailed schematics of all power instrumentation and control equipment and subsystems along with their interconnection diagrams. Schematics shall indicate wiring diagrams, their numbers and quantities, type and ratings of all components and subsystems etc

The contractor can deliver the product to the site only after receiving such approval against their prayer in writing from DPL.

5.12.10.0 DC BATTERY, BATTERY CHARGING EQUIPMENT & DCDB

5.12.10.1 SCOPE

The scope of work under this specification covers the design, manufacture, assembly, testing at manufacturer's works, transportation, transit insurance, delivery at site, storage, installation, testing, and commissioning of D.C equipment comprising of 220 V D.C Battery Bank of suitable designed capacity complete with battery charging equipment, D.C. Distribution Board and other auxiliary equipments.

The scope shall include all associated devices, components, relays, contactors, switches etc. required for satisfactory operation of the DC equipment as per the proposed logic control scheme.

The scope of supply shall also include necessary spares required for normal operation & maintenance of DC equipments for a period of 5 (five) years and special tools & plants required for erection & maintenance.

Corresponding parts of all the equipments & spares shall be of the same material & dimensions, workmanship & finish and shall be interchangeable. All the material & workmanship shall be of suitable commercial quality as have proven successful in their respective uses in similar services & under similar condition.

5.12.10.2 STANDARDS

The equipments covered under this chapter shall comply with the requirement of latest edition of following IS/BS/IEC specifications as amended up to date except where specified otherwise.

Sl. No.	Standards	Description
1	IS: 1651	Stationary cells & batteries, lead acid type (with tubular positive plates)
2	IS: 266	Battery grade Sulphuric Acid. (Battery electrolyte)
3	IS: 1069	Water for storage batteries
4	IS: 1146	Rubber & Plastic containers for lead Acid storage batteries
5	IS: 1248	Electrical Indicating Instruments
6	IS: 13947	Low voltage switchgear and control gear
7	IS: 3895	Mono-crystalline semi-conductor rectifier cells & stacks
8	IS: 8320	General requirement and methods of tests for lead acid storage batteries
9	IS : 6071	Synthetic separators for lead acid batteries
10	IS : 8623	Factory built assemblies of switchyard and control gear for voltage up to including 1000 V AC and 1200 V DC (Part 1 to 3)
11	IS : 4540	Non-crystalline semi-conductor rectifier assemblies &

		equipment

Equipment meeting any other authoritative national or international standards that ensure equal or better quality than the standards mentioned above are also acceptable. Where the equipment conforms to any other standards than those mentioned above, salient points of difference between the standards adopted and standards mentioned above shall be brought out in the tender.

5.12.10.3 GENERAL REQUIREMENTS

Minimum general requirements for the DC Battery, Battery charger and DC Distribution Board are mentioned below.
Lead acid tubular type battery of required rating shall be provided at Main Control Room and each Local Control room. Battery Bank at Main Control Room shall be 110/220 V and Battery Bank at Local Control Rooms shall be selected based on the Control Voltage required for closing and tripping of 6.6 kV Indoor type VCBs. 10 hours continuous discharge shall be considered for sizing the battery.

One float charger and one float cum boost chargers shall be provided to maintain constant voltage at D.C. bus bars while supplying the continuous load in addition to keeping the battery on float charge.

In case of sudden D.C. requirements due to failure of A.C. supply or charger itself, the battery shall be capable of meeting the system load demand. In case of failure of float charger supplying the continuous DC load, the affected battery charger shall get disconnected automatically from the DCDB and the complete D.C. load requirements shall be met by the float charger of float cum boost charger unit.

The charger shall be protected against overloads by having suitable characteristics so that all loads in excess of the capacity of the charger would be transferred to the battery.

In the event of failure of A.C. supply, the battery shall meet the complete D.C. requirements. After the discharge of battery to a considerable extent, the boost charger on restoration of A.C. supply shall recharge the battery in a short period. During the period of boost charging, the D.C. load requirements of power station shall also continue to be met.

The distribution board with necessary switch and interlock, if any, shall be provided for distributing the D.C. power for the control & protection circuits, emergency D.C. supply for essential lighting etc.

The bidder may give his recommendation on the scheme of operation of battery, battery chargers as described in the specifications. However, the decision of the owner in this regard shall be final and bound to the bidder/contractor.

The battery shall be capable of delivering the rated output at the minimum temperature of -3°C and maximum temperature of +40°C.

The battery shall be mounted on the two tier wooden racks supplied along with the battery. Each cell as well as its locations shall be numbered for proper record of maintenance operations. Battery should be placed on the porcelain base kept on the wooden rack.

The battery shall be connected to D.C. distribution board by single core cables laid above ground. Suitable terminal arrangement with glands shall be provided for this purpose.

Battery room shall be painted with acid proof paint. Exhaust fans should be provided in the battery room. Contractor shall submit the details of the same to the owner.

The ripple content in the D.C. current shall be less than 1%.

The float charger unit shall be capable of supplying continuous D.C. load and trickle charge the battery.

Necessary alarm and indication shall be provided with the DC System and also in the annunciation window of the Battery Charger.

Necessary terminals with lugs for earthing the charger panels with two distinct separate earthing for each panel shall be provided. In addition,

separate terminals for earthing of equipment shall be provided. The charger panels shall have space heaters.

Compression type cable glands of suitable rating for PVC unarmoured cable, suitably mounted in the panel for cable entry from the bottom for A.C. & D.C. supplies shall be provided.

Type of cell, cell terminal, containers and installation of battery, chargers, inverter, DC Distribution Board, cables etc. should conform to the latest edition of relevant Indian Standard.

During installation of battery, charging & discharging and charging is to be done proper installation procedure.

5.12.10.4 TECHNICAL REQUIREMENTS

Minimum technical requirements for the DC Battery, Battery charger and DC Distribution Board are as following.

The battery shall be made of lead-acid cells with tubular type plates conforming to latest issue of IS 1651. The battery cells shall be high discharge performance (HDP) type.

The capacity of 110/220 V D.C. batteries based on 10 hours discharge rate shall be selected to fulfill the plant’s requirement. The contractor shall propose the same to the owner and decision of the owner will be final and bound to the contractor.

The battery shall normally remain under floating condition with the charger supplying the normal continuous load. However, the battery shall be capable of supplying the load without fall of terminal voltage per cell below 1.85V (92.5% of rated voltage).

The number of cells of the 110/220 volt battery bank at Main Control Room and required voltage at Local Control Room shall be chosen to suit the following conditions.

- Nominal floating voltage per cell shall be between 2.15 and 2.21 V.

- The voltage of each cell under floating conditions shall be of optimum value for its performance and maintenance in a healthy condition.

- The voltage of the battery after meeting the D.C. load cycle shall not be less than 90% of the rated voltage. The manufacturer shall ensure safe operation of the battery after the aforementioned end voltage.

- The voltage across the load shall not exceed 110% of rated value under charging conditions of the battery. To achieve this condition under quick charging, a blocking diode may be incorporated by the supplier in the charging equipment.

The bidder shall clearly justify the choice of number of cells in the tender on the above lines and furnish any clarifications required by the owner.

All cell terminals shall have adequate current carrying capacity and shall be of lead-alloy or lead-alloy reinforced with copper core inserts. Cell terminal posts shall be equipped with acid resisting connector bolts and nuts.

The electrolyte shall be of battery grade sulphuric acid. The battery shall be transported dry.

The charging equipment shall preferably employ solid state full wave rectifier in a 3 phase full wave bridge circuit with suitable filter circuit of AC ripples, suitable for operation in conjunction with static voltage regulator. A.C. and D.C. Circuit breakers with thermal overload and instantaneous short circuit releases shall be provided on input and output sides of chargers respectively.

Capacity of the float charger and the boost charger in the float cum boost charger shall be sufficient to meet the system requirement. Contractor shall submit the details to the owner.

The charger shall be capable of providing the floating voltage between 2.15 V to 2.21 V per cell with the variation of not more than +1% irrespective of input supply voltage fluctuations within +/-10%, frequency fluctuation within +/-5 % throughout its ampere rating with ambient air temperature range of -3$_{0}$C to 40$_{0}$C.

The DC Distribution Board (DCDB) shall be free standing, self-supporting and floor mounting type. It shall be totally enclosed and compartmentalized. DCDB shall be made as per relevant Indian Standard.

The Emergency Lighting Board supplying the emergency lighting requirement of the power house at A.C shall have an arrangement so that automatic changeover to emergency lighting in case of A.C. failure, is achieved through an inverter of suitable capacity. Normally, the inverter shall run on AC. supply. In the event of failure of AC, the inverter shall automatically switch-over to DC supply and feed the selected emergency loads (lighting loads) at 230 V AC. On restoration of AC supply, the inverters load will automatically return to AC.

The DC system shall have necessary control & protection arrangement which include but not limited to the following.

- Auto/Manual selector switch
- Digital D.C. voltmeter, ammeter
- A.C. failure alarm
- Ground fault relay and its annunciation
- Double pole D.C. contactor of suitable capacity for annunciation
- Triple pole A.C. contactor of suitable capacity for ON/OFF operation
- MCCB and DC contactor of suitable capacity in output circuit of each charger to suit the operation requirements.
- Indicating lamps, as required
- Triple pole, A.C. circuit breaker of sufficient capacity to meet system requirements & capacity with overload and short circuit release for incoming A.C. supply to charger panel
- MCB/MCCBs for A.C. supply to individual chargers
- A.C. under voltage relay
- A.C. voltmeter, ammeter etc.

Nearest local control room from the main control room should be connected with 110/220 V DC from Battery Bank DCDB.

110/220 V AC/DC converter is to be provided in each isolated switchgear for operation of circuit breaker/isolator as and where required. Power required in ACDB/DCDB for illumination, control system etc. for each control room should be collected from 415 V (3phase+N) transmission line with necessary cables and protection.

5.12.10.5 APPROVAL

The Detailed Design Report submitted by the contractor to DPL must contain but not limited to the following details of the DC system:

- Detailed specification of all the items.
- Necessary Drawings

- Test Certificates etc.

Prior to the delivery of the product, the contractor shall submit but not limited to the following documents:

- Guarantees
- Instructions for installation and operation, manual
- Detailed schematic, connection and control wiring diagrams etc.

The contractor can deliver the product to the site only after receiving such approval against their prayer in writing from DPL.

5.12.11.0 PROTECTION SYSTEM

5.12.11.1 SCOPE

The scheme shall consist of design, engineering, quality surveillance, manufacture, tests at manufacturer's works before dispatch, transport, transit insurance, supply, delivery to site, storage at site, erection, testing, trial run and commissioning, handing over to the purchaser of protection system for

- PV Array yard
- Solar Inverter
- Three winding Step up Transformer
- Incomer feeder for 6.6 kV Switchyard
- Station Auxiliary Transformer
- 6.6 kV outgoing feeder

The protection system shall include protection relays, trip relays, relay contacts, trip & alarm circuits, Annunciation system, diagnostic system, other necessary equipment with all accessories, wiring and cubicles for making the protection system complete for 5 MWSolar PV Power Plant in DPL,Durgapur, West Bengal.

5.12.11.2 STANDARDS

All materials and equipments shall conform to latest edition of relevant Indian/IEC Standards unless otherwise specified. Equipment conforming to any other authoritative standard ensuring equal or better quality than the standards indicated below will also be acceptable. However, in such eventuality, the salient points of difference between the standards adopted and the standards mentioned below shall be brought out by the bidder. The list of reference standards is given below:

Sl. No.	Standards	Description
1	IS: 2705	Current Transformers
2	IS: 3156	Voltage Transformer s
3	IEC: 60255 (Part 1 to 23)	Electric Relays
4	IEC: 60337	Control switches and low voltage switching devices

		for control and auxiliary circuits
5	IS: 1885	Electro-technical vocabulary on Electrical relays, Electric Power System Protection and Switchgear & Control
6	IS:13947	Degree of protection provided by enclosures for low voltage switchgear and control gear
7	IS: 3231	Electric relays for Power System protection
8	IS: 5834	Electric Timer relays
9	IS: 8686	Static Protective relays

5.12.11.3 **TECHNICAL REQUIREMENTS**

The technical requirements of the protection system shall be but not limited to the following. Protection shall be designed to ensure reliability, sensitivity and stability under through fault conditions of the system.

The protection system shall be fully integrated with SCADA system.

The protection scheme shall be coordinated with control & protection of solar modules, solar inverters and generator transformers etc. All protection, though not specified but which are recommended for this capacity of the machine as per relevant IEC / other Standards shall be provided.

The protective relays shall be of the numerical, fully tropicalised, plug in type, arranged in protection cubicles including all ancillary devices, such as interposing transformers, tripping matrix and relays, test facilities, power supply units, etc. with all circuits complying to latest editions of IEC 60255-4 recommendation or British Standard 142 and 5992, parts 1, 2 and 3 or relevant Indian Standard.

The relays/protection system shall be of state of the art of technology and only latest proven versions of the relay series shall be offered. If the protection system mentioned in the awarded Contract become obsolete at the time of supply, the Supplier shall offer the latest model with the approval of Employer, without any extra cost.

Protection system shall be provided to prevent operation of protective equipment due to, magnetizing current inrush during switching-in of the transformer from the high voltage side.

Precaution shall also be taken so that the unavoidable inductive and capacitive couplings from the power circuits do not cause disturbances.

Protection relay shall have features but not limited to the following:

- Man machine communication interface with alarm and trip value setting, displaying of alarm/trip set values, alarmed/tripped values, fault current and disturbance values etc.

- Self-supervision and indication of any failure.

- Continuous monitoring of external and internal auxiliary voltages
- Easiness of replacing a set in case of failure.
- Communication interfaces or ports.
- Indication of alarm and trip condition.
- Test facilities etc.

All devices shall remain inoperative during external faults and transient phenomenon. They shall be insensitive to mechanical shocks, vibration and external magnetic fields.

The protection relays, shall be located in conventional panels and shall be flush mounted in dust and moisture proof cases with protection class IP 54 and of the draw out type with rear connections. The protection class of the cover for all relays or protection systems, in which the modules are mounted, shall not be inferior to IP 54.

The protection systems shall be fed by the battery banks installed in the main control room and local control rooms. Relay shall be suitable for operation on DC systems without the use of voltage dropping resistors.

The supplier has to supply the equipments for protection of best quality. The supplier has to maintain control and quality assurance during the manufacture, installation, testing and commissioning of equipments as per approved quality assurance plan.

5.12.11.4 APPROVAL

The Detailed Design Report submitted by the contractor to DPL must contain but not limited to the following details of the protection system:

- Detailed specification of all the items.
- All required drawing etc.

Prior to the delivery of the products, the contractor shall submit but not limited to the following documents:

- Guarantees
- Instructions for installation
- Instruction O&M manual Testing & commissioning manuals
- Detailed BOQ covering protection relays, CTs /PTs, DC Sources and all other devices.

The contractor can deliver the product to the site only after receiving such approval against their prayer in writing from DPL.

5.12.12.0. EARTHING AND LIGHTNING PROTECTION SYSTEM

5.12.12.1 SCOPE

The scope of work under this specification covers the design, supply, transportation, delivery at project site, transit insurance, storage at site, erection, testing & commissioning of electrical grounding and lightning protection system along with necessary materials. All the equipment and building shall be protected from lightning through Lightning Protection System.

5.12.12.2 STANDARDS

The grounding system shall conform to the requirement of following standards.

Sl. No.	Standards	Description
1	ANSI/IEEE: 80 –2000	Guide for safety in AC Substation Grounding
2	CBIP Publication: 223	Design of Earthing Mat for High Voltage substation
3	IS: 3043	Code of Practice for Earthing Indian Electricity Rules

5.12.12.3 OBJECTIVE

The grounding system shall be designed with the following objectives:

- To provide low impedance path to fault currents, during ground faults, to ensure prompt and consistent operation of protective devices to effect isolation
- To keep the maximum voltage gradient during ground faults along the surface inside and around the switchyard, PV array yard, control rooms etc. within safe limits
- To protect the life and property from electrical shocks due to over voltage
- To stabilize circuit potentials with respect to ground and limit the overall potential rise

5.12.12.4 TECHNICAL REQUIREMENTS

Minimum technical requirement of the earthing system is mentioned below.

- The earth resistance should be less than 1 Ω.
- Suitable number of earthing pit shall be provided at the array field.

- Design and installation of the earth mat and other associated system shall confirm IS: 3043 and shall be followed by modern practice.
- The earthing for solar field and power distribution system shall be made with GI pipe of suitable size including accessories, and providing masonry enclosure with cast iron cover plate having locking arrangement, watering pipe using charcoal or coke and salt as required as per provisions of IS: 3043. The Mounting structure shall be grounded properly using GI strips and maintenance free earthing kit.
- Size of ground earth mat shall be 1000mm below FGL and 40 mm dia MS rod
- Necessary provision shall be made for bolted isolating joints of each earthing pit for periodic checking of earth resistance.
- The earth conduction shall run through GI pipe partly buried and partly on the surface of the control room building.
- The complete earthing system shall be mechanically & electrically connected to provide independent return to earth.
- All three phase equipment shall have two distinct earth connections.
- Along the cable trays suitable size of GI Flat shall be provided inside the control room.
- For each earth pit, necessary Test Point shall have to be provided.
- The earthing system shall be connected to the following.
    - Solar modules with suitable number of earthing pit at the solar array field
    - The neutral point of each system/equipment
    - Equipment framework and other non-current carrying parts
    - Frames of panels & cubicles
    - Metallic structures of switchgear, cable racks, casing of cable boxes
    - Equipment supporting Steel structures
    - All extraneous metallic frame work not associated with equipment

  - The earth point of lightning arrestors; voltage transformers and lightning conductors through their permanent independent earth electrodes.
  - Fence
- For equipment connection to mat/riser, 50 mm x 6 mm or higher size GS flat shall be used.
- Each neutral point of transformer shall be provided with two separate treated earth pit through 80 mm dia GS perforated pipe having 3 mtr depth. Necessary charcoal, salt etc. to be provided for earth pit as per relevant standard. Each earth pit shall be connected with Main earth grid through a bolted type test point.
- Separate grounding grid to be provided for electronic earthing for PLC system earth resistance shall be less than 0.5 Ω. Please refer NIT drawing ST-GROUND-DWG-E-008.
- The conductor shall be of adequate cross-section to safely withstand the system fault current for time duration of fault clearance by the remotest/back up protective system.
- Sufficient allowance needs to be provided for corrosion of the embedded conductor on account of chemical properties of soil and also due to galvanic action with other embedded systems.
- For determination of the size of the conductor, the value of fault current may be taken as 40 kA; duration of fault current may be considered as 1 second. The extra allowance of 20% to take care of corrosion shall be added to arrive at final conductor size.
- For designing of the earth mat for 6.6kV switchyard, the material of ground mat conductor shall be 40 mm MS rod and that of risers emanating from ground mat shall be GS flats. Soil resistance of the site is available in the soil report.

5.12.12.5 APPROVAL

The successful bidder shall carry out the earth resistance measurement at the site and they need to submit the measurement report to DPL.

The Detailed Design Report submitted by the contractor to DPL must contain but not limited to the following details of the earthing system:

Detailed specification of all the items.

Soil resistivity measurement data

Necessary calculations and drawings etc.

The successful bidder required to produce schematic diagram of the earthing system and the proposed locations for earth mat as per relevant standard with the Detailed Design Report.
All drawings and calculations submitted by the contractor will be subjected to approval of the DPL.

5.12.13.0 CONTROL, MONITORING AND DATA ACQUISITION SYSTEM – SCADA

5.12.13.1 General Requirements

5.12.13.1.1 ***The scope of work under this specification covers the design, engineering, manufacture, testing at manufacturer's works, transportation, transit insurance, delivery at project site, storage at site, erection, testing at site and commissioning of control, monitoring & data acquisition system comprising of computers, VDU, key board/mouse, PLC's, input and output relays, meters, fields sensors, panels/cubicles for housing above equipment/devices, power supplies, transducers, converters, wiring etc. to make the system complete. Since the Employer intends to extend the project up to 50 MW (AC), provision of future expansion should be kept in general SCADA layout and in control room building design.***

The Contractor shall provide complete SCADA system with all accessories, auxiliaries and associated equipment and cables for the safe, efficient and reliable operation of entire solar plant and its auxiliary systems.
13.1.2 The Contractor shall provide all the components including, but not limited to, Hardware,
Software, Panels, Power Supply, HMI, Laser Printer, Gateway, Networking equipment and associated Cables, firewall etc. needed for the completeness.
13.1.3 SCADA System shall have the provision to perform the following functions.
(i) Real-time acquisition and display of data, status, alarms and trends.
(ii) Display of status of major equipment in Single Line Diagram(SLD) format with string level
monitoring capability
(iii) Display and storage of measured values

(iv) Display and storage of derived/ calculated/ integrated values
(v) Generate, store and retrieve user configurable Sequence of Event (SOE) Reports
(vi) Remote monitoring of essential parameters on the web using standard modem (Internet
connection for transferring data to web shall be taken by Contractor in the name of Employer for O&M period).
13.1.4 It shall be possible to remove/ replace redundant controller or various modules (like any I/O module, interface module, etc.) from its slot for maintenance purpose without switching off power supply to the corresponding rack without releasing any spurious signal to controller and causing disturbance or loss of controller functions for other controller.
13.1.5 The Control system shall be designed to operate in non-air-conditioned area. However, the Contractor shall provide a Package/ Split AC of suitable capacity decided by heat load
requirement in SCADA room at Main Control Room.

5.12.13.2. Programmable Logic based control system at Main Control Room
13.2.1. **Bidder shall provide PLC based SCADA at Main Control Room.** For other locations such as Inverter room, Sub Pooling Switchgear Room (if applicable) bidder may offer IO modules/ RTU/ PLC for completeness of SCADA.
13.2.2. **PLC Processor**
The processor unit shall be capable of executing the following functions.
(i) Receiving binary and analog signals from the field to server.
(ii) Implementing all logic functions for protection and annunciation of the equipment and systems.
(iii) Providing supervisory information for alarm, various types of displays, status information,
trending, historical storage of data etc.
13.2.3. **The SCADA shall be OPC version 2.05a compliant** and implement an OPC- DA 2.05a server as per the specification of OPC Foundation. All data should be accessible through this
OPC server. SCADA shall have OPC connectivity for other systems.

5.12.13.3. Human Machine Interface System (HMIS)

13.3.1 Graphical Interface Unit (GIU) / Operator work station (OWS) shall perform monitoring and operation of all devices interacting with PLC based system.

13.3.2 The system shall have built-in safety features that will allow/ disallow certain functions and entryfields within a function to be under password control to protect against inadvertent and unauthorized use of these functions. The system security shall contain various user levels with specific rights as finalized by the Employer during detailed engineering. However, no. of user levels, no. of users in a level and rights for each level shall be changeable by the programmer(Administrator).

13.3.3 Bidder has to provide suitable hardware and software based firewall for network security to restrict unauthorized access to HMI/Solar SCADA PCs and system.

5.12.13.4 Programming Functionalities

Programming of the PLC Processor / controller as well as programming of HMIS shall be user

friendly with graphical user interface. All programming functionalities shall be password protected to avoid unauthorized modification.

5.12.13.5 Software Requirements

13.5.1 All necessary software required for implementation of control logic, operator station displays / logs, storage & retrieval and other functional requirement shall be provided.

13.5.2 Industry standard operating system like WINDOWS (latest version) etc. to ensure openness and connectivity with other system in industry shall be provided. SCADA system shall support following standard protocols (included but not limited to) to communicate with different sub system/Devices:

(i) Modbus (TCP/IP, RTU, ASCII)

(ii) Sub Station Protocol like IEC-61850

(iii) IEC 60870 – 5 – 101/ 104

(iv) Any other protocol on which the offered equipment (by Contractor) will communicate with SCADA.

13.5.3 The Contractor shall provide software locks and passwords to Employer for all operating & application software. Also, the Contractor shall provide sufficient documentation and program listing so that it is possible for the Employer to carry out modification at a later date.

5.12.**13.5.4 System Spare Capacity**

Over and above the equipment and accessories required to meet the fully implemented system as per specification requirements, Control System shall have spare capacity along with the necessary hardware/equipment/accessories to meet the future expansion requirement.

5.12.13.6. Minimum Parameters to be Measured (for SCADA purposes)

The following **MINIMUM parameters will be measured** and provided to Employer's Interface Panel through agreed communication (either by GPRS with Public IP, or Fiber Optics will be connected with MOICT Grid Later*).*

Station meteorological signals:

a. Temperature (at the met mast thermometer and modules)
b. Relative humidity (at the met mast)
c. Atmospheric pressure (at the met mast)
d. Solar radiation/irradiation
e. Wind speed

Other signals:

f. Sent out total energy at each delivery point MW, MVAR at 6.6 KV side.
g. Exported total energy at each delivery point MW, MVAR at the 6.6 kV side.
h. Sent out energy at each delivery point half hour, hourly, daily, monthly, and MWH, MVARH at the 6.6 kV side.
i. Exported energy at each delivery point, half hour, hourly, daily, monthly, and total MWH, MVARH at the 6.6 kV side.
j. Bus bar KV at the 6.6 kV side.
k. Frequency at the 6.6 kV side.
l. Power Factor at each delivery point, half hour, hourly, daily, monthly at 6.6 KV.
m. Real time measurement at each delivery point for (Current, Voltage, KW, KVAR, Power Factor and Harmonics). The quantities of current, KW, KVAR should be shown bi-directional.
n. Frequency at the MV (6.6 kV) side at the PV Facility.

<u>The following communication hardware must be provided:</u>

a. Remote terminal units and modem
b. DPL interface server and software
c. All requirements to facilitate communication through latest GPRS, wireless communication and fibre optics cable
d. Communication cards for the protection relays
e. The following ports must be provided: (RS232, RS422, RS485 and Ethernet)

<u>The following communication protocols, at least must be provided:</u>

a. DNP3 Serial
b. DNP3 LAN/WAN
c. IEC 60870-5-101
d. IEC 60870-5-104
e. IEC 61850
f. Modbus (ASCII/RTU)
g. OPC AC
h. OPC DA

<u>Alarms</u>

The following alarms in the SCADA system must be provided to the Employer:

(i) Urgent and Non-Urgent and protections alarms for the station transformer and circuit breaker located at the PV Facility.
(ii) Protection Trip.
(iii) Protection Signals.
(iv) Trip signals

<u>General requirements for incorporation in SCADA:</u>

In addition to conventional parameters (V, I, kW, kVA, kVAr etc.) below is the non-exhaustive list of monitoring and recording parameters.

<u>The Plant:</u>

a. String parameters
b. String failure detection
c. Inverters communication bus
d. Low voltage panels (both from field and from electrical room)

e. High Voltage cabinets and positions
f. Fire detection at inverters
g. Water treatment plant
h. General fire detection alarms
i. Power generation at interconnection
j. Daily power generation in kWh
k. Monthly power generation in kWh
l. Annual power generation power in kWh
m. Power generation from the date of commissioning

Plant Performance Ratios (PR):

a. Instantaneous PR (Current value)
b. Day's average PR
c. Month's average PR
d. Quarterly average PR
e. Annual average PR
f. Facility Performance Ratio, since commissioning.

Meteorological:

a. Global horizontal irradiation
b. Global irradiation on tilted plane
c. Ambient air temperature
d. Module temperature
e. Wind speed and direction
f. Precipitation

5.12.13.7 Data Communication System (DCS)

13.7.1 The DCS shall have the following minimum features.

Redundant communication controllers shall be provided to handle the communication between I/O Modules (including remote I/O) and PLCs and between PLCs and operator work station.

13.7.2 The design shall be such as to minimize interruption of signals. It shall ensure that a single failure anywhere in the media shall cause no more than a single message to be disrupted and that message shall automatically be retransmitted. Any failure or physical removal of any station/module connected to the system bus shall not result in loss of any communication function to and from any other station/module.

13.7.3 Built-in diagnostics shall be provided for easy fault detection. Communication error detection and correction facility (ECC) shall be provided at all levels of communication. Failure of one bus and changeover to the standby system bus shall be automatic and completely bump less and the same shall be suitably alarmed/logged.

13.7.4 Data transmitting speed shall be sufficient to meet the responses of the system in terms of displays, control etc. plus 20% spare capacity shall be available for future expansion.

13 .7.5 Contractor shall employ redundant Fiber optic backbone for data communication between Inverter rooms and main control room.

13.7.6 The Contractor shall furnish details regarding the communication system like communication protocol, bus utilization calculations etc.

5.12.13.8 Operator Interface Displays/Logs/Reports

Suitable Operator Interface Displays/Logs/Reports for control operation & monitoring shall be

provided. The details shall be furnished and finalized during detailed engineering stage.

5.12.13.9 Historical Storage and Retrieval System (HSRS)

13.9.1 The HSRS shall collect, store and process system data from man-machine interface system and plant information system (MMIPIS) data base. The data shall be saved online on hard disk and automatically transferred to erasable long-term storage media once in every 24 hours periodically for long term storage. Provision shall be made to notify the operator when hard disk is certain percentage full. The disk capacity shall be sufficient to store at least seven days data.

13.9.2 The data to be stored in the above system shall include alarm and event list, periodic plant data, selected logs/reports. The data/information to be stored and frequency of storage and retrieval shall be finalised during detail engineering stage.

5.12.13.10 Control and Power Supply Scheme

Contractor shall provide the UPS/DC Power supply of suitable rating to cater all the load requirements of SCADA system and its auxiliaries. The power backup for the entire system should be at least for 02 hours.

5.12.13.11 Control Cabinets / Panels / Desks at Main Control Room

The cabinets shall be IP-22 protection class. The Contractor shall ensure that the temperature
rise is well within the safe limits for system components even under the worst condition and
specification requirements for remote I/O cabinets.

The cabinets shall be totally enclosed, free standing type and shall be constructed with minimum 2 mm thick steel plate frame and 1.6 mm thick CRCA steel sheet or as per supplier's
standard practice for similar applications.

5.12.13.12 Software Licences

The Contractor shall provide software license for all software being used in Contractor's System. The software licenses shall be provided for the project and shall not be hardware/ machine-specific.

5.12.13.13 Hardware at Main Control Room

13.13.1 The Hardware as specified shall be based on latest state of the art Workstations and Servers and technology suitable for industrial application & power plant environment.

13.13.2 All the peripherals shall conform to the following minimum requirement but the exact make & model shall be as approved by Employer during detailed engineering. The LAN to be
provided shall support TCP/IP protocol (Ethernet connectivity) with OPC RDI for interface with
PLCs/other systems and shall have data communication speed of min. 100 Mbps. All network
components of LAN and Workstations shall be compatible to the LAN, without degrading its
performance.

- Processor 64 Bit
- Hard disk 1 TB - RAID 1
- Memory 8 GB RAM upgradable to 16 GB
- Monitor Min 22" LED Flat Monitor with non-interfaced refresh rate min. 75 Hz.

Communication port-2 Serial bus, one parallel Dual 10/100/1000 Mbps. Ethernet

- Graphic Memory=16 MB Expansion slot=3
- Screen display unit Min 50" LED Flat Monitor with wall mounted arrangement for the display of SCADA screen
- Removable bulk storage drive(DVD / DAT) 6 GB (minimum)
- Removable Bulk Storage Media for above (with each server/ workstation) 10 nos(minimum).
- DVD R/W 16x or higher Intelligent UPS (on line) with remote monitoring for each
- workstation/ server 1 no. with 30 mins. Battery backup on machine load
- Keyboard ASCII
- Pointing Device Mouse
- Colour Laser Printer Heavy duty type with resolution of 600 dpi Resolution

5.12.13.14 Factory Acceptance Test (FAT)

FAT procedure shall be submitted by bidder for approval. SCADA shall communicate with all
third devices which are part of solar plant and same shall be demonstrated during the FAT.

5.12.13.15 The Detailed Design Report submitted by the contractor to DPL must contain but not limited to the following details of the data acquisition and monitoring system:

- Detailed scheme
- Details of panels, metering system
- Necessary drawings for the scheme etc.

Drawings and scheme submitted by the contractor will be subjected to approval of the owner.

5.12.13.15. CCTV system:

- Bidder shall supply IP based Close Circuit Television (CCTV) System for the total plant area and control room building surveillance.

PTZ (Pan-Tilt-Zoom)/CCTV outdoor/indoor cameras covering the whole plant area (nos. of cameras requirement shall be as per design for well coverage of the plant) and total Inverter cum Control Room building.

- Bidder shall supply CCTV system (with all the items and requisite quantity as mentioned in the relevant technical specification) covering entire Solar Plot area under the scope of the bidder . It shall be centrally monitored from Switchgear cum control room.

- CCTV system shall be powered from UPS ACDB.

- Bidder's scope of supply shall be CCTV system comprising CCTV cameras, CCTV Viewing Station, Management Server, Recording Server, Network Storage Device, L2 Ethernet Switch, OFC, Cat 6 cable, power cables, cable protections, cable conduits & sub-trays, supports, junction boxes, distribution frames, enclosures etc. and grounding connections as required for safe and trouble free operation of IP Based CCTV SYSTEM

- CCTV system network shall be separate from SCADA network.

- Control desk has to be supplied for CCTV monitoring stations, servers, storage device etc. in the control rooms . Also minimum 2 nos. of chairs to be provided for CCTV system control desk in control room.

5.12.14.0 CABLES & CONDUCTOR:

5.12.14.1 SCOPE

The scope of work under these specification covers the Design, Manufacture, Assembly, Shop Testing, Delivery at site, transit insurance, Storage, Erection, Testing & Commissioning of power, control and instrumentation cables (complete with cable terminals and all accessories for making the systems complete and for warranting a trouble free and safe operation).

The scope shall also include supply of all material, fabrication and erection of cable supporting structure, cable trance, cable racks & trays as well as laying of cables on cable racks.

The scope of supply shall also include necessary spares required for a period of 5 (five) years & special tools & plants required for erection & maintenance.

5.12.14.2 STANDARDS

The equipment covered under this chapter shall comply with the requirement of latest edition of following IS/BS/IEC specifications as amended up to date except where specified otherwise.

Sl. No.	Standards	Description
1.	IEC 60529/IEC 60502	All cables that are submerged or in contact with water should be with IP 68 rating.
2.	IS: 7098 – Part 1	Cross linked polyethylene insulated PVC sheathed cables for working voltage up to and including 1.1kV
3.	IS: 7098 – Part 2	Cross linked polyethylene insulated PVC sheathed cables for working voltage from 3.3kV up to and including 33kV
4.	IS 10418	Drums for cables
5.	IS 8130	Conductors for insulated electric cables and flexible cords
6.	IS 8308	Compression type tubular inline connectors for aluminium conductors
7.	IS 8309	Compression type tubular terminals for aluminium conductors
8.	IS 8438	Moulds of cast resin based straight joints of cable up to including 1.1kV
9.	IS 11967	Specifications for co-axial cables
10.	IS : 2062	Structural Steel (Standard Quality)
11.	IS : 513	Cold rolled low carbon steel sheets & strips
12.	IS : 277	Galvanized sheet steel
13.	IS : 808	Rolled Steel Beam, Channels and Angle section
14.	IS : 2629	Recommended practice for hot dip galvanizing of iron and steel.
15.	IS : 2633	Method of testing uniformity of coating on zinc coated articles.

Sl. No.	Standards	Description
16.	IS : 800	Specification for use of structural steel in general building construction.

Cables and other accessories complying with other internationally accepted standards such as IEC, IEEE, BS, etc. will also be accepted if they ensure performance and constructional features equivalent or superior to standards listed above. In such a case the Contractor shall clearly indicate the standard/standards adopted and furnish a copy of English version of the latest revision of the standard(s) along with the Bid and the salient features of comparison shall be brought out.

5.12.14.3 GENERAL REQUIREMENTS

Minimum requirements are mentioned hereunder.

The cables shall be of type and design with proven record of similar power station installations.

The colours of the cables (both AC & DC) should be so selected that there should not be any problem for identification of cables used for various circuits during inspection & testing.

To facilitate easy identification of cores, multi-core control and instrumentation cables shall be colour coded by using PVC insulation of red, black, yellow, blue and grey colours in accordance with IS 1554 (Part I).

Cable lengths shall be considered in such a way that straight through cable joint is avoided Cable terminations shall be made with suitable cable lugs & sockets etc, crimped properly and passed through brass compression type cable glands at the entry & exit point of the cubicles.

The contractor shall ensure that no bimetallic action takes place between the Aluminium conductor of the cable and the cable connecting lugs by filling the lugs with suitable compound.

For the main cable ways, a system of cable racks and trays as well as cable ducts and trenches shall be provided. The power and the control cables will run on separate trays. The cables for emergency lighting, fire alarm systems, etc., shall run on separate trays. The power cables shall be laid on the uppermost rack to prevent spread of fire.

In indoor installations, the cables must be laid through PVC conduit or GI pipe. In case of using metallic pipe as conduit proper grounding of the conduit must be done.

Exposed cables, wherever, used, shall preferable have UV resistant jacket besides being water resistant.

Cables for each equipment must be tagged with permanent metal tag of impregnated cable number as per drawings at MCC/switchgear end and equipment terminal end as well as in the mid portion of the cables at certain distances as instructed by the owner or his authorized representative.

The loop length shall be provided for various cables as per the relevant Indian Standard.

Cables shall be properly clamped at regular intervals with the help of non magnetic/molded fiber glass strip clamps/PVC sleeved clamps, of suitable size.

When power cables are laid in the proximity of communication cables, the minimum horizontal and vertical separation between them may be 300 mm.

5.12.14.4 TECHNICAL REQUIREMENTS FOR CABLES AND CONDUCTOR
Minimum Technical requirements are mentioned below:

All cables and connectors for use for installation of solar field must be of solar grade which can withstand harsh environment conditions including High temperatures, UV radiation, rain, humidity, dirt, burial and attack by moss and microbes for 25 years and voltages as per latest IEC standards. (Note: IEC standards for DC cables for PV systems is under development, the cables of 600 – 1800 volts DC for outdoor installations should comply with the draft EN 50618/ TUV 2PfG 1169/09.07 for service life expectancy of 25 years)

All cables shall be Fire Retardant Low Smoke (FRLS) type. The cables shall be sized based on the following considerations:

- Rated current of the equipment
- The voltage drop in the cable, during motor starting condition, shall be limited to 10% and during running condition, shall be limited to 3% of the rated Voltage

o Short circuit withstand capability De-rating factor for various conditions of installations shall be considered while selecting the cable size

o Variation in ambient temperature for cables laid in air

o Grouping of cable

o Variation in ground temperature and soil resistivity for buried cables

HT cable shall be designed based on the short circuit conditions and LT cable shall be sized based on the voltage drop.

Size of aluminium power cable shall in no case be less than 16 mm^2 and copper power cable shall not be less than 6 mm^2. Where there is requirement of cables less than the above mentioned values, copper cable of appropriate size but not less than 4 mm^2 may be used.

Minimum size of the control cable for CT circuit shall be 4 mm^2 and that for potential circuit shall be 2.5 mm^2.

The cables shall be capable of satisfactory operation under a power supply system voltage variation of ± 10% and frequency variation of ± 5% and a combined frequency voltage variation of 10% (absolute sum). The cables shall have heat and moisture resistant properties.

Conductor size of cables and wires shall be selected based on efficient design criteria. The wiring size of shall be designed such that maximum voltage drop at full power from the PV Array to Inverter(s) should be less than 1.5%. From Inverter to AC Grid interfacing panel should be less than 2.5%.

The continuous withstand temperature shall be 90°C and 70°C for XLPE and PVC cables respectively and the short circuit withstand temperature shall be 250°C and 160°C for XLPE and PVC cables respectively.

The Jointing Boxes shall comply in all aspects with the provision of the latest issue of relevant standards.

The control cables shall be multi-core, colour coded, annealed, stranded high conductivity copper, single conductor, insulated with HR-PVC insulation, PVC

sheathed, unarmoured FRLS type conforming to IS 1554 (part I & II)/relevant IEC. The outer sheath is of specially formulated PVC compound.

The instrumentation cables in addition to meeting the requirements of control cables shall be provided with electrostatic shielding by aluminium tape and screening by annealed tinned copper wire.

For connecting solar modules with solar inverter via array junction box (1.1 kV), solar inverter output with three winding transformer input (1.1 kV), three winding transformer output with 6.6 kV Indoor Switchgear (6.6 kV) and 6.6 kV Indoor Switchgear (6.6 kV) with the 6.6 kV Switchyard (6.6 kV), cables of suitable size shall be provided.

Number of Local Control Rooms with 6.6 kV Indoor Switchgear shall be as per plant design. One number Main Control Room shall be provided adjacent to the 6.6 kV Switchyard. Cable Trench of suitable size as per relevant standard shall be a part of the scope of work.

5.12.14.5 TECHNICAL REQUIREMENTS OF CABLE RACKS AND TRAYS

Minimum technical requirements for cable racks and trays are mentioned below:

The contractor shall fabricate and supply the mounting arrangement for the support and installation of all the cable trays on galvanized steel structure including channels, angles, rods etc at requisite spacing in the suspended cable trays, cable trenches. Supporting structures wherever necessary, shall be provided by the contractor.

The contractor shall provide embedment/anchor fasteners for fixing the supporting structures.

These supporting structures shall be fabricated from structural steel members (channels, angles and rods) of the required size.

The vertical member of the support will be of ISRO12 threaded rod or ISMC100 channel. The horizontal member of the support will be of angle ISA 50X50X6. For the threaded rod support configuration the horizontal member shall be fixed by bolting whereas for channel configuration the horizontal member shall be fixed by welding to the channel.

Trays shall be of ladder type. The trays shall be fabricated from Hot Rolled Carbon Mild Steel (conforming to IS 1079, Grade "O", of chemical composition (C, Si, Mn, S, Ph) sheet of proper thickness as per IS.

Cable trays shall be fixed with support by hold-down clamps. The clamps shall be fabricated from MS sheet of appropriate thickness and Hot Dip Galvanized.

The contractor shall supply various tray fittings and accessories like coupler plate with fasteners, horizontal tees, vertical and horizontal elbows, vertical and horizontal adjustable connectors required for the mentioned trays. All accessories, fittings, elbows and tees shall be Hot Dip Galvanized. The nuts, bolts and washers shall be cadmium plated or electrolytically galvanized.

Proper earthing of the trays and continuity between tray components must be ensured by the contractor.

The contractor shall install the cable trays in accordance with relevant standards.

The cable trays shall conform to bending & galvanization tests as per the relevant standards.

5.12.14.6 APPROVAL

The Detailed Design Report Submitted by the contractor to DPL must contain but not limited to the following details of the Cables and conductor and the accessories for their installation:

- Detailed design and specification of all the items.
- All necessary drawings
- Calculations for choosing cable size
- Type test reports and necessary certificates etc.

Before dispatch, sample pieces of the cable shall be subjected to type, routine, acceptance and FRLS tests at the manufacturer's works as stipulated in IS 1554 (Part I)/IEC in the presence of owner or his representative. Routine tests and acceptance tests as per relevant standards shall be carried out on each type of cable in presence of the owner or his representative.
Before commissioning of complete system all cabling system shall be checked as per cable schedule and complete report shall be prepared by Contractor and shall be submitted.
Prior to the delivery of the product, the contractor shall submit but not limited to the following documents:

- Guarantees
- Cable routing and layout drawings
- Detailed procedure adopted for the earthing of the trays
- Type test certificates for cable trays etc.

The contractor can deliver the product to the site only after receipt of such approval against their prayer in writing from DPL

5.12.15.0 WEATHER MONITORING STATION

Weather Monitoring Station with all the instruments and accessories (with requisite quantity as mentioned below) has to be provided

- As a part of weather monitoring station, Bidder shall provide following measuring instruments with all necessary software & hardware required to integrate with SCADA so as to enable availability of data from meteorological instrument in SCADA. Each instrument shall be supplied with necessary cables, transmitters and accessories (Trackers, Mounting and base stand etc.) provided by OEM of the sensors only.
- Aux. power required by instruments and data logger shall be from UPS only. Data logger shall have provision to receive redundant power supply.
- All the measuring instruments to be supplied shall have valid and traceable calibration certificate.

  Each Pyranometer shall be recalibrated at an interval not more than one year and all other instruments shall be recalibrated at an interval not more than four years.

  During O&M period under bidder, calibration of all the instruments shall be done by the bidder. If calibration of Pyranometer is done at some other places outside the plant, a similar instrument to be provided as temporary replacement until the calibration process is complete.

- Sensors shall be installed at suitable height for which Mast/Structure for the sensor shall be provided by the bidder. Proper fencing shall be provided around meteorological station where the Pyranometer, Wind, Ambient Temp. Sensor, Data logger etc. are installed.
- Single sensor for measuring combination of Wind Speed, Wind Direction, Relative humidity and Rainfall is also acceptable however offered sensor shall meet the specification as mentioned in following sections.

5.12.15.1. Solar Radiation Sensors

Contractor shall provide Solar Radiation Sensors as per specification given in following section.

Contractor has the option to provide these sensors on separate base or on a single base (radiation monitoring station) with tracker, shadow ring and transmitter etc. provided by the OEM.

Calibration certificate with calibration traceability to World Radiation Reference (WRR) or World Radiation Centre (WRC) shall be furnished along with solar radiation sensors. Bidder shall provide Instrument manual in hard and soft form.

Pyranometer:

Bidder shall provide minimum 03 (Three) numbers of Secondary Standard Pyranometers as per ISO 9060 for measuring incident solar radiation as for following:

- Global Horizontal Irradiance (GHI)- 1 Nos.
- Global Inclined Irradiance (GII)-1 Nos.
- Diffuse Horizontal Irradiance (DHI)- 1 Nos.

Technical Requirement of Pyranometer:

Pyranometer shall be Class II or better.

Sl. No.	Description	Requirement
01	Principle	Thermopile

02	Spectral Response	310 to 2800 nm
03	Time response (95%)	Max 15 s
04	Sensitivity	Min 7 micro-volt/w/m2
05	Non linearity	±0.5%
06	Temperature Response	±2%
07	Tilt error:	< ±0.5%.
08	Operating temperature range	0 deg to +80 deg. C
09	Zero offset thermal radiation	±7 w/m2
10	Zero offset temperature change	±2 w/m2
11	Uncertainty (95% confidence Level):	Hourly- Max-3%, Daily- Max-2%
12	Response Time(95% of final value)	<5 sec
13	Non stability	Max ±0.8%

Shadow ring/ball for measuring DHI shall require no regular adjustment for of tracker and shadow ring/ball. Pyranometer shall be shaded throughout the day and shall be exposed to diffuse solar radiation only to provide DHI value without any calculation.

All the Pyranometer have to be mounted at single location at shadow free area. The GII Pyranometer has to be at the same inclination as the angular tilt of module mounting structure.

Bidder shall provide 1 (One) no. Battery powered portable handheld data logger supplied by the OEM of the offered Pyranometer.

5.12.15.2. Temperature Sensors

i) Ambient Air Temperature Sensor (Qty -1 no.)

Sl. No.	Description	Requirement
01	Principle	RTD (Platinum) Resistance proportional to temperature

02	Range	0-50 deg C
03	Accuracy	± 0.2 deg C
04	Operating Temperature	0 to 50 deg C
05	Radiation Shield	Non-aspirated Radiation Shield

ii) Indoor Air Temperature Sensor (Qty – 1 no. at each Inverter room)

Sl. No.	Description	Requirement
01	Principle	RTD (Platinum) Resistance proportional to temperature
02	Range	0-70 deg C
03	Accuracy	± 0.2 deg C
04	Operating Temperature	0 to 70 deg C

iii) Module Temperature Sensor (Qty – 1 no. per 05 MW)

Sl. No.	Description	Requirement
01	Principle	RTD (Platinum) Resistance proportional to temperature
02	Range	0-100 deg C
03	Accuracy	± 0.2 deg C
04	Operating Temperature	0 to 100 deg C

Module temperature sensor shall be fixed on the back of module surface with adhesive or tape without using any mechanical fastener.

5.12.15.3. Wind Sensor

i) Wind Speed Sensor (Qty- 1 no)

Sl. No.	Description	Requirement
01	Principle	Frequency proportional to wind speed/Ultrasonic Sensor
02	Velocity range	0-60 m/ sec
03	Threshold	0.3 m/s
04	Operating Temperature	0 to 50 deg C
05	Accuracy	3% (upto 35 m/s), 5% (Above 35 m/s) RMS

ii) Wind Direction Sensor (Qty- 1no)

Sl. No.	Description	Requirement
01	Principle	Potentiometric type sensor (Resistance proportional to Wind direction) /Ultrasonic Sensor
02	Range	0-360 deg
03	Accuracy	± 5 deg
04	Operating Temperature	0 to 50 deg C

5.12.15.4. Relative Humidity (%) (Qty- 1no)

Sl. No.	Description	Requirement
01	Range	0-100 %
02	Accuracy	± 3 %

03	Resolution	1 %
04	Operating Temperature	0 to 50 deg C

5.12.15.5. Additional Measurement

As per latest regulatory requirement, following measurement for the Solar PV is also included in the scope of bidder.

i. Direct Normal Irradiance (DNI)

ii. Sunrise and Sunset time

iii. Rainfall (mm)

iv. Cloud Cover –(Okta)

Instrument and accuracy for the above-mentioned measurement shall comply with applicable regulation (“Implementation of the framework on forecasting, scheduling and imbalance handling for Renewable Energy (RE) generating stations including Power Parks on Wind and Solar at Inter-State Level”).

5.12.16.0 Data Logger

- Weather Monitoring system shall be provided with standalone Data logger suitable for outdoor application with IP65 Protection and industrial grade hardware suitable for operating temperature up to 55 Deg. C.

- Data logger shall be calibrated and proven in field for at least one year in outdoor environment.

- Data logger shall have following minimum features:

Processor	:	32 bits
Time synchronization	:	With Built in GPS Clock or with Solar SCADA GPS Clock
Data storage	:	Min. 2 MB internal Flash Memory and Min. 8 GB SD card (expandable) for storage of raw and processed data locally at resolution of 1 Second for retrieval whenever required. Data to be stored shall be in unencrypted CSV or equivalent format.
Display	:	LCD screen which should be easily accessible to be provided. It should display relevant details like all sensor values, battery strength, network strength etc. Display shall be for easy maintenance and debugging for site engineer
Scan resolution	:	1 Sec
Analog to Digital Converter (ADC)	:	16 Bit, Sampling -10 Hz (Min)
I/P Channel	:	As required with 20 % spare of each type of channel
Port	:	Provision for RS232 and RS485 serial outputs

- Connectivity and Data transmission:

- Built-in GSM/ GPRS modem for wireless data transmission to SCADA/cloud server (procurement of GPRS enabled SIM Card and connection subscription to be the responsibility of Contractor). It should be possible to remotely communicate with the device for configuration settings.
- Data logger shall be interfaced with Solar SCADA on modbus preferably on TCP-IP for data collection and storage on SCADA
- Web interface with provision for user login to enable viewing and downloading of weather data in XLS/ CSV format
- Communication protocol should support fast data transmission rates, enable operation in different Frequency bands and have an encryption based data security layer for secure data transmission

- Data Logger shall have provision for analog, digital and counter type inputs for interfacing with various type of sensors.
  It shall have facility for arithmetic processing (Time Integration, Simple Average, and Moving Average etc.) of incoming raw data.

- The data logger shall have reliable battery backup and data storage capacity to record all sorts of data simultaneously round the clock.

- Data logger shall be provided with key-locked door access and all the cables (Power and Signal) to the data logger shall be protected with heavy duty HDPE pipes.

- Project file (software, settings and sample reports) shall be handed over to site on permanent storage media (CD/DVD) in two copies after data integrity is verified by site and weather monitoring is commissioned. Any configuration changes shall be possible only with authorized User ID and password.

- Vendor shall submit Factory Acceptance Test (FAT) report and procedure before dispatch of material to site.

5.12.17.0. EXPORT IMPORT ENERGY METER:

3 phase whole current Export Import Energy Meter. The Meter to be supplied must be tested. This Export- Import meter shall be installed to measure the total energy to be imported and exported from the PV power Plant. The export Import Energy meter shall be installed at the following Panels:

- **The export Import Energy meters (Class 2S) shall be installed at Outgoing feeder at 6.6 kV Transformer BCU panel at Switchyard/ or as agreed upon between DPL and Bidder during detailed Engineering at the time of execution of the Project. Summation meter for recording & display of Total generation should also be provided.**

5.12.18.0. ILLUMINATION SYSTEM

5.12.18.1. SCOPE

The scope of work under this specification covers design, manufacture, assembly, shop testing, delivery, site erection, testing & commissioning of Illumination system comprising of main Illumination switchboards, distribution boards, sub distribution boards, switchboards, lighting fixtures, convenience and power outlets, conduits & fittings, cabling, outdoor lighting including mounting structures & poles, lighting for control rooms, security cabin, watch tower, access road, across the peripheral road for Phase I and Phase II (maximum 15 m between two adjacent lamps).

The illumination system shall be designed as per relevant Indian Standard / Guideline for different location of the plant. The lighting arrangement should be LED Based.

The scope of supply shall also include necessary spares required for normal operation & maintenance of illumination equipment for a period of 5 (five) years & special tools & plants required for erection & maintenance. Corresponding parts of all the equipment & spares shall be of the same material & dimensions, workmanship & finish and shall be interchangeable.

5.12.18.2. STANDARDS

The material, equipment and its installation under the scope shall comply with all applicable provisions of the latest Indian standards and codes of practice. Some of the relevant standards are given below:

Sl. No.	Standards	Description
1	IS: 3646	Code of practice for interior Illumination (Part I, II, III)
2	IS: 6665	Code of Practice for Industrial Lighting
3	IS: 732	Code of Practice for Electrical wiring installations
4	IS: 9537	Conduits for Electric installations
5	IS: 2418	Tubular fluorescent lamps for general lighting service
6	EN 61347-2-13	Particular requirements for D.C. or A.C. supplied electronic control gear for LED modules
7	EN 62384	D.C. or A.C. supplied electronic control gear for LED modules
8	EN 61000-3-2	Electromagnetic compatibility (EMC). Limits for harmonic current emissions (Equipment input current < 16 A per phase)
9	EN 61000-3-3	Limitation of voltage fluctuation and flicker in low voltage supply systems for equipment with rated current < = 16 A

The installation shall generally be carried out in conformity with the requirements of Indian Electricity Act 1910 (latest Amendment) & Indian Electricity Rules.

5.12.18.3. REQUIREMENT

The lighting system for outdoor and indoor areas of Solar Power Plant shall be designed in such a way that uniform illumination is achieved.

In outdoor yard equipment / bus bar areas and the peripheral wall are to be illuminated and luminaires shall be aimed for clear view.

5.12.18.4. LIGHTING LEVELS

The complete switchyard shall be lightened with an average illumination level of 100 lux.

Lighting in other areas such as control room, office rooms and battery room & other areas (i.e. street light) shall be such that the average LUX level to be maintained shall be as under:

Sl No.	Area	LUX
1	Control Room and conference rooms	400
2	220 kV, 6.6 kV Switchyard, inside and outside RMU	100
3	Battery & other rooms	150
4	Outdoor areas including peripheral road	20
5	H – pole and metering point	20
6	Equipment room	250
7	Transformer yard	100

5.12.18.5. EMERGENCY LIGHT POINTS

Light points using LED lamps at 220 V shall also be provided as per requirement of the following area:

- All emergency light shall be from 220 V DC Battery.
- Control room and equipment room, Battery room, UPS Room/ Office, Corridor, Local Inverter Room, Main Gate Security Room and any other places where light is required for clear vision.
- These lights shall operate on AC/DC changeover supply from the DC distribution Board. Separate wiring and distribution board shall be provided from these lights.
- Battery room shall be corrosion proof type lamp and fixtures.

5.12.18.6. APPROVAL

The Detailed Design Report submitted by the contractor to DPL must contain but not limited to the following details of the illumination system:

- Detailed scheme and specification

- Illumination calculations for arriving at the number of lighting fixtures for different areas & rooms considering the required lux level as per relevant IS Code.
- Necessary drawings etc.

Drawings and scheme submitted by the contractor will be subjected to approval of the owner.

The contractor can deliver the product to the site only after receipt of such approval against their prayer in writing from DPL.

D. MISCELLANEOUS WORKS:

5.13.0. FIRE PROTECTION SYSTEM

5.13.1. SCOPE

The scope of work under this specification covers design, engineering, quality assurance, manufacture, shop testing, transport, transit insurance, delivery to site, storage at site, site erection, testing & commissioning of fire protection system (fire extinguisher (type shall be selected as per requirement), fire buckets, fire alarms at all control rooms etc.) complete with all accessories.

The scope of supply shall also include necessary spares required for normal operation & maintenance of illumination equipment for a period of 5 (five) years & special tools & plants required for erection & maintenance. Corresponding parts of all the equipment & spares shall be of the same material & dimensions, workmanship & finish and shall be interchangeable.

5.13.2. STANDARDS

All equipment covered under this section will conform to the latest edition of following Indian Standards:

Sl. No.	Standards	Description
1	IS: 3034	Code of Practice for Fire Safety of Industrial buildings: Electrical generating and distributing stations.
2	IS: 3844	Code of Practice for installation of internal fire hydrants in multi-storied buildings
3	IS: 1646	Code of Practice for fire safety of buildings (General) Electrical Installations
4	IS: 2878	Specification for fire Extinguishers – Carbon dioxide type
5	IS: 2171	Specification for fire Extinguishers – Dry Powder type
6	IS: 933	Specification for fire Extinguishers – Foam type
7	IS: 2175	Specification for heat sensitive fire detectors for use in automatic electrical fire alarm system
8	IS: 2189	Code of Practice for installation of automatic fire alarm system using heat sensitive type fire detectors

5.13.3. APPROVAL

The Detailed Design Report submitted by the contractor to DPL must contain but not limited to the following details of the fire protection system:

- Detailed scheme and technical specification
- Placing and type of fire extinguisher with justification
- Necessary drawings related to the system etc.

Drawings and scheme submitted by the contractor will be subjected to approval of the owner.

The contractor can deliver the product to the site only after receiving such approval against their prayer in writing from DPL.

5.14.0.VENTILATION SYSTEM

5.14.1. SCOPE

The scope of work under this specification covers design, manufacture, shop testing, supply, transportation, delivery, storage at site, erection, testing and commissioning of ventilation system complete with all accessories at each Inverter cum control rooms, store room etc.

The Scope shall include supply of all blower fans, GS ducting, air plenum, exhaust fans air dampers etc as required to make the ventilation system complete in all respects for satisfactory operation.

The scope of supply shall also include necessary spares required for normal operation & maintenance of ventilating equipment for a period of 5 (five) years and special tools & plants required for erection & maintenance.

Corresponding parts of all the equipment & spares shall be of the same material & dimensions, workmanship & finish and shall be interchangeable. All the material & workmanship shall be of suitable commercial quality as have proven successful in their respective uses in similar services & under similar condition.

5.14.2.DESIGN:

To prevent the maximum permissible temperature in the inverter room from being exceeded because of internal heat emission of inverters and other auxiliaries in the inverter room, the inverter room in the PV plant shall be adequately ventilated. The Ventilation plant capacity and air quality of inverter room shall be as per inverter and other auxiliary's system manufacturer's recommendations. Filters at the air inlet of the inverter room shall be provided to prevent dust ingress. Bidder shall furnish peak power consumption of cooling system of the PCU along with the data sheet.

Ventilation shall be designed in such a way that the temperature rise of the inverter rooms doesn't exceed the maximum designed temperature of Inverters and other auxiliary equipment's placed inside the inverter room. Accordingly the air velocity through the filter shall be suitably chosen to remove the heat from the inverter room. All exhaust and fresh air fans shall be provided with thermostat control.

5.14.3.STANDARDS

The ventilating equipment shall comply with the requirement of the latest edition of relevant Indian standards or equivalent British Standards. Some of the relevant standards are given below:

Sl. No.	Standards	Description
1	IS : 3103	Code of Practice for industrial ventilation
2	IS : 2312	Specifications for propeller type A.C. Ventilating fans.
3	IS: 4894	Centrifugal fans

5.14.4. APPROVAL

The Detailed Design Report submitted by the contractor to DPL must contain but not limited to the following details of the fire protection system:

- Detailed scheme and technical specification
- Calculations showing air requirements at various locations
- Necessary drawings etc.

The successful bidder required to produce all necessary test certificates and approvals of the product as per relevant standard with the Detailed Design Report.

Drawings and scheme submitted by the contractor will be subjected to approval of the owner. The contractor can deliver the product to the site only after receiving such approval against their prayer in writing from DPL.

- Additional requirement:

  - 100% redundant corrosion proof exhausts fans to be provided at Battery room.
  - Equipment rooms shall be having forced air ventilation.
  - All fans shall be of industrial type.

5.15.0.AIR CONDITIONING SYSTEM

5.15.1.SCOPE

The scope of work under this specification covers design, manufacture, testing, supply, transportation, transit insurance, delivery, storage at site, erection, testing and commissioning of Air conditioning system with control and accessories at the operator's work station, SCADA room and UPS room with 100% redundancy at Control Room building of Phase I and Phase II Plot II. Bidder shall submit heat load calculation before finalization of AC system for control room and conference room considering all parameters

5.15.2 STANDARDS

Equipment shall conform to the latest Indian standards or equivalent British Standards.

Sl. No.	Standards	Description
1	IS: 659	Safety code for Air conditioning
2	IS: 660	Safety code for Mechanical Refrigeration
3	IS: 655	Metal Air ducts

5.15.3. APPROVAL

The successful bidder required to produce all necessary test certificates and approvals of the product as per relevant standard with the Detailed Design Report.

The Detailed Design Report submitted by the contractor to DPL must contain but not limited to the following details of the Air Conditioning system:

- Detailed scheme and technical specification
- Necessary drawings etc.

Drawings and scheme submitted by the contractor will be subjected to approval of the owner. The contractor can deliver the product to the site only after receipt of such approval against their prayer in writing from DPL.

5.16.0. DRINKING WATER

5.16.1. SCOPE

The scope of supply under this section shall cover the design, manufacture, shop testing, supply, transportation, delivery, storage at site, erection, testing and commissioning of drinking water system with Industrial grade water purifier unit and other related plumbing arrangement and pipes and accessories etc. for drinking water supply for the personnel at the Control Buildings of Phase I and Phase II area. A drinking water point will be provided (within 500m from control building) and the contractor to draw pipelines to the requisite location.

5.16.2. STANDARDS

The whole system shall conform to the latest edition of relevant Indian Standard.

5.16.3. APPROVAL

The Detailed Design Report submitted by the contractor to DPL must contain but not limited to the following details of the water purification unit:

- Detailed Technical specification
- Necessary drawings etc.

Specification submitted by the contractor will be subjected to approval of the owner. The contractor can deliver the product to the site only after receiving such approval against their prayer in writing from DPL.

5.17.0. SIGNAGE:

5.17.1. Project information Signage:

The Signage will be made up of metallic base of minimum size 3'x 2'. The Signage provide with detail of the project as approved by DPL. The font size on the signage has to be big enough so that everyone can read it easily. The Signage will be fixed **up two (02)** prominent place of the project area.

5.17.2. SCHEMATIC DIAGRAM SIGNAGE:

Schematic Diagram of Installation must be provided on a display board of minimum size 3'x 2' made up of metallic base. The schematic diagram must be fixed up at any prominent place of installation.

5.17.3. SAFETY SIGNAGE:

Safety Signage must be provided indicating the level and type of voltage and symbols as per IE Rule at different position as may be required. In the safety signage Voltage level and type of voltage must be mentioned

Each set of Safety Signage contain minimum 06 (six) nos. safety signage:

Location	Quantity
PV Array Field	Minimum 2 nos.
On PV Array JB each	01 No. (Sticker)
Near each Inverters	01 No
On Inverter Interfacing HT Panel	01 no. (Sticker)
On Switchyard	01 No. (Sticker)

5.18.0.FIRE BUCKETS AND HOLDING STANDS

Each set of Fire Buckets and Fire Bucket Holding Stand shall have four (04) Fire Buckets and one (01) Double Tier Fire Bucket Holding Stand with an arrangement of holding of minimum four (04) Fire Buckets. The Fire Bucket Stand shall be installed at the suitable location.

The minimum technical specification is a follows:

BIS Specification	IS 2546 (with latest amendments)
Fire Bucket Capacity	10 liters
Fire Bucket Body Material	Mild Steel Sheet

5.18.1. FIRE EXTINGUISHERS

Minimum three (3) nos DCP type dry power portable fire extinguishers of minimum capacity 5 kg shall be provided at each Control room, Switchgear & Inverter room. Standard of Fire Extinguisher BIS 2171 (with latest amendments). Contractor shall periodically replace this Fire Extinguisher during their Comprehensive O&M contract period.

5.19.0 KIOSK

Some small outdoor Panels (if required) shall be installed in suitable locations in Metallic Kiosks. The kiosks of must be of a suitable design, painted for avoiding rust, covered with a door and locking arrangement with good air circulation. The Kiosks must have security arrangement against theft, manhandling etc. Minimum clearance of the lower edge of the equipment from floor should be 1.0 m. There must be a suitable clearance from the front door of the Kiosk with any of the equipment / panel within the Kiosk. Panel area shall be well illuminated.

Necessary civil work as required to construct / fixing the Kiosks (s) shall be done by the contractor. If any civil construction is required for installing the whole arrangement, as and where required, it will be within the cost of contract value.

5.20.0. TOOLS, TACKLES AND SPARES

The Contractor shall supply and keep ready stock of tools, tackles and essential spares at site, that will be needed for the day-to-day maintenance of the solar PV system. This shall include but not be limited to the following:

i. Screw driver suitable for the junction boxes and combiner boxes.
ii. Screw driver and / or Allen key suitable for the connectors, power distribution blocks, Circuit breaker terminals and surge arrestor terminals.
iii. Spanners / box spanners suitable for the removal of solar PV modules from the solar PV module support structure.
iv. Solar panel mounting clamps.
v. Cleaning tools for the cleaning of the solar PV modules.
vi. Panel efficiency measurement tools

These list is not exhaustive, all other tools and instruments required for completion of Project and successful execution of comprehensive O&M are also to be supplied. Month wise ready stock report shall be submitted to DPL site authority.

5.21. Robotic Dry Cleaning

5.21.1 Scope

The scope of work under this specification covers design, engineering, quality assurance, manufacture, shop testing, transport, transit insurance, delivery to site, storage at site, site erection, testing & commissioning of a semi-automated/automated robotic dry cleaning system complete with all accessories and suitable for cleaning of 50 MW_p of PV modules.

The scope of supply shall also include necessary spares required for normal operation & maintenance of the robotic dry cleaning system for a period of 5 (five) years & special tools & plants required for erection & maintenance. Corresponding parts of all the equipment & spares shall be of the same material & dimensions, workmanship & finish and shall be interchangeable.

The respective technical measures planned by the Contractor shall be described in detail in the Detailed Design Report.

Remark: Bidders shall offer standard robotic cleaning systems available on the national/international market, suitable for high wind speeds (170 km/h or higher). After the conclusion of the EPC contract, the successful bidder shall verify in close cooperation with the selected manufacturer of the robotic dry cleaning system whether measures can be taken to increase the resilience of the systems at wind speeds typical for the site (280 km/h). Based on the cost/effort ratio, WBSEDCL will decide whether these systems shall ultimately be used.

5.21.2 Standards

All materials and equipment shall conform to latest edition of relevant Indian/IEC Standards unless otherwise specified. Equipment conforming to any other authoritative standard ensuring equal or better quality than the standards indicated below will also be acceptable. However, in such eventuality, the salient points of difference between the standards adopted and the standards mentioned below shall be brought out by the Bidder. The list of reference standards is given below:

Sl. No.	Standards	Description
1.	IEC 61724-1: 2017	Photovoltaic system performance Part 1: Monitoring
2.	IEC 62446-2:2020 ED1	Photovoltaic (PV) systems - Requirements for testing, documentation and maintenance - Part 2: Grid connected systems - Maintenance of PV systems

A complete set of documents and standards in English shall be supplied by the Bidder without any extra charge. It shall, however, be ensured that equipment offered comply with one consistent set of standards except in so far as they are modified by the requirement of this specification.

5.21.3 Technical Requirements

a. All offered system components and materials must be suitable for operation under the extreme climatic conditions prevailing at the site (refer to chapter 2.3).

b. The specific soiling present at the site needs to be well understood and characterized by the Contractor.

c. As there is no identified water source nearby which can be used for cleaning of PV modules, robotic dry-cleaning of solar modules is to be applied to 50 MW_p of the solar modules.

d. The weight of the cleaning device on top of the PV module must be within the allowable loads given by the PV module manufacturer.

e. The PV module manufacturer should provide explicit approval for each cleaning system to be deployed, including confirmation that warranty coverage of the PV module will not be voided by use of the system.

f. The Contractor must ensure that no abrasion of the PV module´s anti-reflective coating and scratching of the glass surface will occur. To evaluate the abrasion risk, cleaning system suppliers must use an independent third party to test the devices.

g. No cell or PV module breakage or damage must occur which could impair the safety, mechanical or electrical integrity of PV modules.

h. The selected dry-cleaning system must be technically robust for the respective operating environment. No sensitive parts should be exposed in desert conditions.

i. Minimal maintenance of the cleaning device shall be required. Ideally unskilled, low-cost labor can be used to perform maintenance on the cleaning devices. Maintenance should be easy, fast and require few tools.

j. In the case of semi-automated, autonomous and fully-automated devices batteries are to be supplied with the cleaning devices. The batteries are to be provided with charging infrastructure that is permanently installed. Ideally this charging infrastructure shall be made via solar modules that are permanently installed. The number of hours required for battery charging and the battery lifetime must be considered and clearly stated in the technical bid. In this case, remote battery status monitoring is required.

k. The cleaning system supplier must provide operator training on the correct use of the system.

l. Given that the economic and operational life-time of the SPVPP is expected to be at least 25 years, the cleaning system supplier also needs to guarantee the supply of spare parts for the same time period.

m. All cleaning devices must be supplied with a safe storage location that the cleaning devices can be placed in or to which the cleaning devices move automatically when not in use.

5.21.4 Approval

The Detailed Design Report submitted by the contractor to the Employer must contain but not limited to the following details of the robotic dry-cleaning system:

a. Detailed scheme and technical specification

b. Test certificates and approval of PV modules supplier and independent testing institute

c. Necessary drawings etc.

d. Operation and maintenance plan

The Contractor is required to produce all necessary test certificates and approvals of the product as per relevant standard with the Detailed Design Report.

Drawings and scheme submitted by the contractor will be subject to approval by the Employer. The Contractor can deliver the product to the site only after receiving such approval in writing from the Employer.

5.22.0. OTHER CONDITIONS

The work includes necessary excavation, concreting, flooring, platform, necessary finishing, painting, back filling, shoring & shuttering, cable laying, location of installation of different component of PV Power Plant etc. if any , required for completion of the project in all respect shall be as per direction of Engineer-in-Charge.

NOTE:

Any item/equipment not mentioned in the Technical Specification, but required for successful completion of the project shall be deemed to be a part of the scope of the work and the same shall be included by the bidder in their Billing Break Up (BBU).

E. Approved List of Vendors

5.23.0. APPROVED VENDOR LIST FOR BOIs:

Equipment	List of Vendor for various BOIs
SOLAR PANELS	Any Solar PV Manufacturer in India (domestic manufacturer only for module and cell) having MNRE Certification (as per ALMM and its all terms and conditions)
INVERTERS	ABB/ HITACHI/SMA/ DELTA /SUNGROW
TRANSFORMER	BHEL / GE / AREVA T & D INDIA LIMITED NAINI /ABB
INVERTER AND AUX, TRANSFORMER	SUDHIR/VOLTAMP/BHEL/AREVA
POWER CABLES	KEI/ FINOLEX /POLYCAB /APAR
CONTROL CABLES	KEI / DELTON/ FINOLEX /POLYCAB /APAR
LT SWITCHGEAR	L&T / SIEMENS / SCHNEIDER/ABB
STEEL MEMBERS	TATA / VIZA STEEL/SAIL (GI coating done) or any equivalent ISI Mark
EARTHING/ LIGHTNING	CG Power/ELPRO INT. LTD/OBLUM
WEATHER MONITORING STATION	KIPP & ZONNEN / EPPLEY / EKO INSTRUMENTS /SOLAR L /GREEN POWER Monitoring
LT POWER PANEL	L&T / SIEMENS / SCHNEIDER/GE POWER

Equipment	List of Vendor for various BOIs
JUNCTION BOX	L&T / PYROTECH / SCHNEIDER
ENERGY METER	SECURE METERS/RISHABH(L&T)
PLC and SCADA System	BHEL, ABB, SCHNEIDER, GE, ROCKWELL AUTOMATION (ALLENBRADLEY), SIEMENS
6.6 kV INDOOR SWITCHGEAR	BHEL / SCHNEIDER / SIEMENS
UPS	EMERSON / HITACHI-HIREL / MERLINEGERINE / AEG (SAFT)
220 V DC Battery	EXIDE
220 V DC Battery Charger	CHLORIDE POWER SYSTEMS AND SOLUTIONS LTD. / STATCON POWER LTD./ CHHABI / HBL POWER
UPS BATTERY (Ni-Cd)	HBL POWER SYSTEMS LTD.
INDUSTRIAL PC ALONGWITH CRT (EWS/ OWS/ SERVER/ HISTORIAN)	DELL / HP (Pavilion)
PRINTER	HP
ETHERNET SWITCH	CISCO / MOXA
CONTROL DESK/ LOCAL PANEL	PYROTECH / RITTAL / SCHEINDER / CONTROL & SWITCHGEAR / CONTROL DEVICE / SWITCHING CIRCUIT
OPTICAL FIBRE CABLE	TYCO / MOLEX / BIRLA ERICSSION / HFCL
CCTV CAMERA	BOSCH/ PELCO/ HONEYWELL
VCB	SIEMENS LTD. / ABB INDIA LTD. / SCHNEIDER ELECTRIC / TOSHIBA T&D SYSTEMS (INDIA) / CG POWER & INDUSTRIAL SOLUTION LTD. (FORMERLY KNOWN AS CROMPTON GREAVES

Equipment	List of Vendor for various BOIs
	LTD.)
ISOLATOR	SIEMENS / ELPRO
PLC BASED PANEL	PYROTECH / RITTAL
INSTRUMENTATION CABLE	UNIVERSAL CABLES/ DELTON/ RELIANCE/ NICCO/ POLYCAB/ KEI INDUSTRIES/ CORDS
COUPLING RELAY	ELESTA, OEN, OMRON, MAGNETRUFF, ROCKWELL
TERMINAL BLOCK	WAGO/ WIED MULER/ PHOENIX

F. List of Mandatory Spares

1.00.00	Inverter Transformer (33/0.38 kV)
1.01.00	Bushing for each type and rating

(i)	HV Bushing with metal parts, connectors and gaskets	1No.	
(ii)	LV bushing with metal parts, connectors and gaskets	1No.	
(iii)	Neutral bushing with metal parts, connectors and gaskets	1No.	
(iv)	CT at Transformer each type and rating	1No.	
1.02.00	6.6 kV Surge Arrestor complete with insulating base and surge monitor	1Set.	
1.03.03	6.6 kV CT	1No.for each type, rating &application	
1.04.00		6.6 KV Switchyard	
1.04.01		6.6KV VCB Breaker	
(i)	Closing Coil with resistor	6Nos.	
(ii)	Tripping Coil with resistor	12Nos.	
(iii)	Breaker Auxiliary Contact	2Sets	
(iv)	Set of O-ring	3Sets	
(v)	Set of Seals	3Sets	
1.04.03	6.6kV CT	1No.for each type, rating & application	
1.04.04		6.6 kV Switch yard Protection and Substation Automation System	
(i)	Numerical Relays	1 No of each make, type & range &rating.	
1.05.00		6.6 kV INDOOR SWITCHGEAR	
1.05.01	Pole of breaker of each type & rating	1 set (1set is complete for 3 phases)	
1.05.02	Spring charging motor complete	2 nos of each type	
1.05.03	Trip coil	5 nos of each type	
1.05.04	Closing Coil	3 nos of each type	

1.05.05	Current transformer	1 nos of each type and ratio	
1.05.06	Fuse for Potential transformer of each type & ratio	3 no of each type	
1.05.07	Relays	2 no of each type	
1.05.08	Limit switches of each type	5 nos.	
1.05.09	Operating mechanism rod for each rating	2 nos	
1.05.10	Ammeter of each type & range	1 no of each type & range	
1.05.11	Voltmeter of each type & range	1 no of each type & range	
1.05.12	Indicating lamps	3 nos. each type	
1.06.00		Inverter	
1.06.01	Control Unit	2 nos. Each Type and Rating and make.	
1.06.02	Protection Switch	2 nos. Each Type and Rating and make.	
1.06.03	Control Panel	2 nos. Each Type and Rating and make.	
1.06.04	AC Fuse	5 nos. Each Type and Rating and make.	
1.06.05	Fuse link	2 nos. Each Type and Rating and make.	
1.06.06	I/O module	2 nos. Each Type and Rating and make.	

1.06.07	AC Breaker	1 no. Each Type and Rating and make.	
1.06.08	AC and DC Contactor	2 nos. Each Type and Rating and make.	
1.06.09	ARRESTER	2 nos. Each Type and Rating and make.	
1.06.10	FAN	2 nos. Each Type and Rating and make.	

1.06.11	SPD	2 nos. Each Type and Rating andmake.	
1.07.00		PV Module	
1.07.01	PV Module	100 nos each type and rating.	
2.00.00		**CONTROL & INSTRUMENTATIONITEMS**	
2.01.00	**Control System**		
2.01.01	**Multifunction Processor Unit**	**10% of total nos. used in the systemor minimum 1(one) no. whichever is more.**	
2.01.02	**Binary Input Module**	**10% of total nos. used in the system or minimum 2(two) nos. whicheveris more.**	
2.01.03	**Pulse Input Module**	**10% of total nos. used in the systemor minimum 1(one) no. whichever is more.**	
2.01.04	**Analog Input Module (4 to20mA input type)**	**10% of total nos. used in the system or minimum 2(two) nos. whicheveris more.**	
2.01.05	**Analog Input Module (Thermocouple input type)**	**10% of total nos. used in the systemor minimum 1(one) no. whichever is more.**	
2.01.06	**Analog Input Module (3-wireRTD input type)**	**10% of total nos. used in the systemor minimum 1(one) no. whichever is more.**	
2.01.07	**Analog Output Module (4 to20mA output type)**	**10% of total nos. used in the systemor minimum 1(one) no. whichever is more.**	
2.01.08	**Pulse Output Module**	**10% of total nos. used in the systemor minimum 1(one) no. whichever is more.**	
2.01.09	**Binary Output (contact) Module**	**10% of total nos. used in the system or minimum 2(two) nos. whichever is more.**	
2.01.10	**Binary Output (Voltage) Module**	**10% of total nos. used in the system or minimum 2(two) nos. whichever is more.**	

2.01.11	**Interposing Relays**	**10% of total nos. used in the system or minimum 5(five) nos. whichever is more.**	
2.01.12	**Output Relay modules/ Relay Board & Auxiliary Relay**	**10% of total nos. used in the system or minimum 1(one) no. whichever is more.**	
2.01.13	**Rack Mounted Power Supply Unit (SMPS or otherwise)**	**10% of total nos. used in the system or minimum 1(one) no. whichever is more.**	
2.01.14	**MCB (Moulded case circuit breaker)**	**10% of total nos. used in the system or minimum 2(two) nos. whichever is more for each type and rating.**	
2.01.15	**Racks for housing I/O & Processor Modules**	**1(One) no. each type used in the system**	
2.01.16	**Bus extension Module**	**10% of total nos. used in the system or minimum 1(one) no. whichever is more.**	
2.01.17	**Bus end module**	**10% of total nos. used in the system or minimum 1(one) no. whichever is more.**	
2.01.18	**Bus extension cable with end connectors.**	**1(one) no. each type used in the system**	
2.01.19	**Termination Unit/Card**	**1(one) no. each type used in the system**	
2.01.20	**Backup cable for Redundant Processor Modules with end connectors**	**1(one) no. each type used in the system**	
2.01.21	**Network communication cable with end connectors (from the Processor module)**	**10% of total nos. used in the system or minimum 2(two) nos. whichever is more for each type.**	
2.01.22	**I/O Connector with prefab cable**	**10% of total nos. used in the system or minimum 2(two) nos. whichever is more for each type.**	
2.01.23	**Communication Processor /**	**10% of total nos. used in the system or minimum 1(one) no. whichever**	

	Card	**is more for each type.**	
2.01.24	**Network Interfacing card**	**10% of total nos. used in the system or minimum 1(one) no. whichever is more.**	
2.01.25	**Signal Isolator Module/Card**	**10% of total nos. used in the system or minimum 1(one) no. whichever is more.**	
2.01.26	**Any other system specific Module/Cards used in the system but not mentioned in this list**	**10% of total nos. used in the system or minimum 1(one) no. whichever is more for each type and rating.**	
2.01.27		Network Items for Control System	
i)	Managed Switch	10% of total nos. used for each type and model in the system or minimum 1(one) no. whichever is more.	
ii)	Un-managed Switch	10% of total nos. used for each type and model in the system or minimum 1(one) no. whichever is more.	
iii)	Transceiver (Optical to UTP converter)	10% of total nos. used for each type and model in the system or minimum 1(one) no. whichever is more.	
iv)	Fibre-optic Patch Cords	10% of total nos. used for each type and model in the system or minimum 1(one) no. whichever is more.	
v)	Fibre-optic Patch Box/Patch Panel	10% of total nos. used for each type and model in the system or minimum 1(one) no. whichever is more.	
vi)	Other associated Hardware	10% of total nos. used for each type and model in the system or minimum 1(one) no. whichever is more.	

vii)	RS232 to RS485/RS422 Converter or any other media Converter (if applicable)	10% of total nos. used for each type and model in the system or minimum 1(one) no. whichever is more.	

Bidder shall arrange and supply all the items during execution of the project to their site store and under Bidder's custody. Finally, all the above items shall be handed over to DPL store after completions of defect liability period i.e. after the end of the O&M contract.

SECTION – VI

FORMS

Sl No	Form Name	Form No
01.	Check List	Form-1
02.	Forwarding Letter for submission of Bid Security and Tender Fee	Form -2
03.	Bid Form	Form-3
04.	Bid security (Bank Guarantee format)	Form- 4
05.	Summary Statement Of Yearly Turnover And Net Worth	Form-5
06.	Capability status	Form-6
07.	Statement of similar type of order orders executed as on date of issuance of the NIT	Form-7
08.	Curriculum Vitae Of Key Personnel	Form-8
09.	Format For Submission Of Pre-Bid Queries	Form- 9
10	Proposed modifications	Form-10
11	Joint Venture/Consortium Agreement	Form-11
12	Power of Attorney	Form-12
13	Declaration for Net Minimum Guaranteed Generation	Form-13

FORM-1: CHECK LIST: FORM

1.	Tender Fee **(Scanned copy)**	Drafts	NA		
2.	Bid Security **(Scanned copy)**	Drafts	Statutory Cover (Technical		
3.	Check List **(Form – 1)**	Forms	Statutory Cover (Technical		
4.	Forwarding Letter for submission of Bid Security **(Form – 2)** (**Scanned copy)**	Forms	Statutory Cover (Technical proposal)		
5.	Bid Form/Undertaking **(Form – 3)**	Forms	Statutory Cover (Technical		
6.	Summary statement of yearly turnover and net worth **(Form 5)**	Forms	Statutory Cover (Technical proposal)		
7.	Capability Status **(Form 6)**	Forms	Statutory Cover (Technical		
8.	Statement of similar type of order orders executed as on date of issuance of the NIT **(Form 7)**	Forms	Statutory cover (Technical proposal)		
9.	Curriculum Vitae of Key Personnel **(Form 8)**	Forms	Statutory Cover (Technical		
10.	Net Minimum Generation Guarantee **(Form 13)**	Forms	Statutory Cover (Technical		
11.	Copy of the GST / VAT / TIN Certificate	Certificates	Non-Statutory cover (Technical		
12.	Copy of the Service Tax Registration Certificate	Certificates	Non-Statutory cover		
13.	Copy of the PAN certificate/ PAN Card	Certificates	Non-Statutory cover (Technical		
14.	Declaration of PF Registration Number **or** Proof of PF Registration	Certificates	Non-Statutory cover (Technical		

15.	MNRE Chanel partner certificate under *"Grid Connected Ground Mounted and Small Solar Power Plants Programme"* / Website	Certificates	Non-Statutory cover (Technical proposal)		
16.	Copy of the Registration Certificate under Company Act (Company Incorporation Certificate) o**r** copy of the Registered Deed for Partnership Firm	Company Details	Non-Statutory cover (Technical proposal)		
17.	Copy of the Order(s)/ Contract Agreement(s) with the Purchaser / any other Proof of Purchase, as primary agency **AND** Corresponding Copy of the Completion Certificate(s) /Commissioning report signed by the Purchaser / Ordering Authority to substantiate the proof of	Credential (Technical)	Non-Statutory cover (Technical proposal)		
18.	Copy of the Audited Balance Sheet and Statement of Profit and Loss Account / Tax Audit report as per NIT	(Financial)	Non-Statutory cover (Technical proposal)		
19.	Copy Income Tax return Acknowledgement for assessment years as per NIT	(Financial)	Non-Statutory cover (Technical		
	Finance Proposal				
20.	BOQ (Financial proposal)	Bill of Quantities (BOQ)			
21.	Mode Of Transaction Statement Of Materials and Equipment	Mode of Transaction	Financial Proposal		

Date : (Printed Name)...

Place : (Designation)..

Signed and Upload

FORM-2: FORWARDING LETTER FOR BID SECURITY AND TENDER FEE

Date:

Bidder's Name and address

To

The General Manager (O&M)

The Durgapur Projects Limited

Durgapur – 713201, W.B.

Subject : Design & Engineering, Manufacture / Procurement, Supply, Erection, Testing and Commissioning of 7 MW Grid Connected Ground Mounted Solar Photovoltaic Power Plant in The Durgapur Projects Limited (DPL) , Durgapur, West Bengal including warrantee obligation with 05 (Five) years comprehensive Operation and Maintenance.

Reference : NIT No:

Dear Sir,

We are enclosing the following:

1. Account payee Demand draft / Bankers Cheque (No. ……[insert No.]…………. dated …………… [insert date]………. drawn on …………….[insert name of the bank on which drawn]………… for …….. [insert amount in Rs. and words]………., drawn in favor of" **The Durgapur Projects Limited(DPL)**." payable at Kolkata **towards Tender Fee.**
2. Account payee Demand draft / Bankers Cheque (No. ……[insert No.]…………. dated …………… [insert date]………. drawn on …………….[insert name of the bank on which drawn]………… for …..[insert amount in Rs. and words]………., drawn in favor of "DPL" payable at Kolkata **towards Earnest Money (Bid Security) Deposit.**

Or

Bank Guarantee in the prescribed proforma from a scheduled commercial bank in India in favour of DPL for Rs. ……[Insert value]…………../-valid for a period of one hundred and eighty days (180) days from the bid opening date with a further claim period of thirty (30) days.

[*Strike out whichever is not applicable*].

Enclosures:

1. …………………………… (Tender fee : DD / BC in original)
2. …………………………… (Bid security : DD / BC / BG in original)
3. ……………………………

Thanking You,

……………………………

(Signature of authorized signatory)

Name: ……………………………

Designation: ……………………………

Date: ……………………………

Place: ……………………………

FORM-3: BID FORM/UNDERTAKING

(To be executed on non-Judicial stamp paper of requisite value)

(For genuineness of the information furnished on-line and authenticity of the documents produced before Tender Committee for verification in support of his eligibility)

Reference : NIT No:

I, the undersigned, being the authorized signatory of ………………………(Name of the Bidder), having read and examined in detail the NIT including minimum eligibility criteria in particular, instruction to Bidders, general terms & conditions, special terms & conditions and specification, do hereby submitting our offer to execute the contract as per terms & conditions as said forth in your Tender document.

1. We are submitting Tender for the Work _______________against Tender NIT. No. ____

2. We confirm having submitted the eligible criteria as required by you in your Tender Document along with this proposal. In case you require any further information or clarification in this regard, we agree to furnish the same in time.

3. We have submitted the requisite amount of Earnest Money (Bid Security) of **……….(Bid Security Amount)……….. (Rupees. In Word………….)** only in form of …………. (DD/BC/BG). Detailed as follows :

Mode of Financial Instrument (DD/DC/BG)	Details (No., Name of the Bank etc.)

4. We hereby furnish the following:

1.	Company / Partnership/LLP regeneration i)Registration No: ii) Place of registration:	
2.	i) Central Sales Tax Regn. No: ii)VAT Regn. No:	
3.	Excise Regn. No	
4.	Service Tax Regn. No	
5.	PAN No	
6.	PF A/C No	
7.	Channel partner Certificate No of MNRE Government of India under **"Grid Connected Ground Mounted**	

	and Small Solar Power Plants Programme".	

5. Our contact details related to this tender are as follows:

Information	**Local office (In West Bengal)**	**Head office**
Name of the Contact Person		
Designation		
Telephone No		
Fax No		
Mobile No		
Email Address		

6. We confirm that our bid in response to the NIT is consistent with all the requirements of submission as stated in the Tender Document and subsequent communications from DPL.
7. We confirm none of the Partners of our firm is relative of employee of The Durgapur Projects Limited (DPL) .
8. All information furnished by us in respect of fulfillment of eligibility criteria and qualification information of this Tender is complete, correct and true.
9. All documents/ credentials submitted along with this Tender are genuine, authentic, true and valid.
10. If any information and document submitted is found to be false/ incorrect any time, department may cancel my Tender and action as deemed fit may be taken against us, including termination of the contract, forfeiture of all dues including Earnest Money and banning / delisting of our firm and all partners of the firm etc.

11. Should this Bid be accepted, I/We* also agree to abide by and fulfill all the terms and conditions of provisions of the above mentioned Bidding Documents.

12. We have neither made any statement nor provided any information in this Bid, which to the best of our knowledge is materially inaccurate or misleading. Further, all the confirmations, declarations and representations made in our Bid are true and accurate.

13. We declare that the submitted our offer is without any deviations and are strictly in conformity with the documents issued by DPL.

14. We declare that content of the Tender Document including NIT, ITB, BDS ,GCC, SCC, Technical Specification and subsequent corrigendum, addendum, if any, are acceptable to us and we have not taken any deviation in this regard. This is to expressly certify that our offer contains **no deviation** either in direct or indirect form.

15. We also declare that in case any deviations are noticed which might have crept inadvertently, that such deviations without reservation of any kind are automatically deemed to have been withdrawn by us.

16. If you accept our offer, we agree to complete the entire work in accordance with work completion time given in the Tender document. We fully understand that the work completion time stipulated in is the essence of the contract, if awarded.

17. We offer to execute the work in accordance with the conditions of the NIT document as available in the website.

18. This Bid and your subsequent Letter of Acceptance / Work Order /agreement shall constitute a binding contract between us.

19. We hereby confirm our acceptance of all terms and conditions of the NIT document unconditionally.

20. We also declare that, we have never been blacklisted and / or there were no debarring actions against us as on date due to any reason what-so-ever, by any Government or Government Agencies. In the event of any such information pertaining to the aforesaid matter found at any point of time either during the course of the contract or at the bidding stage, our bid/contract will be liable for

truncation / cancellation / termination without any notice at the sole discretion of DPL.

Date : (Printed Name)…………………………………………

Place : (Designation)……………………………………………

Signed and Upload

FORM-4: BID SECURITY

FORM FOR BID SECURITY

BANK GUARANTEE FORMAT FOR EMD/ BID SECURITY

(To be stamped in accordance with Stamp Act, if any, of the Country of the issuing Bank)

Bank Guarantee No.: __________________

Date: ______________________

To

The General Manager (O&M)

The Durgapur Projects Limited

Durgapur – 713201, W.B.

Dear Sir,

In accordance with your NIT No. ___________M/s XXX (Name of Participating Contractor) having its Registered Head Office at ___________ (hereinafter called the Bidder) wish to participate in the said RFP/NIT for ____________________ (Name of Job).

As an irrevocable Bank Guarantee against Bid Security for an amount of ____ is required to be submitted by the Bidder as a condition precedent for participation in the said Tender, which amount is liable to be forfeited on the happening of any contingencies mentioned in the Tender Document, we, the ______________________ Bank at ________________ having our Head Office / Registered Office at _________________ (Address of Bank) guarantee and undertake to pay immediately on demand by the The Durgapur Projects Limited (DPL) the amount of __________ (in words and figures) without any reservation, protest, demur and recourse. Any such demand made by said Procuring Entity shall be conclusive and binding on us irrespective of any dispute of difference raised by the Bidder.

This Guarantee shall be irrevocable and shall remain valid up to *Date*___________ (six months from the Closing date of submission of bid) with a claim period of another 3(three) months. If any further extension of this guarantee is required, the same shall be extended to such required period on receiving instructions from M/s XXX (Participating Bidder) on whose behalf this Guarantee is issued.

All rights of the The Durgapur Projects Limited (DPL) under this Guarantee shall be forfeited and the Bank shall be relieved and discharged from all liabilities there under unless the DPL brings any suit or action to enforce a claim under this Guarantee against the Bank within ninety (90) calendar days from the above mentioned expiry date of validity or, from that of the extended date.

In witness whereof the Bank, through its authorised Officer, has set its hand and stamp on

this __________ day of __________ Year ……………. at ____________.

WITNESS:

____

(Signature and Name) (Signature and Name)

__________________________ __________________________

(Engineer / Officer address) (Designation with Bank Stamp)

Attorney as per Power of Attorney No. ___________________________

Date:

FORM-5: SUMMARY STATEMENT OF YEARLY TURNOVER AND NET WORTH

NIT No:

Bidder's Name & Address:

This is to certify that the following statement is the summary of the Audited /Tax audited Accounts of our Company/firm (The Bidder) arrived for the three consecutive years or for such period since inception of the Firm, if it was set in less than such three year's period as follows:

Sl.	**Financial**		
		Turnover rounded up to in lakh	
1.	20.. - 20..		
2.	20… - 20…..		
3.	20… - 20…..		
5.	Net Worth as at last financial Year		

Note:

1. Average turnover is to be expressed in rupees in lakh, rounded up to two digits after decimal.
2. The statement must be the individual bidder's turnover and not the consolidation as the result of accounts of group and associates.
3. Average turnover for 3 years is to be obtained by dividing the total turnover by 3.0. If the bidder was set up in less than 3 year's period, consider the turnover for the period from inception to the Year-1. It may be either 1.0 or 2.0. Average turnover is to be obtained by dividing the total turnover by 1.0 or 2.0, as the case may be.
4. In case, the bidder was set up in less than 3 year's period, mention the year of inception in the 'Remarks' column.

Signed and Upload

FORM-6: CAPABILITY STATUS

NIT No:

Bidder's Name & Address:

To

The General Manager (O&M)

The Durgapur Projects Limited

Durgapur – 713201, W.B.

Subject : Design & Engineering, Manufacture / Procurement, Supply, Erection, Testing and Commissioning of 7 MW Grid Connected Ground Mounted Solar Photovoltaic Power Plant in The Durgapur Projects Limited (DPL) Durgapur, West Bengal including warrantee obligation with 05 (Five) years comprehensive Operation and Maintenance.

We provide the following details to conform that we have sufficient capacity to execute the supply of Goods covered in the NIT:

A	Manufacturing Capacity (applicable in case of original manufacturers only)	
B	Orders in Hand	
i	Total value of Orders	
ii	Value of work completed out of above value upto 31.01.2018.	

Details of orders in hand are as follows:

Sl. No	Purchaser / Client	Scope of works	Order Value	Schedule Time of Completion	Value of Outstanding work	Estimated Completion date

Note:

- Continuation sheets of like size and format may be used and annexed to this format if required.

Date : (Printed Name)...

Place : (Designation)..

Signed and Upload

FORM-7: SIMILAR TYPE OF ORDERS

STATEMENT OF SIMILAR TYPE OF ORDERS EXECUTED AS ON DATE OF ISSUANCE OF THE NIT

[Applicability up to the extent of meeting Technical QR].

NIT No:

Bidder's Name & Address:

To

The General Manager (O&M)

The Durgapur Projects Limited

Durgapur – 713201, W.B.

Subject : Design & Engineering, Manufacture / Procurement, Supply, Erection, Testing and Commissioning of 7 MW Grid Connected Ground Mounted Solar Photovoltaic Power Plant in The Durgapur Projects Limited (DPL) Durgapur, West Bengal including warrantee obligation with 05 (Five) years comprehensive Operation and Maintenance.

Sl No	Name of the Installed Plants/ Project	Financial year	Order No. and date	Name of Purchaser / order issuing authority	Cumulative capacity of the order (Considering similar type of work) (scanned copy of certificate to be Submitted in non-statutory cover) (kWp)	Ordered Value/Time (extended time, if any) of Completion	Cumulative capacity installed (Considering similar type of work) (kWp)	Completion report of installed systems (scanned copy of certificate to be Submitted in non-statutory cover)	Remarks

- Continuation sheets of like size and format may be used and annexed to this format if required.

Similar type of work means Solar PV power plant each of minimum capacity as per QR of the tender

Date : (Printed Name)……………………………………………

Place : (Designation)………………………………………………

Signed and Upload

FORM-8:CURRICULUM VITAE OF KEY PERSONNEL

NIT No:

Bidder's Name & Address:

To

The General Manager (O&M)

The Durgapur Projects Limited

Durgapur – 713201, W.B.

Subject : Design & Engineering, Manufacture / Procurement, Supply, Erection, Testing and Commissioning of 7 MW Phase 1 and Phase 2 Grid Connected Ground Mounted Solar Photovoltaic Power Plant in The Durgapur Projects Limited (DPL) Durgapur, West Bengal including warrantee obligation with 05 (Five) years comprehensive Operation and Maintenance.

S.No	Proposed Position	Name	Position Held since	Professional Qualification	Experience in relevant Field	Any other Information

Date : (Printed Name)...

Place : (Designation)..

Signed and Upload

Note:

- Continuation sheets of like size and format may be used and annexed to this Form if required.

FORM-9: FORMAT FOR SUBMISSION OF PRE-BID QUERIES

FORMAT FOR SUBMISSION OF PRE-BID QUERIES			
NIT No.			DTD.
Subject : Design & Engineering, Manufacture / Procurement, Supply, Erection, Testing and Commissioning of 7 MW Phase 1 and Phase 2 Grid Connected Ground Mounted Solar Photovoltaic Power Plant in The Durgapur Projects Limited (DPL) , Durgapur, West Bengal including warrantee obligation with 05 (Five) years comprehensive Operation and Maintenance.			
NAME OF THE BIDDER:	<To be filled in by the bidder>		Work name : <To be filled in by the bidder>
PART A - TECHNICAL QUERIES			
Sl no	GCC Clause reference (if any)	BIDDER'S QUERY	DPL's REPLY
1			
2			
3			
PART B: COMMERCIAL/GCC RELATED/CONTRACTUAL QUERIES			
Sl no	GCC Clause reference (if any)	BIDDER'S QUERY	DPL's REPLY
1			
2			
3			
4			
5			
• Continuation sheets of like size and format may be used as per Bidders requirements and shall be annexed to this Form.			

Note:

i. To be submitted before Pre- bid meeting.

ii. This sheet must not the part of the offer submitted by the bidder and not to be upload

iii. This sheet to be mailed in Excel Format at email address : contractcelldpl@gmail.com

Date :	(Signature)………………………………………………
Place :	(Authorized Representative of bidder)..
	(Designation)……………………………………………
	Name of the bidder: ………………………………….

FORM-10: PROPOSED MODIFICATIONS

(To be submitted before Pre-bid meeting)

Ref:

Bidder's Name & Address:

To

The General Manager (O&M)

The Durgapur Projects Limited

Durgapur – 713201, W.B.

Subject : Design & Engineering, Manufacture / Procurement, Supply, Erection, Testing and Commissioning of 7 MW Phase 1 and Phase 2 Grid Connected Ground Mounted Solar Photovoltaic Power Plant in Santaldih Thermal Power Station , Durgapur, West Bengal including warrantee obligation with 05 (Five) years comprehensive Operation and Maintenance.

Reference : NIT No: _______________

We have carefully gone through the Technical Specifications and the General Conditions of Contract and we have satisfied ourselves and hereby propose certain modifications as mentioned below:

S.No.	Sec./Clause & Page No.	Existing Clause	Modified clause (proposed by Bidder)	Reasons for modification

Note:

i. To be submitted before Pre- bid meeting.

ii. This sheet must not the part of the offer submitted by the bidder and not to be upload

iii. This sheet to be mailed in Excel Format at email address : contractcelldpl@gmail.com

Date : **(Signature)**...

(Authorised Representative of bidder)

Place :

(Designation)……………………………………………

Name of the bidder: …………………………………

FORM-11: PROFORMA FOR JOINT VENTURE/CONSORTIUM AGREEMENT

(ON NON-JUDICIAL STAMP PAPER OF APPROPRIATE VALUE)

This Joint Venture/Consortium Agreement made and entered into on ________ day of………. (year)

BY AND BETWEEN ___________________________ (Name of the Lead Member), a

Company/Firm registered under the laws of __________________________ (Name of the Country) with its Head/Registered Office at __________________________ (Address of the Head/Registered Office) and a place of business in __________________________ (Address of place of business) (hereinafter referred to as "The Lead Member") and represented by Mr/Mrs/Ms. _____________________________________ (Name of Authorized Signatory).

AND

______________________________ (Name of the other Member), a Company/Firm registered under the laws of __________________________ (Name of the Country) with its Head/Registered Office at __________________________ (Address of the Head/Registered Office) and a place of business in __________________________ (Address of place of business) (hereinafter referred to as "The Member") and represented by Mr/Mrs/Ms. _____________________________________ (Name of Authorized Signatory).

WITNESSETH

WHEREAS The Durgapur Projects Ltd. (DPL) (hereinafter referred to as "The Procuring Entity") has issued RFP/ Notice Inviting Tender No. ____________________ Dated ________________ for (……………….Name of the Job……….). The Procuring Entity intends to select the suitable Bidder through competitive bidding process for the aforesaid Project/Works / Services.

WHEREAS the Parties are interested in jointly preparing and submitting an Application to Bid for the Project/ Works / Services mentioned above as a Joint Venture/Consortium.

1.0 PURPOSE OF THIS AGREEMENT

1.1. The purpose of this Agreement is to define the principles of collaboration among the Parties to:

1.1.1 Submit an Application jointly to Bid for the (……………….Name of the Job……….) as a Joint Venture /Consortium.

1.1.2 Negotiate and sign Contract in case of award.

1.1.3 Provide and perform the supplies / works / services / EPC etc. which would be ordered by the Procuring Entity pursuant to the Contract.

1.2. For the purpose of participating in the Bid, the name of the Joint Venture /Consortium shall be "________________".

2.0 LEGAL RELATIONSHIP OF THE MEMBERS

2.1. This Agreement shall not be construed as establishing or giving effect to any legal entity such as, but not limited to, a company, a partnership, etc. It shall relate solely towards the Procuring Entity for (……………….Name Of the Job……….) and related execution works to be performed pursuant to the Contract and shall not extend to any other activities.

2.2. The Parties shall be jointly and severally responsible and bound towards the Procuring Entity for the performance of the Job in accordance with the terms & conditions of the Tender Document and Contract.

3.0 LEAD MEMBER

__________________________ (Name of Member) shall act as Lead Member of the Joint Venture /Consortium. As such, it shall act as the coordinator of the Members' combined activities and shall carry out the following functions:

3.1. To ensure the technical, commercial and administrative co-ordination of the Project/Works / Services.

3.2. To lead the contract negotiations of the Project/ Works / Services with the Procuring Entity.

3.3. The Lead Member is authorized to submit bills, receive payments and instructions and incur liabilities for and on behalf of Joint Venture /Consortium Members.

3.4. In case of an award, act as a channel of communication between the Procuring Entity and the Joint Venture /Consortium to execute the Contract.

4.0 SCOPE OF WORK AND SERVICES OF EACH MEMBER

4.1 Scope of Work and Services: The Scope of Work and Services for each Member shall be defined as follows:

4.1.1. _________________ (Name of Member) shall be responsible for the following (Define the Scope of Work):

a)

b)

4.1.2. _________________ (Name of Member) shall be responsible for the following (Define the Scope of Work):

a)

b)

4.1.3. _________________ (Name of Member) shall be responsible for the following (Define the Scope of Work):

a)

b)

4.2 Participation Share of each Member:

Lead Member_______ %

Other Member______ %

Other Member______ %

4.3 Financial Commitment of each Member in terms of Contract Value:

Lead Member_______ %

Other Member______ %

Other Member______ %

5.0 SECURITIES

Securities, in the form of Bank Guarantees or any other mode as required under the Tender Document and/or Contract shall be provided in the following manner.

Lead Member_______ Rs.

Other Member______ Rs.

Other Member______ Rs.

6.0 LIABILITY

Liability of the Parties with respect to Claims of the Procuring Entity: All the joint venture/consortium members are jointly and severally liable to the Procuring Entity for the Performance in terms of Scope of Work under the Contract in its entirety.

7.0 DURATION OF THE AGREEMENT

The present Agreement is valid until successful completion of the Contract including defect liability period, if any, and full and final settlement of all accounts and disputes, if any, between the Parties and the Procuring Entity, except if the Procuring Entity has decided not to award the Contract to the Parties, in such case the Parties are free from any obligation under this Agreement.

IN WITNESS WHEREOF, this agreement executed on the ______ day of ________ (month)………… (Year) by the duly Authorized Representatives of the Parties hereto.

For and on behalf of M/s. ___________________________

(Lead Member)

Name:

Company Seal:

For and on behalf of M/s. ___________________________

(Other Member)

Name:

Company Seal:

For and on behalf of M/s. ___________________________

(Other Member)

Name:

Company Seal:

Notary Seal:

FORM-12: POWER OF ATTORNEY

Format for Power of Attorney to be provided by each of the other members of the Consortium in favour of the Lead Member

(To be on non-judicial stamp paper of appropriate value as per Stamp Act relevant to place of execution)

Whereas the The Durgapur Projects Ltd. (DPL) (the “Procuring Entity”) has invited Bids from bidders for “ EPC Contract for Design & Engineering, Manufacture / Procurement, Supply, Installation, Testing and Commissioning of 7 MW Phase 1 and Phase 2 Grid Connected Ground Mounted Solar Photovoltaic Power Plant in The Durgapur Projects Ltd. (DPL) , Durgapur, West Bengal, India in turnkey basis including warrantee obligation with 05 (Five) years comprehensive Operation and Maintenance”

Whereas, M/s……………………., and M/s……………………. (collectively the “**Joint Venture/ Consortium**”) being Members of the Joint Venture/ Consortium are interested in bidding for the Project in accordance with the terms and conditions of the tender document and other Bid documents including agreement in respect of the Project/works/services,

AND

Whereas, it is necessary for the Members of the Joint Venture/ Consortium to designate one of them as the Lead Member with all necessary power to do for and on behalf of the Joint Venture/ Consortium, all acts, deeds and things as may be necessary in connection with the Joint Venture’s/ Consortium’s Bid for the Project/Works/Services and its execution.

NOW THEREFORE KNOW ALL MEN BY THESE PRESENTS

We, M/s……………………. having our registered office at ……………………., and M/s……………………. having our registered office at ……………………., (hereinafter collectively referred to as the “**Principals**”) do hereby irrevocably designate, nominate, appoint and authorize M/s ……………………. having its registered office at ……………………., being one of the Members of the Joint Venture/ Consortium, as the Lead Member and true and lawful attorney of the Joint Venture/ Consortium (hereinafter referred to as the “**Attorney**”). We hereby irrevocably authorize the Attorney (with power to sub-delegate) to conduct all business for and on behalf of the Joint Venture/ Consortium and any one of us during the Bidding process and, in this regard, to do on our behalf and on behalf of the Joint Venture/ Consortium, all or any of such acts, deeds or things as are necessary or required or incidental to the Bid, including but not limited to signing and submission of all applications, Bids and other documents and writings, participate in meetings, respond to queries, submit information/ documents, sign and generally to represent the Joint Venture/

Consortium in all its dealings with the Procuring Entity, in all matters in connection with or relating to or arising out of the Joint Venture's/ Consortium's Application.

AND hereby agree to ratify and confirm and do hereby ratify and confirm all acts, deeds and things done or caused to be done by our said Attorney pursuant to and in exercise of the powers conferred by this Power of Attorney and that all acts, deeds and things done by our said Attorney in exercise of the powers hereby conferred shall and shall always be deemed to have been done by us(Joint Venture/ Consortium).

IN WITNESS WHEREOF WE THE PRINCIPALS ABOVE NAMED HAVE EXECUTED THIS POWER OF ATTORNEY ON THIS …………………. DAY OF ………. 20…

For …………………….

(Signature)

…………………….

(Name & Title)

For …………………….

(Signature)

…………………….

(Name & Title)

For …………………….

(Signature)

…………………….

(Name & Title)

(Executants)

(To be executed by all the Members of the Joint Venture/ Consortium)

Witnesses:

1.

2.

Notes:

1. The mode of execution of the Power of Attorney should be in accordance with the procedure, if any, laid down by the applicable law and the charter documents of the executants (s) and when it is so required, the same should be under common seal affixed in accordance with the required procedure.

2. Also, wherever required, the Bidder should submit for verification the extract of the charter documents and documents such as a board or shareholders' resolution/ power of attorney in favour of the person executing this Power of Attorney for the delegation of power hereunder on behalf of the Bidder.

FORM-13: DECLARATION FOR NET MINIMUM GUARANTEED GENERATION

DECLARATION SHEET – FOR NET MINIMUM GUARANTEED GENERATION

(To be submitted on official letter head by the bidder)

We .... declare that the (project description) offered by us within the scope of this tender, will generate minimum 1.60 MU/MW/year measured in the Net Meter installed at the outgoing feeder. The Net Minimum Guaranteed Generation shall be calculated after deducting the Auxiliary Consumption for the plant.

The Net Minimum Guaranteed Generation shall be reduced @ 0.7% per year.

In case we fail to produce the Net Minimum Guaranteed Generation as stated above, the provisions of penalty according to Clause No. 4.3.5 of the Special Conditions of Contract of this tender shall be applicable.

I hereby certify that I am duly authorized representative of the Bidder whose name appears above my signature.

Bidder's Name: ………………………………………………………………………….......

Authorised Representative's Signature……………………………………………….

SECTION – VII

ANNEXURES

09	No-Claim Certificate	Annexure-9
10	Indemnity bond to be executed by the contractor for the equipment handed over by the Purchaser for performance of its contract (Entire Equipment Consignment in one lot)	Annexure-10A
11	Application for Material Gate Pass	Annexure-10B
12	Authorization letter	Annexure-11
13	Material Receipt Certificate	Annexure-12

NOTE: AS THE LOA'S ARE DIFFERENT, TWO PROJECTS PHASE 1 AND PHASE 2 SHALL BE TREATED SEPARATELY. HENCE ALL THE ABOVE ANNEXURES ARE APPLICABLE SEPARATELY FOR PHASE-1 & PHASE-2. SUCCESSFUL BIDDER SHALL SUBMIT THE DOCUMENTS DURING EXECUTION STAGE INDIVIDUALLY.

ANNEXURE-1: CONTRACT AGREEMENT

PROFORMA OF CONTRACT AGREEMENT

(To be executed on Non-Judicial Stamp Paper of Rs. 100/-)

THIS AGREEMENT made at this ____________ day of _________________, _____, between **The Durgapur Projects Limited, Durgapur - 713201** (hereinafter called "the Procuring Entity"), of the one part, and __________________________________of _____________________________ (hereinafter "the Contractor"), of the other part:

WHEREAS the Procuring Entity invited bids "<Tender Description> in DPL" and has accepted the Bid offered by the Bidder/Contractor for the same in the sum of ____ (hereinafter "the Contract Price"). After due consideration, the procuring entity has decided to entrust to the contractor with the job/ work/ supply of **"Design & Engineering, Manufacture / Procurement, Supply, Erection, Testing and Commissioning of 7 MW Phase **______ Grid Connected Ground Mounted Solar Photovoltaic Power Plant in The Durgapur Projects Limited (DPL), Durgapur, West Bengal including warrantee obligation with 05 (Five) years comprehensive Operation and Maintenance"**

FOR THE CONSIDERATION payable under this agreement, the contractor hereby agrees to complete the execution of job/ work/ supply in a satisfactory manner following scope of Work within the specified period.

NOW THIS AGREEMENT WITNESSETH AS FOLLOWS:

1. In this Agreement words and expressions shall have the same meanings as are respectively assigned to them in the Contract referred to.
2. The following documents shall be deemed to form and be read and construed as part of this Agreement, viz.:
   a) the Procuring Entity's Notification (LoA) to the Contractor of Award of Contract;
   b) the Bid Forms (including Price Bid) submitted by the Contractor;
   c) the Special Conditions of Contract;

d) the General Conditions of Contract;

e) the Technical Conditions of Contract with all Pre-bid clarification, amendment (if any) etc.

f) ____________________________ ** specify I or II.

g) ____________________________

This Contract shall prevail over all other Contract documents which are not covered under Clause 2 above. In the event of any discrepancy or inconsistency within the Contract documents referred under Clause 2, then the contract shall be governed by the documents in the order listed above.

3. In consideration of the payments to be made by the Procuring Entity to the Contractor as indicated in this Agreement, the Contractor hereby covenants with the Procuring Entity to provide the goods and services / to execute works and to remedy defects therein in conformity with the provisions of the Contract in all respects.
4. The Procuring Entity hereby covenants to pay the Contractor in consideration of the provision of the goods and services / execution of works and the remedying of defects therein, the Contract Price or such other sum as may become payable under the provisions of the Contract at the times and in the manner prescribed by the Contract.

IN WITNESS WHEREOF, the parties through their duly authorized representatives have executed these presents (execution whereof has been approved by the competent authorities of both the parties) on the day, month and year first above mentioned at Kolkata.

Signed by for and on behalf of the Procuring Entity

…………………………………………………………………………

[Signature]

…………………………………………………………………………

[Title]

In the presence of

……………………………………………… (Signature, Name and Title)

Signed by for and on behalf of the Contractor/ Lead Member

………………………………………………………………………

[Signature]

………………………………………………………………………

[Title]

In the presence of

……………………………………………… (Signature, Name and Title)

ANNEXURE-2: BG (CONTRACT PERFORMANCE)

PROFORMA OF BANK GUARANTEE FOR CONTRACT PERFORMANCE

(To be executed in non-judicial stamp paper of Rs. 100/-)

(To be stamped in accordance with Stamp Act, if any, of the Country of the issuing Bank)

Bank Guarantee No.: _________________

Date: _____________________

To,

The Durgapur Projects Limited,

Durgapur - 713201

Dear Sir,

In consideration of the West Bengal Power Development Corporation Limited (Hereinafter referred to as the 'Procuring Entity' which expression shall unless repugnant to the context or meaning thereof, include its successors, administrators and assigns) having awarded to M/s..... *[Contractor's Name]*............. With its Registered/Head Office at............................ (Hereinafter referred to as the 'Contractor', which expression shall unless repugnant to the context or meaning thereof, include its successors administrators, executors and assigns), a Contract by issue of *Procuring Entity's* Letter of Acceptance No. .... dated .... and the same having been acknowledged by the Contractor, for .... [Contract sum in figures and words] for .... *[Name of the work]* and the Contractor having agreed to provide a Contract Performance Guarantee for the faithful performance of the entire Contract equivalent to ...... (*).......... of the said value of the aforesaid work under the Contract to the Procuring Entity.

We.............. *[Name & Address of the Bank]*............ having its Head Office at.................. (hereinafter referred to as the 'Bank', which expression shall, unless repugnant to the context of meaning thereof, include its successors, administrators, executors and assigns) do hereby guarantee and undertake to pay the *Procuring Entity* on demand any and all monies payable by the Contractor to the extent of .... (*) .... as aforesaid at any time upto............... (@) .... *[days/month/year]* without any demur, reservation, contest, recourse or protest and/or without any reference to the Contractor. Any such demand made by the *Procuring Entity* on the Bank shall be conclusive and binding notwithstanding any difference between the *Procuring Entity* and the Contractor or any dispute pending before any Court, Tribunal, Arbitrator or any other authority. The Bank undertakes not to revoke this guarantee during its currency without previous consent of the *Procuring Entity* and further agrees that the guarantees herein contained shall continue to be enforceable till the *Procuring Entity* discharges this guarantee or till ....*[days/month/year]* whichever is earlier.

The *Procuring Entity* shall have the fullest liberty, without affecting in any way the liability of the Bank under this guarantee, from time to time to extend the time for performance of the Contract by the Contractor. The *Procuring Entity* shall have the fullest liberty, without affecting this guarantee, to postpone from time to time the exercise of any powers vested in them or of any right which they might have against the Contractor, and to exercise the same at any time in any manner, and either to enforce or to forbear to enforce any covenants, contained or implied, in the Contract between the Procuring Entity and the Contractor or any other course or remedy or security available to the Procuring Entity. The Bank shall not be released of its obligations under these presents by any exercise by the Procuring Entity of its liberty with reference to the matters aforesaid or any of them or by reason of any other act or forbearance or other acts of omission or commission on the part of the *Procuring Entity* or any other indulgence shown by the *Procuring Entity* or by any other matter or thing whatsoever which under law would, but for this provision have the effect of relieving the Bank.

The Bank also agrees that the Procuring Entity at its option shall be entitled to enforce this Guarantee against the Bank as a principal debtor, in the first instance without proceeding against the Contractor and notwithstanding any security or other guarantee the Procuring Entity may have in relation to the Contractor's liabilities.

Our liability under this Bank Guarantee shall not exceed …………………..

This Bank Guarantee shall be valid up to and including ………………………

We are liable to pay the guaranteed amount or any part thereof under this Bank Guarantee only and only if Procuring Entity serve upon Bank a written claim or demand on or before …………….@…………. Dated this………………. Day of ………………(Month)………………… (Year)……….. at………….

WITNESS

……………………………… ……………………………

(Signature) (Signature)

……………………………… . ………………………………

(Name) (Name)

……………………………. ……………………………………………

(Engineer / Officer Address) (Designation with Bank Stamp)

…………………………………………

Attorney as per Power of Attorney No……………….

Dated………………………

Notes:

1. (*) This sum shall be 3% (ten percent) of the Project cost plus GST.
2. (@) This date will be Ninety (90) calendar days beyond the defects liability period as specified in the Contract.
3. The stamp papers of appropriate value shall be purchased in the name of guarantee issuing Bank.

ANNEXURE-3: BG (MOBILISATION ADVANCE)

PROFORMA OF BANK GUARANTEE FOR MOBILISATION ADVANCE

(To be executed in non-judicial stamp paper of Rs. 100/-)

(To be stamped in accordance with Stamp Act, if any, of the Country of the issuing Bank)

Bank Guarantee No.: ________________

Date: ____________________

To,

The Durgapur Projects Limited,

Durgapur - 713201

We have been informed that ____________________________ (hereinafter called "the Contractor") has entered into Contract No.: ___________________________ dated ______________ with you, for the execution of ________________________________ (hereinafter called "the Contract").

Furthermore, we understand that, according to the conditions of the Contract, an advance payment in the sum of Rs.__________________/- (___________________) is to be made against an advance payment guarantee.

At the request of the Contractor, we ______________________ hereby irrevocably undertake to pay you any sum or sums not exceeding in total an amount of Rs. _______________/- (_____________) upon receipt by us of your first demand in writing accompanied by a written statement stating that the Contractor is in breach of its obligation under the Contract because the Contractor used the advance payment for purposes other than toward the execution of the Works.

It is a condition for any claim and payment under this guarantee to be made that the advance payment preferred to above must have been received by the Contractor on his account number _____________________ at ______________________________.

The maximum amount of this guarantee is valid shall be progressively reduced in proportion to the value of each part-shipment or part-delivery of plant and equipment to the site, as indicated in copies of the relevant shipping and delivery documents that shall be presented to us. This guarantee shall expire, at the latest, upon our receipt of documentation indicating full repayment by the Contractor of the amount of the advance payment, or on the ______ day of __________, 2____, whichever is earlier. Consequently, any demand for payment under this guarantee must be received by us at this office on or before that date.

This guarantee is subject to the Uniform Rules for Demand Guarantees, ICC Publication No.: 458.

____

[Signature(s) name of bank or financial institution]

ANNEXURE-4: FOR EXTENSION BG

PROFORMA FOR EXTENSION OF BANK GUARANTEE

Ref......................................

Date......................................

To

…………

……………

……………… West Bengal

Sub: Extension of Bank Guarantee No...for Rs.......................... favoring yourselves, expiring on……………………………………………on account of M/s...in respect of Contract No..dated.................... (hereinafter called original Bank Guarantee).

Dear
Sirs,

At the request of M/s...,We..........................Bank, branch office at...and having its Head Office at.. .do hereby extend our liability under the above mentioned Bank Guarantee No............................... dated........................ for a further period of..............................(Years/Months) from....................to expire on……………………Expect as provided above, all other terms and conditions of the original bank guarantee No..........................dated......................shall remain unaltered and binding.

Please treat this as an integral part of the original bank guarantee to which it would

be attached. Yours Faithfully,

For ....

Manager/Agent/Accountant... Power of Attorney

No... Dated..

SEAL OF BANK

NOTE: The non-judicial stamp paper of appropriate value shall be purchased in the name of the bank who has issued the Bank Guarantee.

ANNEXURE-5: INDEMNITY BOND

PROFORMA OF INDEMNITY BOND

(To be executed on Non-Judicial Stamp Paper of Rs. 100/-)

BY THE PRESENT INDEMNITY BOARD EXECUTED by me/us on this……………………..Day of…………….20…….,I/We having Registered Office / residing at………………………………………………………………………….(herein after called "OBLIGOR/ OBLIGORS" which expression shall mean and includes my/our successors legal representatives, assigns) do hereby binds myself/ourselves and also our company/firm………………………………………………. after having the power to bind so with the promise and undertaking in favour of '**The Durgapur Projects Limited, Durgapur - 713201** (hereinafter called as OBLIGEE, which expression shall mean and include it's legal representative, administrators assigns.

Whereas OBLIGOR/OBLIGORS has/have been awarded to execute the job/works under letter no……………………………………………….dated………………………issued by the OBLIGEE after having observing necessary formalities, the details of which is described in the schedule given here under as per letter mentioned herein-above and whereas the said job/works will be/likely to be done in places covered under Employees' State Insurance Act (ESI) and/or the Employee Compensation Act, 1923 (W.C. Act) and/or other laws relating to the Labour Management and Welfare.

And whereas according to the condition of the contract the OBLIGOR/OBLIGORS is/are under obligation to execute this Indemnity Bond before the commencement of actual execution and OBLIGOR/OBLIGORS is/are aware that unless this Indemnity Bond is executed in accordance with the condition of contract before the actual execution in accordance with law the OBLIGEE shall have the power to deem that actual work has been stated within the meaning of the contract before the execution of this Indemnity Bond.

Now this indenture witnesses that I/we the OBLIGOR/OBLIGORS do hereby undertake:

1. THAT the OBLIGEE shall not be held responsible for any type of accident which may take place during the course of work undertaken by the OBLIGOR/OBLIGORS.

2. THAT the OBLIGOR/OBLIGORS will take/adopt all safety norms in respect of each and every workmen labour personnel according to the rules or to the satisfaction of the OBLIGREE IN ALL CASES.

3. That the OBLIGOR/OBLIGORS undertakes/undertake to engage only those labour worker or any other personnel whether skilled or unskilled or any other person whether in technical management or non-managerial or any other capacity in the area covered under Employee' State Insurance Act, 1948 who has/have insurance coverage within the meaning of Employees' State Insurance Act and further undertakes NOT to engage any person in the area covered under Employees State Insurance Act., who does/do not has/have insurance coverage within the meaning of Employees' State Insurance Act,1948.

4. That the OBLIGOR/OBLIGORS further undertakes/undertake to engage only those labour, worker, or any other personnel, whether skilled or unskilled, whether in technical, managerial or non-managerial or any other capacity in the area NOT covered under Employees' State Insurance Act 1948, who has life insurance for the sum assured equivalent to the amount of Compensation under the Employees Compensation Act in case of accidental death or inquiry and such insurance has been effected by the OBLIGOR/OBLIGORS.

5. THAT the OBLIGOR/OBLIGORS undertakes/undertake to indemnify and keep harmless the OBLIGEE from all claims, action, proceedings and of risk, damage, danger to any person whether belonging to/or not belonging to OBLIGOR/OBLIGORS.

6. THAT the OBLIGOR/OBLIGORS shall keep harmless the OBLIGEE from all claims, compensation, damages, any proceedings in respect of any of its employee/workmen under the Employee Compensation Act or any other laws for the time being in force.

7. THAT, if during the course of execution of work as stated in the letter mentioned hereinabove issued by the OBLIGEE, it is found that the OBLIGOR/OBLIGORS has/have not complied with guidelines/formalities within the meaning of Employees' State Insurance Act or Employee Compensation Act or any other laws relating to the Labour Welfare for the time being in force, and also has not observed the safety norms

in accordance with the law to the satisfaction of the OBLIGEE, the OBLIGEE shall have the right to stop the execution of work/job and the period of such stoppage shall continue till adequate safety and other compliance mentioned hereinabove under the labour welfare legislation have been observed and such period of stoppage shall not be taken into account for the calculation of the total period of completion of work for which the OBLIGOR/OBLIGORS is responsible to complete the work/job and it will be deemed that discontinuance was due to default of OBLIGOR / OBLIGORS.

8. THAT, if at any time, due to exigency, the OBLIGEE i.e. the **The Durgapur Projects Limited, Durgapur - 713201** (DPL) as the Principal Employer, becomes liable to pay any such compensation mentioned hereinabove, whether on failure of the OBLIGOR/OBLIGORS or for any other reason, the OBLIGEE shall have the right to recover the said amount from any amount receivable by OBLIGOR/OBLIGORS or any bank guarantee deposited or anything payable whether in connection with this contract or other contract by the OBLIGEE to the OBLIGOR/OBLIGORS.

9. THAT the OBLIGOR/OBLIGATOR is/are aware and accept that for the persistent or repeated violation of any condition mentioned in this Indemnity Bond, the OBLIGEE shall have right to terminate the contract of work issued by the OBLIGEE to OBLIGOR/OBLIGATOR.

SIGNED AND DELIVERED	
Signature	....
WITNESS:	
1) Name & Designation	....
Signature	....
1) Name & Designation	....
Signature	....

ANNEXURE-6: COMPLETION CERTIFICATE

Completion Certificate

(On the letter head of the Purchaser as per provisions of GCC 3.32)

Date: ………………

Contract No.: ………………

[*Name of Contract]*

To: *[Name and address of Contractor]*

Dear Sirs,

Pursuant to Clause GCC 3.32 of the General Conditions of the Contract entered into between yourselves and the DPL dated *[date]*, relating to the *[brief description of the Works],* we hereby notify you that the following part(s) of the Works was (were) complete on the date specified below, and that, in accordance with the terms of the Contract, the Purchaser hereby takes over the said part(s) of the Works, together with the responsibility for care and custody and the risk of loss thereof on the date mentioned below.

1. Description of the Works or part thereof: *[description]*

Billing Schedule Sl. No.	Description of item	Total	Quantity/Percentage (%)		
			Cumulative achieved up to last bill	Claimed in this bill	Cumulative achieved up to date

2. Date of Completion: *[date]*

3. Defects to be rectified, if any:

However, you are required to complete the outstanding items listed in the enclosure hereto as soon as practicable.

This letter does not relieve you of your obligation to complete the execution of the Works including Guarantee Test(s) in accordance with the Contract nor of your obligations during the Defects Liability Period.

Very truly yours,

……………………………………….

Title

(Project Manager)

Encl: List of outstanding items to be completed

ANNEXURE-7: APPLICATION FOR PAYMENT

Application for Payments

Site	:		Lot No	:	
Name of the Package	:		Date	:	
Name of Contractor	:		Contract No.	:	
Contract Value	:		Application Serial Number	:	

To

…………………………………………………… *

Dear Sir,

1. Pursuant to the above referred contract dated …………. the undersigned hereby submit claim for payment of the sum of …………. (Specify amount for which claim is made)

2. The above amount is on account of : (Check whichever is applicable)
   Advance payment (Schedule**)
   Interim payment as advance (Schedule**)
   Progressive payment against receipt of equipment at site (Schedule**)
   Progressive payment against erection (Schedule**)
   Transportation
   Insurance
   Price adjustment
   Extra work not specified in contract
   (Ref. Contract change order No……………….)
   Others (specify)
   Final payment (Schedule**)

   as detailed in the attached Schedule(s) which form an integral part of this application

3. The payment claimed is as per item(s) No(s) ………………… of payment schedule annexed to the above mentioned Contract.
4. This application consist of this page, a summary of claim statement (**) and the following signed schedules:
   1) ………………………
   2) ………………………
   3) ………………………
   The following documents are also enclosed:
   1) ………………………
   2) ………………………
   3) ………………………

Signature and Seal of Contractor / Authorised Signatory

* Application for payment will be made to "Project Manager" designated for this purpose at the time of Award of Contract.

** Proforma for the summary of claim will be finalized during the finalization of the Contract Agreement.

ANNEXURE-8: TAKING OVER CERTIFICATE

Taking-Over Certificate

(On the letter head of the Purchaser as per provisions of GCC 3.33)

Date: ………………

Loan/Credit No: ………………

Tender Notice No: ………………

[*Name of Contract]*

To: *[Name and address of Contractor]*

Dear Sirs,

Pursuant to clause GCC 3.33 of the General Conditions of the Contract entered into between yourselves and the Purchaser dated [date], relating to the [brief description of the Works], we hereby notify you that the Functional Guarantees of the following part(s) of the Works were satisfactorily attained on the date specified below.

1. Description of the Works: [description]

2. Date of Take-Over: [date]

This letter does not relieve you of your obligation to complete the execution of the Works in accordance with the Contract nor of your obligations during the Defects Liability Period.

Very truly yours,

…………………………………………

Title

(Project Manager)

ANNEXURE-9: NO-CLAIM CERTIFICATE

No-Claim Certificate

(To be issued by the Contractor)

Name of the Package: …………………… LoA No. /Contract No……………………..

Name of the Contract: …………………… Date…………………………………………

This is to certify that we have received all payments due to us in respect of the above referred LoA/Contract and we have no claims whatsoever pending with DPL for this Contract.

We further confirm that we shall have no claim against this Contract in future also.

Date :	(Signature)…………………………………………………
Place :	(Name)……………………………………………
	(Designation)………………………………………………

ANNEXURE-10A: INDEMNITY FOR EQUIPMENT

Indemnity bond to be executed by the contractor for the equipment handed over by the Purchaser for performance of its contract (Entire Equipment Consignment in one lot)

(To be executed on non-judicial stamp paper of appropriate value as per provisions of clause GCC 33.5)

INDEMNITY BOND

THIS INDEMNITY BOND is made this ……………………………………………day of ………………………….. 20 ………………. by ………………………(Contractor's Name) a Company registered under the Companies Act, 1956/Partnership firm/Proprietary concern having its registered office at ………………………………. (hereinafter called as '**Contractor**' or "Obligor" which expression shall include its successors and permitted assigns) in favour of **The Durgapur Projects Limited, Durgapur - 713201** and its project at ………………………… (hereinafter called "**DPL**" which expression shall include its successors and assigns) :

WHEREAS DPL has awarded to the Contractor a Contract for ....vide its Letter of Award/Contract No................dated .... and its Amendment No. .... and Amendment No.................................., (applicable when amendments have been issued) (hereinafter called the "Contract") in terms of which DPL is required to hand over various Equipment to the Contractor for execution of the Contract.

And WHEREAS by virtue of Clause No........................... of the said Contract, the Contractor is required to execute an Indemnity Bond in favour of DPL for the Equipment handed over to it by DPL for the purpose of performance of the Contract/Erection portion of the contract (hereinafter called the "Equipment")

AND THEREFORE, This Indemnity Bond witnesseth as follows:

1. That in consideration of various Equipment as mentioned in the Contract, valued at (Currency and amount in Figures)................................. (Currency and amount in words) .... handed over to the Contractor for the purpose of performance of the Contract, the Contractor hereby undertakes to indemnify and shall keep DPL indemnified, for the full value of the Equipment. The Contractor hereby acknowledges actual receipt of the Equipment etc. as per dispatch title documents handed over to the Contractor as detailed in the Schedule appended hereto. The Contractor shall hold such Equipment etc. in trust as a "Trustee" for and on behalf of DPL.

2. That the Contractor is obliged and shall remain absolutely responsible for the safe transit/protection and custody of the Equipment at DPL project site against all risks whatsoever till the Equipment are duly used/erected in accordance with the terms of the Contract and the Works duly erected and commissioned in accordance with the terms of the Contract is taken over by DPL. The Contractor undertakes to keep DPL harmless against any loss or damage that may be caused to the Equipment.

3. The Contractor undertakes that the Equipment shall be used exclusively for the performance/execution of the Contract strictly in accordance with its terms and conditions and no part of the equipment shall be utilised for any other work of purpose whatsoever. It is clearly understood by the Contractor that non-

observance of the obligations under this Indemnity Bond by the Contractor shall inter-alia constitute a criminal breach of trust on the part of the Contractor for all intents and purpose including legal/penal consequences.

4. That DPL is and shall remain the exclusive Purchaser of the Equipment free from all encumbrances, charges or liens of any kind, whatsoever. The Equipment shall at all times be open to inspection and checking by the Project Manager or other employees/agents authorised by him in this regard. Further, DPL shall always be free at all times to take possession of the Equipment in whatever form the Equipment may be, if in its opinion, the equipment are likely to be endangered, misutilised or converted to uses other than those specified in the Contract, by any acts of omission or commission on the part of the Contractor or any other person or on account of any reason whatsoever and the Contractor binds himself and undertakes to comply with the directions of demand of DPL to return the Equipment without any demur or reservation.

5. That this Indemnity Bond is irrevocable. If at any time any loss or damage occurs to the Equipment or the same or any part thereof is misutilised in any manner whatsoever, then the Contractor hereby agrees that the decision of the Project Manager of DPL as to assessment of loss or damage to the Equipment shall be final and binding on the Contractor. The Contractor binds itself and undertakes to replace the lost and/or damaged Equipment at its own cost and/or shall pay the amount of loss to DPL without any demur, reservation or protest. This is without prejudice to any other right or remedy that may be available to DPL against the Contractor under the Contract and under this Indemnity Bond.

6. NOW THE CONDITION of this Bond is that if the Contractor shall duly and punctually comply with the terms and conditions of this Bond to the satisfaction of DPL, THEN, the above Bond shall be void, but otherwise, it shall remain in full force and virtue.

IN WITNESS WHEREOF, the Contractor has hereunto set its hand through its authorised representative under the common seal of the Company, the day, month and year first above mentioned.

<table>
<tr><td colspan="6">Schedule</td></tr>
<tr><td rowspan="2">Particulars of
Equipment handed over</td><td rowspan="2">Quantity</td><td colspan="2">Particulars of Dispatch title Documents</td><td rowspan="2">Value of the Equipment</td><td rowspan="2">Signature of
the Attorney in token of receipt</td></tr>
<tr><td>RR/GR/ Bill of lading No & Date</td><td>Carrier</td></tr>
<tr><td></td><td></td><td></td><td></td><td></td><td></td></tr>
</table>

For and on behalf of

………………………

(Contractor's Name)

1	1.	Signature……………………	Signature……………………………
	2.	Name………………………	Name……………………………
	3.	Address………………………	Designation of …………………………
			Authorized representative*
2	1.	Signature……………………	
	2.	Name………………………	(Common Seal)
	3.	Address………………………	(In case of company)

*Indemnity Bond are to be executed by the authorised person and (i) in case of contracting Company under common seal of the Company or (ii) having the Power of Attorney issued under common seal of the company with authority to execute Indemnity Bond, (iii) In case of (ii), the original Power of Attorney if it is specifically for this Contract or a photostat copy of the Power of Attorney if it is General Power of Attorney and such documents should be attached to Indemnity Bond.

ANNEXURE-10B: APPLICATION FOR MATERIAL GATE PASS

(On Company letter head of Contractor)

Application for Material Gate Pass

To,

(Project In-Chrge)
DPL,

Ref. No.

i. LOA no. ____________________________ Dated: _______________________

ii. MDCC No.__________________________ Dated:_________________________

iii. Invoice Details

No.______________________________ Dated: ____

Quantity _______________________ Cost:_____________ Dated:_________________________

Subject: *Name of the work*

Dear Sir,

1. Pursuant to the above referred LOA/contract, the undersigned hereby submit request for gate pass for the above referred invoice materials.

2. As per the clause no. 3.39 of the GCC of the contract document, *name of the contractor* is fully responsible for care of the materials after the entry of the materials in side DPL premises until handover of the plant.

3. *Name of the contractor* is fully responsible for storage and safety security of the materials.

4. If at any time any loss or damage occurs to the Equipment or the same or any part thereof is misutilised in any manner whatsoever, then name of the contractor hereby agrees that DPL will not be responsible in any way

Very truly yours,

(Project Manager)

ANNEXURE-11: AUTHORISATION LETTER

Authorization letter

(On the letter head of Purchaser)

Ref No: ……………………..

Date: ……………………

To,

M/s ………… (Contractor's Name)…………….

Ref: Contract No ………………Dated ………………………
for …………………….. awarded by DPL

Dear Sirs,

Kindly refer to Contract No……………. Dated …………… for………………………. (Contract Name).You are hereby authorised on behalf of DPL having its registered office at ………………………………………………………………………… to take physical delivery of materials/equipment covered under dispatch Document/ Consignment Note no…………………[mention LR/RR No]………….dated ……………………………. and as detailed in the enclosed Schedule for the sole purpose of successful performance of the aforesaid contract and for no other purposes, whatsoever.

(Signature of Project Manager)

Designation: ……………….

Date: ………………..

ENCL: As above

ANNEXURE-12: MATERIAL RECEIPT CERTIFICATE

Material Receipt Certificate

Name of the Work: ....

LoA No. /Contract No..........................

Name of the Contractor: ....

Material Receipt Certificate No.__________________ Date:_____________

Sl.	CR	No.	No. &	Oy	In r	Descr	Quantity	mar e

							UOM	As per BS	Claim up to previous MRC		Claim in this MRC	Cuml. Received	Balance	
									MRC No	Qty				

Note: Above Materials received and verified and handed over the above to *name of the contractor* for their safe custody, erection and commissioning.

Contractor DPL

ANNEXURE-13: RECORD BOOK FOR GRID OUTAGE

Record Book for Grid Outage

Name of the Work: …………………..

LoA No. /Contract No.…………………….

Name of the Contractor: …………………..

Table 1 : Record of individual Grid outage where contractor is not responsible

Sl. No.	Date & Time from (Reference of WMS)	Date & Time to (Reference of WMS)	Reason for Grid Outage	Time not to be considered for NMGG calculation ***	Signature of Contractor with stamp	Signature of DPL authority with stamp
. . . .						

***

i)**If grid outage for few hours :**

The average generation of that particular day (MWHr/generation Hr) X number of hours of Grid outage (supporting document)

ii) **If Grid outage for days:**

Average generation of that week (MW Hr/days generation) X number of days grid outage (supporting document)

Table 2 : Record of individual Grid outage where contractor is responsible

Sl. No.	Date & Time from (Reference of WMS)	Date & Time to (Reference of WMS)	Reason for Grid Outage	Signature of Contractor with stamp	Signature of DPL authority with stamp
. . . .					

This Book shall be kept at the Custody of the Contractor. Record of outages and the period of outages shall be registered immediately after the outage.

Contractor

Figure 1: Geographical location of the Proposed Project

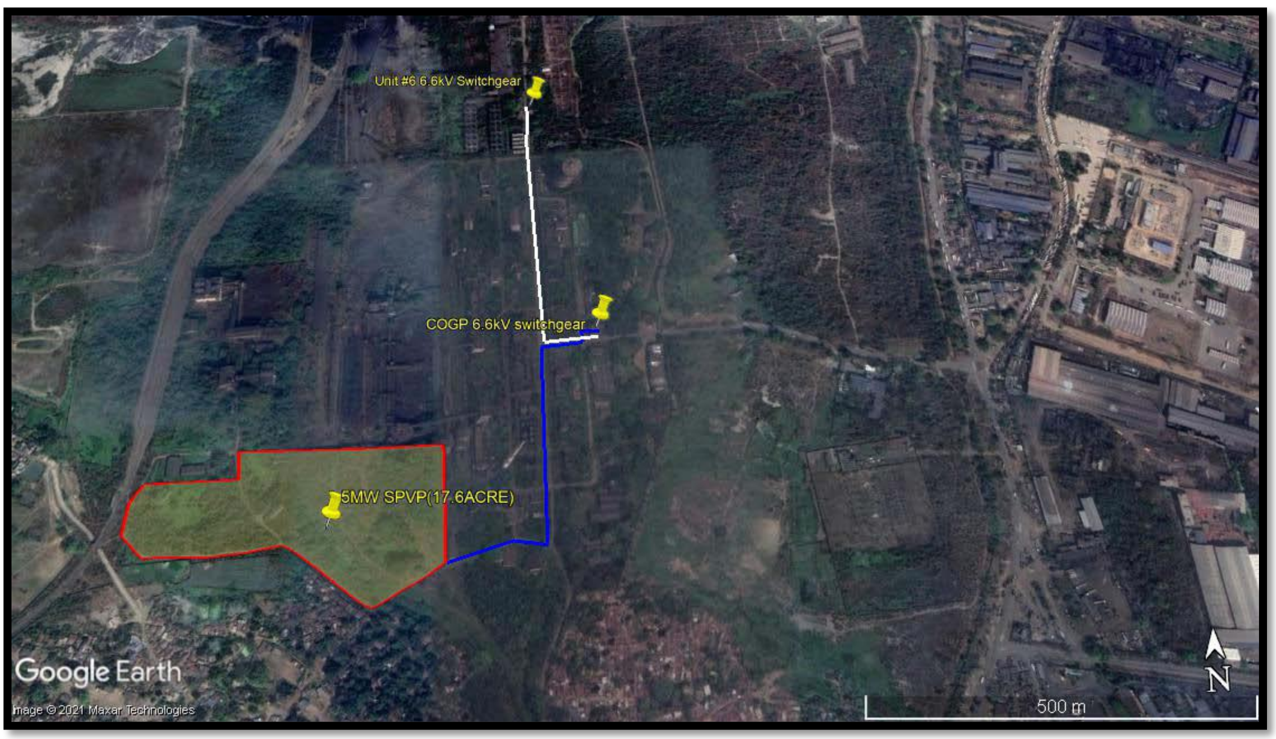


Figure 2: Layout of Location for Solar Plant at DPL

Figure 3: Proposed 20 MW(7 MW + 8 MW +5 MW) Solar PV Project Site

Figure 4: Proposed Solar Project Site Photograph

Total area which has been identified for installation of 5 MWAC Solar PV Project is 17.6 Acre

Figure 5: Proposed 6.6kV cable laying for 5MWAC Solar Plant

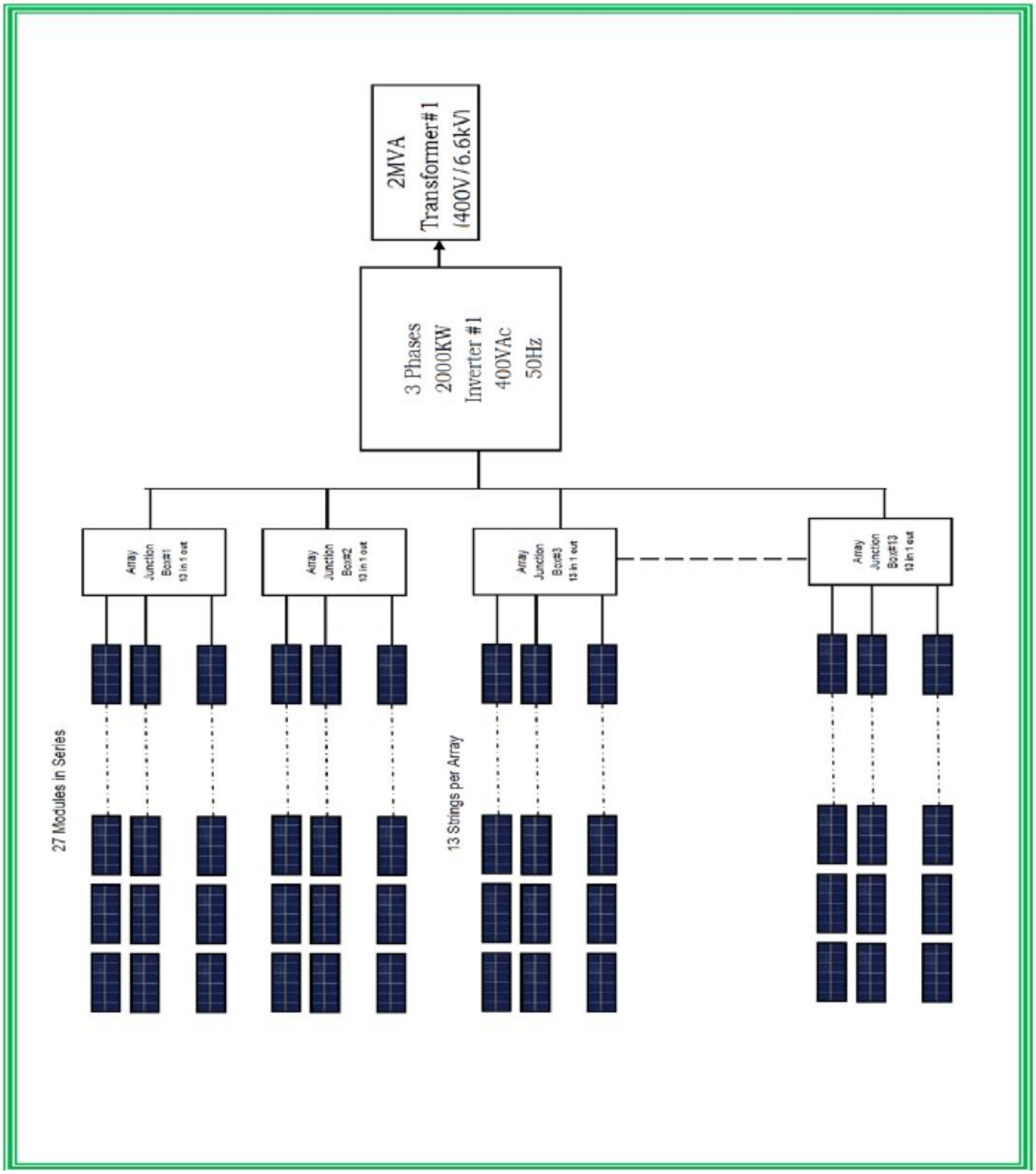


Figure 6 : Solar Panel Layout Indicative

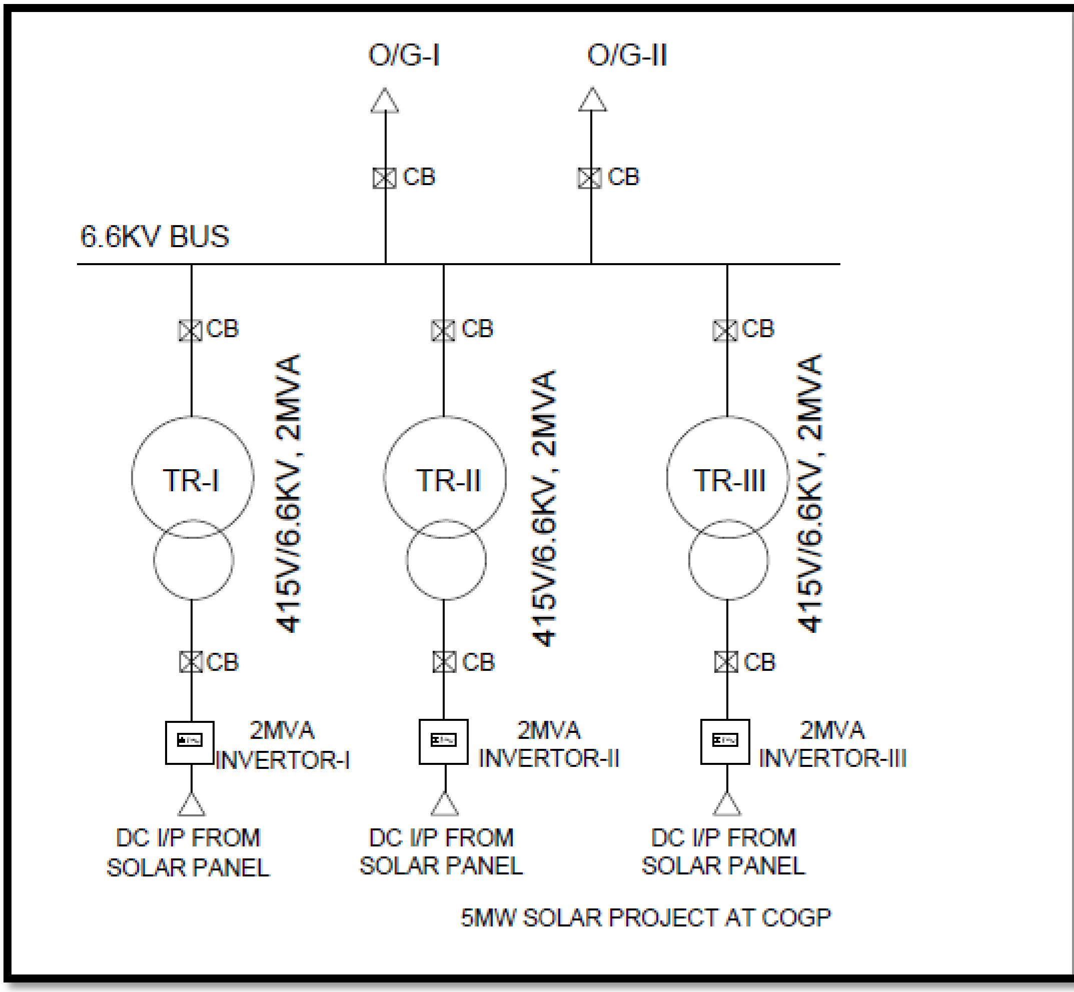


Figure 7 : Solar Panel Layout Indicative

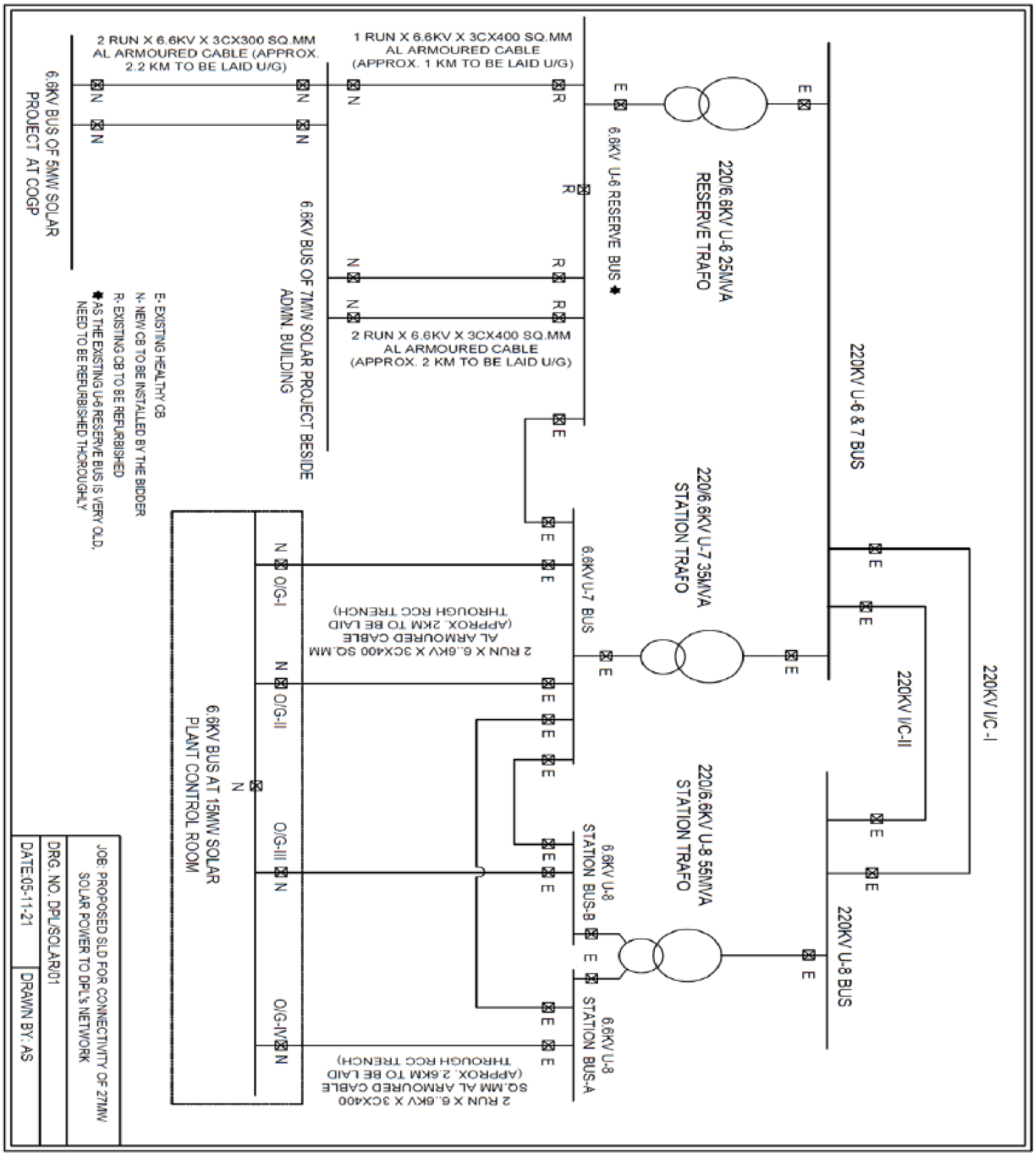


Figure 8: Electrical layout Diagram Indicative

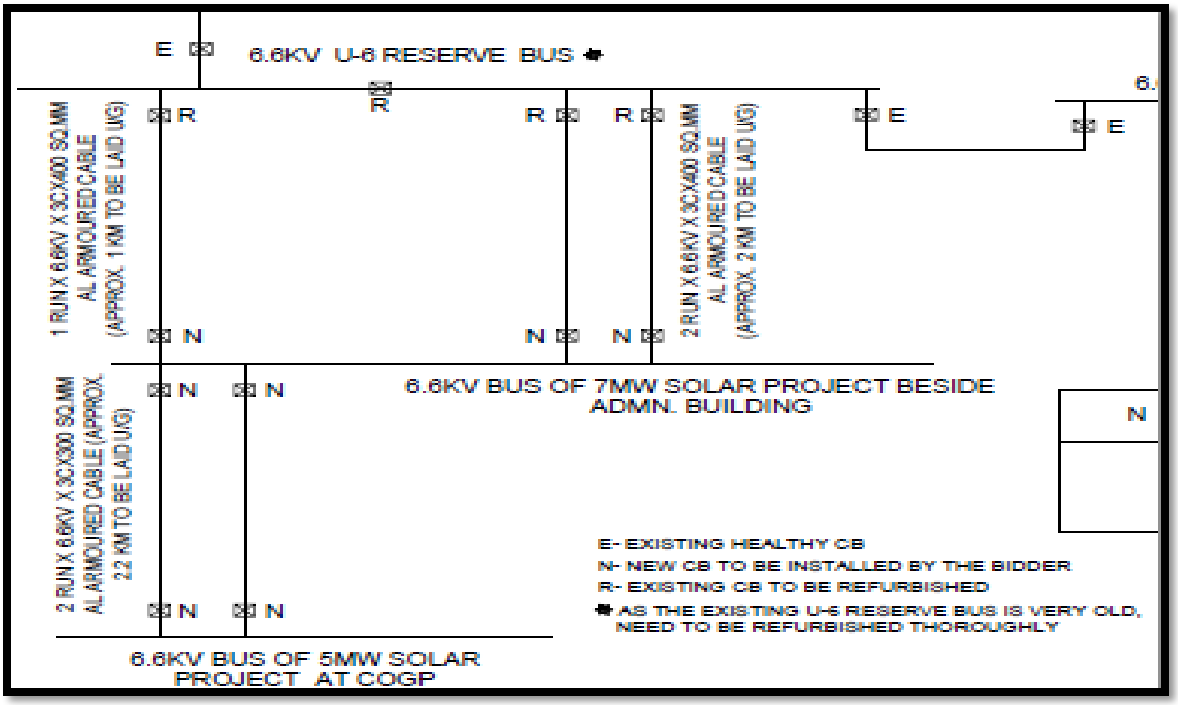


Figure 10: Electrical layout Diagram Indicative

Area	Description	Unit	Amount
Plant	Capacity	MWac	7
Module	Mono Crystalline Module Capacity	Wp	530
	Sub-Array#1		
	Module in Series	Module	27
	Module Parallel	String	335
	Sub-Array#2		
	Module in Series	Module	19
	Module Parallel	String	119
	Total Module	Nos	11306
	Total Module Area	m^2	28899
AJB	Array Junction Box	No	36
Inverter	Capacity	kWAc	2000
	Total Inverter	Nos	2
	Capacity	kWAc	1000
	Total Inverter	Nos	1
	Operating Voltage	V	575-1000
	Total Power	kWac	7000

Area	Description	Unit	Amount
Transformer	Capacity	MVA	2
	Transformer quantity	No	3
	Capacity	MVA	1
	Transformer quantity	No	1
	Operating Voltage	V	400V/6.6kV
	Evacuation Voltage Level	V	6.6kV

Table 1 : Indicative design of 8 MW AC Plant

No. of AJB	Inverter No		
	Inv#1	Inv#2	Inv#3
	27 Module in Series		19 Module in Series
12	13	13	
1	12	11	
9			13
1			2
Total String	168	167	119

Table 2 : Tentative distribution of String in AJB

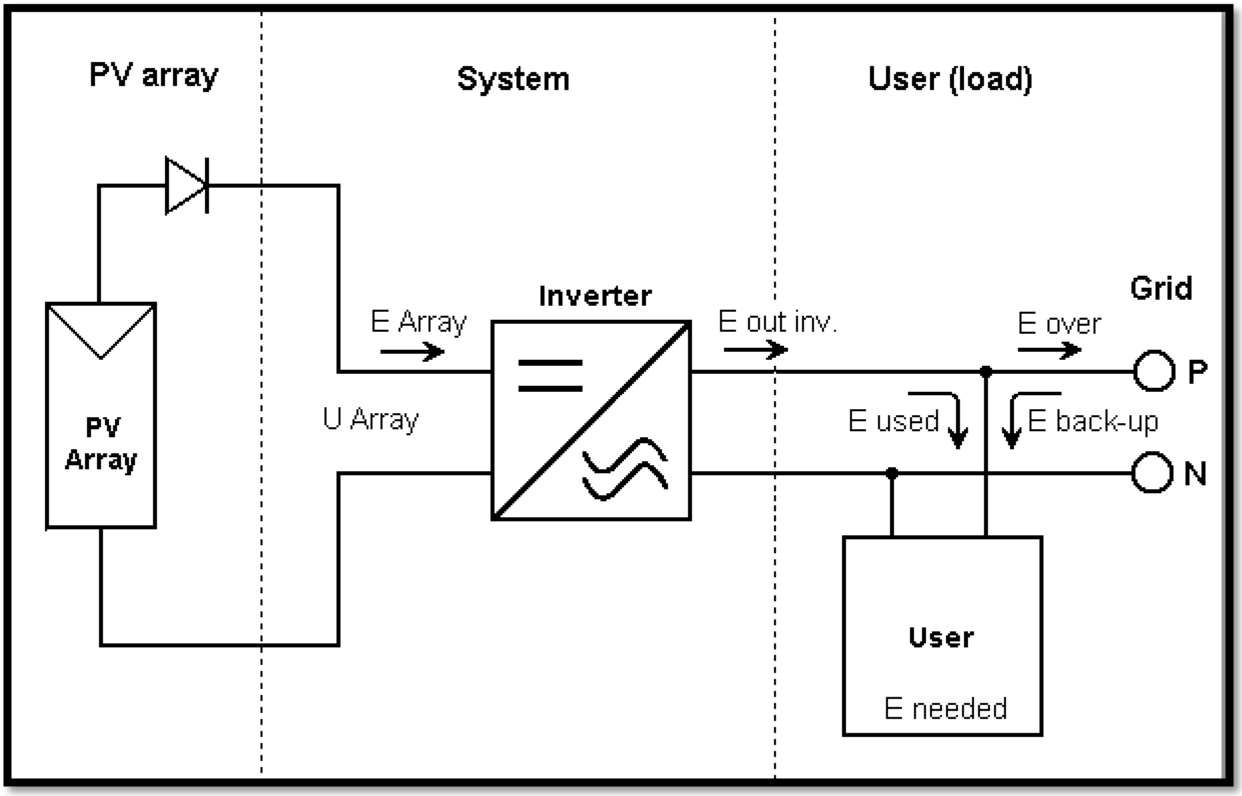


Figure 10: System Simplified Sketch

Installation of 8 MW Phase-II Grid connected Ground Mounted Solar PV Power Plant at DPL, Durgapur, WB.													
Sl No	**ACTIVITY**	**Months**											
		1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
1	Site Survey												
2	Order Finalisation												
3	Preparation & Finalisation of Detailed Design Report												
4	Arrangement of construction power and water												
5	Site Preparation & Land Development												
6	Building of Control Room & Switchyard												
7	Other Civil Works												
8	Foundation for Module Mounting structure												
9	Supply of Support Structures												
10	Erection of Module Mounting Structure												
	Supply of Module												
	Installation of PV Modules												
	Supply of Cables, Conductors etc.												
	Installation of Wiring & Cabling												
	Supply of Solar Inverter & other equipments												
	Installation of Solar Inverter & Other equipments												
	Commissioning												
	Load Trials & Data Collection												

Table 3 : Tentative L1 network